

USER MANUAL

Remodel Studio

The complete guide to every page, tool and report

Version 2.3 · Windows and web



Remodel Studio

User manual · Version 2.3

Remodel planning for one designer: measure, draw, design kitchens and baths, check against the 2021 IRC and IBC, the 2010 ADA Standards and the NKBA guidelines, size the structure and the systems, price the work, print the plan set and run the build.

Prepared for A&A Remodeling.

Remodel Studio runs on your own computer; no account, subscription or internet connection is needed. The program's calculations are planning numbers and do not replace a licensed engineer, trade or contractor.

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PART

Getting started

Installing the program, finding your way around the window, light and dark mode, and a first project.

CHAPTER 1

Welcome

Remodel Studio is a complete remodel-planning program for one designer. It takes a job from the first site visit to the hand-over binder: measure the house from photos and a tape, draw the existing building and the proposed remodel, design kitchens and baths with real cabinet lines, check the design against the 2021 IRC/IBC, the 2010 ADA Standards and the NKBA Kitchen and Bath Planning Guidelines, size the structure and the systems, price the work, print a plan set and a presentation, and run the build with a schedule, orders, inspections and a punch list.

Everything runs on your own computer. Projects are ordinary files, nothing is sent anywhere, and no account or subscription is needed.

This manual covers every page, tool, dialog, setting and report in the program, version 2.3. It is written to be read front to back once, then kept as a reference: the table of contents, the glossary and the index at the back take you straight to any feature.

1 How this manual is organized

Part	What it covers
1 Getting started	Installing the program, the layout of the window, light and dark mode, and a 20-minute first project.
2 The window	The menus, the tab bar, the command palette, standards, lessons and keyboard shortcuts.
3 Designing	The Designer, every drawing and placement tool, selecting and editing, the 3D view, levels, roofs and site.
4 The Library and vendors	The catalog, custom items, vendor catalogs and price books, appliances, finishes and paint.
5 Kitchens and baths	Cabinets, countertops, tile, hardware, NKBA guidelines and the kitchen drawings.
6 Measuring	Photos, tape measurements, scans and plan images, making walls, field notes.
7 Checking	The Check page and every code, accessibility, NKBA, structure, clash and energy rule.
8 Structure and systems	Spans and loads, framing, wall removal, electrical, plumbing, heating and cooling, energy, lighting.
9 Documents	The plan set, schedules, keynotes, details, revisions, presentation, finish board, exports.
10 Cost	Builder's list, estimate, change orders, shopping, selections and orders, budget, bids, cut plan.
11 Building	The dashboard, job schedule, field notes, wall photos, inspection prep, homeowner binder.
12 Site and settings	Site and landscape tools and every project setting.
Appendices	File formats, troubleshooting, where to get help, the glossary and the index. Keyboard shortcuts are listed in Part 2.

2 Conventions

- **Bold** names are things you see on the screen: buttons, menu items, tabs, fields.
- *Design › Model* means the Design tab in the tab bar, then its Model page. *File › Import › Vendor catalog…* means a menu, a submenu and an item.
- Keys are shown as **Ctrl+K**, **Shift**, **Enter**. Single-letter tool keys (**W** for Wall) work with no modifier while the plan has focus.
- Lengths are entered the way you say them: `12'6"`, `12-6`, `150"`, `6 ft 3 in`, `1.9 m` or `45 cm`. A bare number in a length box is read as feet where a wall or room length is expected and as inches for small sizes (the hint beside the box tells you which).
- A **note** box like the one below marks a tip, a limit or a thing that is easy to miss.

> **Note.** The program calculates planning numbers: spans, loads, panel sizes, heat loss, prices and schedules are there to design and compare, not to replace an engineer's stamp, a licensed trade's design or a contractor's quote.

CHAPTER 2

Installing and starting

1 The Windows program

Remodel Studio for Windows is a single installer, **Remodel Studio Setup 2.3.0.exe**. Double-click it, pick where to install (the default is your user folder, so no administrator password is needed), and finish. A **Remodel Studio** shortcut is placed on the Start menu and the desktop, and `.rsp` project files are associated with the program so double-clicking one opens it.

The first start opens the **sample house** so there is something to look at. Use **File › New...** to start your own project.

To update, run the newer installer over the old one; projects, standards and vendor catalogs are kept. To remove the program use *Apps* in Windows Settings; your project files are never deleted.

2 The web version

The same program is also a single web page that runs in Chrome or Edge, with no installation:

- **Remodel Studio offline.html** works with no internet connection at all (every library it needs is inside the file). Keep it anywhere, for example `Documents\Remodel Studio\app`, and open it in the browser.
- **Remodel Studio.html** is the small copy; it needs an internet connection the first time it loads its libraries.

In the browser, projects are saved in the browser's own storage and can also be saved to or opened from files on your computer (Chrome and Edge ask for permission the first time). Browser storage belongs to that browser on that computer; clearing site data clears the projects, so keep files of anything that matters.

3 Where things live

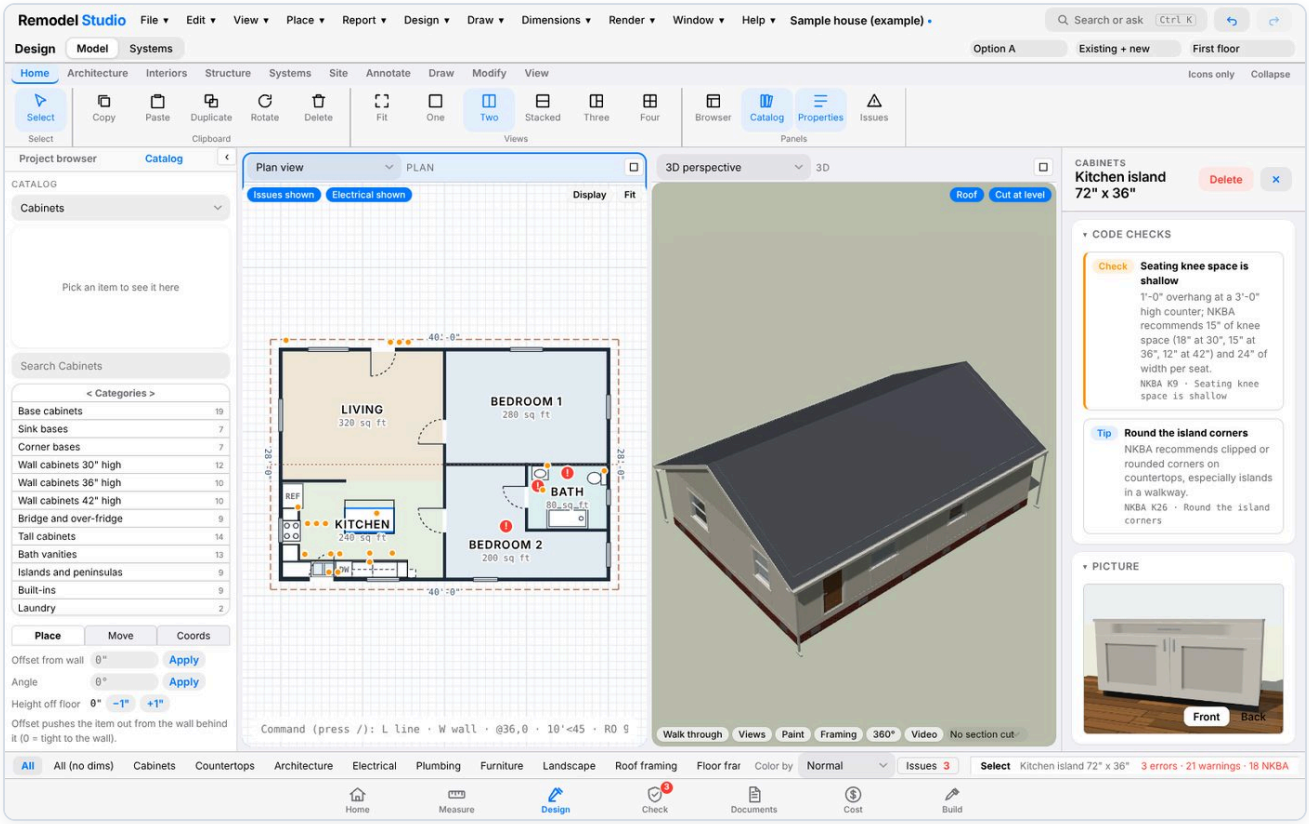
What	Windows program	Web version
Projects	`.rsp` files wherever you save them (**File › Show file in folder** opens the place).	The browser's project list (**File › Open...**) and/or `.json` files you download or save.
Backups	Automatic copies in the program's backup folder (**File › Backups...**).	Named **Versions** in browser storage (**File › Versions...**).
Vendor catalogs, store prices, custom items, standards, appearance	On this computer, shared by every project.	In this browser's storage.
Photos, scans and renders	Stored on this computer; the project remembers where each is pinned.	Stored in the browser; the project remembers the pins.

4 System requirements

Windows 10 or 11, 64-bit, 8 GB of memory and a graphics driver that supports WebGL 2 (any computer from the last ten years). A mouse with a scroll wheel makes drawing much faster; a touchpad works. A second monitor is useful for the plan on one and the 3D view on the other (**Window › Two views side by side** also does this on one screen).

CHAPTER 3

A tour of the window

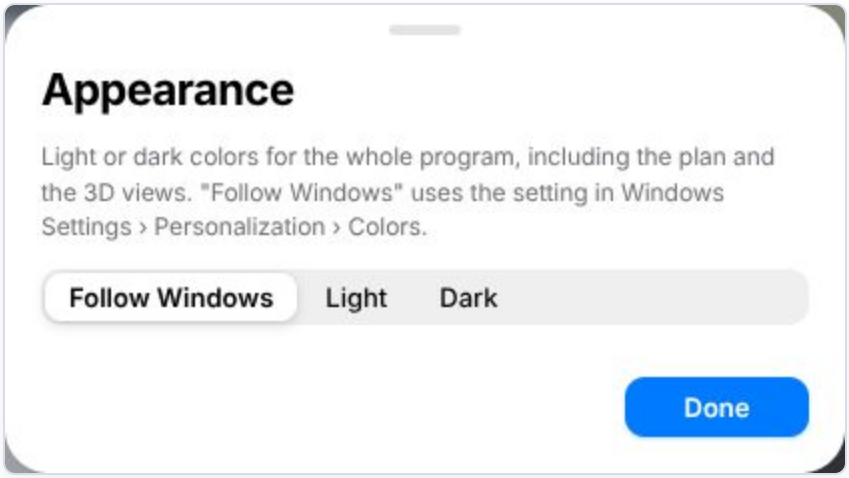


The Designer: tab bar at the bottom, title bar with the page control, ribbon, catalog, plan and 3D panes, Inspector.

The window is laid out like a modern iOS or iPad app:

1. **Menu bar** (top). The brand mark, the **File**, **Edit** and **Help** menus (on the Design › Model page the Designer adds View, Place, Report, Design, Draw, Dimensions, Render and Window), the project name, **Search or ask** (the command palette, **Ctrl+K**) and **Undo** / **Redo**.
2. **Title bar**. The name of the current workflow, a **page control** for its pages (for example *Model · Systems* on the Design tab), and on the right the **design option**, the **remodel layer** (*Existing + new*, *Existing only*, *New only*), the **level** and, outside the Designer, the **Library** button.
3. **Content**. The page itself: the Designer, the dashboard, the Check page, the sheets, the cost tabs and so on.
4. **Tab bar** (bottom). Seven tabs in job order: **Home**, **Measure**, **Design**, **Check**, **Documents**, **Cost**, **Build**. The Check tab wears a red badge with the number of errors (or an amber one with the number of warnings). Each tab remembers the page you used last.

1 Light and dark mode

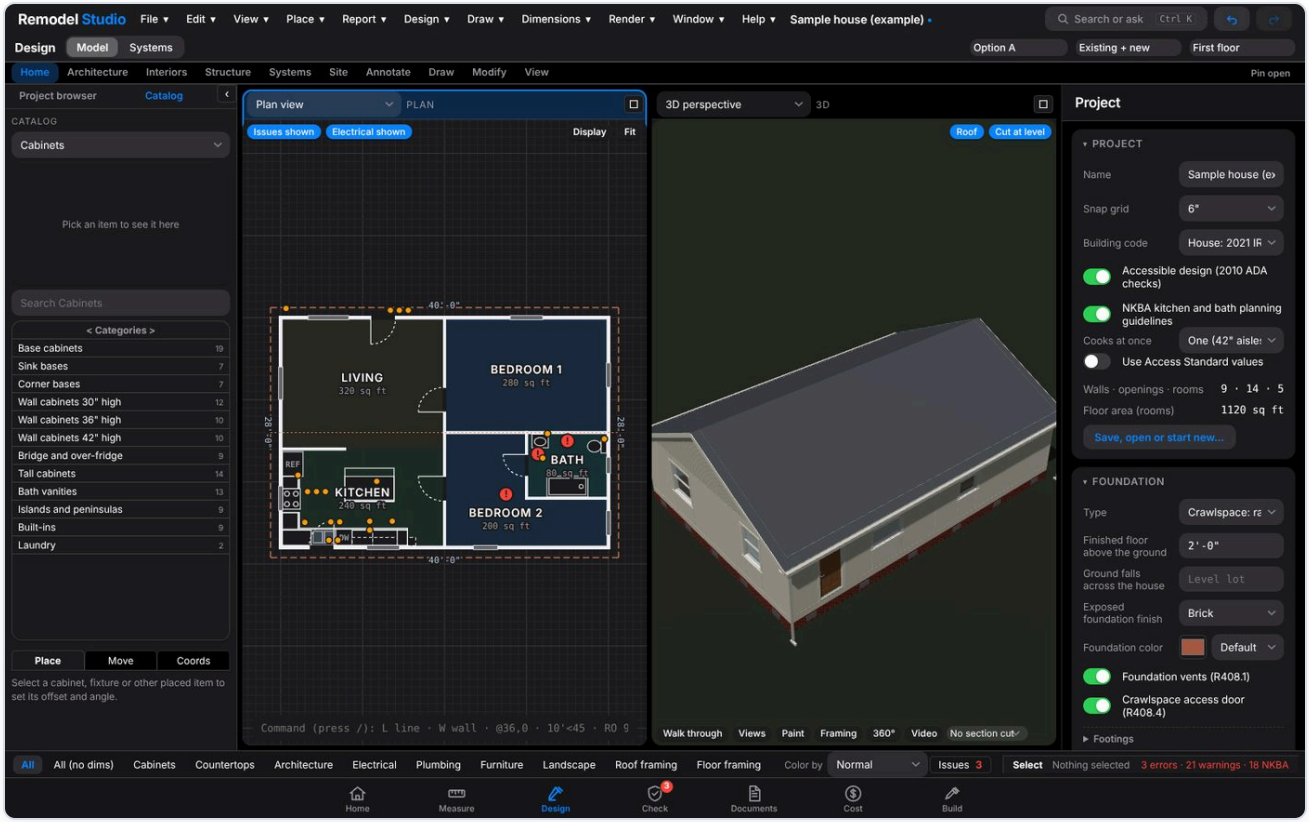


Help › Appearance: Follow Windows, Light or Dark.

Open **Help › Appearance (light or dark)…** (or press **Ctrl+K** and type *appearance*). Choose:

- **Follow Windows** — the program uses the color setting in *Windows Settings › Personalization › Colors* and changes when Windows does. This is the default.
- **Light** — always light.
- **Dark** — always dark. The plan, the 3D view, the sheets preview and every page follow; printed sheets and PDFs always print on white.

The choice is kept on this computer and applies to every project. Press **Done** to close the dialog.



The Designer in dark mode.

CHAPTER 4

Your first project in twenty minutes

This walk-through draws a small addition, puts a kitchen in it, checks it, and prices it. Everything it uses is explained in full later in the manual.

1. **Start a project. File › New… › Blank project**. Name it with **File › Rename project…** and fill in the address and client in the Project panel (select nothing in the Designer and the Inspector shows the **Project** panel).
2. **Draw the walls.** On the Design tab, press **W** (or **Architecture › Wall**). In the option bar pick *Exterior 2×6*. Click a corner, type `24'` and press **Enter**, type `16'` **Enter**, `24'` **Enter**, then click the first corner to close. Press **Esc**. Walls are **New** work; existing walls you measured are set to *Existing* in the Inspector.
3. **Make the room.** Press **R** and click inside the walls. Name it *Kitchen* and set **Use** to *Kitchen* in the Inspector; the floor finish, paint and ceiling height are there too.
4. **Doors and windows.** Press **D**, hover a wall and click where the door goes; press **N** for windows. Each one's size, sill height, swing and style are in the Inspector; the Library has more types.
5. **Cabinets.** Open the **Catalog** in the left dock, choose **Cabinets › Base cabinets**, click *B36* and click along the wall; the back snaps to the wall. Use **Interiors › Base cabinet run** to fill a whole wall at once, then **Wall cabinet run** above it. Pick the sink base, the range and the refrigerator from the **Appliances** category. Set the door style, finish and hardware once in the **Kitchen style (all cabinets)** section of any cabinet's panel.
6. **Countertop and splash.** Select any base cabinet: the **Countertops (all)** section of its panel sets the material, color, edge, thickness, overhang and backsplash for the whole job. Select the room for the **Tile** section: a 4" or full-height splash, a tub surround or a wainscot.
7. **Look at it.** Press the **3D** tab in the pane header, or **Home › Views › Two** to see the plan and the model side by side. Drag to orbit, scroll to zoom, right-drag to pan. **Walk through** puts you at eye level.
8. **Check it.** Press the **Check** tab. Errors are red, warnings amber, NKBA tips blue. Click an issue, then **Show it in Design** to jump to it. Fix the design and the list updates by itself.
9. **Price it.** Press the **Cost** tab. The **Builder's list** is the take-off; **Estimate and labor** adds labor, tax and markup; set your rates once in **My standards** so every project starts with them.
10. **Print it.** Press **Documents**. Tick the sheets you want and **Print plans (PDF)**. The **Board presentation** makes a slide deck and the **Finish board** a sample page.
11. **Save. Ctrl+S**. The dashboard on the **Home** tab now shows the job at a glance.

> **Tip.** Press **Ctrl+K** at any time and type what you want to do (*wall*, *joists*, *finish board*, *how many bedrooms*). It finds commands, pages, tools, levels and catalog items, and answers questions about the project.

PART

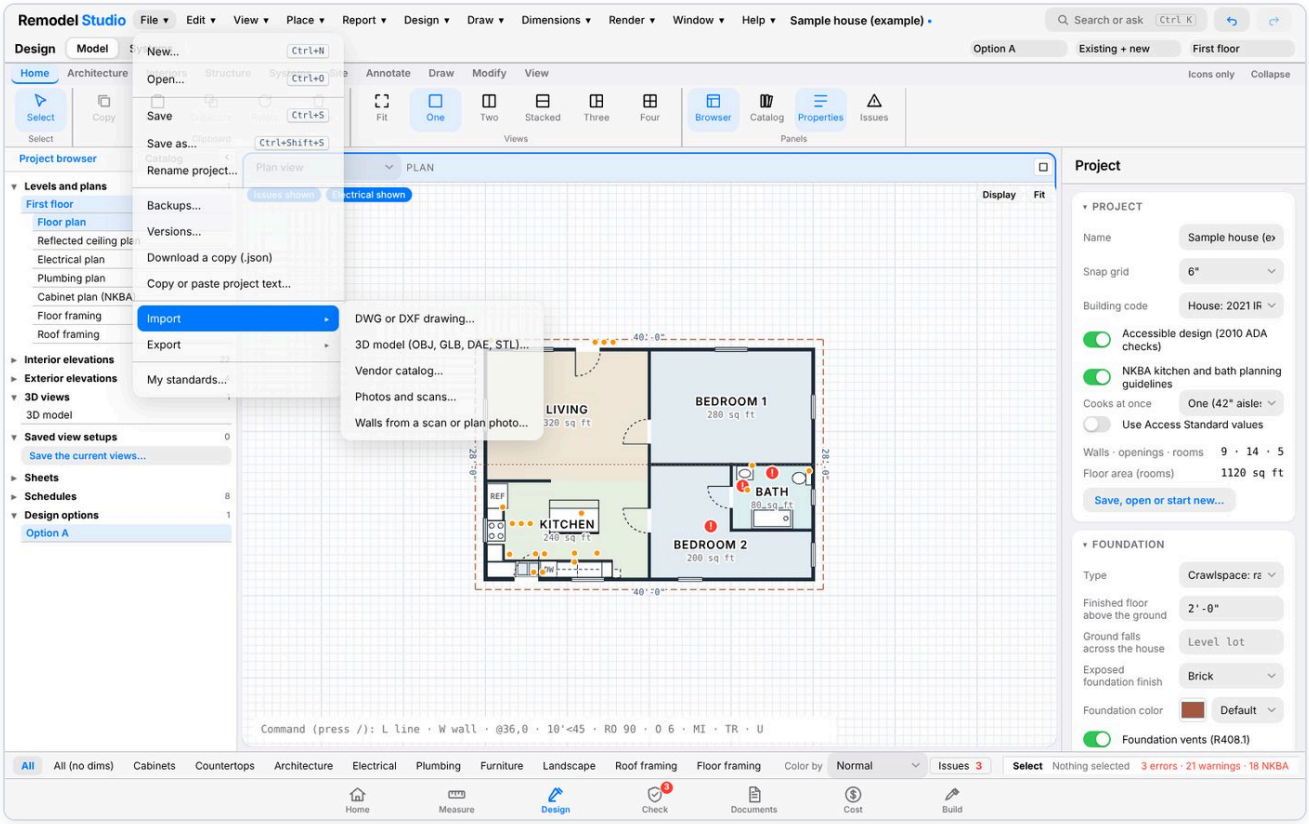
The window

The menus, the tab bar, the command palette, standards, lessons, shortcuts and project files.

CHAPTER 5

The window chrome

1 Header bar

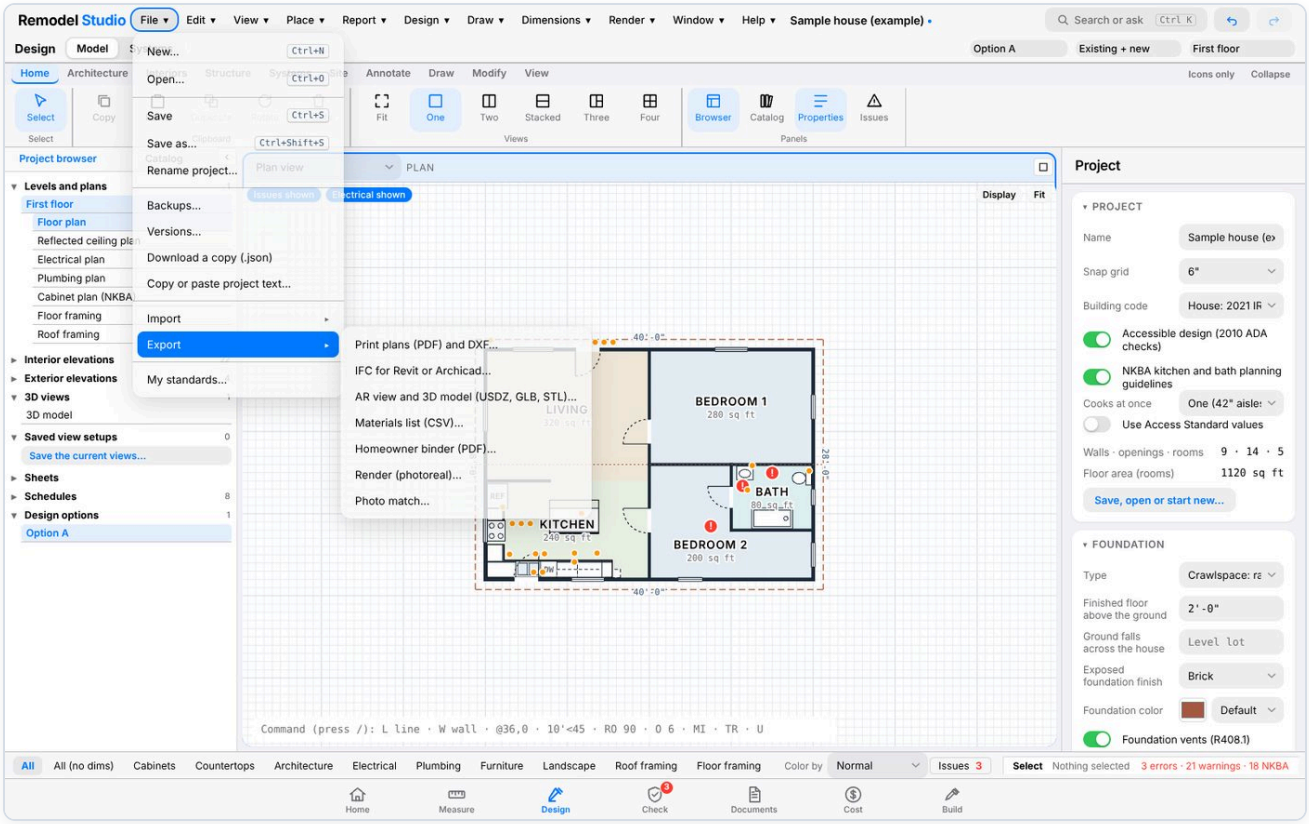


The File menu with the Import submenu.

Left to right:

- **Remodel Studio** brand mark.
- **File**, **Edit** and (outside the Design › Model page) **Help** menus. On the Design › Model page the Designer adds its own menu set in this same slot (View, Place, Report, Design, Draw, Dimensions, Render, Window, Help — see §7.2).
- **Project name**. A bullet (•) after the name means there are unsaved changes; hovering shows " (not saved)".
- **Search or ask** button with `Ctrl K` badge — opens the command palette (§5.8).
- **Undo** and **Redo** arrow buttons (greyed out when there is nothing to undo or redo). Undo history holds the last 200 changes.

2 File menu

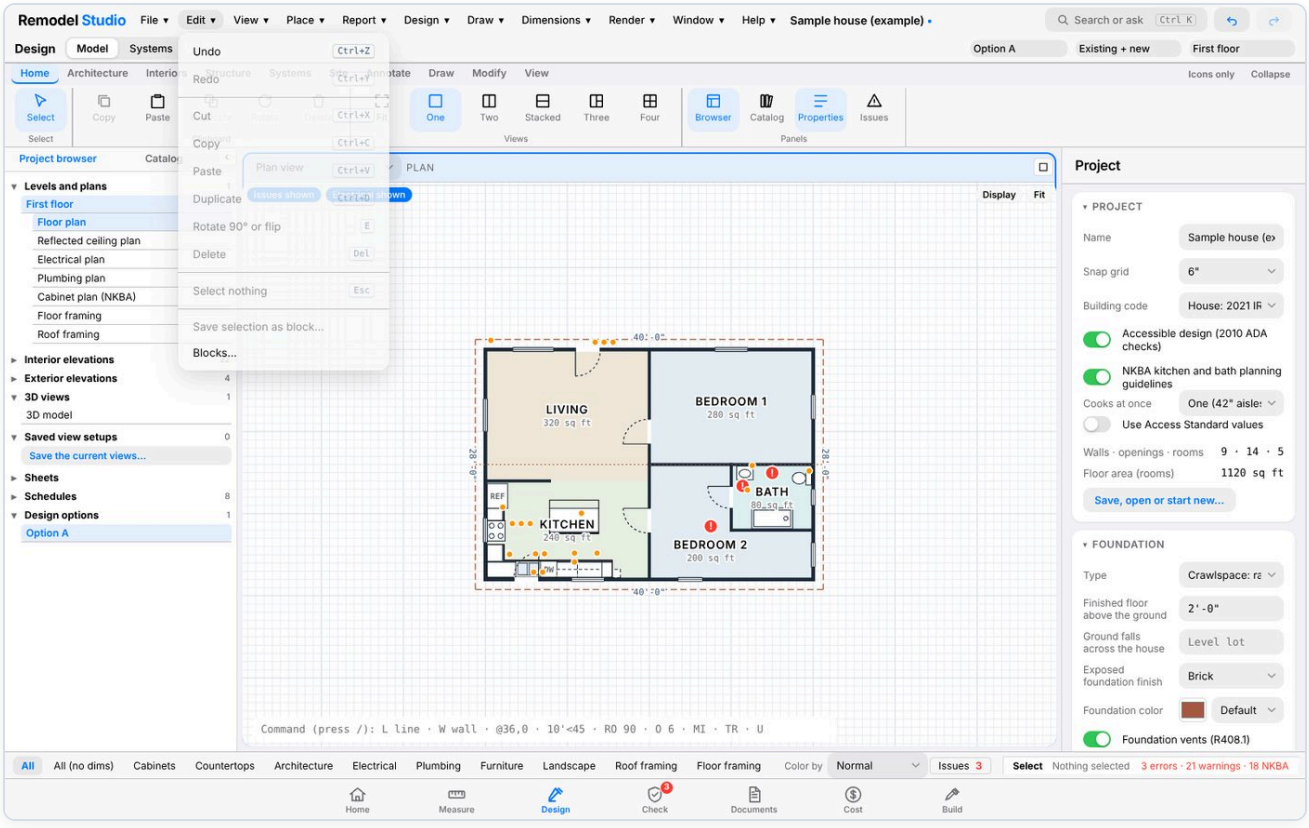


File › Export.

Item	Shortcut	What it does
New...	Ctrl+N	Opens the **New project** dialog: *Blank project* or *Sample house*. If the current project has unsaved changes a warning shows with a *Save current project* button. A blank project starts from **My standards** (§5.10) when any are saved.
Open...	Ctrl+O	Desktop: Windows Open dialog filtered to `.rsp` (and older `.json`) files. Browser: the **Open** dialog listing projects saved in this browser (name, square footage, saved date; *Open* / *Delete* per row) plus *Open a file on this computer...* (Chrome/Edge) or *Open a .json file...*.
↻ *recent files*	—	Desktop only: up to 8 recently opened files, each reopens that file.
Save	Ctrl+S	Desktop: writes the current `.rsp` file (or asks where to save the first time). Browser: saves to this browser's project list and, if the project was opened from or saved to a file handle, rewrites that file too.
Save as...	Ctrl+Shift+S	**Save as** dialog: *Project name* box. Desktop: *Choose where to save...*. Browser: *Save in this browser* and *Save to a file on this computer...* (or *Download as a .json file* where file pickers are not available). The copy becomes the open project and gets a new project id.
Rename project...	—	Renames the open project (Enter confirms).

Item	Shortcut	What it does
Backups…	—	Desktop backups dialog (§6.4).
Versions…	—	Named snapshots kept in this computer's browser storage (§6.5).
Show file in folder	—	Desktop only; opens Explorer at the saved file (disabled until the project has a path).
Download a copy (.json)	—	Browser only; downloads the project as a `.json` file.
Copy or paste project text…	—	Opens the **Save or open** text dialog: the whole project as JSON text with *Copy project*, *Download .json*, *Open pasted text*, *Open .json file*, and *New project…* (blank or sample, with a confirm step).
Import ▸		*DWG or DXF drawing…* (import dialog) · *3D model (OBJ, GLB, DAE, STL)…* · *Vendor catalog…* · *Photos and scans…* (goes to Measure › Photos and measurements) · *Walls from a scan or plan photo…*
Export ▸		*Print plans (PDF) and DXF…* (goes to Documents › Sheets) · *IFC for Revit or Archicad…* · *AR view and 3D model (USDZ, GLB, STL)…* · *Materials list (CSV)…* (goes to Cost) · *Homeowner binder (PDF)…* · *Render (photoreal)…* (Render studio) · *Photo match…*
My standards…	—	§5.10.

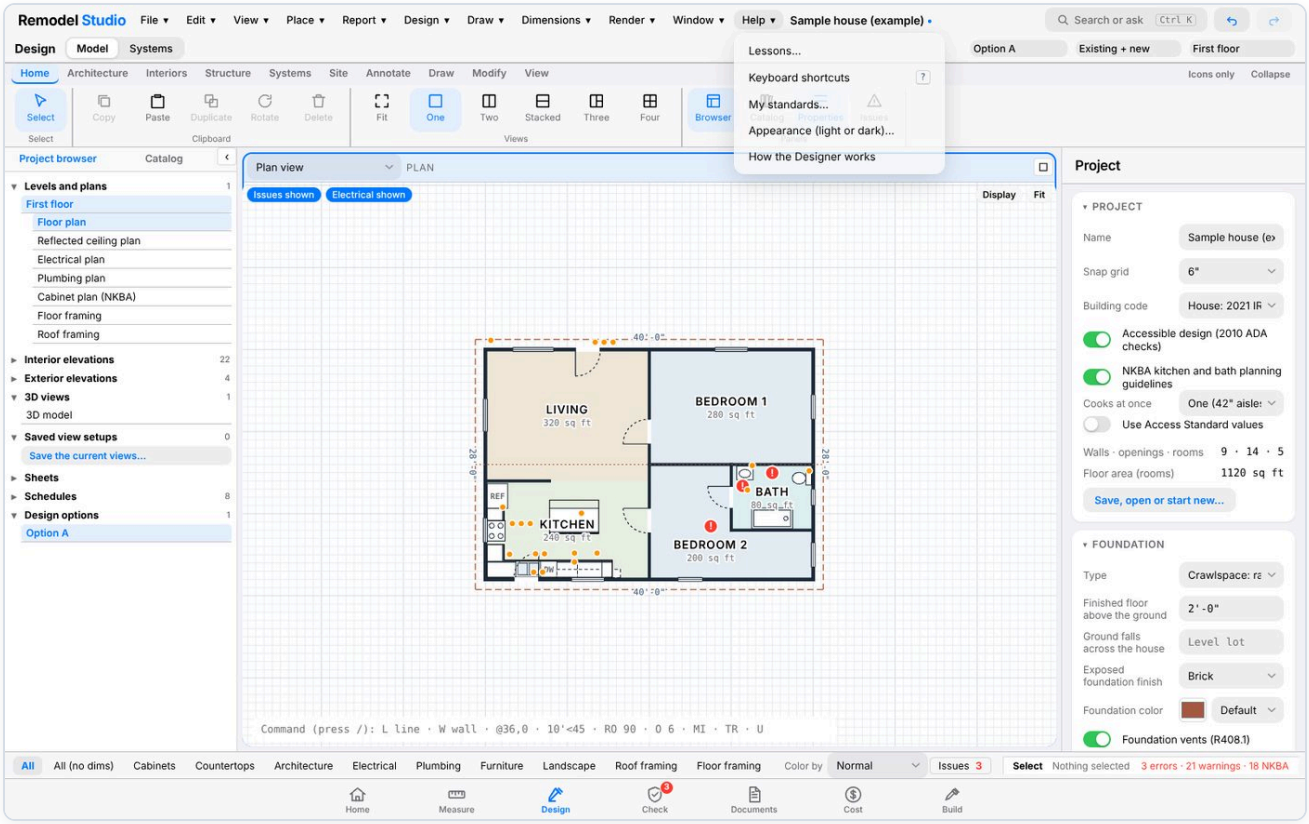
3 Edit menu



The Edit menu.

Item	Shortcut	What it does
Undo	Ctrl+Z	Steps back one change.
Redo	Ctrl+Y (also Ctrl+Shift+Z)	Steps forward.
Cut	Ctrl+X	Copies the selection to the program clipboard and deletes it.
Copy	Ctrl+C	Copies the selection (walls bring their doors and windows along).
Paste	Ctrl+V	Pastes on the current level, offset 24" right and down per paste (the second paste 48", and so on). A copied door or window pastes further along the same wall; if there is no room the message "There is no room to paste that here" shows. Rooms are pasted as "*name* copy".
Duplicate	Ctrl+D	Copy and paste in one step, 24" offset.
Rotate 90° or flip	E	Fixtures and cabinets turn 90°; stairs turn 90° (or step N→E→S→W); roofs swap ridge direction; a door swaps hinge side then swing side; section cuts, dormers, chimneys, MEP items and other parts turn as their type allows.
Delete	Del	Deletes the selection (every item of a multi-selection). Deleting a wall removes its doors and windows.
Select nothing	Esc	Clears the selection.
Save selection as block...	—	Opens **Blocks** with the selection ready to name (§5.12).
Blocks...	—	Opens the Blocks dialog.

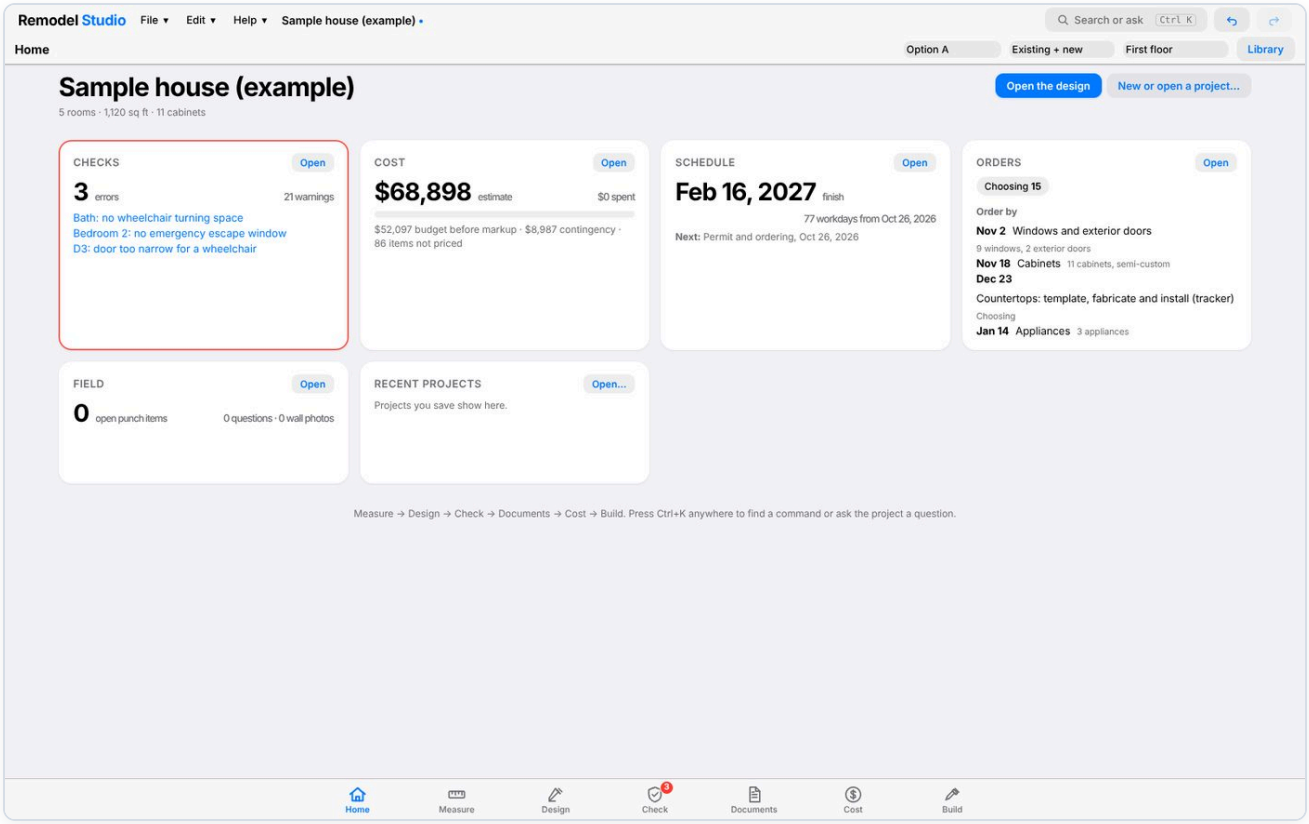
4 Help menu



The Help menu (outside the Designer).

Lessons… (§5.11) · Keyboard shortcuts (`?`, §5.13) · Appearance (light or dark)… (§5.9) · My standards… (§5.10). On the Design page the Designer's own Help menu adds *How the Designer works* (shows a one-line tip).

5 Tab bar and page control



The Home tab with the tab bar along the bottom.

The seven workflows sit in the **tab bar** along the bottom of the window, in job order, each with an icon. Clicking a tab opens the page you used last on that tab (remembered per tab). The **Check** tab wears a red badge with the error count or an amber one with the warning count. Under the menu bar, the **title bar** names the current workflow and, when it has more than one page, shows a **page control** (a segmented control) to switch between them.

Tab	Pages	Purpose
Home	Dashboard	The job at a glance: Checks, Cost, Schedule, Orders, Field and Recent projects cards, *Open the design*, *New or open a project...*.
Measure	Photos and measurements	Photos, tape measurements, scans, underlays and field notes.
Design	Model · Systems (on narrow screens: Plan · 3D · Systems)	Draw and design the model; systems calculations.
Check	Issues	Code, NKBA, structure, clash and installation checks.
Documents	Sheets · Schedules	Plan set, presentation, finish board and schedules.
Cost	Cost	Builder's list, estimate, change orders, shopping, selections and orders, budget vs actual, bids, cut plan, cabinet order.

Tab	Pages	Purpose
Build	Job	Job schedule, field notes and punch list, wall photos, inspection prep, homeowner binder.

Check page layout: left column has severity chips (Errors / Warnings / Tips with counts), category chips (IRC, IBC, ADA, NKBA, Structure, Framing, Electrical, Plumbing, Appliances, Cabinets, Energy, Zoning, Landscape, Model), a search box, *Show reviewed*, and the issue list grouped by level ("Whole job" for project-wide items). The middle is the plan with markers; the right column shows the chosen issue (severity pill, title, code reference, detail, where), *Show it in Design*, *Select it*, and a *Reviewed, leave as is* checkbox. Reviewed issues hide from the list and the plan until something about them changes but still print in the permit checklist. With nothing chosen it shows a summary table and *Wall framing and cut list…* / *Floor and ceiling joists…* buttons.

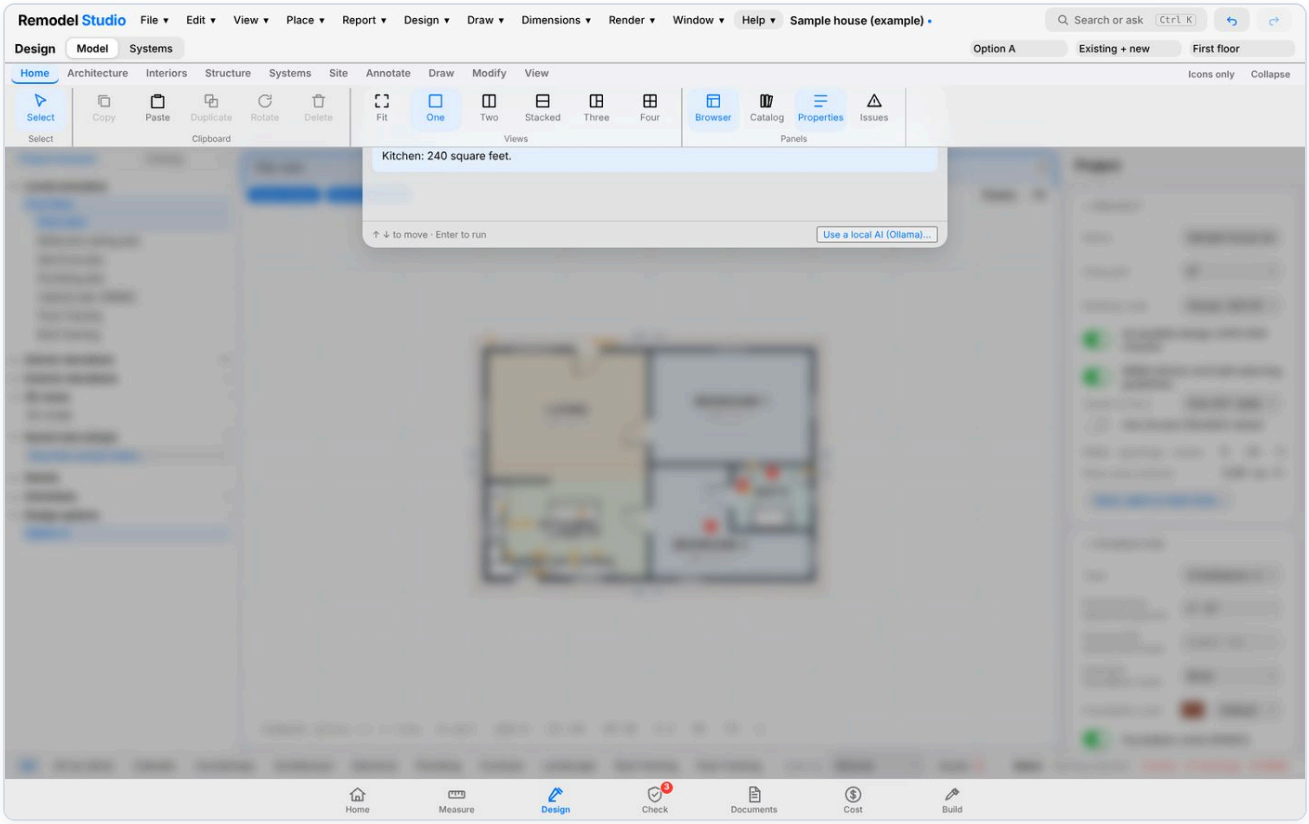
6 Title bar controls (right side)

- **Design option** dropdown: lists the options (the first is "Option A") plus **+ New option (copy)**, which snapshots the current model as a new option (named Option B, C…) and switches to it. Switching options swaps the whole model (walls, rooms, fixtures and so on); the active option is the one drawn, checked and printed. Options can be renamed and deleted in the Project panel (§9.6).
- **Compare** dropdown (shows only when there are 2+ options): *Compare: off* or *vs Option X*. The other option's walls that differ are drawn in magenta on the plan.
- **Remodel layer**: *Existing + new* (default), *Existing only*, *New only*. Filters walls, doors/windows and fixtures by their remodel status in both the plan and 3D.
- **Level** dropdown: the level you are working on, sorted by elevation.
- **Library** button (hidden on the Design › Model page, which has the Catalog dock instead): opens the Library panel on the right (doors, windows, cabinets, fixtures, trim, finishes and colors). If you are on a non-drawing page it also returns you to the Design page.

7 Narrow-screen (phone/tablet) layout

Below 820 px the Design tab offers Plan, 3D and Systems pages, the classic left **tool rail** with fly-out groups replaces the ribbon, and the Inspector collapses to a drawer with a ▲/▼ button. Two-finger pinch zooms and pans the plan.

8 Command palette — Search or ask (Ctrl+K)



Search or ask: a question about the project.

Press **Ctrl+K** or click **Search or ask**. One text box; ↑ ↓ move, Enter runs, Esc closes. It works offline.

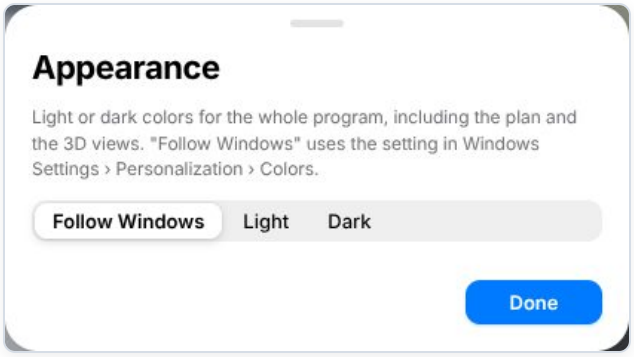
What you can type:

- **Command names** – every page ("Go to Check › Issues"), Save / Open / New / Save as / Backups / Versions, Undo / Redo, every dialog (Roof wizard, Floor and ceiling joists, Wall framing and cut list, Remove or open a wall, Change cabinet door style and finish, Countertop slab layout, Lessons, 360° panorama, Walk-through video, Photo match, Export IFC, Store prices, Import DWG or DXF, Import 3D model, Import vendor catalog, Keyboard shortcuts, Blocks, Walls from a scan or plan photo, Homeowner binder, My standards, Appearance), Render studio, AR view and 3D export, specific Cost/Documents/Build tabs (Cut plan for sheet goods, Bids and quotes, Actual costs, Change orders, Selections and orders, Finish board, Board presentation, Inspection prep, Homeowner binder, Wall photos, Job dashboard), every **Tool:** (switches to Design and picks the tool), every **Schedule:**, and every **Level:**.
- **"Place: ..." catalog items** – type part of a catalog code or name (for example `B36`, `sink base`, `pendant`). Up to eight matching library items appear as *Place:* with their catalog category;

choosing one switches to Design and arms the placing tool so your next click in the plan places it.

- **Questions and simple commands**, answered from the model itself (an **Answer** box appears; some answers add an action row):
- *how many / count / number of …* (`how many base cabinets in the kitchen`, `how many lights on the second floor`) → count and list, with *Select them in Design*. Scope words "in the " or "on the " are understood. `how many errors` gives the error and warning counts.
- *select / find / show / where is / highlight …* → matches with a select action.
- *area / square feet / how big of* → square footage.
- *linear feet / how long / length of …* → total length of matching walls or fixtures; `linear feet of counter` totals countertop runs.
- *cost / price / estimate / budget* → estimate total, materials, labor, unpriced count and $/sq ft, with *Open the estimate*.
- *errors / issues / what's wrong* → error list with *Open the Check page*.
- *move the 6 in left* (also `move it 6 in left` for the selection; directions left/right/up/down/north/south/east/west/forward/back) → *Move it* action.
- *set the lights in the kitchen to 3000K* → *Apply* action that sets color temperature.
- **Local AI (optional)** – the footer shows *Use a local AI (Ollama)…*. Turning it on looks for Ollama at `127.0.0.1:11434`; if it is running, a model dropdown appears and any question longer than three characters gets an extra row *Ask the local AI (model): "…"*. The answer (with a *Copy* button) is produced from a plain-text summary of the project (levels, rooms, placed items, electrical counts, openings, estimate, check results). Nothing leaves the computer. If Ollama is not running the footer says so with a *Turn off* link. The chosen model and the on/off choice are remembered.

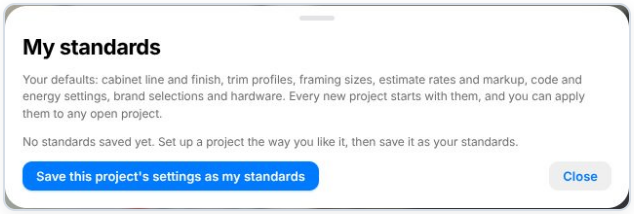
9 Appearance dialog (Help › Appearance)



Help › Appearance.

Three radio buttons: **Follow Windows** (default; uses Windows Settings › Personalization › Colors), **Light**, **Dark**. Applies to the whole program including the plan and 3D views, immediately. *Done* closes.

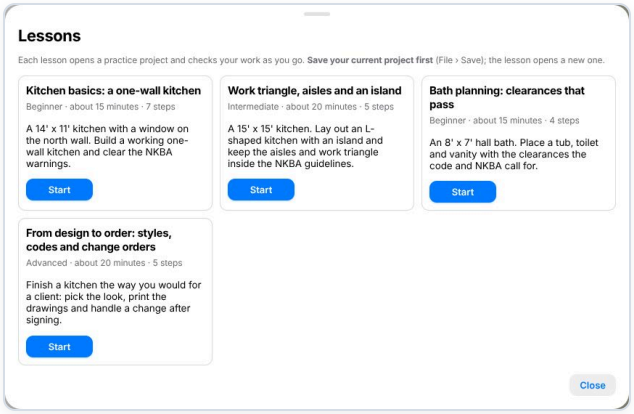
10 My standards (File › My standards / Help › My standards)



My standards.

Your personal defaults that every new blank project starts with. The dialog shows a table of what is saved: Cabinets (door style, finish, line, vendor), Trim profiles, Framing (floor joist size and spacing, ceiling joists, foundation type), Estimate (markup %, contingency %, tax %, own-labor flag), Code (IRC or IBC, ADA on, NKBA on), Energy (climate zone) and the saved date. Buttons: **Save this project's settings as my standards**, **Apply my standards to this project** (cabinets without their own style follow the new defaults), **Clear**, Close. Standards also carry the snap grid, waste %, hardware finish and brand, selections and the designer name. The sample house and lesson projects do not receive standards.

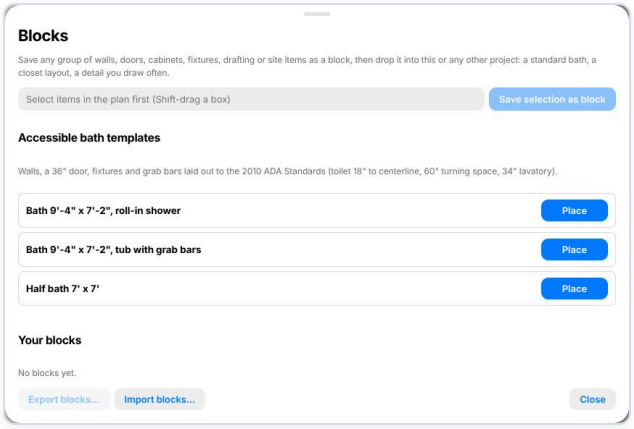
11 Lessons (Help › Lessons)



Help › Lessons.

Cards for each lesson with level, minutes and step count; **Start** opens a fresh practice project (your own project stays available under File › Open; save it first if it has changes). Lessons: *Kitchen basics: a one-wall kitchen* (Beginner, 15 min), *Work triangle, aisles and an island* (Intermediate, 20 min), *Bath planning: clearances that pass* (Beginner, 15 min), *From design to order: styles, codes and change orders* (Advanced, 20 min). While a lesson is open a **Lesson** panel floats over the workspace listing the steps; each step checks itself off as the model meets it, the current step shows its hint, other steps have a *Hint* link, and there are *All lessons*, *Close lesson* and a collapse (–/+) button.

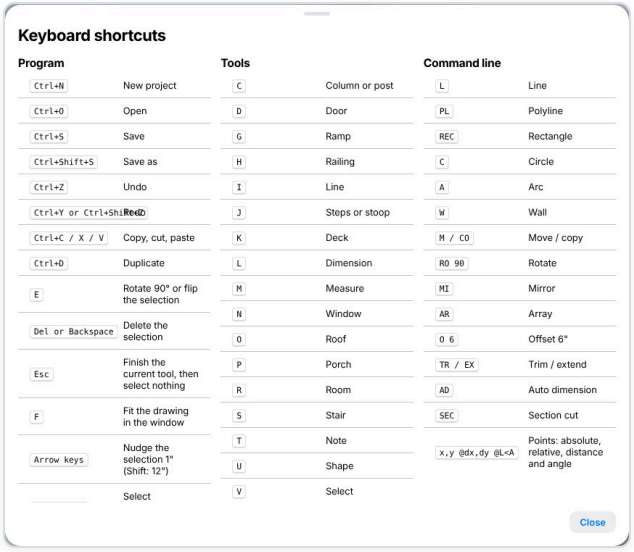
12 Blocks (Edit › Blocks)



Edit › Blocks.

A block is any group of walls, doors, cabinets, fixtures, drafting or site items saved for reuse in this or any other project. Select items first (Shift-drag a box), name them, **Save selection as block**. The list shows *Accessible bath templates* (walls, a 36" door, fixtures and grab bars to the 2010 ADA Standards) and *Your blocks*, each with **Place** (arms the Block tool: click to drop a copy, keep clicking for more, Esc when done) and delete. *Export blocks…* / *Import blocks…* move them as a `.json` file. Blocks live in this computer's browser storage.

13 Keyboard shortcuts (Help › Keyboard shortcuts, or `?`)



The Keyboard shortcuts dialog.

Program

Keys	Action
Ctrl+N / Ctrl+O / Ctrl+S / Ctrl+Shift+S	New / Open / Save / Save as
Ctrl+Z · Ctrl+Y or Ctrl+Shift+Z	Undo · Redo
Ctrl+C / Ctrl+X / Ctrl+V	Copy / cut / paste
Ctrl+D	Duplicate
Ctrl+K	Search or ask
E	Rotate 90° or flip the selection
Del or Backspace	Delete the selection
Esc	Finish the current run (wall, room, line…), then put the tool away (back to Select), then select nothing; also closes dialogs and menus
Enter	Finish a wall/room/shape run or polyline; with a typed length pending, applies that length
F	Fit the drawing (plan and 3D) in the window
Arrow keys	Nudge the selection 1" (Shift: 12")
Shift-drag	Box-select everything inside; Shift-click adds or removes one item
Alt-drag or middle-drag	Pan the plan
Right-click or double-click	Finish the current run
/ or ;	Focus the command line
?	Keyboard shortcuts card
0-9 ' " / - (while drawing)	Type a length for the next segment (bare numbers are feet)
Shift (while drawing or dragging)	Free angle instead of 45° snaps; drag fixtures freely away from walls

Tool keys (single letter, no modifier): V Select · W Wall · D Door · N Window · R Room · X Fixture · S Stair · O Roof · Y Electric · G Ramp · U Shape · L Dimension · T Note · M Measure · I Line · K Deck · P Porch · J Steps · C Column · H Railing.

3D walk-through: W/S or ↑/↓ walk, A/D step sideways, ←/→ turn, Shift hurry, drag to look, Esc stop.

Command line (see §8.2): L, PL, REC, C, A, W, M / CO, RO 90, MI, AR, O 6, TR / EX, AD, SEC, and points as `x,y`, `@dx,dy`, `@L<A`.

CHAPTER 6

Project files, saving and the desktop program

1 Where projects live

- **Desktop (Windows) program**: projects are `.rsp` files (JSON text inside). Older `.json` project files also open. The default folder is **Documents\Remodel Studio\Projects**; the last folder used is remembered. Double-clicking an `.rsp` file in Explorer opens it in Remodel Studio (file association); a second copy of the program hands the file to the running one.
- **Browser version**: *File › Save* stores the project in this browser's project list (name, area, saved date); *Save as* can also write a `.json` file on your computer (Chrome/Edge file picker, which is then rewritten on each Save) or download one. *Download a copy (.json)* and *Copy or paste project text* move projects between computers.
- In both versions a working copy is written to browser storage about 0.3 s after every change (crash recovery); the program reopens it on the next start. On the desktop the window title shows the project name with a • when unsaved.

2 New, open, save, recent

See §5.2. On the desktop, closing the window with unsaved changes asks **Save / Don't save / Cancel**. The Home dashboard's *Recent projects* card and File menu list recent files (desktop: last 12 kept, 8 shown in the menu). Opening a project lands on the Home dashboard.

3 File format

A project is one JSON document holding the name, levels, walls, openings, rooms, stairs, roofs, fixtures, electrical, drafting, site items, settings (grid, code, ADA, foundation, show/snap/layer settings, standards-derived defaults), sheet setup, project info, design options, saved 3D views, view setups and keynotes. Blocks, My standards, theme and layout preferences are kept per computer, not in the file. Photos and Versions are kept in the browser's asset store on the computer.

4 Backups (desktop)

File › Backups… Every save writes a dated copy to **Documents\Remodel Studio\Backups** (at most one per 3 minutes; the newest 25 kept). While there are unsaved changes an **autosave copy** is written every 5 minutes (newest 40 kept). The dialog groups copies by project with time, Saved/Autosave, size and **Open** (opens as an unsaved copy named "… (restored)" — save it to keep it); *Open the backups folder*. In the browser version the dialog points to Versions instead.

5 Versions

File › Versions… Named snapshots stored in this computer's browser storage: name it (default "Version "), **Save this version**; the table compares each version's rooms, area and estimate with the current project ("same" or old → now), **Restore** (keeps the current drawing in Undo) and delete.

6 Desktop program specifics

- Window opens at 1600×1000 (minimum 900×600), remembers size and position, hides its own menu bar (Alt shows a small **View** menu — Reload the program, full screen, zoom in/out/reset, Developer tools — and **Help** menu — Open my projects folder, Open backups folder, Open renders folder, *Use the built-in program version* (ignore an updated program page placed in Documents\Remodel Studio\app), About Remodel Studio).
- Exports and downloads prompt a Windows Save As dialog (default: Documents or the last export folder). Renders go to Documents\Remodel Studio\Renders. Web links open in the normal browser.
- The program uses the graphics card for the built-in renderer; Blender is found automatically (or picked) for photoreal rendering.

PART

Designing

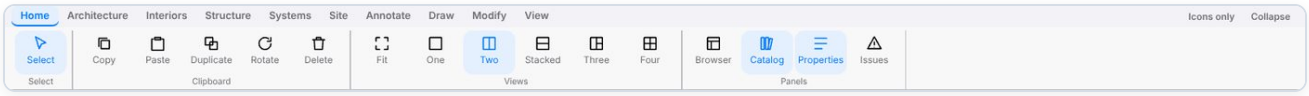
The Designer, every drawing and placement tool, selecting and editing, and the 3D view.

CHAPTER 7

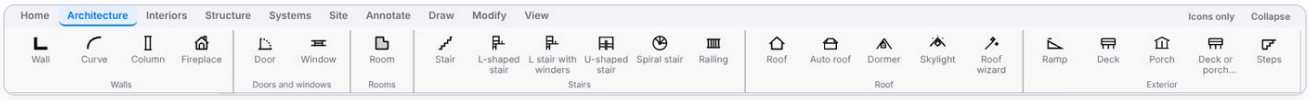
The Designer (Design › Model)

The Designer is laid out like a kitchen-and-bath design program: menus and a ribbon on top, a dock on the left, one to four view panes in the middle, the Inspector (properties) on the right, an optional Issues dock under the views, and a strip of plan-display tabs with the status readout along the bottom. Layout, pane views, dock choice, ribbon tab, pin state and icons-only state are all remembered between sessions.

1 Ribbon



The Home ribbon tab.



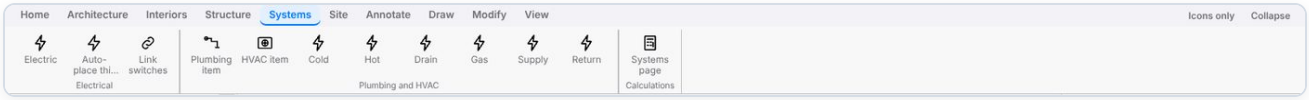
Architecture.



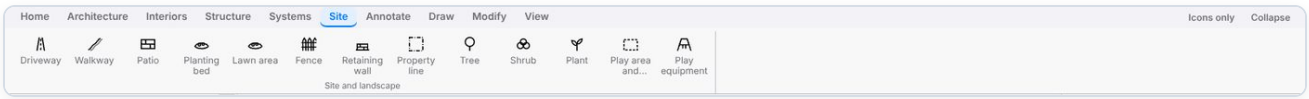
Interiors.



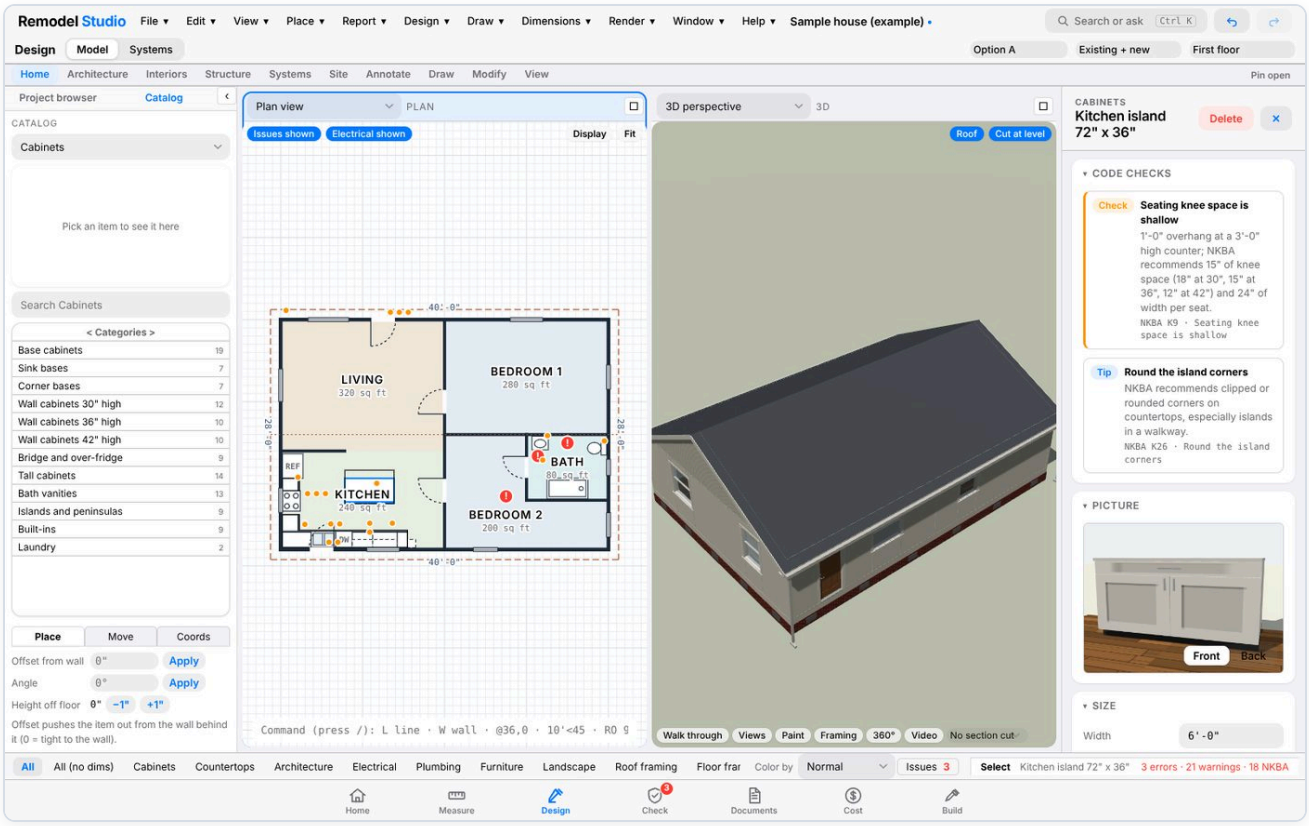
Structure.



Systems.



Site.



The ribbon collapsed to its tabs; a tab opens as a popover.

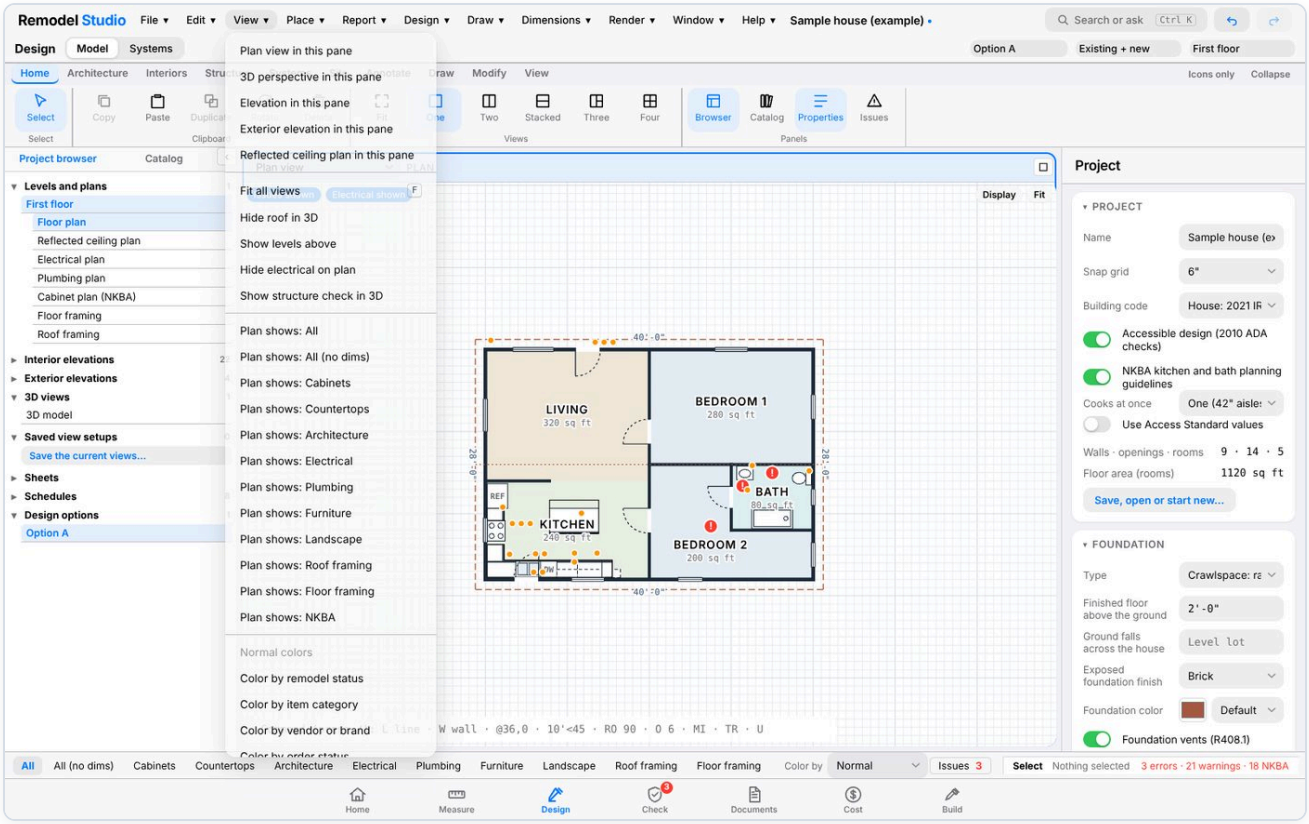
Ribbon tabs run across the top of the ribbon; click a tab to open its tool groups. Each group is a row of buttons with the group name underneath.

- **Pin open / Collapse** button at the right end of the tab row keeps the ribbon body open or collapses it to just its tabs (more room to draw). When collapsed, clicking a tab drops the body down temporarily; it closes after you pick a tool, click elsewhere or press Esc.
- **Icons only / Show labels** toggles button labels (visible only while the body is open).
- Picking a tool from another tab makes the ribbon jump to that tool's tab.

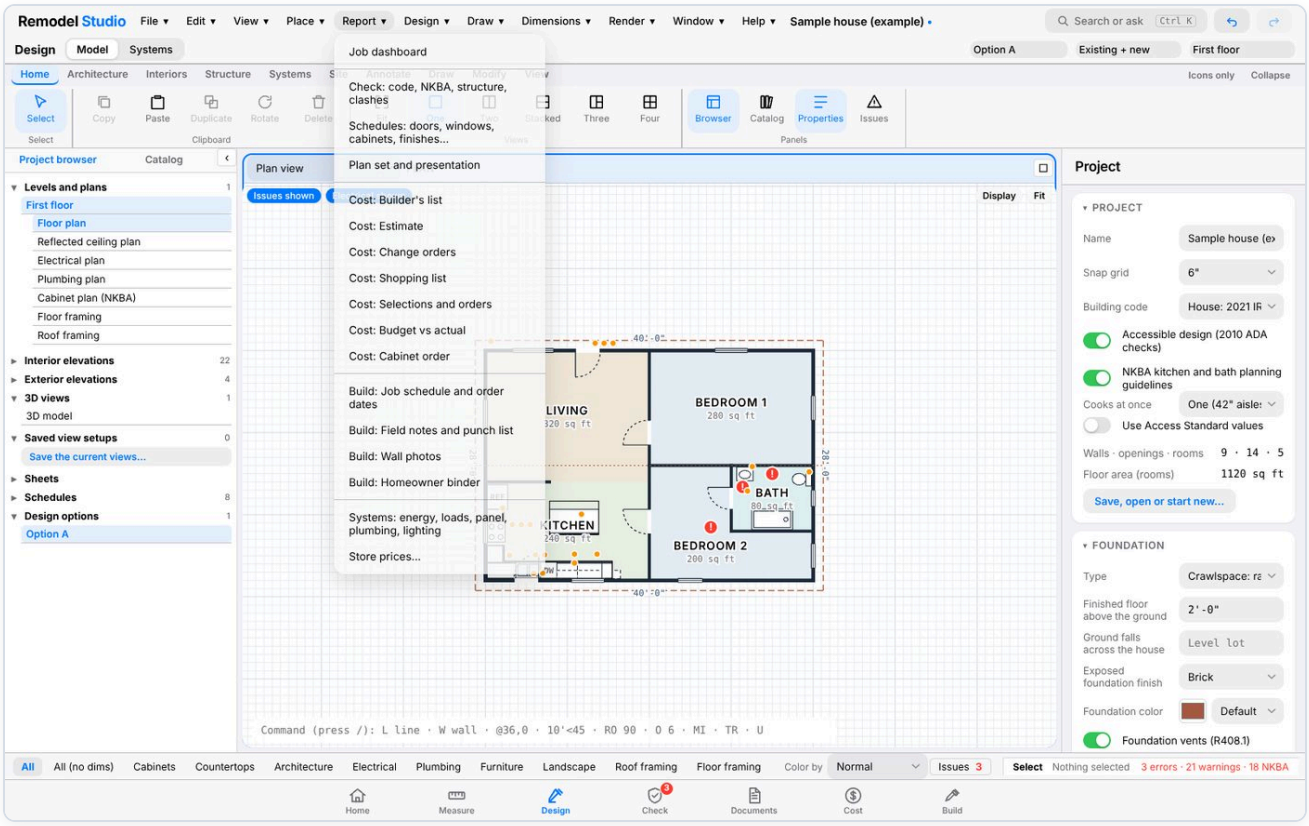
Tab	Group	Buttons
Home	Select	Select
	Clipboard	Copy, Paste, Duplicate, Rotate, Delete
	Views	Fit; One, Two, Stacked, Three, Four (pane layouts)
	Panels	Browser, Catalog (left dock tabs; clicking the open one hides the dock), Properties (right panel on/off), Issues (dock on/off)
Architecture	Walls	Wall, Curved wall, Column, Fireplace
	Doors and windows	Door, Window
	Rooms	Room
	Stairs	Stair, L-shaped stair, L stair with winders, U-shaped stair, Spiral stair, Railing

Tab	Group	Buttons
	Roof	Roof, Auto roof, Dormer, Skylight, Roof wizard (dialog)
	Exterior	Ramp, Deck, Porch, Deck or porch outline, Steps
Interiors	Place	Fixture, Base cabinet run, Wall cabinet run, Catalog
	Kitchen and bath	Layout templates menu (One-wall kitchen, L-shaped kitchen, L-shaped kitchen with island, U-shaped kitchen, Galley kitchen, Bath: tub, toilet and vanity on one wall — applied to the selected room or the first kitchen/bath on the level), Fill run (fills the selected cabinet's run with stock widths and a filler), Door style and finish (dialog), Countertop slabs (dialog), Under-cabinet lights (adds one LED strip under every wall cabinet on the level that has none)
Structure	Framing	Roof wizard, Floor and ceiling joists, Wall framing, Remove or open a wall (needs a wall selected)
	Show	Floor framing plan, Roof framing plan (plan display tabs), Structure check in 3D (colors members by load)
	Posts	Column
Systems	Electrical	Electric, Auto-place this floor (lays out receptacles, lights, switches and alarms to IRC rules; replaces earlier auto-placed devices, keeps manual ones), Link switches (runs a switch leg from each unswitched light to the nearest switch in its room)
	Plumbing and HVAC	Plumbing item, HVAC item, Cold water pipe, Hot water pipe, Drain pipe, Gas pipe, Supply duct, Return duct
	Calculations	Systems page
Site	Site and landscape	Driveway, Walkway, Patio, Planting bed, Lawn area, Fence, Retaining wall, Property line, Tree, Shrub, Plant, Play area and surfacing, Play equipment
Annotate	Dimensions	Dimension, Automatic dimensions, Angle dimension, Radius dimension
	Text and tags	Note, Keynote, Tag by example
	Markup	Revision cloud, Section cut, Field pin, Wall photo
	Measure	Measure, Area takeoff
Draw	Lines and shapes	Line, Polyline, Rectangle, Circle, Arc, Spline, Ellipse, Polygon, Construction line, Filled region, Shape, Block
Modify	Modify	Move, Copy, Rotate, Mirror, Mirror copy, Array, Offset, Trim, Extend, Fillet, Chamfer, Scale, Break, Join, Explode, Polar array, Grid array, Match properties
View	Render	Render, 360° panorama, Video, Photo match
	Color by	*Color the views by* dropdown (§9.7)
	Export	Export 3D, IFC for Revit or Archicad, Print plans
	Saved views	Save view setup

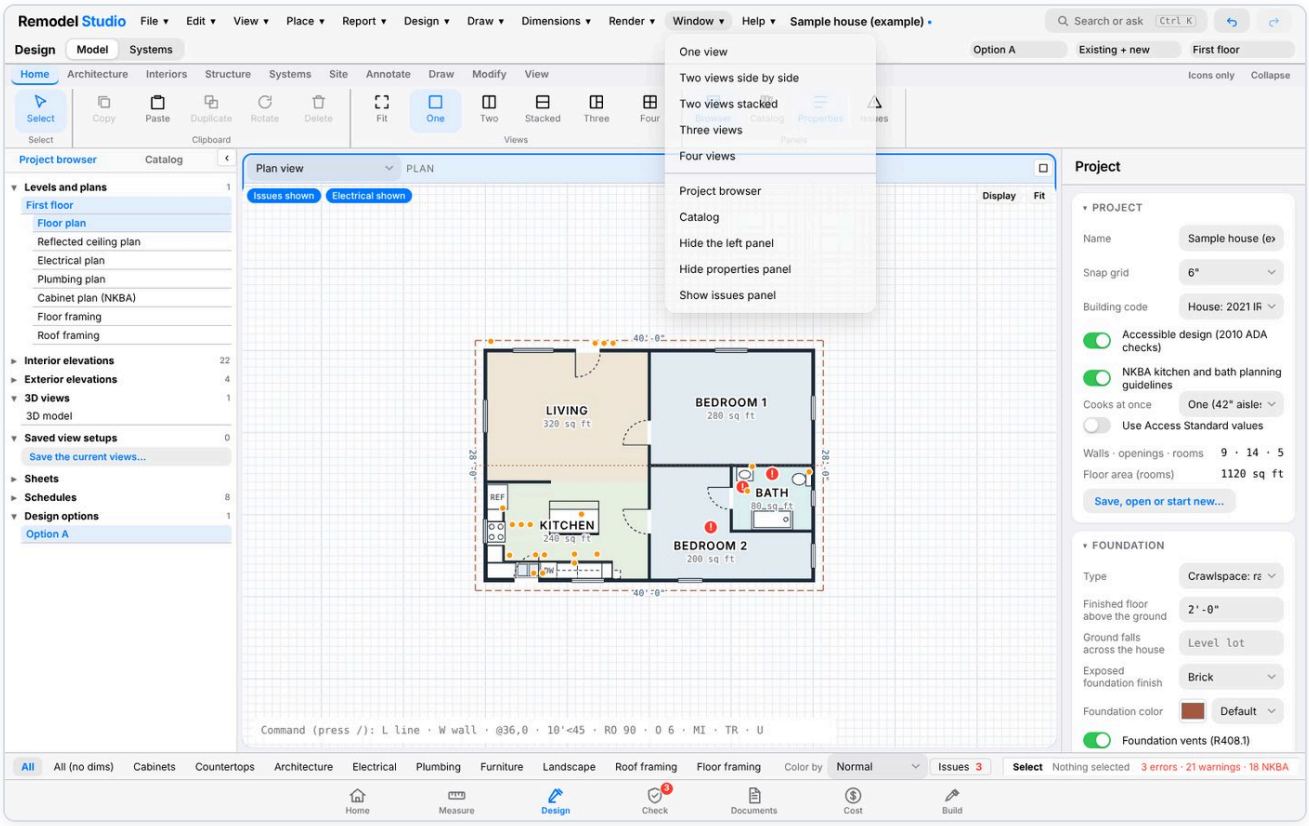
2 Designer menus (in the header menu slot)



The View menu.



The Report menu.

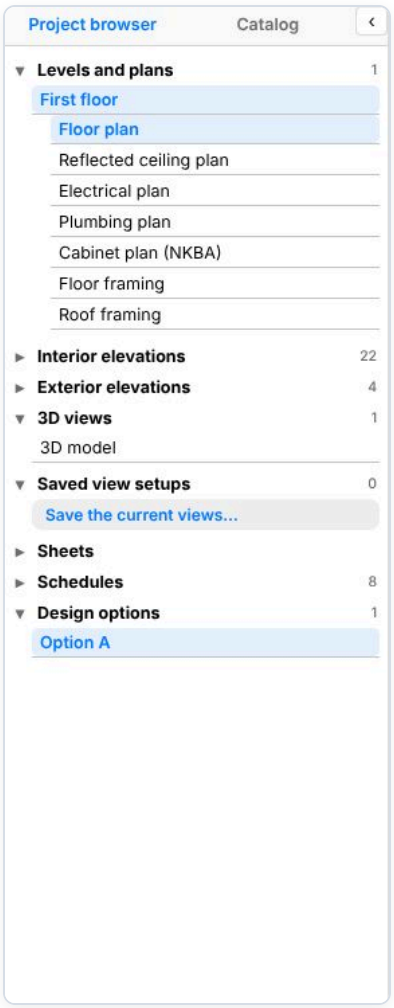


The Window menu.

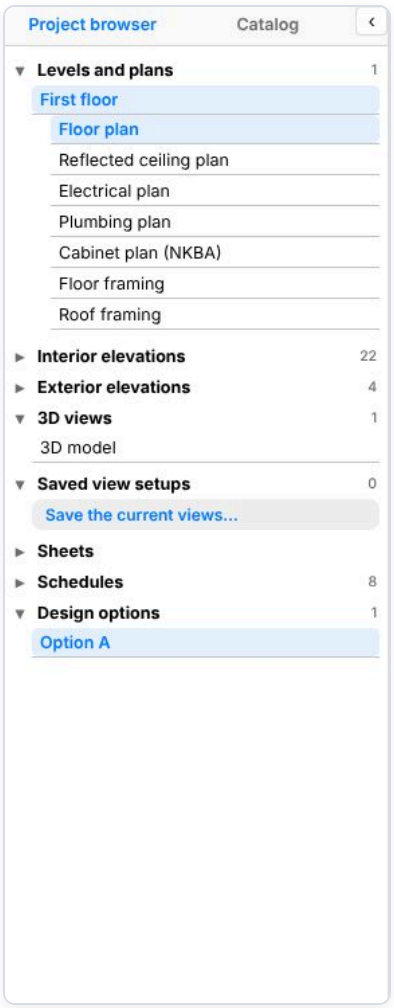
- **View** – Plan view / 3D perspective / Elevation / Exterior elevation / Reflected ceiling plan *in this pane*; Fit all views (F); Hide/Show roof in 3D; Cut away/Show levels above; Hide/Show electrical on plan; Hide/Show structure check in 3D; *Plan shows:* one entry per plan display tab (All, All (no dims), Cabinets, Countertops, Architecture, Electrical, Plumbing, Furniture, Landscape, Roof framing, Floor framing, NKBA); Normal colors and *Color by …* entries.
- **Place** – Select, Wall, Door, Window, Room, Fixture, Stair, Roof, Electric (with keys); Roof wizard...; Floor joists...; Catalog item (from the left dock)...; Column, Fireplace, Curved wall, Deck, Deck or porch outline, Porch, Steps, Railing, Skylight, Dormer, Plant, Play equipment, Play area, Block, Field pin.
- **Report** – Job dashboard; Check; Schedules; Plan set and presentation; Cost: Builder's list / Estimate / Change orders / Shopping list / Selections and orders / Budget vs actual / Cabinet order; Build: Job schedule and order dates / Field notes and punch list / Wall photos / Homeowner binder; Systems; Store prices....
- **Design** – Cabinet job settings (door, finish, species)... (opens the Project panel); NKBA checks on or off (opens the Plan check panel); Lighting, energy and systems; Layout: one entry per template; Change door style and finish...; Countertop slab layout...; Roof wizard...; Floor framing (joists)...; Wall framing and cut list...; Show roof framing; Show floor framing; Fill the cabinet run; Under-cabinet lights for this level; Remove or open a wall...; Library and customize items.

- **Draw** – every drawing tool, then every Modify tool.
- **Dimensions** – Dimension, Automatic dimensions, Angle dimension, Radius dimension, Measure, Area takeoff; Show all dimensions; Hide dimensions.
- **Render** – Render studio..., Photo match..., 360° panorama..., Walk-through video..., Export 3D model....
- **Window** – One view, Two views side by side, Two views stacked, Three views, Four views; Project browser; Catalog; Hide/Show the left panel; Hide/Show Inspector; Hide/Show issues panel.
- **Help** – Lessons..., Keyboard shortcuts, My standards..., Appearance..., How the Designer works.

3 Left dock



The Catalog dock.



The Project browser dock.

Two tabs: **Project browser** and **Catalog**; a ‹ button hides the dock (a › tab on the edge brings it back).

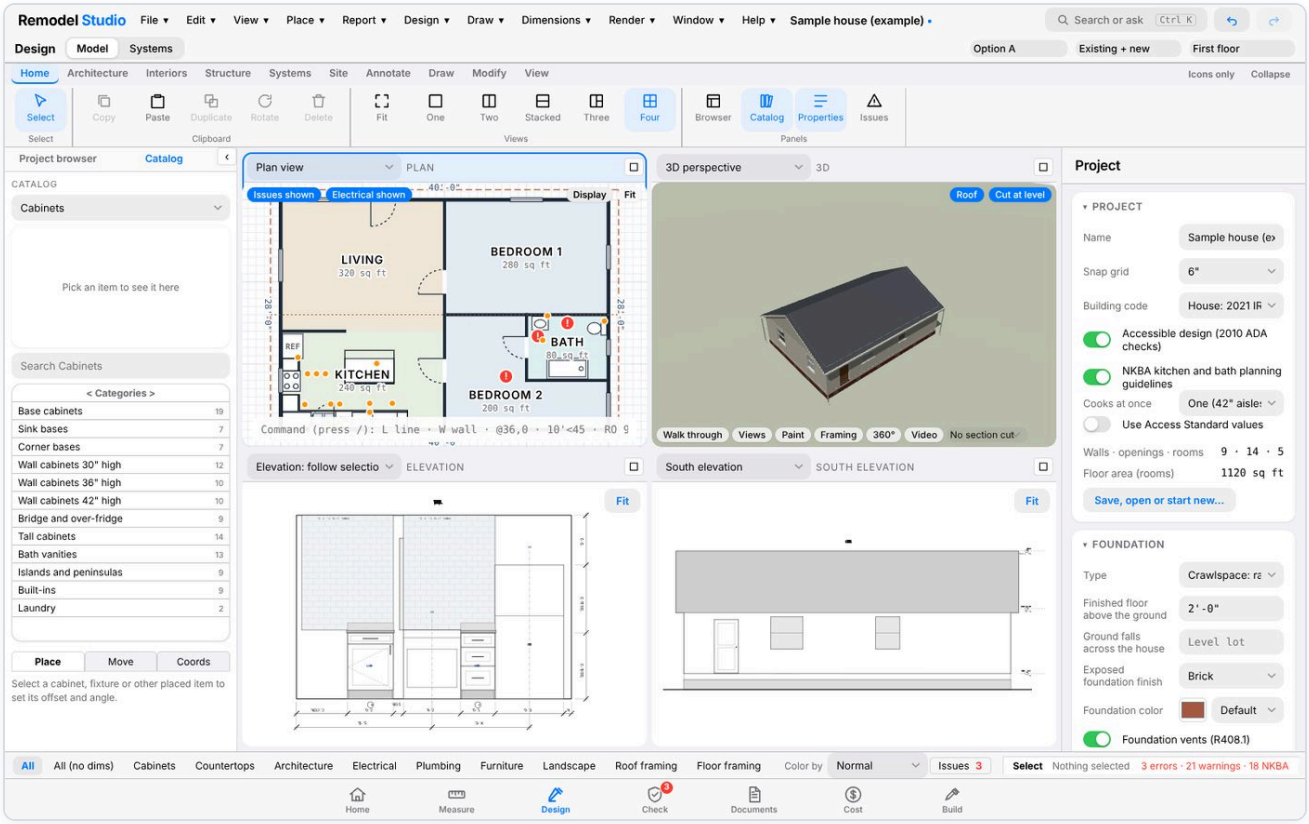
Project browser (a tree; open/closed state of each branch is remembered):

- **Levels and plans** – each level; the current level expands to *Floor plan*, *Reflected ceiling plan*, *Electrical plan*, *Plumbing plan*, *Cabinet plan (NKBA)*, *Floor framing*, *Roof framing* (each sets the active pane's view and plan display tab).
- **Interior elevations** – one entry per room, expandable to each wall ("North wall · 12'-0"") which opens that wall elevation in the active pane.
- **Exterior elevations** – North/South/East/West views.
- **3D views** – *3D model* plus every saved camera view.
- **Saved view setups** – saved pane layouts (apply by clicking; × deletes) and *Save the current views…* (names and saves the layout, pane views, plan tab, level and color-by).
- **Sheets** – the plan-set sheets by number; clicking opens Documents › Sheets at that sheet.
- **Schedules** – each schedule type; opens Documents › Schedules.
- **Design options** – switch options by clicking.

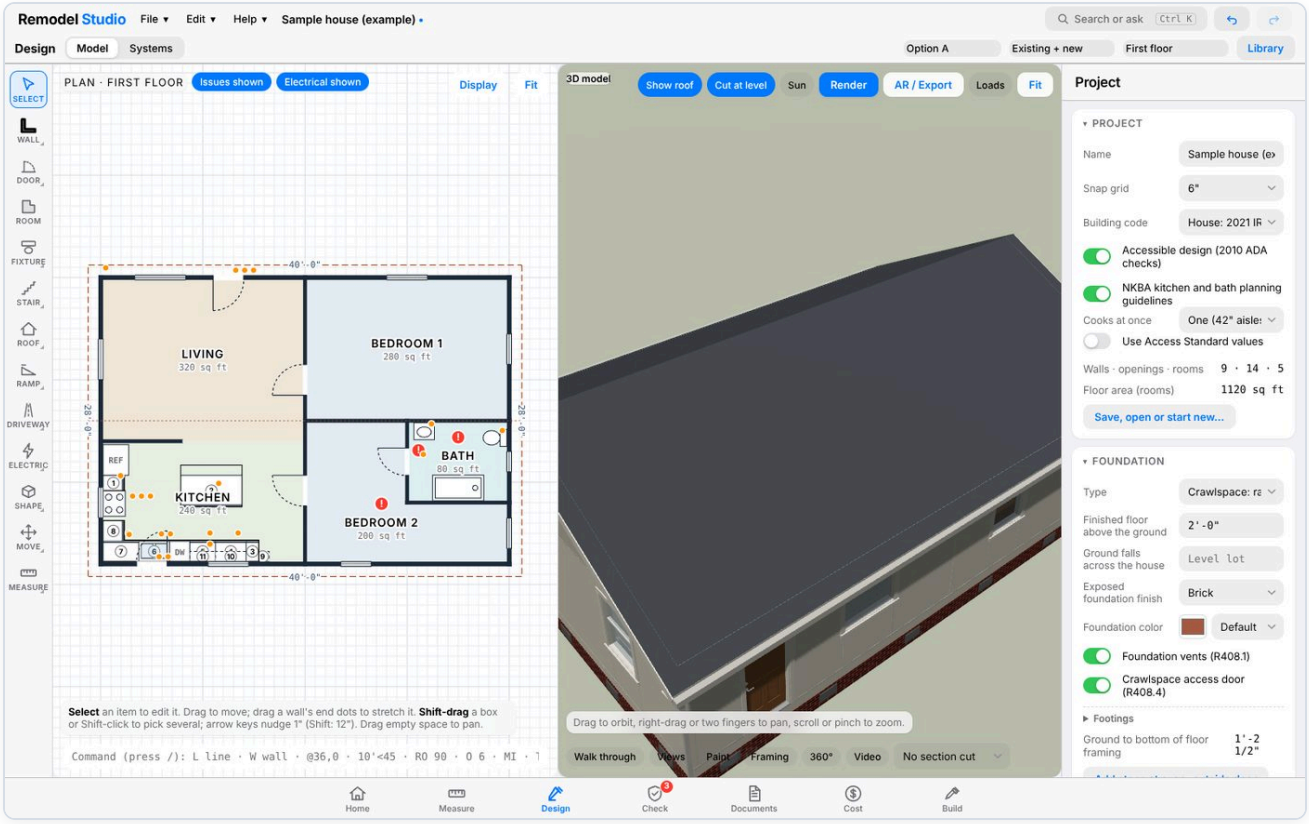
Catalog (the classic catalog list):

- *Catalog* dropdown: **All catalogs (search)** plus grouped catalogs — *Kitchen and bath* (Cabinets, Appliances, Plumbing, Bath accessories, Countertops), *Rooms and furniture* (Living room, Bedroom, Dining room, Office, Kids room, Laundry and utility, Garage and storage, Games and fitness, Electronics, Furniture), *Lighting and decor* (Lighting and fans, Decor and window treatments, Indoor plants, People and pets), *Building* (Doors, Windows, Wall assemblies, Trim and molding, Floors, Wall finishes, Ceilings, Roofing, Siding, Colors), *Outdoor* (Outdoor living, Plants, Landscape, Playground, Hardscape), *More*, and *Cabinet vendor catalogs* (for example KraftMaid).
- Preview picture and info (code, name, dimensions, note) of the picked item.
- Search box (searches the chosen catalog, or every catalog).
- List: `< Categories >` with counts; click a category to list its items as **CODE > Description**. Single-click picks (shows the preview); **double-click** or the primary action button (for example *Place*) arms the placing tool — then click in the plan. Up to three secondary actions show as buttons.
- **Place / Move / Coords** tabs act on the selected placed item:
- *Place*: **Offset from wall** (0 = tight to the wall; pushes the item out from the wall behind it) with Apply; **Angle** with Apply; **Height off floor** with −/+ step buttons (for items that have an elevation).
- *Move*: **Step** (1/8", 1/4", 1/2", 1", 3", 6", 1', 2'); arrow pad (↑ ← → ↓), ⟳ rotate 90°, *Forward*/*Back* (along the item's facing), *Turn 15°* / *Turn −15°*.
- *Coords*: X, Y, Z (height off floor), Width, Depth, Height, Angle — type and press Enter.

4 View panes



Four views: plan, 3D, elevation and exterior.



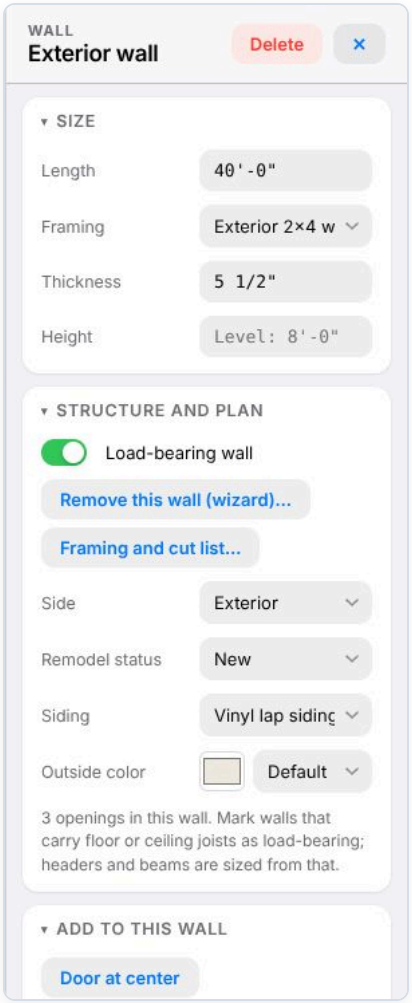
Plan and 3D side by side.

- **Layouts**: One, Two side by side (default), Two stacked, Three, Four. Each pane has a header with a **view dropdown**, the view name, and a ☐ maximize / ❐ restore button. Click a pane to make it active (the active pane gets the option bar, command line and keyboard focus).
- **View choices** per pane: *Plan view*, *Reflected ceiling plan* (ceiling lights, fans, alarms, wall and tall cabinets; no dimensions), *3D perspective* (only one pane may be 3D at a time), *Wall elevations* ▸ *Elevation: follow selection* (shows the wall the selected item sits on, else the first wall with cabinets) and one entry per room wall, *Exterior elevations* ▸ North, South, East, West.
- **Elevation panes** are drawn at 1/2" = 1'-0" (exterior at 1/4" = 1'-0"), scroll-wheel zoom, drag to pan, double-click or **Fit** to refit; clicking a fixture in an interior elevation selects it.
- The 3D pane has **Roof** and **Cut at level** chips in its corner (plus the 3D toolbar, §9.8).
- Picking a drawing tool activates the first plan/ceiling pane (or turns the active pane into a plan).

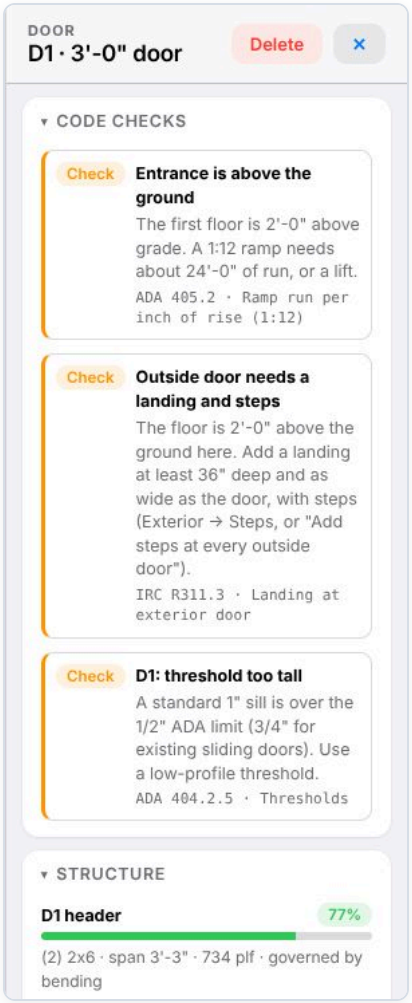
5 Plan display tabs (bottom strip)

All · All (no dims) · Cabinets (cabinets and appliances only, with cabinet callouts) · Countertops · Architecture (no fixtures or electrical) · Electrical · Plumbing (plumbing fixtures only) · Furniture · Landscape (site items, decks; fits to the lot) · Roof framing · Floor framing · NKBA (shows the NKBA issue list beside the plan, with a *Turn on NKBA checks* button if they are off) · Legend (counts of everything placed on the level by category) · Print (jumps to Documents › Sheets). Also in the strip: **Color by** dropdown (§9.7), **Issues** button with the error count (toggles the Issues dock), and the **status strip**: current tool, selected item name (or "Nothing selected"), and "N errors · N warnings (· N NKBA)" which opens the Check page.

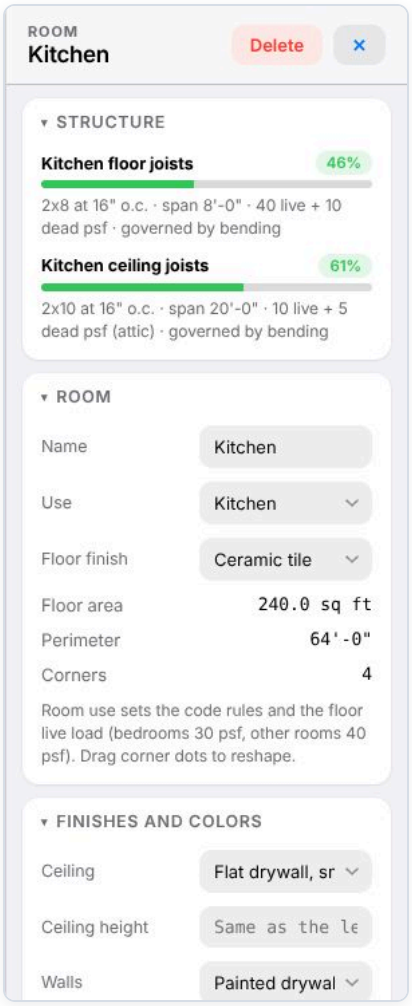
6 Inspector (right panel, "Properties")



The Inspector for a wall.



A window.



A room.



A cabinet.

Shows the properties of the selected item, or the **Project** panel when nothing is selected. Header: kind, title, **Delete** and × (deselect). Sections have headings; **click a heading to fold or unfold** that section, and the folded set is remembered by heading name across projects. The panel starts with any *Code checks* and *Structure* cards that apply to the selected item, and ends with pinned photos. Section contents per item type are listed with the tools in §8 and §9.

Project panel (nothing selected): Project — Name, **Snap grid** (1", 3", 6" default, 1'), **Building code** (House: 2021 IRC / Commercial or city building: 2021 IBC; IBC adds *Occupancy group* and *Fire sprinklers throughout*; IRC adds *Accessible design (2010 ADA checks)*), NKBA settings, counts (walls · openings · rooms, floor area), *Save, open or start new…*; **Foundation** (§8.11); **Landscape** (USDA zone, Show plants: At planting / In 5 years / Mature, Show flowers in 3D); **Trim (whole house)** per trim slot plus trim color; **Design options** (cards with name box, Active pill or *Open*, delete ×, stats: area, cost, fails; *New option from this one*); **Levels** (§8.10); **Typing lengths** help note.

7 Issues dock

A strip under the panes with Errors / Warnings / Tips / All filter buttons (with counts), *Open the Check page*, × to close, and up to 200 issues; clicking one goes to the level, selects the item and opens its properties.

CHAPTER 8

Drawing and placement tools

1 How drawing works

- The plan is a top-down drawing. **Scroll** zooms (Ctrl+scroll zooms faster; range 0.08× to 20×), **Alt-drag**, **middle-drag** or dragging empty space with Select pans, **F** or the **Fit** button fits the building (the Landscape tab fits the lot). A faint grid is drawn at 1 ft (closer zoom), 5 ft or 10 ft; heavier lines every 10 ft.
- A **hint** at the top of the plan explains the current tool's next click. **Coordinates** of the cursor (and the active snap name) show in the corner; a typed length appears as you type.
- **Snapping** (§8.3) is on by default. Hold **Shift** for free angles.
- **Click-click tools** (walls, rooms, lines, polylines): click points; a wall run continues from the last point. Finish with **Enter**, **Esc**, **double-click**, **right-click** or the **Finish** / **Close room** button. Clicking the first point again closes a room or polygon.
- **Typing a length** while a run is active: type `12'6"`, `150"` or a bare number (feet) and press Enter; the next point is placed that far along the cursor direction. Backspace edits.
- **Two-point tools** (circle, rectangle, offset…): first click, second click (rectangles may also be dragged).
- **Click-to-place tools** (fixtures, stairs, electrical, trees…): one click. Many stay armed so you can keep placing; others return to Select after one placement (noted below).
- **Esc** ends the run, a second Esc puts the tool away (back to Select), a third clears the selection.
- A **Display** button (plan corner) opens the Display popover: *Show in the plan* checkboxes (Room names and areas, Wall lengths, Fixtures and cabinets, Electrical, Roof lines, Dimensions, Notes, Drafting lines, Site and landscape, Decks porches and steps, Code-check markers, Floor below (ghosted), Cabinet numbers, Switch legs), the **Snaps** chips, and **Drafting layers** (radio = current layer, color, On/Off, Lock, Print, *New layer*). Default layers: Drafting (blue), Dimensions, Text and notes, Details (purple), Site lines (green). These settings are saved with the project.
- The **option bar** above the plan (badge "Plan · ") changes with the tool: Wall → wall type dropdown and **45° snap on/off** chip; Door/Window → type dropdown and *More in the Library…*; Fixture → fixture dropdown (Bath, Kitchen, Laundry, Furniture) and *More in the Library…*; Electric → device dropdown, **Auto-place this floor**, **Link switches to lights**; otherwise **Issues shown/hidden**, **Electrical shown/hidden**, and **Move underlay** (when an underlay image or scan exists on the level; drag it with Select).
- The **ghosted floor below** is drawn under the current level's plan.

2 Command line

A text box under the plan (press **/** or **;** to focus it; Esc clears and leaves). Type and press Enter:

- **Tool aliases**: L/line, PL/pline/polyline, REC/rect/rectangle, C/circle, A/arc, SPL/spline, H/hatch/fill, W/wall, R/room, D/door, N/window, DIM/dimension, T/text/note, V/select, S/stair, ROOF, DECK, PORCH, STEPS, COL/column, RAIL, M/move, CO/CP/copy, RO/rotate, MI/mirror, MC (mirror copy), AR/array, O/offset, TR/trim, EX/extend, AD/autodim, SEC/section, ME/measure, CAB (base cabinet run), CURVE (curved wall), DORMER, AUTOROOF, EL/ellipse, POL/polygon, RC/revcloud, XL/xline, F/fillet, CHA/chamfer, SC/scale, BR/break, J/join, X/explode, APO (polar array), AG (grid array), MA/matchprop, AA/area, DAN/dimang, DRA/dimrad. Any registered tool id also works.
- **Points**: `12',8'` absolute; `@36",0` relative to the last point; `@10'<90` distance and angle (degrees, counter-clockwise). A bare length (`12'6"`) applies to the active run.
- **Values**: `GRID 3"` snap grid; `OSNAP on|off|all|end mid int…` snap modes (bare `OS` opens Display); `LAYER name` / `LA name` switches to or creates a drafting layer (bare opens Display); `RO 90` rotates the selection 90° (about the first clicked point if the Rotate tool is active, else its center); `O 10'` sets the offset distance and arms Offset; `AR 5` sets the array count and arms Array; `F 6` fillet radius; `CHA 12` chamfer distance; `SC 0.5` scale factor; `POL 8` polygon sides; `APO 8` polar array count; `AG 4x5` grid array rows × columns; `SS` select similar (same kind; drafting: same layer and shape; fixtures: same type).
- **Actions**: `U` /undo, `REDO`, `Z` /zoom/fit, `E` /erase/del/delete. A short reply ("Snap grid 6"", "Rotated 90°", "Unknown command…") appears beside the box.

3 Snapping and grid

- **Snap grid** default **6"** (Project panel: 1", 3", 6", 1'). Free clicks round to the grid.
- **Object snaps** (Display popover or `OSNAP`): Endpoint ■, Midpoint ▲, Intersection ×, Center ○, Perpendicular ⊥ (needs a previous point), Nearest (off by default) and Grid. On by default: end, mid, int, cen, perp, grid. Snaps work on wall centerlines and faces, wall ends, door/window jambs, room corners and edges, drafting lines (ends, centers of circles and arcs), decks, fixture corners, columns and site lines. Snap tolerance is about 12 screen pixels. A glyph at the cursor shows which snap is active and the coordinate readout names it.
- **45° snap** (wall option bar chip, default on): while a wall, room, shape, dimension or drafting run has a previous point, the next point snaps to 45° increments and whole grid steps; Shift inverts it for that click.
- Room drawing also snaps to other room corners and back to the room's first corner.

4 Tool table

Keys are single letters with no modifier. "Stays armed" means you can keep placing; "→ Select" means the tool returns to Select after one placement.

Tool	Key	What it does	How to use
Select	V	Select, move, stretch and edit	Click an item. Drag to move (snaps to the grid; cabinets and plumbing fixtures snap their backs to the nearest wall unless Shift is held; wall-mounted electrical snaps to the wall face). Drag a wall's end dot to stretch it (joined wall ends move together); drag a room's corner dot to reshape. Shift-click adds/removes items; Shift-drag an empty area box-selects. Drag empty space to pan. Clicking a code-check marker opens that issue.
Wall	W	Straight walls, continuous run	Pick the wall type in the option bar. Click corner after corner (or type lengths + Enter). Closing on the first point ends the run. Enter/Esc/double-click/right-click/Finish stop. Walls are status **New**.
Curved wall	—	A curve built from short straight walls (so doors and windows still fit)	Three clicks: start, end, a point the curve passes through (its bulge). Uses the current wall type. Selects all the segments.
Door	D	Door in a wall	Hover a wall; a ghost shows the door hinged at the end nearest the pointer, swinging to the side you are on. Click to place. Option bar: *Standard 36" door* or any library door (Interior swing, Exterior entry, French and double, Sliding patio, Pocket, Bifold closet, Barn, Garage and overhead, Commercial storefront, Accessible…). Default 36" × 80".
Window	N	Window in a wall	Hover a wall and click. Option bar: *Standard 36" × 48" window* (sill 36", double-hung) or a library window (Bedroom egress, Transom, …).
Room	R	Room polygon (name, use, finishes, area)	**Click inside closed walls** and the room draws itself (auto-detect, snapped to wall centerlines), or click each corner in order and close on the first corner / Enter. New rooms are "Room N", use Living.
Fixture	X	Cabinets, plumbing fixtures, appliances, furniture (basic set)	Pick a type in the option bar (Bath: Toilet, Lavatory sink, Vanity cabinet, Bathtub 60", Shower 36", Roll-in shower 60"×30"; Kitchen: Base cabinet, Sink base, Wall cabinet, Tall pantry, Range, Refrigerator, Dishwasher, Island; Laundry: Washer, Dryer; Furniture: Queen bed, Sofa, Dining table) or any Library/Catalog item. A ghost follows the cursor; click to place. Wall-backed items turn to put their back against the nearest wall. Stays armed.
Base cabinet run / Wall cabinet run	—	Fills a stretch of wall with stock cabinets	Click where the run starts against a wall (a corner or a door edge), then where it ends along the same wall. Fills with widths 36" down to 9" and a filler for the remainder; a ghost previews.
Stair	S	Straight stair	Click where the bottom step starts (36" wide, 10" treads, runs up the page). → Select. Set shape, direction, width, rise in the panel.

Tool	Key	What it does	How to use
L-shaped stair / L stair with winders / U-shaped stair / Spiral stair	—	Stairs of that shape	One click at the bottom step (spiral: 60" diameter). Press E to turn. → Select.
Railing	H	Guard/handrail along any edge	Click points along the edge; Enter/Esc finishes. 36" high, balusters.
Roof	O	Rectangular roof	Drag a rectangle over the walls (ghost shows size), or click once to roof the whole level from its outline. New roofs: gable, 6/12, 12" overhang, ridge along the long side. → Select.
Automatic roof	—	Roofs the level from its outside walls	One click anywhere. L, T and U footprints get a roof per wing with valleys. Existing roofs on the level are replaced. Style/pitch/overhang come from *Automatic roof settings* in the Project panel. → Select.
Dormer	—	Dormer on a roof slope	Click on a roof slope where the dormer's front wall goes; it faces down the slope. Press E to turn. → Select.
Skylight	—	Skylight on the roof plane under the click	Click on a roof (show roof lines). Default 2×4 fixed.
Fireplace and chimney	—	Fireplace with chimney	Click to place; turn it so the firebox faces the room. Chimney height is figured from the roof.
Column or post	C	Column	Click. 8" square by default; style, size, height, bottom in the panel.
Ramp	G	ADA ramp from an outside door	Click on the outside wall at a door; the ramp runs away from the house at 1:12 with 60" landings, 48" wide. → Select.
Deck / Porch	K / P	Rectangular deck or porch against the house	Drag a rectangle (or click two corners); the edge touching the house snaps to the wall face and becomes the ledger. Porches come covered with columns. Minimum 24" each way. → Select.
Deck or porch outline	—	Any-shape deck	Click each corner; close on the first or Enter. → Select.
Steps or stoop	J	Landing and steps	Click an outside wall at a door (the landing centers on the door) or the edge of a deck/porch. Sized to code from the floor height. → Select.
Electric	Y	Electrical device	Pick the device in the option bar (Receptacles: Duplex, GFCI, Countertop GFCI, Weatherproof GFCI, 240-volt, Floor, USB; Switches: Single-pole, 3-way, Dimmer, Occupancy sensor; Lighting: Ceiling light, Recessed, Pendant, Ceiling fan with light, Wall sconce, Exterior wall light, Under-cabinet, Exit sign; Ventilation and safety: Bath exhaust fan, Smoke alarm, CO alarm, Smoke/CO combo; Power and data: Panel, Data or TV jack, Thermostat, Doorbell or camera). Click to place; wall devices snap to the wall face on the side you click. Stays armed.
Plumbing item / HVAC item	—	Plumbing or HVAC symbol	Click to place (water heater by default for plumbing, floor supply register for HVAC); each click repeats the last type used; change the type in the panel. Wall items snap to the wall.

Tool	Key	What it does	How to use
Driveway / **Patio** / **Planting bed** / **Lawn area** / **Property line**	—	Site areas	Click the corners; close on the first point or Enter. Placed on the ground level. → Select.
Walkway / **Fence** / **Retaining wall**	—	Site lines	Click points; Enter/Esc finishes. Walk 36" wide; fence 72" high; retaining wall 30" high × 12" thick. → Select.
Tree / **Shrub**	—	Generic tree (20' canopy) or shrub (42")	Click; keep clicking. Kind and size in the panel.
Plant	—	A specific plant species	Click; places the last plant chosen (boxwood by default). Pick species in the Library › Landscaping or in the panel.
Play equipment	—	Playground item (GameTime catalog)	Click; places the chosen item with its dashed use zone. Pick items in the Library › Playground.
Play area and surfacing	—	Safety surfacing outline	Click the corners; close on the first. → Select.
Shape	U	Custom solid (built-ins, porches, planters, angled counters)	Click the corners of any outline; close/Enter. Makes a 36"-high extrusion.
Line	I	Drafting line(s)	Click points; each segment is its own line. Enter/Esc stops.
Polyline	—	Connected line, one object	Click points; click the first point to close; Enter/Esc finishes.
Rectangle	—	Rectangle	Drag or click two corners; `@w,h` for an exact size.
Circle	—	Circle	Click the center, then a point on the circle (or type the radius).
Arc (3 points)	—	Arc	Click start, a point it passes through, end.
Spline (smooth curve)	—	Smooth curve	Click 3+ points; Enter finishes; click the first point to close.
Ellipse	—	Ellipse	Click the center, then a corner of the box it fits in.
Polygon (equal sides)	—	Regular polygon (6 sides default; `POL 8`)	Click the center, then a corner.
Construction line	—	Endless guide line that never prints	Click a point, then a direction point. Stays armed.
Filled region (hatch)	—	Hatched/filled area	Click the corners; close/Enter. Default diagonal pattern; pick the pattern in the panel.
Revision cloud	—	Scalloped cloud around a changed area	Click around the area; click the first point to close. Tag it with a note.
Block	—	Place a saved block	Pick a block in Edit › Blocks first, then click to drop copies (center at the click). Esc when done.

Tool	Key	What it does	How to use
Automatic dimensions	—	Exterior dimension strings (overall, corners, door and window centers)	One click. Earlier automatic dimensions on the floor are replaced. → Select.
Angle dimension	—	Angle between two legs	Click the corner (vertex), a point on the first leg, a point on the second leg.
Radius dimension	—	Radius/diameter note with arrow	Click a circle or arc, then where the label goes. → Select.
Note	T	Text note (prints in capitals)	Click where it goes; type the text in the panel. → Select.
Keynote	—	Numbered note tag (diamond) tied to a project-wide keynote list	Click to place; reuses the last keynote; pick another or make a new one in the panel.
Tag by example	—	Labels every item like the one you click	Click a cabinet, appliance, door, window, light or room; every item of that kind on the level gets a tag (cabinet code, appliance model, door/window mark, light mark, room name and area). Stays armed.
Section cut	—	Section line that becomes a building-section sheet (A3.x)	Click the start and end; it snaps square to the plan. Letters A, B, C… assigned automatically. → Select.
Field pin	—	Pin for an existing condition, punch item, question or photo spot	Tap where it is; add a note and photos in the panel.
Wall photo (before drywall)	—	Camera mark pinned to a wall	Click the wall on the side you stood; add photos in the panel.
Measure	M	Temporary distance readout	Click two points (45° snap from the first). A third click starts over. Nothing is added to the model.
Area takeoff	—	Tagged outline showing square feet and perimeter	Click around any area; close on the first point. Stored as a dashed orange drafting outline.
Move	—	Move the selection by two points	Select first. Click a base point, then the destination (or type `@x,y` / `10'<90`). Stays armed.
Copy	—	Copy the selection	Select first; base point, destination. Stays armed for more copies.
Rotate	—	Rotate the selection	Select first; click the pivot, then move the mouse (snaps to 15°; Shift for any angle) and click — or type an angle such as `30` or `-90` (`RO 45` rotates about the center).
Mirror / Mirror copy	—	Flip the selection across a line (copy keeps the original)	Select first; click two points of the mirror line (45° snap). Door swings flip with their walls.
Array (row of copies)	—	N copies in a row (3 default; `AR 5`)	Select first; base point, then the spacing point.
Polar array	—	Copies around a center (6 default; `AP0 8`)	Select first; click the center. → Select.
Grid array	—	Rows × columns (3×3 default; `AG 4x5`)	Select first; base point, then the diagonal spacing point. → Select.

Tool	Key	What it does	How to use
Trim	—	Cut away part of a drafting line	Click the part to remove; it is cut back to the nearest crossing lines or wall faces on each side.
Extend	—	Stretch a drafting line to the next edge	Click near the end of the line; it extends to the next line or wall face it points at.
Fillet	—	Join two lines (or two walls) at a corner, rounded by the fillet radius (0 = square; `F 6`)	Click the first line, then the second. For walls, both are trimmed/extended to meet.
Chamfer	—	Cut a corner back (6" default; `CHA 12`)	Click two lines.
Scale	—	Scale the selection (×2 default; `SC 0.5`)	Select first; click the base point.
Break at a point	—	Split a drafting line in two	Click the line where it should split.
Join lines	—	Join selected end-to-end lines into one polyline	Select the lines; click anywhere. → Select.
Explode into lines	—	Break selected polylines/rectangles into lines	Select; click. → Select.
Match properties	—	Copy layer, line type, weight, color and fill between drafting items	Click the source item, then each item to change. Esc when done.

5 Walls

- **Wall types (presets)**: Exterior 2×4 (5 1/2" thick, bearing) — the default; Exterior 2×6 (7 1/2", bearing); Interior 2×4 (4 1/2"); Interior 2×6 (6 1/2"); Pony wall 42" (half wall, 4 1/2" thick, 42" high); Knee wall 36" (2×4, 36" high). Walls without their own height use the level's wall height (default **96"**).
- **Wall panel (Inspector)**: *Size* — Length (editing stretches the wall), Framing preset (plus Custom), Thickness, Height (blank = level height). *Structure and plan* — Load-bearing wall checkbox; *Remove this wall (wizard)…* / *Removal: beam, posts and footings…*; *Framing and cut list…*; Side (Exterior/Interior); **Remodel status** (Existing / New / Demolish); Siding (exterior walls: Vinyl, Fiber cement lap, Brick veneer, Wood lap, Poured concrete or block); Outside color / Color (both sides); a note with the opening count. *Add to this wall* — Door at center, Window at center.
- Plan drawing: new walls, existing walls and demolished walls are styled differently; wall lengths are labeled along exterior walls (interior walls when zoomed in or selected). Joined walls clean up at corners.

6 Doors and windows (openings)

- **Door panel**: Door editor (Door type: Swing, Double swing, Sliding, Pocket, Bifold, Barn (surface slider), Overhead garage; Door style: Flush, 6-panel, 2-panel, Shaker 1-panel, 5-panel shaker, Louvered, Half-lite glass, Full-lite glass, French, 6-lite craftsman, Plank/barn, Hollow metal, Aluminum storefront, Raised-panel garage, Garage with window row; Slab material, Thickness, Glass, Sidelites and width, Transom and height, Frame, Door color, Outside color, Threshold, Door stop, 20-minute rated, Self-closing, Kick plate, Peephole/viewer; Hinge side and Handing tiles; **Hardware**: Handle type, Function, Finish, Brand, Rose, Lever shape, Hinge size, Hinge corners, Hinges per leaf, Screen); *Size* — Type (Door/Window), Width, Height; *Position* — From wall start (edge of the opening from the wall's starting corner), **Flip hinge side**, **Flip swing**; a warning pill if under 36" (ADA 32" clear); *Outside trim and shutters* on exterior walls (Outside casing: None/1×4/1×6/1×8 craftsman; Trim color); *Header* on bearing/exterior walls (Size automatically or a beam size).
- **Window panel**: Window editor (operation picture, lock and crank finish, screen), Width, Height, **Sill height** (default 36"), Projection (Flat in the wall / Bay 45° / Bay 30° / Bow / Garden or box window, with *Projects out* depth), Grilles (None, Colonial 6 over 6, Prairie, Craftsman 3 over 1, Diamond), Top (Square / Half-round transom / Eyebrow arch — exterior, non-bay), Opens as (Double-hung, Casement, Slider, Awning, Fixed), Net clear opening readout, Position, Frame color, outside casing, Shutters (None, Louvered, Raised panel, Board and batten, Bahama) with shutter color, trim color, header.
- Doors and windows move along their wall when dragged; copies paste further along the wall. A window whose sill is below grade on an exterior basement wall gets a window well automatically.

7 Rooms

- Room panel: Name, **Use** (Living, Bedroom, Kitchen, Dining, Bath, Hall, Laundry, Closet, Garage, Office, Meeting or assembly, Classroom, Retail or sales, Storage, Shop or work bay, Corridor, Other — sets code rules and floor live load), **Floor finish** (Carpet, Vinyl plank, Hardwood, Tile, Sealed concrete; default tile for bath/kitchen/laundry, concrete for garage, vinyl plank otherwise), Floor area, Perimeter, Corners. **Finishes and colors**: Ceiling (Flat drywall smooth, knockdown, orange peel, popcorn, and vaulted/tray/coffered and other kinds), Ceiling height (blank = level height; for vaulted: *Flat at (collar ties)* height), Walls finish (Painted drywall eggshell, Level 5 smooth, Orange peel, Knockdown, Shiplap, Beadboard…), Wall color, tile settings, Ceiling color, Floor color, and *Trim in this room* (per slot, with *Use the whole-house trim*).
- Auto-detect: with the Room tool, click inside enclosed walls (doors and windows count as wall) and the outline is traced and snapped to wall centerlines.
- Drag a selected room's corner dots to reshape; drag the room to move it.

8 Stairs

Stair panel: *Layout* — Shape (Straight, L-shaped with landing, L-shaped with winders, U-shaped (switchback), Spiral), Direction up (Up the page, Right, Down the page, Left, Angled), Turns (right/left), Risers before the turn, Winders in the turn (2–4), Gap between flights (U), Turns through degrees (spiral, 180–720), Width (spiral: Diameter), Tread depth (10" min), Total rise (blank = auto from the level above, else level height + 12" floor depth). *Rails and finish* — Handrails (When required (4+ risers), Right, Left, Both, None), Open sides with balusters, Closed underneath, Tread color. *Result* — risers count and height, Riser/Tread/Width OK pills (IRC R311.7: 7 3/4" max riser, 10" min tread, 36" wide). The floor above gets an opening over the stair automatically.

9 Roofs

Roof panel: *Shape* — Style (Gable, Hip, Dutch gable, Gambrel, Mansard, Shed, Flat or low slope), Pitch x/12 (flat: drainage slope and parapet height; gambrel/mansard: upper and lower pitch and break-in distance; dutch: gablet set-in), Overhang (default 12"), Ridge runs (Left to right / Top to bottom), Slope the other way (shed), Wall height under it. *Eaves and gutters* — Fascia board (1×4/1×6/1×8/1×10), Boxed soffit, Gutters and downspouts with gutter color (white/bronze/black/copper), Rake trim, Trim color. *Footprint* — Width, Depth, Peak above floor. *Framing and covering* — Covering, Roof color, Framing (Rafters and ceiling joists / Engineered trusses) with rafter size, spacing (12/16/19.2/24), lumber species, or truss type, truss spacing, heel height, hip girder setback and plies; *Roof wizard…* button. Pressing **E** on a roof swaps its ridge direction.

10 Levels and stories

In the Project panel › **Levels** (listed top to bottom): a radio to work on that level, Name, Floor elevation, Wall height, *Floor and ceiling framing* (Foundation type on the ground level: Crawlspace / Slab on grade / Basement; Floor joists size and Spacing; Joists run; Girders every; Lumber; on the top level Ceiling joists and Attic storage; *Joist layout, I-joists and floor trusses…*), Delete level. Buttons: **Add level** (new level above the top one: elevation = top + height + 12" floor framing depth, same wall height; named "Second floor" then "Level N"), **Add basement** (8"-thick concrete walls inside the ground-floor footprint, 8 ft tall, a Basement room; sets the foundation to Basement), **Copy exterior walls up** (copies the exterior walls to the level above, creating it if needed), **Make attic rooms** (finds the space under the roof with 4 ft or more of headroom and puts knee walls and a room there), **Roof this level**, and *Automatic roof settings* (Style: Gable/Hip/Dutch gable/Gambrel, default Hip; Pitch default 6; Overhang default 12"; *Roof this level automatically*).

11 Foundation, grade and exterior (Project panel › Foundation)

Type (Crawlspace, Piers and beams, Stem wall with filled slab, Slab on grade, Basement); **Finished floor above the ground** (grade offset, default 24"); **Ground falls across the house** (fall) with *Downhill toward* (N, NE, E, SE, S, SW, W, NW); Exposed foundation finish (Brick, Painted block, Parged, Stone veneer, Siding carried down); for piers: Skirting (Lattice, Vinyl/board, Brick curtain wall, Open), Pier size, Piers every; Foundation color; Foundation vents and Crawlspace access door (crawl); Footings (Width, Thickness, Bottom below grade); *Add steps at every outside door*.

Deck/porch panel: Name, Type, Surface (Treated wood, Composite, Cedar, Concrete, Brick pavers, Stone or tile), color, Top below the floor, Boards run, Joists/Spacing, Posts every, Skirting, area and height readouts; *Railing* (Where: only where required over 30" / all open sides / none; Style: Wood balusters, Metal balusters, Cable, Glass, Solid board; Height; color); *Roof over it* (Covered checkbox; Shed/Gable/Hip, pitch, overhang, beam height, Columns style and size, column spacing, Ceiling beadboard/open rafters, roof color). Steps panel: Landing depth (0 = none) and width, Steps go (Straight out / Down the left / Down the right), Tread depth, Landing top below floor, Total drop, Built from (Concrete, Brick, Stone, Treated wood, Composite), color, Handrails, Rail style, riser/tread/landing pills. Column panel: Style (Square, Round, Tapered craftsman, Turned, Plain 6×6 post), Size, Height, Bottom above the floor (negative to reach the ground), Color. Railing panel: Style (adds Pipe handrail only), Height, Bottom above the floor, Color.

12 Site, landscape, drainage and playground

- **Site item panel**: Surface/Kind/Style (Driveway: Concrete, Asphalt, Gravel, Pavers; Patio: Pavers, Concrete, Flagstone, Brick; Walkway: Concrete, Pavers, Stepping stones, Gravel, Brick; Planting bed: Mulch, Pine straw, River rock; Lawn: Sod, Seed; Fence: Wood privacy, Picket, Chain link, Split rail, Ornamental iron, Vinyl privacy; Retaining wall: Segmental block, Stone, Timber, Concrete; Tree: Shade, Evergreen, Ornamental, Palm; Shrub: Rounded, Boxwood, Ornamental grass), Color, Canopy spread/Height (tree), Width (walk), Height (fence), Height above grade and Thickness (retaining wall), Area or Length. Beds add Mass planting species with spacing and plant count, Edging (None, Steel, Aluminum, Plastic, Brick soldier course, Natural stone), Landscape fabric; lawns add Grass (Bermuda, St. Augustine, Zoysia, Centipede, Tall fescue); both add Irrigation (None, Spray heads, Rotors, Drip line).
- **Plant panel**: picture, Plant species (grouped by kind), botanical/size/sun/water/zone note, zone warning, Size to buy, Mature spread, Mature height, Show size (At planting / In 5 years / Mature), "Shown now" size.
- **Drainage checks** run automatically: ground sloping toward the house (IRC R401.3), walkway slope (ADA 403.3/405.2), steep driveways, and low window sills high above grade (R312.2.1). Downspouts and drainage arrows appear on the site sheet when gutters are on.
- **Play equipment panel**: picture, Item (GameTime catalog by category), maker info, Width, Depth, Fall height, Rotation, Colors scheme, Seats (swing frames), play-event counts for structures. **Play area panel**: Surfacing (Engineered wood fiber, Wood chips, Wood mulch, Shredded rubber mulch, and unitary surfaces), Installed depth or Tested critical fall height and Thickness, Border, GameTime product link, Area.

13 Electrical details

Device panel: Type, Height to center (or ceiling), light output lumens, Color temperature, Dimmable, Fixture mark, Remodel status (Existing/New/Remove), Room, "Placed by Auto-place" note; switches list *Controls these lights* checkboxes (switch legs draw on the plan as curves; faint for unselected switches). Auto-place follows IRC E3901 spacing (general receptacles so no wall point is more than 6 ft from one, counter receptacles every 4 ft, bath GFCI near basins, exhaust fans, 240-V for ranges, and so on).

14 Drafting item panel

Line/Polyline/Rectangle/Circle/Arc/Spline/Filled region: Layer, Line type (Solid, Dash, Hidden, Center, Dot, Phantom), Line weight (Fine 0.25, Thin 0.5, Medium 0.8, Heavy 1.2, Extra heavy 2), Color on screen, Radius (circle), Fill (No fill, Solid, Diagonal lines, Crosshatch, Concrete, Earth, Gravel, Batt insulation, Brick, Wood grain, Tile grid) with Pattern scale, Length and Angle readouts, Closed (polyline). Drafting prints on plan sheets in black at the chosen weight. Dimension panel: Offset from the points, *Flip to the other side*, Text (blank = measured). Note panel: text, Text size (Small/Normal/Large/Title), *Add a leader arrow* with arrow x/y. Angle dimension: Arc radius. Section cut: Label (up to 3 letters), *Look the other way*. Shape panel: Name, Kind (Outline pushed up / Round: column, post, cone), Bottom and Top diameter (round), Height, Bottom above floor, Color, Footprint, *Cut a hole using* another shape inside it, Remove holes. Ramp panel: Rise, Runs down toward, Width (outside of rails), Slope 1 in N (8–20), Built from (Treated wood/Concrete), Edge curb or guard, runs and total length, compliance pills. Keynote panel: Note picker (+ New keynote), Text, Number, Builder's list item link. Tag panel: *Remove every tag made with this one*. Field pin panel: Type (Existing condition, Punch item, Question, Photo spot), Note, Status/Trade/Due/Assigned to (punch), Severity (condition), *Take or add photo*. Wall photo panel: Stage (Before drywall, Rough-in inspection, During work, Finished), camera mark height (48" default), Date, What it shows, *Other side of the wall*, *Add photos…*, photo grid.

CHAPTER 9

Selecting and editing

1 Selection

Click with Select. The Inspector shows the item; the status strip names it. **Shift-click** adds or removes an item; **Shift-drag** on empty space box-selects everything inside (the Inspector then lists counts by type with *Delete all*). Clicking in 3D also selects (§9.8). Hit order favors small items (notes, electrical, dimensions, registered parts) over fixtures, stairs, walls, roofs and rooms.

2 Moving and nudging

Drag to move (grid steps; multi-selections move together). Walls drag with their joined ends; dragging a wall end dot stretches every wall sharing that corner. Arrow keys nudge **1"**, Shift+arrow **12"** (a single selected wall nudges both ends with joined walls). The Catalog dock's *Move* tab offers steps from 1/8" to 2' and Forward/Back along the item's facing. `move the island 6 in left` in Ctrl+K also works.

3 Rotate, copy, paste, duplicate, delete, undo

See §5.3. Fixtures also have *Turn 90°*, *Turn around*, *Add fillers* (cabinets) and *Add one beside* buttons in their panel. Undo keeps 200 steps; restoring a Version keeps the previous drawing in Undo.

4 Remodel status

Walls, doors/windows, fixtures, electrical and most parts have **Existing / New / Demolish (Remove)**. Everything you draw is New. Demolished walls and fixtures draw in the demolition style, are left out of 3D, elevations and the new-work takeoff, and show in red under *Color by › Remodel status*. The *Existing + new / Existing only / New only* layer filter (§5.6) hides the others while you work; hidden items are kept safe when you save a change. The *Remove or open a wall* wizard (Structure ribbon / wall panel) handles beams, posts and footings for a removed bearing wall.

5 Design options

See §5.6 and the Project panel. Each option is a separate copy of the model; the Project panel shows area, priced cost and fail count per option, lets you rename the active one, open another, delete one (not the last), or *New option from this one*. The Compare dropdown overlays another option's differing walls in magenta.

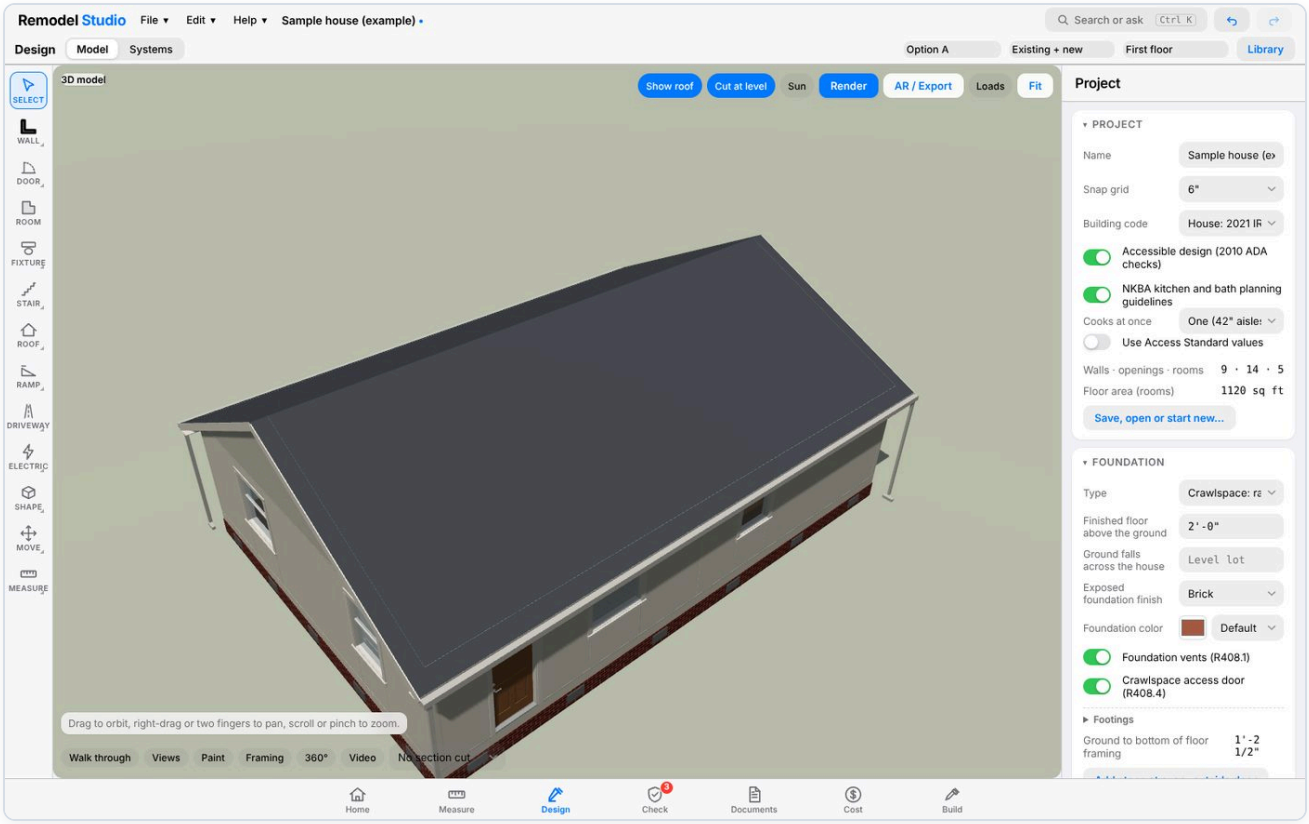
6 Revisions and revision clouds

Draw revision clouds by hand with the **Revision cloud** tool. In addition, when the contract is locked, each change order records which parts changed; the plan sheets then draw red clouds around those areas with a revision triangle carrying the revision number, and the title blocks list the revision. Void change orders lose their clouds; clouds can be turned off in settings.

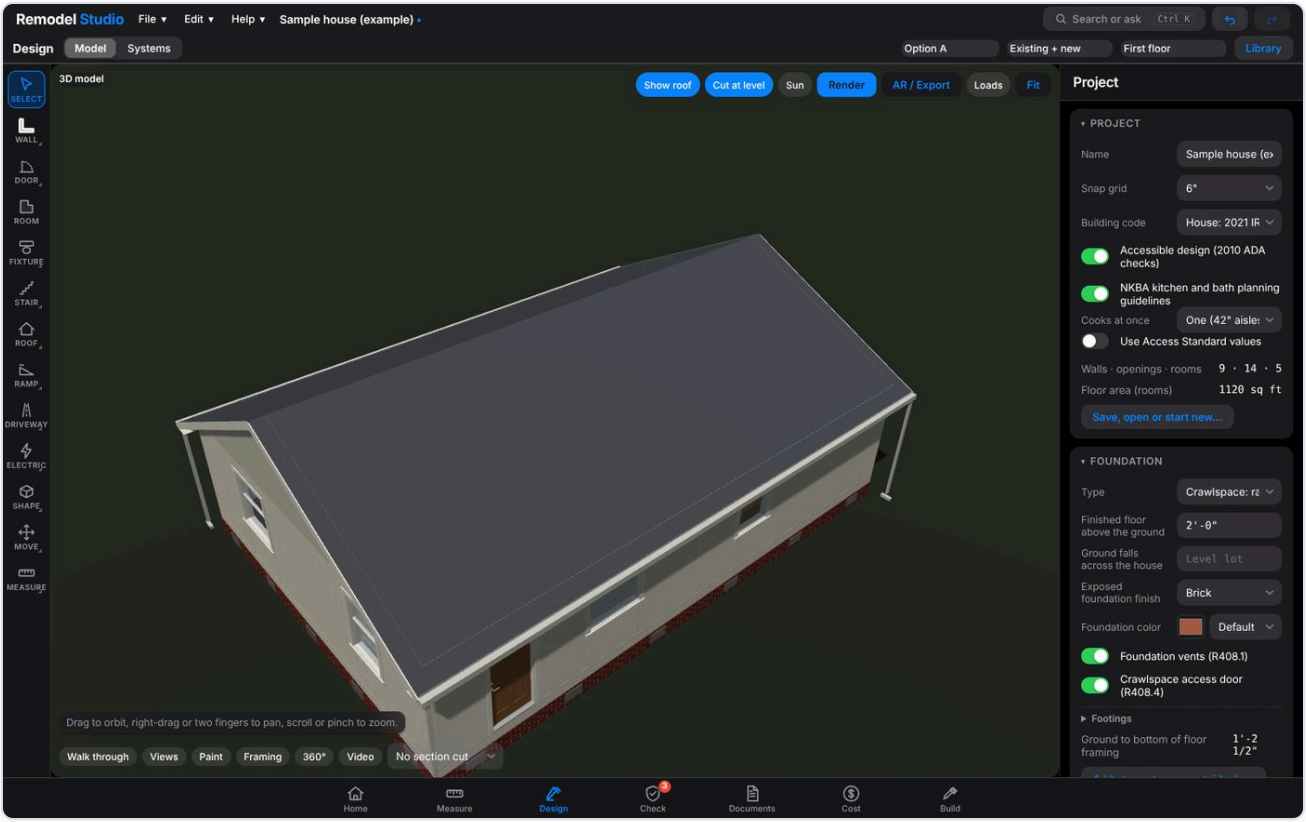
7 Color by

Dropdown in the plan tab strip and the View ribbon: Normal colors, **Remodel status** (Existing grey, New green, Remove red), **Item category**, **Vendor or brand**, **Order status** (Choosing, Approved, Ordered, Delivered, Installed), **Lead time** (≤1 week … over 6 weeks), **Cost** (under $250 … over $3,000), **Issues** (error/warning/tip), **Bearing walls**, **Room use**, **Floor finish**. Plans draw a tinted overlay, 3D swaps materials, and a legend card lists each value with a count (× returns to normal colors). Color-by is saved with view setups.

8 The 3D view



The 3D view.



The 3D view in dark mode.

- **Navigation**: drag to orbit, right-drag or two fingers to pan, scroll or pinch to zoom; the camera cannot go below the ground plane. **Fit** (F) frames the model.
- **Corner chips**: *Show roof* / *Roof*, *Cut at level* (hides levels above the one you are working on), *Sun* (sun study), *Render* (Render studio), *AR / Export*, *Loads* (colors members by capacity: green under 85%, amber 85–100%, red over, blue bearing walls), *Fit*.
- **3D toolbar**: **Walk through** (eye height 64"; W/S or ↑/↓ walk, A/D sidestep, ←/→ turn, Shift hurries, drag to look, Esc stops; doors let you through and walls stop you; on-screen arrow pad for touch; the badge reads "Walking · "), **Views** (name and *Save view*; saved views, including walk views 🚶, replay the camera and feed the board presentation; delete ×), **Paint** (click any wall, floor, ceiling, roof, cabinet, counter, foundation or other part: a paint card offers the relevant finish dropdown and color swatches — Siding + *All outside walls*; Wall color; room walls finish + color; Floor finish + color; Ceiling kind + color; Roof covering + color + *All roofs*; Foundation finish/skirting + color; Cabinets door style + finish + *Whole kitchen*; Countertop material; fixture or object color), **Framing** (shows studs, plates, headers, joists and rafters), **360°** (panorama dialog), **Video** (walk-through or fly-around recording), **Section cut** dropdown (No section cut, Cut height, Cut left to right, Cut front to back, or any drawn Section A/B/...) with a position slider and **Flip**.
- **Clicking** a part selects it (and opens its properties). Lighting scenes (Cooking, Dinner, Night light, Everything on — per switch group dimmer levels) are set up in Systems › Lighting and

used by the render studio.

- **Sun study** (Sun chip): panel with date, time slider (5 AM–9 PM), *Play day*, quick dates (Mar 20, Jun 21, Sep 22, Dec 21), sun altitude/bearing readout, and *Location and orientation* (Place, Top of plan faces compass direction, Latitude, Longitude, UTC offset, US daylight saving). The north arrow on sheets follows the orientation.
- Ceilings are modeled per room (flat, vaulted following the roof, tray, coffered…) and appear when a section cut is active.

PART

The Library and vendors

The catalog, custom items, vendor catalogs and price books, appliances, finishes and paint.

CHAPTER 10

The Library

1 Opening the Library

- Click the **Library** button in the title bar (tooltip: "Doors, windows, cabinets, fixtures, trim, finishes and colors"). It replaces the Inspector on the right side. Clicking it again returns to the Inspector. The button is not shown while the Designer (design mode) is active; from the Designer menu use **Library and customize items**.
- If you open the Library while on a page that has no drawing (Dashboard, Sheets, Schedules, Cost, Measure, Check, Build, Systems), the program switches to the plan (on a narrow screen) or the design view so there is a drawing to place into.
- When the Fixture, Door or Window tool is active, the options bar shows a **More in the Library...** button that opens the panel.
- On a phone-sized screen the panel header has a show/hide (▲/▼) button.

2 Panel layout

- Header: item count ("N items") and the title **Library**, with a close (✕) button.
- **Search box** (placeholder "Search: 36 door, vanity, crown, sage..."). Typing searches every category at once; each word you type must appear in the item's name, subcategory or category. Results are ranked (exact phrase in the name first, then names that start with your first word, then shorter names). Up to 150 results are listed, grouped as "Category: Subcategory"; a footer says "Showing 150 of N. Type more words to narrow it down." when there are more. "Nothing matches. Try fewer words." appears when nothing is found.
- **Category chips** (shown only when the search box is empty). The chosen category is remembered on this computer between sessions (default: Doors).
- A note line tells you what is selected in the plan ("Selected: Kitchen. Buttons apply to it where they fit.") or, with nothing selected, "Select a room, wall, door, window, cabinet or roof in the plan to apply finishes and colors to it."
- With the **Cabinets** category open (and no search), a **Manufacturer price books** box appears above the list (see Price books, section 2).
- The list is grouped by subcategory. Each row shows a picture or icon, the name (with a green **ADA** pill for accessible items), a dimension/description line, and one or more action buttons.

3 Pictures

Doors, windows, fixtures, cabinets, appliances, furniture, floors, wall finishes, ceilings, roofing, siding, plants and play equipment show a small rendered picture made from the real 3D model (a door is shown standing in a wall, a cabinet against a wall in a corner of a room, siding on a small house). Pictures follow the current kitchen style, trim and hardware finish of your project, so a cabinet picture changes when you change the door style. Pictures render as they scroll into view. Walls show a cut section with the layers colored (brick, sheathing, insulation, drywall and so on); trim shows a profile sketch; colors show a swatch.

4 Dimension line

Item kind	What the line shows
Fixture, cabinet, appliance, furniture	W x D x H, plus "· N" up" if it mounts above the floor
Door	Door type · door style
Window	Operation · sill height · square feet of clear opening (for operable windows)
Wall assembly	Thickness · "can carry load" or "partition"
Trim	Face height (wainscot: height)
Color	Hex value
Plant	Height x spread (feet) · sun · USDA zones · "native" where it applies
Playground piece	Model number · ages · fall height · use zone

5 Action buttons (what an item can do)

Item kind	Buttons
Fixture, cabinet, appliance, furniture, decor	**Place** (primary); **Customize...** (or **Edit...** for a My items entry); **Swap selected** when a fixture is selected in the plan (keeps position, takes the new size, name and type)
Door, window	**Place on a wall** (primary); **Customize...**; **Apply to selected** when a door or window is selected (refuses with "That is wider than the wall." if it does not fit; otherwise swaps size, type and style in place)
Wall assembly	**Draw with this** (primary; becomes the Wall tool's preset); **Apply to selected** when a wall is selected (sets thickness, kind, bearing, material; a knee wall sets height 42")
Trim	**Whole house** (primary); **This room only** when a room is selected
Ceiling, wall finish, floor	**Selected room** (primary when a room is selected); **All rooms**
Roofing	**Selected roof**; **All roofs**
Siding	**Selected wall** (exterior wall selected); **All exterior walls**

Item kind	Buttons
Color	With a room selected: **Room walls**, **Room ceiling**, **Room floor**. With a wall: **Wall** or **Wall outside**. With a door/window: **Door** / **Window frame**. With a fixture: **Selected item**. With a roof: **Roof**. Always: **All interior walls**, **All trim**, **House siding**, **All cabinets**, **All doors**
Landscape plant	**Place**; **Swap selected** (plant selected); **Fill selected bed** (a planting bed is selected and the plant is a groundcover, perennial, grass or shrub)
Playground piece	**Place**
Safety surfacing	**Draw** (draws a play area with that surface)
KraftMaid spec-book accessory	**Add to selected cabinet** (goes on that cabinet's order line); **Add to order** (loose item on the cabinet order)

6 Placing an item

1. Click **Place** (or **Place on a wall**). The program switches to the Fixture (or Door/Window) tool and, if you were in the 3D view, to the split view. The options bar shows a drop-down with the chosen item (you can switch among basic fixture types there).
2. Move over the plan: a ghost outline follows the cursor. Cabinets, appliances, plumbing fixtures and other wall items snap their back to the nearest wall face (within about the item depth plus 18") and turn to face into the room, sliding along the wall on a 6" step. Furniture, islands, freestanding tubs, water heaters, condensers and anything marked free-standing place exactly where you click.
3. Click to place. The new item is selected and its panel opens on the right. Press **Esc** to finish the tool.
4. Afterwards, drag an item to move it (wall items re-snap to walls; hold **Shift** while dragging to move it away from walls). **E** rotates 90°. The panel offers **Turn 90°**, **Turn around**, **Add one beside** (duplicates next to it along the wall) and, for cabinets, **Add fillers**.

There is no drag-and-drop from the panel into the plan; placement is click **Place**, then click in the plan.

7 Item properties (the Inspector for a placed item)

Every placed fixture shows: **Picture** (front/back toggle; refrigerators add a "Door hinged on the left / right" choice, tubs "Drain on the left / right"), **Size** (Width, Depth, Height; the height field is labelled Counter height for counter items and Rim height for lavatories; wall cabinets add Bottom above floor), rotate buttons, **Color or finish** (color picker plus a drop-down of library colors by family, with a reset ↺), **Remodel status** (Existing / New / Remove), a hint line (toilet: "ADA: centerline 16" to 18" from the side wall, with a 60" x 56" clear space."; counter items: "Standard counters are 36" high; ADA kitchens need a 30" section at 34" maximum."), and pinned photos. Cabinets add the cabinet panels (section 3). Brand-name appliances add the spec block (below).

8 Category tree (every category and subcategory)

Categories, in panel order: **My items** (appears first once you have saved one), **Doors, Windows, Cabinets, Appliances, Plumbing, Furniture, Wall assemblies, Trim and molding, Ceilings, Wall finishes, Floors, Roofing, Siding, Colors**, then the furnishing categories **Living room, Bedroom, Kids room, Dining room, Office, Bath accessories, Garage and storage, Laundry and utility, Lighting and fans, Decor and window treatments, Indoor plants, Outdoor living, Games and fitness, Electronics, People and pets**, then **Landscaping, Playground**, **KraftMaid (Lowe's 2022)**, and any imported catalog you add.

Doors (all 80" high unless noted) - Interior swing: Flush, 6-panel, 2-panel, Shaker 1-panel and 5-panel shaker doors in 24, 28, 30, 32, 36; Louvered 24, 30, 32, 36. - Exterior entry: 6-panel, Shaker, Half-lite glass, Full-lite glass, 6-lite craftsman, 2-panel in 32 and 36; Tall entry door 36 and 42 × 96. - Accessible: Accessible door 36" x 80" (lever, 1/2" max threshold); Accessible wide door 42"; Accessible pocket door 36" with D-pull. - French and double: French double door 48, 60, 72; Double interior door 48, 60, 64, 72 (closet or pantry). - Sliding patio: 60, 72, 96 × 80; Multi-slide 144" x 96". - Pocket: 24, 28, 30, 32, 36. - Bifold closet: 24, 30, 36, 48, 60, 72 (48 and wider are louvered). - Barn: 36, 42, 48 × 84 on surface track. - Garage and overhead: 8'x7', 9'x7', 16'x7', 18'x7', 8'x8', 16'x8', 10'x10', 12'x12'; 16'x7' with window row. - Commercial: Hollow metal door 36, 42 × 84 in steel frame; pair 72 × 84; Storefront entry 36 × 84; Storefront pair 72 × 84; Storefront entry 36 × 96. - Door types: Swing, Double swing, Sliding, Pocket, Bifold, Barn (surface slider), Overhead garage. Door styles: Flush, 6-panel, 2-panel, Shaker 1-panel, 5-panel shaker, Louvered, Half-lite glass, Full-lite glass, French (10- or 15-lite), 6-lite craftsman, Plank / barn, Hollow metal, Aluminum storefront, Raised-panel garage, Garage with window row (plus Carriage house for overhead doors in the editor).

Windows (width x height; default sills: 36", 24" for units 60" or taller, 48" for sliders 36" or shorter, 60" for awning and hopper, 84" for transoms, 12" for storefront) - Double-hung: 24×36, 24×48, 28×54, 30×48, 30×54, 30×60, 32×60, 34×62, 36×48, 36×54, 36×60, 36×72, 40×60, 44×62. - Single-hung: 24×36, 24×48, 30×48, 32×60, 36×48, 36×60, 36×72. - Casement: 18×36, 24×36, 24×48, 24×60, 30×48, 30×60, 36×48. - Sliding: 36×24, 48×24, 48×36, 60×36, 60×48, 72×48. - Awning: 24×20, 36×20, 36×24, 48×24, 60×24. - Basement hopper: 32×16, 32×20,

32×24. - Picture (fixed): 24×48, 36×36, 48×48, 60×48, 60×60, 72×48, 72×60, 96×60. - Transom: 36×12, 36×18, 60×18, 72×18, 72×24. - Bedroom egress: Egress casement 30×48, 24×60, 36×48 (sill 30); Egress double-hung 36×60 (sill 24) — labelled "meets IRC R310 at 44" sill". - Commercial storefront: 48×72, 60×84, 72×84, 96×96.

Cabinets (generic stock sizes; see section 3 for nomenclature) - Base cabinets: Base cabinet 9, 12, 15, 18, 21, 24, 27, 30, 33, 36, 42, 48 (B-codes), 24 deep x 34-1/2 high; Drawer base (3 drawers) 12, 15, 18, 21, 24, 30, 36. - Sink bases: 30, 33, 36, 42, 48 (SB-codes); Accessible sink base 30, 36 (open knee space, 34" top; 32-1/2 box). - Corner bases: Blind corner base 36, 39, 42, 45, 48; Lazy Susan corner base 33×33, 36×36. - Wall cabinets 30" high / 36" high / 42" high: widths 9, 12, 15, 18, 21, 24, 27, 30, 33, 36, 12 deep, bottom at 54". Diagonal corner wall cabinet 24, 27 (in the 30" group). - Bridge and over-fridge: Bridge wall cabinet 30×12, 30×15, 30×18, 33×15, 36×12, 36×15, 36×18, 36×24 (the 24-high unit is 24 deep; all hang so the top is at 84"); Microwave wall cabinet 30×18 (bottom at 66"). - Tall cabinets: Pantry 18, 24, 30, 36 wide x 84, 90, 96 high; Wall oven cabinet 30, 33 × 84. - Bath vanities: Vanity 24, 30, 36, 42, 48, 60, 72 (60 and 72 are double sink), 21 deep; Accessible vanity 24, 30, 36 (open below, 34" max top); Vanity drawer base 12, 18; Linen tower 18 × 84. - Islands and peninsulas: Kitchen island 48×36, 60×36, 72×36, 84×42, 96×42, 108×48 (36 high, free-standing); Peninsula 60, 72, 84 × 30. - Built-ins: Built-in desk 60, 72, 96 × 30 (30 high); Mudroom locker bench 36, 48, 60; Built-in bookcase 36, 48, 72 × 84. - Laundry: Laundry wall cabinet 30×30 over washer; Laundry sink base 24. - Price-book subcategories (": Base / Sink bases / Wall / Tall / Vanities / Fillers") appear here after you import a price book.

Appliances - Cooking: Range 24, 30, 36, 48; Cooktop 30, 36 in base cabinet; Double wall oven 27, 30; Range hood or microwave 30, 36. - Refrigeration: Top-freezer 28, 30; Bottom-freezer 33; French-door 36; Side-by-side 36; Column 30; Built-in 42, 48; Under-counter beverage cooler 15; Chest freezer 48×30. - Dishwashers: 18, 24. - Laundry: Front-load washer 27; Front-load dryer 27; Top-load washer 28; Stacked washer and dryer 27. - Mechanical: Tank water heater 40 gal (22" dia.), 50 gal (24" dia.); Tankless water heater (wall-hung); Furnace or air handler 21×28; Outdoor A/C or heat pump unit 30×30; Electrical panel 14" x 4" (36" clear in front). - Fireplaces: Fireplace insert 36, 42; Wood stove on hearth 48×48. - Brand models: "Ranges · brand models", "Cooktops · brand models", "Wall ovens · brand models", "Microwaves · brand models", "Range hoods · brand models", "Dishwashers · brand models", "Refrigerators · brand models", "Laundry · brand models" (see Appliance catalog below).

Plumbing - Toilets: round bowl; elongated bowl; Accessible toilet 17"-19" seat height; Wall-hung toilet with in-wall carrier; Wall-hung urinal; Bidet. - Sinks: Wall-hung lavatory 18, 20; Pedestal sink 24; Accessible wall-hung lavatory 34" rim; Trough sink 48; Commercial lavatory counter 36; Laundry tub 24×22; Mop (service) sink 24×24; Hi-lo drinking fountain pair. - Tubs: Bathtub 60×30, 60×32, 60×36, 66×32, 72×32, 72×36; Corner soaking tub 60×60; Freestanding tub 67×32; Walk-in tub 52×30. - Showers: 32×32, 36×36, 42×36, 48×36, 48×48, 60×32, 60×36, 60×42, 72×36; Accessible transfer shower 36×36; Accessible roll-in shower 60×30; Roll-in shower 60×36 (alternate).

Furniture (simple blocks) - Bedroom: Twin, Full, Queen, King, California king, Twin XL, Bunk bed; Nightstand, Dresser, Chest of drawers, Wardrobe. - Living: Loveseat; Sofa 72, 84, 96; Sectional 110; Recliner; Armchair; Coffee table; TV console; End table; Bookcase; Piano bench area. - Dining: Dining table 36 square (4 seats), 48 round (4), 60×36 (6), 72×40 (6), 96×42 (8); Dining chair; Bench. - Office, school and shop: Desk 48, 60; Executive desk 72; Office chair; Filing cabinet 4 drawer; Lateral file; Conference table 8', 12'; Reception desk; Student desk; Folding table 6'; Workbench 5'; Shelving unit 8'. - Garage: Car (parking space 9' x 18'); Pickup truck; Riding mower; Golf cart.

Wall assemblies (name, thickness, bearing) - Exterior walls: Exterior 2×4 wood frame 5-1/2; Exterior 2×6 wood frame 7-1/2; 2×4 with brick veneer 10-1/2; 2×6 with brick veneer 12-1/2; 2×4 with stone veneer 11-1/2; 8" CMU block 8; 8" CMU with furring and drywall 9-3/4; 12" CMU 12; 8" CMU with brick veneer 13; ICF wall 6" core 11; SIP panel 6-1/2" (7); Exterior metal stud 6" with panels 8 (not bearing); Post-frame (pole barn) wall with metal panels 6. - Interior walls: Interior 2×4 partition 4-1/2; Interior 2×4 bearing wall; Interior 2×6 plumbing wall 6-1/2; Interior 2×3 closet partition 3-1/2; Interior metal stud 3-5/8" (4-7/8); Interior metal stud 6" (7-1/4); Interior 2×4 with 5/8" Type X both sides (fire-rated) 4-3/4; Staggered 2×4 on 2×6 plate (sound wall) 7-1/2; Interior 8" CMU (bearing); Knee wall 2×4 (half-height, draws at 42"); Glass block wall 4. - Each wall assembly is also a preset in the Wall tool's drop-down.

Trim and molding - Baseboard: Ranch 2-1/4; Colonial 3-1/4, 4-1/4, 5-1/4; Craftsman flat 1×6 (5-1/2); Craftsman with cap 7-1/4; Modern square 3-1/2; Tall colonial 7-1/4; Rubber cove base 4" (commercial); Tile base 4". - Door and window casing: Ranch 2-1/4; Colonial 2-1/4, 3-1/4; Fluted 3-1/4 with rosettes; Craftsman flat 1×4 with head cap; Modern square 2-1/2; Backband 3-1/2; Hollow metal frame (commercial). - Crown: Cove 3-5/8; Colonial 3-5/8, 4-5/8, 5-1/4; Large crown 6-5/8; Built-up 7-1/4 (frieze + crown); Craftsman flat crown with cap 5-1/2; No crown molding. - Chair and picture rail: Chair rail 2-5/8" at 32"-36"; Picture rail 1-3/4" at 84"; No chair rail. - Shoe and quarter round: Base shoe 3/4; Quarter round 3/4; No base shoe. - Wainscot: Beadboard 32" with cap; Board and batten 48"; Raised-panel 36"; Shiplap 42"; No wainscot. - Window stool: Window stool and apron; Picture-frame casing (no stool). - Whole-house defaults: Ranch baseboard 2-1/4, Colonial casing 2-1/4, no crown, no chair rail, Base shoe 3/4, no wainscot, Window stool and apron. A room's panel ("Trim in this room", under Finishes and colors) can override each slot; "Use the whole-house trim" clears the override.

Ceilings - Drywall: Flat drywall smooth; knockdown texture; orange peel; Popcorn texture (existing; test for asbestos before removing if built before 1980). - Feature: Tray ceiling 12" step; Coffered; Vaulted or cathedral; Beadboard; Tongue-and-groove pine; Wood plank; Exposed beams over drywall; Shiplap. - Commercial: Suspended 2×2 acoustic tile; Suspended 2×4 acoustic tile; Tin (metal) tiles; Open to structure, painted.

Wall finishes - Paint and drywall: Painted drywall eggshell; Level 5 smooth drywall; Orange peel texture; Knockdown texture; Plaster; Painted CMU. - Wood and paper: Shiplap; Beadboard full height; Wood paneling; Wallpaper. - Tile, stone and wet areas: Subway tile 3×6; Large-format tile 12×24; Mosaic tile accent; Stone veneer accent; Thin brick accent; FRP panels (commercial, wet areas); Cement board with tile (wet area).

Floors - Wood and plank: Luxury vinyl plank; Laminate plank; Oak hardwood 3/4"; Engineered wood; Heart pine. - Tile and stone: Ceramic tile; Porcelain tile 12×24; Natural stone tile; Brick pavers. - Resilient and carpet: Carpet; Sheet vinyl; Cork. - Commercial and garage: Vinyl composition tile (VCT); Rubber flooring; Sealed concrete; Polished concrete; Epoxy coating. - Floor patterns (planks, tile grids, speckle) are drawn on the 3D floors and in swatches.

Roofing - Shingles: Architectural asphalt shingles; 3-tab asphalt shingles; Cedar shake. - Metal: Exposed-fastener metal panels; Standing seam metal; 5V crimp metal. - Tile and slate: Clay tile; Concrete tile; Slate. - Low slope: TPO membrane; Modified bitumen; EPDM rubber.

Siding - Vinyl: Vinyl lap; Vinyl Dutch lap. - Wood, fiber cement and metal: Fiber cement lap; Fiber cement board and batten; Engineered wood lap; Wood lap; Cedar shingle; Log siding; Metal wall panels. - Masonry and stucco: Brick veneer; Stone veneer; Stucco; Painted CMU; Split-face block.

Colors (generic names, not brand colors) - Whites: Pure white, Soft white, Warm white, Antique white, Linen, Bright white, Cool white, Ivory. - Greiges and beiges: Agreeable greige, Light greige, Sand, Khaki, Taupe, Mushroom, Oatmeal, Wheat. - Grays: Pale gray, Silver gray, Repose gray, Pewter, Charcoal, Slate gray, Graphite, Iron ore. - Blues: Sky blue, Powder blue, Coastal blue, Denim, Navy, Teal, Slate blue, Porch ceiling blue. - Greens: Sage, Olive, Hunter green, Mint, Celadon, Forest, Eucalyptus, Moss. - Reds and oranges: Barn red, Brick red, Terracotta, Burgundy, Coral, Rust, Pumpkin. - Yellows: Butter, Mustard, Pale yellow, Gold. - Browns and wood tones: Natural oak, Honey oak, Walnut, Espresso, Cherry, Maple, Gray wash, Chocolate. - Blacks and darks: Tricorn black, Soft black, Bronze, Black. - Metals and stone: Brushed nickel, Stainless steel, Matte black metal, Galvalume, Copper, White quartz, Black granite, Butcher block. - The same list fills every color drop-down in the item panels (room wall/ceiling/floor color, door color, cabinet custom color, countertop color, tile color, window outside color).

Furnishings and decor (every item draws as a shaped 3D model and a proper plan symbol; most come in several fabric, wood or metal colors; codes follow a "CODE > description" catalog style) - Living room: Sofas and loveseats (Track-arm, Rolled-arm, Tuxedo in loveseat 60, sofa 72, 84, 96; gray, navy, cream, charcoal, sage, leather); Sectionals (110×84 and 124×94, right- or left-arm chaise, 4 colors; Chaise lounge); Chairs (Club, Wingback, Slipper, Barrel swivel, Accent, Rocking; Recliner and Recliner reclined for clearance); Ottomans and benches (rectangle, tufted, round pouf, round ottoman 36, Bean bag); Tables (coffee tables 48×24, 54×28, glass, round 36, round 42, trunk; end tables; console 48, 60); Media (Media console 48/60/72/84 with 43/55/65/75" TV; Entertainment wall unit 120"); Storage (Bookcases 30×72, 36×72, 36×84, 48×84, 24×48; Cube organizers 2×2, 3×3, 4×2, 4×4, 2×4; Leaning ladder shelf); Music (Upright piano, Baby grand piano). - Bedroom: Beds (Panel, Platform, Upholstered, Sleigh, Four-poster, Storage, each in Twin, Twin XL, Full, Queen, King, California king; mattress sizes in the name); Daybed; Nightstands (1, 2, 3 drawers, with lamp); Dressers (60" 6-drawer, 66" 9-drawer, 60" with mirror, Chest 36" 5-drawer, Lingerie chest 22"); Storage (Armoire, Makeup vanity with mirror and stool); Benches (Bed-end bench 48, 60 storage). - Kids room: Bunk bed twin over twin; Nursery (Crib, Changing table, Nursery glider); Storage (Toy chest); Play (Play kitchen, Kids table and 4 chairs, Rocking horse). - Dining room: Dining sets (table with chairs: 36 square/4, 60×36/6, 72×40/6,

84×40/8, 96×42/8 trestle, 108×44/10 trestle, round 42/4, 48/4, 60/6, oval 72×42/6); Tables (without chairs: 36 square, 60×36, 72×40, trestle 96×42, round pedestal 48, 60, metal-base 72×36); Chairs (Side, Upholstered, Windsor, Parsons); Stools (Counter stool 24", Bar stool 30"; square, with back, round, round upholstered); Storage (Sideboard 60, 72; China cabinet with hutch 60, 72; Bar cart). - Office: Desks (Writing 48, Pedestal 60, Executive 72, Desk 60 with hutch, L-shaped 66×60, Standing 60); Chairs (Office task chair); Storage (File cabinet 2 and 4 drawer, Lateral file 36" 2 and 4 drawer); Tables (Conference table 8', 12' with chairs); Equipment (Desktop computer and monitor, Laptop, Printer). - Bath accessories: Towel bars and rings (18, 24, 30 bars; towel ring; five metal finishes: Matte black, Brushed nickel, Brass, Chrome, Bronze); Paper holders and hooks (toilet paper holder, robe hook); Mirrors and cabinets (Medicine cabinet 16×26, 20×26, 24×30 surface mount; Vanity mirror 24×30, 30×36, 36×36, 48×36, 60×36 in black, brass, nickel frame); Shower doors and rods (Hinged shower door 30, 36; Sliding glass tub door 60; Shower rod and curtain 60); Rugs and hampers (Bath mat 21×34; Laundry hamper); Storage (Over-the-toilet shelf). - Garage and storage: Shelving (Wire shelving 36×14, 48×18, 48×24, 60×24 × 72); Workshop (Workbench 60, 72, 96 with pegboard; Rolling tool chest 41" red/black/blue); Vehicles and sports (Bicycle, Motorcycle, Kayak 10'); Bins (Trash cart 96 gal, Recycling cart 96 gal). - Laundry and utility: Laundry (Ironing board, Folding drying rack, Laundry hamper); Shelving (Utility wire shelf 36×14, 48×18). - Lighting and fans: Ceiling fans (42" 4-blade, 44" 5, 52" 5, 60" 3, 72" 6, with light, in matte black, brushed nickel, bronze, white; 52" no light); Chandeliers (Candle 6-arm, Candle 8-arm, Drum, Linear 40", Lantern); Pendants (Globe, Cone, Lantern, Drum); Flush mounts (11", 13", 16"); Lamps (Floor lamp drum shade, Arc, Tripod, Torchiere; Table lamp 26" in blue, white, green, amber); Sconces and vanity lights (Wall sconce; Vanity light 2, 3, 4 lights). - Decor and window treatments: Rugs (5'x3', 8'x5', 10'x8', 12'x9', 8'x2.5' runner; round 5', 6', 8'; cream, gray, navy, rust, sage); Curtains and blinds (Curtain panels and rod for 36, 48, 60, 72, 96 windows; Horizontal blinds 24, 30, 36, 48, 60 × 60); Wall art (16×20 matted, 24×36, 36×48, canvas 48×60, panoramic 60×24; Coastal, Warm abstract, Botanical, Modern palettes); Mirrors (Wall mirror 30×40, Round 30, Round 36, Arched 24×40, Floor mirror 24×66; gold, black, wood frame); Accents (Wall clock 24"; Floating shelf 36" single or x3; Floor vase 24"; Vase with flowers 14"). - Indoor plants: Fiddle-leaf fig (6' and 5'), Olive tree, Ficus, Areca palm, Kentia palm, Bird of paradise, Monstera, Snake plant, ZZ plant, Pothos hanging, Boston fern hanging, Peace lily, Orchid, Succulent bowl, Herb pot, Rubber plant, Cactus; each in terracotta, white or black pot. - Outdoor living: Patio dining (48 round/4, 60 round/6, 72×40/6, Bistro set; black or teak); Patio seating (Outdoor sofa 80", Outdoor lounge chair, Chaise lounger, Adirondack chair, Porch swing 60", Hammock with stand, Garden bench 60" with or without back); Shade (Patio umbrella 8', 9', 11'; Pergola 10×10, 12×10, 12×12, 16×12); Cooking (Gas grill 3 and 5 burner; Outdoor kitchen island 8', 10', 12'); Fire features (Fire pit 36, 48 round, 42 square, Fire table 48); Spa (Hot tub 7'x7', 8'x8'); Planters and water (Planter boxes, Tiered fountain, Birdbath). - Games and fitness: Pool table 7', 8' (allow 5' cue room each side; the plan symbol shows the cue zone); Table tennis table; Card table 36" with 4 chairs; Treadmill; Exercise bike; Weight bench with rack. - Electronics: TVs 43, 50, 55, 65, 75, 85 on stand and wall mounted; Soundbar 36"; Tower speaker 36, 42. - People and pets: Man 5'10", Woman 5'5", Man 6', Woman 5'4", Child 4', Person in wheelchair (30" x 48" clear space); Large dog, Small dog, Cat.

Landscaping: every plant species in the site tools, grouped Shade trees, Ornamental trees, Evergreen trees and screens, Palms, Shrubs, Ornamental grasses, Groundcovers, Perennials and color.

Playground: every piece from the playground catalog grouped "GameTime: Play structures / Swings / Slides / Climbers / Spinners / Rockers and spring riders / Freestanding play / Musical and sensory / Inclusive play / Fitness / Shade / Site furnishings / Other", plus **Safety surfacing** (engineered wood fiber, wood chips, wood mulch, shredded rubber mulch and the other surfaces; "(accessible)" marks surfaces that meet accessibility).

9 Finish and color pickers in room panels

Selecting a room shows **Finishes and colors**: Ceiling (the ceiling list), Ceiling height (or, for a vaulted ceiling, "Flat at (collar ties)"), Walls (wall finish list), Wall color, the **Tile and wainscot** group (section 3), Ceiling color, Floor color (default follows the floor finish), and the "Trim in this room" overrides.

10 The item editor (Customize...)

Click **Customize...** on any door, window, cabinet, appliance, fixture or furniture item. A dialog titled with the item name opens ("Change anything, then place it. Save it to My items to reuse it in any project.").

Left side, by item kind:

- **Door**: Picture (Front/Back). Door type (the seven types; changing type picks a matching style). Door style grid with a picture of each style allowed for that type (sliders: full-lite, French, 6-lite, plank, flush; bifold: 6-panel, 2-panel, shaker, 5-panel, louvered, flush, full-lite, plank; barn: plank, shaker, 5-panel, full-lite, flush; overhead: Raised panel, Panels with window row, Carriage house; swing/double/pocket: all non-garage styles). Slab: Hollow-core molded, Solid-core molded, Solid wood, Solid MDF, Fiberglass, Steel, Aluminum and glass. Thickness: 1-3/8", 1-3/4", 2-1/4". Glass (shown for glass styles, sliders, or when sidelites/transom are on): Clear, Low-E tempered, Frosted (privacy), Rain glass, Reeded, Decorative leaded. Sidelites (exterior or 48"+ swing doors): None / One, latch side / Both sides, with Sidelite width (default 12). Transom: None / Glass transom above, with Transom height (default 14); a note gives the slab size when the opening is a whole unit. Threshold: None (interior), Standard sill about 1", Low profile 1/2" max (ADA), Saddle. "20-minute fire rated" checkbox (a note explains the IRC R302.5.1 garage-to-house requirement when it applies). Door color. **Hardware**: Handle (Lever, Knob, Handleset with thumb latch, Pull handle, Flush pull (pocket door), Barn door pull, Push bar (exit device), No hardware); Function (Passage, Privacy, Keyed entry, Dummy, Storeroom); Rose (Round, Square modern, Long backplate traditional); Lever shape (Straight, Curved, Round tube); Brand (Any brand, Kwikset, Schlage, Baldwin, Emtek, Weiser, Defiant, Yale, Delaney, Nostalgic Warehouse); Deadbolt; finish swatches (Satin nickel, Matte black, Oil-rubbed bronze, Aged bronze, Polished chrome, Stainless steel, Satin brass, Polished brass,

Antique brass, White); Hinges per leaf 2/3/4; Hinge size 3-1/2", 4", 4-1/2" (heavy); Hinge corners 5/8" radius, 1/4" radius, Square; Door stop (Baseboard, Hinge-pin, Floor, Wall bumper, None); Self-closing; Kick plate; Peephole viewer. A warning appears for knobs when accessible design is on ("ADA 309.4 calls for levers or pulls"). "Reset to the usual hardware for this door" clears overrides. (On a placed door the same editor also shows **Handing** with four pictured choices and residential/ANSI names.)

- **Window**: Picture; Opens as (Double-hung, Single-hung, Casement, Slider, Awning, Hopper, Fixed); Grilles (No grilles, Colonial grid 6 over 6, Prairie border lites, Craftsman 3 over 1, Diamond upper sash); Hinged on (casements: left/right seen from outside); Frame (Vinyl, Wood, Aluminum-clad wood, Fiberglass, Composite (Fibrex), Aluminum); Glass (Double pane low-E, Triple pane low-E, Tempered low-E, Obscure (bath), Laminated (impact)); Sill height; Frame color. (A placed window also offers Screen: Half, Full, None; Lock and crank finish; Outside color on exterior walls.)
- **Fixture, appliance, furniture**: Picture with hand choice where it applies; a cabinet shows the full cabinet panel (layout, style, countertop); anything else shows a Color picker.

Right side: **Size** (fixtures: Width, Depth, Height, Off the floor; doors/windows: Width, Height), **Name**, then **Place** / **Place on a wall** (places a temporary customized copy), **Save to My items** (or **Save changes** when editing a My items entry), **Remove from My items**, **Cancel**.

11 My items

- Saved items appear in the **My items** category (first chip), grouped by their original category (fixtures) or as Doors / Windows. They carry every option you set (size, style, hardware, color, cabinet layout).
- My items are stored in this computer's browser storage and are available in every project on this computer. The same storage limit applies as for blocks; a failure shows "Could not save on this computer."
- **Edit…** on a My items row reopens the editor; **Save changes** overwrites it; **Remove from My items** deletes it.
- Placing a My items entry copies all of its settings onto the new object.
- There is no separate export for My items (Blocks have export/import; see below).

12 Blocks (Edit › Blocks)

A block is a saved group of walls, doors, cabinets, fixtures, drafting or site items that you can drop into this or any other project.

- Select items in the plan (Shift-drag a box), open **Edit › Blocks…** (or **Edit › Save selection as block…**, enabled when something is selected), type a name and click **Save selection as block**. Openings in selected walls come along; the block is stored centered on the selection.
- **Accessible bath templates** are built in: "Bath 9'-4" x 7'-2", roll-in shower", "Bath 9'-4" x 7'-2", tub with grab bars", "Half bath 7' x 7'" (walls, a 36" door, fixtures and grab bars to the 2010

ADA Standards: toilet 18" to centerline, 60" turning space, 34" lavatory).

- **Your blocks** lists each saved block with its item count and door/window count; **Place** starts the Block tool, ✕ deletes.
- Block tool: "click to place a copy (its center goes where you click). Keep clicking to place more; Esc when done." The tool is also in the drafting tool group.
- **Export blocks…** saves all blocks as "Remodel Studio blocks.json"; **Import blocks…** reads such a file and adds its blocks to the list.
- Blocks are kept in this computer's browser storage.

13 Appliance catalog (brand models)

- 67 real models with published sizes, cut-outs and clearances, gathered October 2026 from manufacturer and retailer spec pages. Brands: GE (15), Whirlpool (10), Thermador (7), GE Profile (6), Samsung (5), Bosch (4), LG (3), Frigidaire (3), Frigidaire Gallery (3), Zephyr (2), Sharp (2), KitchenAid (2), Cafe (2), Broan-NuTone (2), Maytag (1). By type: refrigerators 14, ranges 12, wall ovens 8, cooktops 7, microwaves 7, range hoods 6, dishwashers 6, washers 4, dryers 2, stacked washer/dryer 1.
- Each appears under Appliances as "Brand Model: description" (for example "Whirlpool WFES3330RS: 30" freestanding electric range") with the model number as its code. Its footprint on the plan comes from the published size: ranges take the larger of the width and the cut-out width; cooktops take a base cabinet rounded up to a 3" stock width; wall ovens a tall cabinet 3" wider than the cut-out; over-the-range microwaves a 30 × 15 × 17 box at 66"; microwave drawers a base; range hoods a wall item at 66" (chimney hoods up to 36" high); dishwashers the cut-out width; refrigerators their real size (15" beverage coolers become under-counter units); washers and dryers their real size.
- With a brand appliance selected, the panel adds a block headed "Brand Model" with: Size (W x H x D), Cut-out (W x H x D) and its note, Above cooking surface clearance and note, Airflow (CFM), Electrical, Fuel, List price, notes, and a **Spec page** link ("confirm with the current install guide before cutting cabinets").
- Checks run automatically: "opening too narrow" (error) when the gap between neighboring floor cabinets is smaller than the cut-out (refrigerators: width + 1/4" + side clearance each side); "gap beside it" (warning) when the opening is more than 0.6" wider than the cut-out ("Close it with a filler"); "cabinet too close above" (warning) when a wall cabinet over a range or cooktop is lower than the published clearance; "makeup air" (tip) for hoods over 400 CFM (IRC M1503.6).
- List prices from the catalog feed the estimate's Appliances pricing.
- The appliance catalog is also kept as a data file that lists every model with its brand, type, size, cut-out, clearances, fuel, electrical need, price and sources; the program's copy is built from it, and a vendor catalog import of the *Appliances* kind adds to it.

CHAPTER 11

Vendors

1 Where vendor choices live

- **Cost › Selections and orders** — the top half is **Selections** (brands and products for the job); the bottom half is the orders tracker. Header: "Pick the brands and products for this job. They go on the materials list, the schedules and the proposal. N cabinet makers and N product brands are in the library." with an **Import vendor catalog…** button.
- **Cabinet panel › Cabinet maker** (select any cabinet) — brand, line, door style, finish and price book, with an **Import catalog…** button.
- **File › Import › Vendor catalog…** opens the import dialog directly. The Designer assistant also lists "Import vendor catalog".

2 Selections page, section by section

1. **Door hardware, whole house**: Finish (Satin nickel default; Matte black, Oil-rubbed bronze, Aged bronze, Polished chrome, Stainless steel, Satin brass, Polished brass, Antique brass, White) and Brand (Any brand, Kwikset, Schlage, Baldwin, Emtek, Weiser, Defiant, Yale, Delaney, Nostalgic Warehouse). "Every door uses these unless you change it in the door's own panel. Hinges, stops and kick plates follow the finish."
2. **Cabinets**: shows the current maker · line · door · finish ("Generic cabinets." if none) and points to the cabinet panel's Kitchen style section.
3. One section per product category, each with **Brand** (Not chosen + brands; "(confirm with dealer)" marks entries not re-checked against the maker's current catalog), then **Product line** (Series / Lines / Collections / Product for appliances / Color for countertops), then **Color** where the line or brand lists colors. A note shows the line's frame, material or notes, and catalog links. Windows show the ENERGY STAR 7.0 Southern-zone targets (U-factor ≤ 0.32, SHGC ≤ 0.23; "Forest, MS is in the ENERGY STAR Southern zone").
4. **Paint colors**: Paint brand, then a per-room table (Wall color; Trim and ceiling) and a swatch strip of the brand's colors (first 48). Picking a color writes its hex to the room's wall color (or ceiling color) so the 3D view updates. "Screen colors are close, not exact; check a physical sample."

Selected brands are stamped onto the Builder's list rows ("Selected: Andersen 400 Series White") for roofing, siding, flooring, countertops, windows, exterior doors, interior doors; the cabinet rows carry the maker label.

3 Built-in vendor list

Cabinet makers (5, compiled October 2026):

Maker	Tier	Made by	Sold at	Lines (frame)	Door styles	Finishes
KraftMaid	semi-custom	Cabinetworks Group	dealers, Home Depot, Lowe's	KraftMaid (framed, half/full overlay); KraftMaid Vantage (framed); KraftMaid 200 Professional Series (framed); KraftMaid 400 Designer Series (framed); KraftMaid 600 Luxury Series (framed, frameless, inset); plus "KraftMaid (Lowe's)" from the spec book	209 (+29 from the Lowe's book)	53 paint/stain/specialty (+56 from the book)
Merillat (incl. Masterpiece)	stock (Basics, Classic) / semi-custom (Autograph, Masterpiece)	Cabinetworks Group	dealers, lumberyards/builder channel	Merillat Basics; Merillat Classic; Merillat Autograph; Masterpiece (all framed)	70	42
Diamond	semi-custom	MasterBrand	dealers, Lowe's	Diamond Vibe; Diamond Distinction; Diamond Edge (framed); Diamond Full Access (frameless)	44	42
Schrock	semi-custom	MasterBrand	dealers	Schrock Entra; Trademark; Boutique (framed); Schrock Full Access (frameless)	43	23
Mid Continent	semi-custom (builder-focused)	MasterBrand	dealers, builders	Mid Continent Signature (framed); Mid Continent Pivot (frameless)	34	38

Door constructions recognized: shaker, slim shaker, raised, cathedral, arch, mitered, flat/slab, inset, beadboard, louver, glass-ready, other.

Product brands by Selections category:

Category	Brands and lines
Windows	Andersen (100 Series, 200 Series, 400 Series, A-Series, E-Series, Andersen Aluminum); Pella (250 Series, Encompass, Lifestyle Series, Impervia, Reserve Traditional, Reserve Contemporary, Vista Series); Marvin (Ultimate, Elevate, Essential, Modern, Vivid); JELD-WEN (Premium Vinyl V-4500, Builders Vinyl V-2500, Premium Atlantic Vinyl, DF Hybrid Vinyl, Brickmould Vinyl, Flat Casing Vinyl, Siteline Clad-Wood, EpicVue Clad-Wood, W-2500, W-5500, Custom Wood, Auraline Composite, Builders Atlantic Aluminum); Simonton (5500, ComfortSmart, 5050, HarborLight, StormBreaker Plus, 6150, 6200, 6550, Brickmould 300, Brickmould 600, Contractor, DaylightMax, Madeira, Premium, Pro); Ply Gem (1100, 1500, 3700, 4700, 4800, 5000, Classic, MIRA aluminum-clad, East Premium/Pro/2000/5100, West Pro 200/700, West 400)
Exterior doors	Therma-Tru (Classic-Craft, Fiber-Classic, Smooth-Star, Pulse, Profiles and Traditions, Benchmark, Veris); Masonite (Belleville Smooth, Belleville Mahogany Textured, Belleville Oak Textured, Heritage Series Winslow, VistaGrande, Masonite HD Steel, Sta-Tru HD Steel, Utility Steel, Performance Door System Plus+); JELD-WEN (Custom Wood, Authentic Wood, Aurora Custom Fiberglass, Architectural Fiberglass, Design-Pro, Smooth-Pro, FiberLast, Steel); Pella (clad wood with glass, solid clad wood, solid fiberglass, fiberglass with glass, solid steel, aluminum-clad wood)
Interior doors	Masonite (Logan, Lincoln Park, Livingston, Riverside, Cheyenne, Winslow, Berkley, Melrose Flush, 2 Panel Roman, 2 Panel Arch-Top, 2 Panel Square, 4 Panel, 6 Panel, Flush); JELD-WEN (Birkdale, Caiman, Camden, Carrara, Colonist, Conmore, Craftsman III, Madison, Monroe, Rockport, Santa Fe)
Roofing	GAF (Timberline HDZ, UHDZ, NS, Camelot II, Grand Sequoia, Slateline, Royal Sovereign); Owens Corning (TruDefinition Duration, Oakridge, Duration Designer, Duration FLEX, Duration STORM, Supreme); CertainTeed (Landmark, Landmark Pro, Landmark Premium, XT 25, Presidential Shake, Grand Manor); Atlas (Pinnacle Pristine, StormMaster Shake, ProLAM, Briarwood Pro, Pinnacle Sun, Pinnacle Impact, Castlebrook); IKO (Dynasty, Cambridge, Cambridge Cool Colors, Nordic, Dynasty Cool Plus, Royal Estate, Crowne Slate, Armourshake, Marathon); TAMKO (Heritage, Titan XT, StormFighter FLEX, HailGuard, Elite Glass-Seal)
Siding	James Hardie (HardiePlank, HardiePanel, HardieShingle, HardieTrim; 13 colors); LP SmartSide (Lap, Nickel Gap, Shake Lap, Panel & Vertical, Trim & Fascia, Soffit; 16 colors); Ply Gem / Mastic / Variform; Royal; CertainTeed vinyl
Flooring (unverified)	LifeProof (Sterling Oak, Seasoned Wood); Pergo (Outlast+); COREtec (Plus, Pro Plus); Shaw (Floorte); Mohawk (RevWood); Armstrong (vinyl, LVT)
Countertops (unverified)	Silestone, Caesarstone, Cambria, MSI Q Quartz, Formica, Wilsonart, each with a short color list (Calacatta Gold, Calacatta Nuvo, Brittanicca, Calacatta Laza, etc.)
Plumbing fixtures (unverified)	Kohler (Cimarron, Highline, Wellworth, Santa Rosa, Memoirs); Moen (Arbor, Align, Brantford, Adler, Genta); Delta (Trinsic, Cassidy, Leland, Essa, Windemere); American Standard (Cadet, Champion, Studio); Gerber (Viper, Maxwell); Pfister (Pasadena, Ladera)
Appliances (unverified)	GE, GE Profile, Café, Whirlpool, KitchenAid, Frigidaire, Frigidaire Gallery, LG, Samsung, Bosch, Maytag — each with product types: range 30"/36", cooktop 30"/36", wall oven 27"/30", dishwasher 18"/24", refrigerator 30"/33"/36", over-the-range microwave 30", range hood 30"/36"
Paint	Sherwin-Williams (41 colors with SW codes, hex and LRV); Benjamin Moore (43 colors with codes); Behr (41 colors with codes)

4 Choosing a cabinet vendor, line, door and finish

In any cabinet's panel, **Cabinet maker**:

- **Brand**: Generic (no brand) or a maker (with its tier). Choosing a brand selects its first line.
- **Line**: the brand's lines with frame type.
- **Door style**: doors available in that line (with construction kind). Choosing a door sets the 3D door style from its construction (slab → flat, raised/cathedral/arch/mitered → raised panel, inset → inset, beadboard, louver, otherwise shaker).
- **Finish**: the brand's finishes with type (paint/stain). The finish name sets the screen color by its words (whites, linens, greiges, grays, greens, blues, blacks, espresso/walnut browns, cherry/cognac reds, honey/caramel ambers, natural/maple, oak/hickory/driftwood), unless you set a custom color.
- **Price book**: "Planning prices" or any imported price book (shows the SKU count). The line's frame (framed/frameless) is also applied.
- A note shows "Made by …; sold at …", the line's overlay and notes, the door material, catalog links, and an **Open** button for a catalog file you imported.

5 File › Import › Vendor catalog… (the import dialog)

"Bring in a manufacturer's catalog or price list as Excel, CSV, PDF or JSON. Door styles, finishes, lines and colors join the brand lists; SKUs with prices go into a price book that prices the cabinets you draw. Catalogs stay on this computer and show up in every project."

Fields - **What is it?** — Cabinet catalog (lines, door styles, finishes, SKUs); Cabinet price list (SKUs and prices only); Windows catalog; Exterior doors catalog; Interior doors catalog; Roofing catalog; Siding catalog; Flooring catalog; Countertops catalog; Plumbing fixtures catalog; Appliances catalog; Paint colors (name, code, hex). - **Brand name** — filled from the file name (with "price list", "price book", "catalog" and years removed); edit it. - **Choose catalog file…** — accepted: `.csv`, `.tsv`, `.txt` (comma- or tab-separated), `.xlsx`, `.xls`, `.xlsm`, `.ods`, `.pdf`, `.json`. - **Download a template** — saves a starter CSV for the chosen type. - **Sheet** (spreadsheets with several sheets) and **Header row** (auto-detected among the first 15 rows by best column match; change it if wrong). - **Match the columns** — one drop-down per field, pre-guessed from the header words; "(not in this file)" leaves a field out. A live count shows what was found ("3 lines · 12 door styles · 8 finishes · 140 priced SKUs found in 160 rows") and a preview table shows the first 8 rows. - **Add to library**, **Done**. - **Your imported catalogs** — each imported cabinet brand (door and finish counts, "priced" if it has a book) and product brand, with **Open file** (reopens the original file, which is kept on this computer) and ✕ to remove.

Columns by catalog type (header words that are recognized automatically; matching is case-insensitive, exact word first, then contains):

Type	Field	Recognized header words
Cabinet catalog	Line or series	line, series, collection, product line
	Door style	door, style, door style, door name
	Door type	door type, construction, panel, kind
	Material or species	material, species, wood
	Finish or color	finish, color, colour, stain, paint
	Finish type	finish type, paint/stain, type of finish
	SKU	sku, item #, item, code, model, part
	Description	desc, description, name
	Cabinet type	cabinet type, category, class, type
	Width / Height / Depth	width, w / height, h / depth, d
	Price	price, list, cost, msrp, amount, retail
Cabinet price list	SKU, Description, Cabinet type, Width, Height, Depth, Price	as above (type: type, category, class)
Paint colors	Color name	name, color name, color, colour
	Color code	code, number, #, id
	Hex color	hex, html, rgb hex, web
	LRV	lrv, reflectance
Any product catalog	Product line or style	line, series, collection, style, product, model, name
	Color	color, colour, finish
	Notes	notes, description, desc, material, type, size
	SKU	sku, item, model #, part
	Price	price, cost, msrp

Templates (what Download a template gives you):

```
Cabinet catalog:
Line,Door style,Door type,Material,Finish,Finish type,SKU,Description,Cabinet
type,Width,Height,Depth,Price
Classic,Hampton,Shaker,Maple,Linen White,Paint,B15,Base 1 door,Base,15,34.5,24,289
Classic,Hampton,Shaker,Maple,Espresso,Stain,W3030,Wall 2 door,Wall,30,30,12,325

Cabinet price list:
SKU,Description,Type,Width,Height,Depth,Price
B15,Base 1 door,Base,15,34.5,24,289
SB36,Sink base,Sink,36,34.5,24,412
W3030,Wall 2 door,Wall,30,30,12,325

Paint colors:
Color name,Code,Hex,LRV
Agreeable Gray,SW 7029,#d1cbc1,60

Product catalog:
Product line,Color,Notes,SKU,Price
400 Series Double-Hung,White,Fibrex and wood,TW2846,612
```

How rows become a catalog - Lines: every distinct Line value. Door styles: every distinct Door style, tagged with the lines it appears in, its material, and a construction guessed from the Door type or name (shaker; slab/flat/plain/euro/modern → flat; inset; bead → beadboard; mitered; cathedral/arch; louver; raised; recess → shaker; otherwise other). - Finishes: every distinct Finish. Type comes from the Finish type column (paint/opaque/solid → paint; stain/glaze/natural/clear → stain); without it, names containing white, gray, grey, graphite, charcoal, slate, blue, navy, green, sage, black, onyx, cream, linen, ivory, alabaster or paint are treated as paint, everything else as stain. - Price book SKUs: rows with both a SKU and a price. Cabinet type from the type column (sink → sink base; wall/upper → wall; tall/pantry/util/oven → tall; vanit → vanity; fill → filler; base/drawer → base) or inferred from the SKU: `SB36` / `SBA36` sink base, `DB18` / `BD18` drawer base, `B36` / `BB` / `BBC` / `BLB` / `BWB` base, `W3030` or `W3030-24` wall (width, height, optional depth), `U248496` / `T` / `P` / `PC` / `OC` + width/depth/height tall, `V3021` / `VSB` vanity (width, optional depth), `BF3` / `WF3` / `F3` filler. Missing heights default to 34.5 (base, sink, vanity, filler), 30 (wall), 84 (tall); depths 24 (base/sink/tall/filler), 12 (wall), 21 (vanity). Rows without a width are skipped; duplicate SKUs keep the first. A "Type" containing "drawer" sets a 3-drawer layout. - Cabinet catalogs are stored as a brand named from the Brand field (tier "imported"); its price book is " price list". A price-list import makes only the book, named for the brand. - Paint: rows with a name; hex must be `#rrggbb` (a missing `#` is added; anything else is dropped so the color has no swatch); code and LRV optional. - Other product catalogs: each distinct Product line becomes a series carrying the colors and the first notes seen with it; with no lines, the colors apply to the brand.

Excel files are read sheet by sheet (empty rows dropped). **PDF price books**: text is read page by page (up to 400 pages) and rebuilt into rows; runs of text more than about 14 points apart become separate columns, so tabular pages come through best ("Read the text of the PDF. Tables come through best; check the columns below. The PDF itself is kept so you can open it from the cabinet panel."). The paint PDF sample reads as columns Color name / Code / Hex with rows such as "Harbor Fog, HF 101, #9aa7ad".

JSON files use the program's own format and are added as-is after pressing Add to library:

```
{
  "brands": [
    {
      "id": "kraftmaid", "name": "KraftMaid", "parent": "...", "tier": "semi-custom",
      "soldAt": ["dealers", "Home Depot", "Lowe's"],
      "lines": [{"name": "KraftMaid Vantage", "frame": "framed", "overlay": ["half","full"],
"notes": "..."}],
      "doors": [{"name": "Amhurst", "lines": ["KraftMaid","KraftMaid Vantage"], "kind":
"shaker", "material": "", "notes": "Full Overlay; ..."}],
      "finishes": [{"name": "Dove White", "type": "paint", "code": "..."}],
      "url": ["https://..."], "sources": ["https://..."]
    }
  ],
  "windows":   [{"brand": "Andersen", "series": [{"name": "400 Series", "frame": "...",
"ops": ["double-hung","casement"], "notes": "..."}], "url": "..."}],
  "extDoors":  [{"brand": "...", "series": [...]}],
  "intDoors":  [{"brand": "...", "series": [...]}],
  "roofing":   [{"brand": "...", "series": [...] }],
  "siding":    [{"brand": "...", "series": [...], "colors": ["Arctic White", "..."]}],
  "flooring":  [...], "countertops": [...], "plumbing": [...], "appliances": [...],
  "paint":     [{"brand": "Sherwin-Williams", "colors": [{"name": "Agreeable Gray", "code":
"SW 7029", "hex": "#d1cbc1", "lrv": 60.4}]}]
}
```

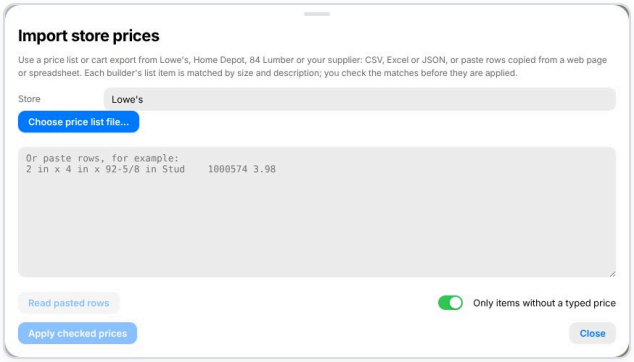
A bare array of cabinet brands also works. Each cabinet brand needs `name` and `doors` or `lines`; an `id` is made from the name if missing. Product categories use the keys windows, extDoors, intDoors, roofing, siding, flooring, countertops, plumbing, appliances, paint; each entry needs a `brand`. Sub-list keys per category: windows/extDoors/intDoors use `series`, roofing/siding/flooring use `lines` (series also accepted), countertops use `colors`, plumbing `collections`, appliances `types`; entries can be plain strings or `{name, colors[], notes, frame, material, type, unv}`. A door's `kind` uses the construction names above; a finish `type` is paint or stain. On load the dialog reports "Found N cabinet brands and N product brands in the file. Press Add to library."

After import - "Added : 3 lines, 12 door styles, 8 finishes, 140 priced SKUs. It is in the brand lists now, and its SKUs are in the Library and price the cabinets." - Imported cabinet brands appear in the **Cabinet maker › Brand** list and the Selections cabinet note; imported product brands in their Selections section; paint brands in Selections › Paint colors. A brand with the same id/name as a built-in or earlier import replaces it. - Price-book SKUs appear in the Library under **Cabinets › ": Base / Sink bases / Wall / Tall / Vanities / Fillers"**, labelled "SKU · description · $price". Placing one creates a cabinet carrying that SKU, price and book; any cabinet whose generated SKU matches a book entry (case- and space-insensitive) is priced from the chosen book. - **Remove**: ✕ in "Your imported catalogs" removes the brand and, for a cabinet brand, its price book; price books alone can be removed in Library › Cabinets › Manufacturer price books. - **Update**: import again with the same brand name; it replaces the earlier catalog. - Storage: catalogs live in this computer's browser storage; an oversize file gives "The catalog is too big to keep on this computer. Split it into smaller files." or "The price list is too big to keep on this computer."

6 Manufacturer price books (Library › Cabinets)

A simpler CSV-only path: name the book (e.g. "Wolf Classic 2026"), click **Import CSV...** (.csv or .txt, comma or tab). "Columns it looks for: SKU (required), Description, Type, Width, Height, Depth, Price (required). Sizes are read from SKUs like B36, SB36, DB18, W3030 and U248496 when the columns are missing." Errors: "Could not find a SKU or item column. Name the first column "SKU"." / "Could not find a price column. Name it "Price"." / "No cabinets were recognized. Check that the SKU, width and price columns have values." Success: "Imported N cabinets from . Find them in the Library under Cabinets." Each book is listed with its item count and a remove ×.

7 Store prices (Cost › Estimate › Store prices, or Designer menu "Store prices...")



Store prices.

"Use a price list or cart export from Lowe's, Home Depot, 84 Lumber or your supplier: CSV, Excel or JSON, or paste rows copied from a web page or spreadsheet. Each builder's list item is matched by size and description; you check the matches before they are applied."

- **Store**: type any name; suggestions Lowe's, Home Depot, 84 Lumber, Menards, Local supplier.
- **Choose price list file...**: `.csv`, `.txt`, `.xlsx`, `.xls`, `.json`.
- **Paste rows** (tab-separated or comma-separated, one per line, e.g. `2 in x 4 in x 92-5/8 in Stud 1000574 3.98`), then **Read pasted rows**.
- Header words: SKU — sku, item #, item#, item number, model, model #, store sku, internet #, part, part #, product id, upc; Description — description, desc, product, product name, name, item, title, item description; Price — price, unit price, each, cost, sale price, your price, amount, price each; Unit — uom, unit, unit of measure, um, sold by. Without a header row, the longest text cell is the description and the last money-looking cell the price. JSON: an array or `{"items":[...]}` with `sku` / `model` / `id`, `description` / `name` / `title`, `price` / `cost`, `unit` / `uom`.
- The table lists each builder's-list item with a drop-down of up to six store items scored by word match (shown as a percentage); matches of 55% or better are pre-selected. **Only items without a typed price** hides items you already priced. **Apply checked prices** writes the prices

(and the store, SKU, unit and date as the source) to the Builder's list, estimate and shopping list.

8 KraftMaid catalog specifics

- The **KraftMaid (Lowe's 2022)** Library category holds every SKU from the KraftMaid (Lowe's) 2022 Specification Book (462 pages; 703 products, 7,158 unique SKUs; 29 door styles; 56 finishes; 149 construction options). "The book prints no prices. Confirm SKUs and options with the dealer before ordering."
- Subcategories follow the book's sections: Wall Cabinets (Wall Cabinets, Wall Corner, Wall End Run, Wall Specialty, Wall Open, Wall Microwave, Wall Hoods, Peninsula Wall and Corner, Wall Fillers, Wall Panels, Wall Shelving, Wall Accessories, Wall Decorative Accents, Valances, LED Lighting); Base Cabinets (Base Cabinets, Base Corner, Base Sink & Appliance, Base Specialty, Base End Run, Base Island, Base Peninsula, Peninsula Base Corner, Base Fillers, Base Panels, Base Accessories, Base Decorative Accents, Countertop & Shelf Supports); Tall Cabinets (Tall Cabinets, Tall Appliance, Tall Fillers, Tall Panels, Tall Accessories, Tall Decorative Accents); Vanity Cabinets (Vanity Cabinets, Vanity Sink, Vanity Tall, Vanity Wall, Fillers, Panels, Accessories, Decorative Accents); Office & Furniture (Office Cabinets, Bookshelves, Wood Tops, Furniture Fillers, Panels & Shelves, Accessories, Decorative Accents); Passport (accessible line: Passport Cabinets, Base Sink & Appliance, Passport Accessories); Panels, Shelving & Trays (Decorative Appliance Panels, Paneling, Roll-Out Trays, Shelf Kits); Molding, Accents & Hardware (Molding, Decorative Accents, Hardware).
- Cabinets (wall, base, tall, vanity, hood, office, filler SKUs with a size) place as cabinets with the real SKU; their label is "Title · W x H x D" and the note gives the book page ("Spec book page 212.") and "Some sizes are standard sizes, not printed on the page." where the book omitted a dimension. Wall cabinets hang at 54" (or so the top is at 90" for short ones); hoods at the height that puts the top at 84" plus 12"; vanity wall cabinets at 48".
- Panels, moldings, hardware, trays and accessories are spec items: **Add to selected cabinet** attaches the SKU to that cabinet (listed in its panel with Remove), **Add to order** adds a loose line; both appear under "Moldings, panels and hardware" on the Cabinet order.
- A placed KraftMaid cabinet shows a **KraftMaid** block: Hinge (Automatic / Right (R) / Left (L)) for SKUs the book hands (the order code substitutes R/L, RR/RL or LR/LL as the book does); **Order code** (SKU plus any option codes); **Options on the book's chart for this cabinet** (checkboxes from the book's option-availability chart, e.g. APC All Plywood Construction 20%, CH Change Height 30%, with the upcharge shown); **Accessories**; and **Features from the book**.
- The book's 29 door styles join the KraftMaid brand as " (Lowe's)" (Bagley, Brenton, Briarwood, Carolton, Cayden, Dakota, Deveron, Durham, Elliot, Fox Ridge, Frederickton, Gentry, Gresham, Hartwell, Hoffman, Jamestown, Jaycey, Kirkland, Malibu, Meredith, Montclair, Parkhurst, Ruxton, Salem, Sonata, Stilton, Tuxbury, Vienna, Merrill) under a "KraftMaid (Lowe's)" line, with their door codes and page; its 56 finishes (32 paints, 21 stains, 2 glazes, 1 other, e.g. Aged Bourbon E75, Canvas G68, Cannon Grey H01) join the finish list with their codes.

9 Paint brand color books

- Built in: Sherwin-Williams, Benjamin Moore and Behr (see counts above), each color with name, code and hex (SW colors also LRV).
- Import your own with **What is it? = Paint colors** from a CSV/Excel (columns Color name, Code, Hex, LRV) or a PDF color list laid out in columns, or in JSON (`"paint": [{"brand": "...", "colors": [{"name","code","hex","lrv"}]}]`).
- Apply: Cost › Selections and orders › Paint colors › choose the brand › pick a wall color and a trim/ceiling color per room. Wall choices set the room's wall color; trim/ceiling choices set the ceiling color (the trim color itself is the whole-house trim color set from the Library's Colors). The brand's colors show as swatches; "Default" clears a room back to the generic color.

PART

Kitchens and baths

Cabinets, countertops, tile, hardware, the NKBA guidelines and the kitchen drawings.

CHAPTER 12

Cabinets and kitchen/bath design

1 Cabinet types and nomenclature

Cabinet types: base, sink base, wall cabinet, tall, island, vanity, filler. Every cabinet gets a stock-style SKU in its panel header and on drawings, schedules and the order:

Cabinet	SKU pattern	Examples
Base	B + width; drawer bases DB + width; pull-out trash BWB + width	B36, DB18, BWB18
Sink base	SB + width; open-knee (ADA) SBA + width	SB36, SBA36
Wall	W + width + height, "-24" when 24" deep, "-GD" for glass doors	W3030, W3618-24, W3030-GD
Tall	U + width + depth + height; oven cabinet OC + …	U248496, OC309684
Vanity	V + width + depth; vanity drawer base VDB + width	V3021, VDB18
Island	B + width + "-ISL"	B72-ISL
Filler	BF + width (base), WF + width + height (wall)	BF3, WF330
Corner and specialty	Blind corner BBC + width + -L/-R; Lazy Susan LS + width; Diagonal wall corner WDC + width + height; Microwave WMC + width + height; Peninsula PEN + width	BBC42-L, LS36, WDC2430, WMC3018, PEN72

Single-door cabinets 24" or narrower carry the hinge side (B15L / B15R). The full order SKU adds modification codes (e.g. "B36 ROT2 SC"). You can type your own **SKU** and a **Price override** in the cabinet panel.

Cabinets are **numbered** automatically per level, room by room, clockwise around each room with base cabinets before wall cabinets, so numbers read in order across the elevations ("Cabinet #4 · B36").

2 Cabinet panel (select a cabinet)

- Header: Cabinet #n · SKU; **Price** (from the price book, or "(planning price)").
- **Layout** by type: base — Drawer over doors, Full-height doors, 3 drawers, 4 drawers, Open shelves, Pull-out trash; sink base — False front over doors, Full-height doors, Open knee space (ADA); wall — Doors, Glass doors, Open shelves, Lift-up door; tall — Pantry doors (upper and lower), Wall oven opening, Full-height door; vanity — Drawer over doors, Doors, 3 drawers; island — Drawers over doors, 3 drawers; filler — Filler.
- **Hinge side** Left/Right (single-door cabinets up to 24").
- **Finished left end**, **Finished right end**; islands: **12" seating overhang on the back**.
- **SKU**, **Price override**.
- **Cabinet maker** (section 2).

- **Kitchen style (all cabinets)**: Line — Stock (builder grade), Semi-custom (default), Custom (price factors 1, 1.6, 2.6); Door style — Shaker (default), Raised panel, Slab (flat), Beadboard, Inset shaker, Louvered; Finish — Painted white (default), linen, greige, gray, sage, navy, black; Natural maple, Natural oak, Rustic hickory, Cherry, Walnut, Espresso; Custom color; Hardware — Bar pulls (default), Cup pulls and knobs, Knobs, Edge pulls, None (push to open); Hardware finish — Brushed nickel (default), Matte black, Satin brass, Oil-rubbed bronze, Polished chrome; **Crown on uppers** (on by default); **Light rail** (off). **Different style for this one** / **Match the kitchen** set or clear a per-cabinet override.
- **Countertops (all)**: Material — Quartz (default), Granite, Marble, Laminate, Butcher block, Solid surface, Concrete; Color; Edge — Eased (default), Bevel, Full bullnose, Ogee, Mitered 3"; Thickness — 2 cm (3/4"), 3 cm (1-1/4", default), 1-1/2"; Overhang (default 1"); Backsplash — 4" splash (default), 6" splash, Tile to the uppers, None. "Countertops build themselves over base cabinets, sink bases, vanities and dishwashers, with sink cutouts and a backsplash wherever they meet a wall." Islands with seating get a 12" overhang on the back; sink bases get an undermount cutout (up to 33 × 18); counters are grouped by top height (a cabinet can be given its own counter height).
- **Modifications** (20/20-style) with code and planning price at the stock line (multiplied by the line factor): Reduced depth RD $35 (added automatically when a cabinet is shallower than standard); Roll-out tray ROT $95 (quantity 1-6); Tray divider TD $45; Spice or utensil pull-out SPO $140; Waste basket kit (double bin) WBK $150; Cutlery divider CD $40; Peg drawer system PEG $110; Blind corner pull-out BCO $350 (blind corner cabinets); Sink tip-out tray TO $35; Matching interior MI $60 (wall/tall); Under-cabinet light prep ULP $25; Extended stile for scribing XS $30; Glass-ready mullion doors GFD $90; Soft-close upgrade SC $30 (only when soft-close is not standard); Appliance garage AG $260 (low wall cabinets). Finished ends add FEL/FER $45. **Attribute group**: Perimeter, Wall cabinets, Island, Tall cabinets, Vanities.
- **Job attributes (all cabinets)**: Wood or door material — Maple (x1), Oak (0.97), Cherry (1.12), Hickory (1.05), Birch (0.98), Walnut (1.25), Paint-grade MDF doors (0.95); Door overlay — Full (x1), Standard (partial) (0.95), Inset (1.2); Drawer box — Standard, Dovetail hardwood (1.06), Metal box (frameless) (1.04); **Soft-close doors and drawers standard** (on). **Two-tone and attribute groups**: a finish and door style for each zone (Wall cabinets, Island, Tall cabinets, Vanities) or "Same as the kitchen".
- **Moldings and decorative pieces (all)**: Crown profile — Cove crown 2-1/4", Crown 3-1/2" (default), Stacked crown 2-piece 5-1/2", Flat batten 1-1/2"; Toe kick skins (on); Scribe at walls (on); Valance over sink window; Panel-ready dishwasher; Refrigerator end panels; Island legs or posts — None, 2 turned legs (open end), 4 turned legs, 2 square posts, 4 square posts; Island back panel — Flat panel, Beadboard, Raised panels, Shaker panels (match doors, default). Toe kick is 4-1/2". These are counted on the Builder's list and the Cabinet order (crown and light rail in 8' sticks with 15% waste, toe kick skin, scribe molding, dishwasher end panels, refrigerator end panels 1-1/2 × 27 × 96, island back and end panels, corbels every 36" under seating, legs/posts, valance; hardware one pull per door/drawer; hinges and slides counted at no charge).
- **Frame type** follows the vendor line (framed/frameless; default framed).

3 Cabinet tools

- **Base cabinet run** and **Wall cabinet run** (interior tool group): click where the run starts against a wall and where it ends; the run fills with stock widths (36" down to 9") and a filler for what is left (wall cabinets 30" high at 54"). The ghost shows the layout before you click.
- **Add fillers** (cabinet panel) / **Fill the cabinet run** (Designer menu): closes gaps between 0.4" and 6" with filler cabinets.
- **Under-cabinet lights for this level** (Designer menu) adds a light under every wall cabinet.
- Corner checks: "Add a filler at the blind corner" (a cabinet butts straight into a blind corner; put a 3" filler and set the pull distance); "Dishwasher is tight to an inside corner" (keep a 3" filler or a cabinet between); "Doors are blocked at the inside corner" (use a blind corner or lazy Susan, or add fillers); "Refrigerator is tight to a wall" (leave 2"-4" or a filler and panel); "Exposed end ... is unfinished" (order FEL/FER or an applied panel).

4 Tile and wainscot (room panel)

- **Backsplash**: None; 4" splash; Up to the wall cabinets (18" or to the wall cabinet bottom); Full height to the ceiling — runs over continuous runs of base cabinets and ranges against a wall, starting at 36".
- **Tub or shower walls** (surround): None; 60" above the floor; 72"; 84"; 96"; To the ceiling — on the wall edges a tub, shower or roll-in shower sits against.
- **Wainscot**: None; 36"; 42"; 48"; 54"; All walls, floor to ceiling — around the room except at doors and surrounds.
- **Tile**: Subway 3" x 6"; Ceramic 4" x 4"; Porcelain 12" x 24"; Mosaic 1" sheets; Hexagon 2"; Stone 12" x 12"; wainscot can also be Beadboard (wood). **Wainscot material** can differ from the tile. **Tile color**.
- Windows inside a zone are deducted. The note reports square feet; zones show in 3D, on interior elevations (with tile grid and a label) and on the materials list (tile with 10-15% waste, thinset, grout, cement board and waterproofing for wet walls, edge trim; beadboard panels and chair rail cap).

5 NKBA kitchen and bath planning guidelines

Turned on in the project panel: **NKBA kitchen and bath planning guidelines** (on by default), **Cooks at once** — One (42" aisles) or Two or more (48" aisles), **Use Access Standard values**. Kitchens and baths are found by room use or by their fixtures. Messages appear in the checks list with the guideline number. Rule texts as the program states them:

Code	Finding	Text
K1 Door/entry	Kitchen doorway is narrow	About X clear opening; NKBA recommends at least 32" clear (a 2'-10" door) — 34" (a 3'-0" door) with the Access Standard.
K2 Door interference	Door swings into the ... opening	An entry door and an appliance door should not run into each other. Move the door, change its swing, or use a pocket door.
K3 Work triangle	Work triangle is long	Sink, cooking surface and refrigerator are X ft apart in total; NKBA recommends 26 ft or less.
K3	Work triangle leg short/long	X ft; NKBA recommends 4 to 9 ft for each leg.
K3	Island cuts into the work triangle	A triangle leg crosses the island for X; NKBA allows no more than 12".
K4	Tall cabinet separates two work centers	NKBA recommends no full-height, full-depth tall cabinet between two primary work centers.
K5 Work triangle traffic	Traffic path cuts through the work triangle	The straight path between two kitchen doors crosses the work triangle. Rearrange so through-traffic passes outside it.
K6	Work aisle is narrow	X between facing counters or appliances; NKBA recommends 42" for one cook (48" for two).
K7	Walkway is narrow	X from the ... to the wall; NKBA recommends a 36" walkway.
K8	Not enough room behind island seating	X from the seating edge to the nearest obstruction; NKBA recommends 32" with no traffic behind, 36" to edge past and 44" to walk past (Access: 36"/60"/60").
K9	Seating knee space is shallow	X overhang at a Y high counter; NKBA recommends 18" of knee space at 30", 15" at 36", 12" at 42", and 24" of width per seat.
K11	Sink landing area is short	About X on one side and Y on the other; NKBA recommends 24" on one side and 18" on the other.
K12	No 36" prep area next to the sink	The longest counter beside the sink is X; NKBA recommends a continuous 36" wide by 24" deep work area next to it.
K13	Dishwasher is far from the sink	X between them; NKBA recommends the dishwasher edge within 36" of the sink edge.
K13	No standing room beside the open dishwasher	X in front of it; leave 21" of standing space beside the open door.
K14	Second sink needs counter beside it	About X and Y; NKBA recommends 3" on one side and 18" on the other.
K15 (tip)	Plan two waste and recycling bins / one more bin	NKBA recommends two waste receptacles: one near the cleanup sink and one for recycling. Use a pull-out trash base or a waste-basket kit.

Code	Finding	Text
K16	Refrigerator needs a landing area	Provide 15" of counter on the handle side, or across from the front within 48".
K17	Cooking surface needs landing area	About X and Y; NKBA recommends 12" on one side and 15" on the other.
K17	Island cooktop needs counter behind it	NKBA recommends 9" of countertop behind a cooking surface on an island or peninsula, at the same height.
K18	Cabinet too close above the cooking surface	X to the cabinet above; allow 30" to unprotected combustibles (24" with a listed hood), per the appliance instructions and IRC M1901.1.
K19	No hood over the cooking surface	NKBA recommends a ducted hood of at least 150 cfm. Add a range hood or microwave hood above it.
K20	Cooking surface under a window	NKBA recommends no cooking surface below an operable window: curtains and reaching over burners are fire risks.
K20 (tip)	Fire extinguisher	Keep a fire extinguisher near the kitchen exit, 15" to 48" off the floor, away from the cooking surface.
K21	Microwave is mounted high / too low	Bottom at X; NKBA recommends 3" below the main cook's shoulder and no more than 54" off the floor / an under-counter microwave should sit at least 15" off the floor.
K23	Wall oven needs a landing area	NKBA recommends 15" of counter next to the oven, or across from it within 48" (not opening into a walkway).
K25	Not much countertop	About X" of 24"-deep countertop; NKBA recommends at least 158" of frontage.
K25	Wall cabinet is low over the counter	X from the countertop; NKBA recommends 15" of clearance above countertops.
K26 (tip)	Round the island corners	NKBA recommends clipped or rounded corners on countertops, especially islands in a walkway.
K27	Storage is short of the NKBA target	About X" of shelf and drawer frontage (estimated); NKBA recommends 1,400" for a small (under 150 SF), 1,700" for a medium (151 to 350 SF), 2,000" for a large (over 350 SF) kitchen.
K29 (tip)	Blind corner without a storage device	NKBA recommends at least one corner cabinet with a functional storage device. Add a blind-corner pull-out or use a lazy Susan.
K30	No countertop receptacles	Countertop receptacles need GFCI protection, and no point along a counter wall may be more than 24" from one (IRC E3901.4). Use the Electric tool to add them.
K31	No task lighting under the wall cabinets	NKBA recommends task lighting at every work surface. Add under-cabinet lights.
K31 Lighting	Kitchen window area is small	X SF of glass for Y SF of floor; NKBA recommends window area of at least 8% of the floor area (IRC R303.1).

Bath guidelines:

Code	Finding	Text

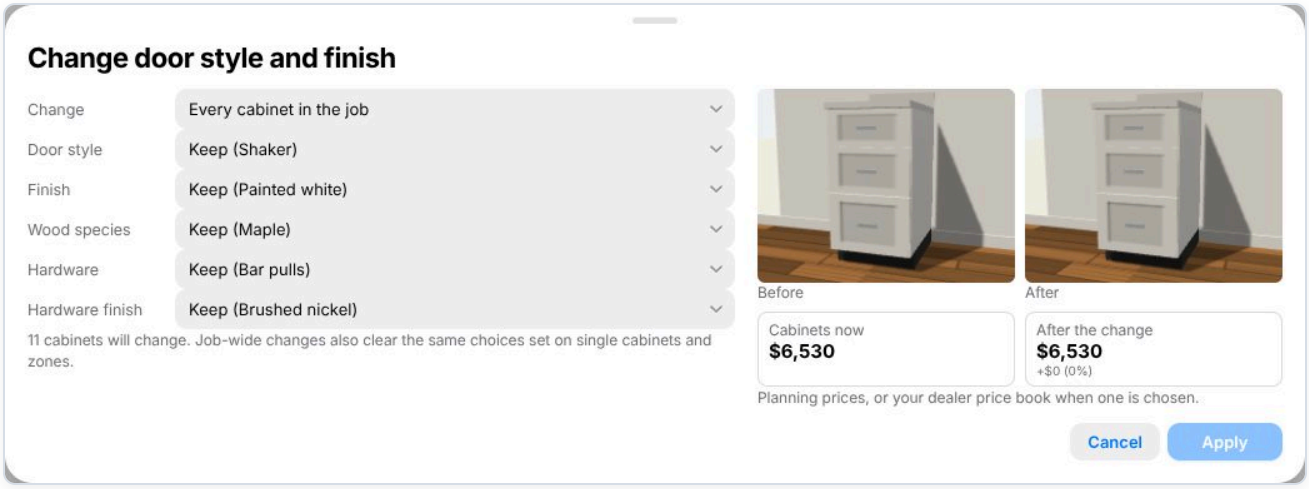
Code	Finding	Text
B-door	Bathroom doorway is narrow	About X clear; NKBA recommends at least 32" clear (34" Access).
B-door	Door swings into a fixture	NKBA recommends that no entry door interfere with a fixture or its clear floor space. Swing the door out, or use a pocket door.
B-clear	Clear floor space in front of the ...	X in front; NKBA recommends 30" (code minimum 21", showers 24").
B-wc	Toilet is close to a side wall or fixture	X from the toilet centerline; NKBA recommends 18" (code minimum 15").
B-wc	Toilet compartment is small	X x Y; NKBA recommends 36" x 66" (code minimum 30" x 60").
B-lav	Sink is close to a side wall	X from the sink centerline to the nearest wall or tall object; NKBA recommends 20" (code minimum 15").
B-lav2	Double sinks / two sinks are close together	X between bowl centers; NKBA recommends 36" (code minimum 30").
B-lav (tip)	Vanity height	Counter at X; NKBA suggests 32" to 43", about 3" below the user's elbow.
B-shower	Shower is small	X x Y inside; NKBA recommends at least 36" x 36" (code minimum 30" x 30", 900 sq in).
B-backing (tip)	Block the walls for grab bars	Install solid blocking at the toilet, tub and shower walls now, so grab bars can be added later (NKBA; ICC A117.1 604/607/608).
B-valve (tip)	Shower and tub controls	Place controls 38" to 48" high, offset toward the room, reachable from outside the spray; use pressure-balance or thermostatic valves (IRC P2708.4).

6 NKBA/KCD-style drawings

- Interior elevations draw cabinets with fronts, hinge marks, handles, the number bubble, crown on uppers and an appliance front labelled OVEN; a dimension string runs across the wall cabinets; each cabinet carries its code (B36, W3030...) in blue; centerlines (CL) of sinks, cooking surfaces, hoods, dishwashers, refrigerators, washers, dryers, toilets, lavatories, vanities, tubs and showers are drawn with a centerline dimension row from the wall end. These NKBA graphics standards can be turned off with the drawing-standard setting in My standards.
- The plan overlay shows the cabinet codes on each cabinet and "CL X" dimensions from the nearer wall end for each fixture centerline.
- Sheet **A5.0 Cabinet plan and schedule**: the cabinet area with numbered cabinets, counters, REF/DW/RANGE labels, a schedule table (#, SKU, Type, W x H x D, Style, Price) and notes (manufacturer, style, hardware, countertops, planning total).
- **Countertop shop drawing** (an A5 sheet): each counter piece (C1, C2...) with edge dimensions (W marks wall sides), SINK cutouts with centerline from the end, suggested SEAM lines at inside corners and where a run is longer than a slab, a pieces table (size, SF, cutouts, seams),

material: quartz 120 × 55, granite 110 × 62, marble 108 × 60, laminate 144 × 25, butcher block 96 × 25.5, solid surface 144 × 30, concrete 96 × 40.

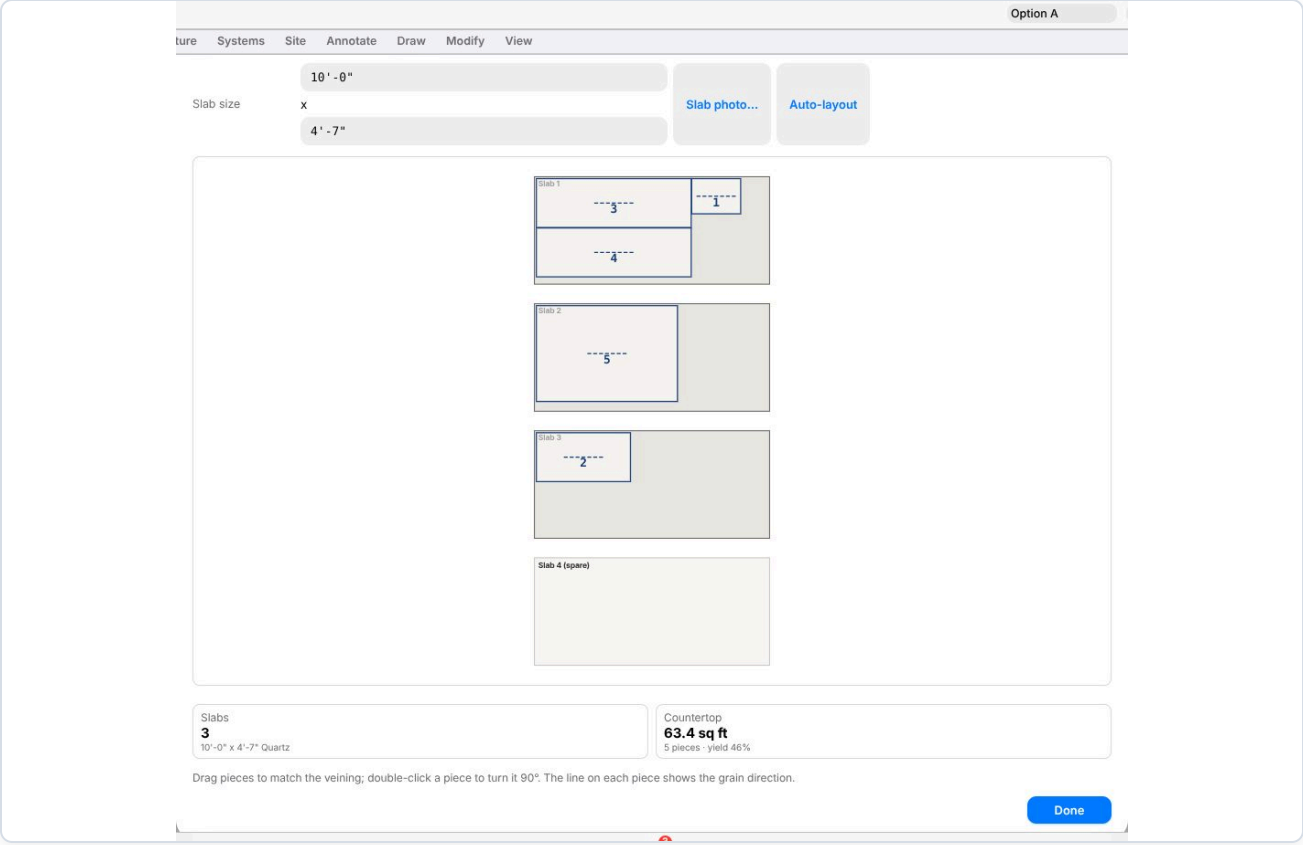
7 Change door style and finish (Style swap dialog)



Change cabinet door style and finish.

Designer menu › **Change door style and finish...** (also from the assistant). **Change**: Every cabinet in the job; Selected cabinets (when cabinets are selected); One room (pick the room); Perimeter base cabinets; Wall cabinets; Island; Tall cabinets; Vanities. Then Door style, Finish, Wood species, Hardware, Hardware finish (each defaults to "Keep (current)"); for the whole job with a vendor chosen, also the vendor's door and finish. A **Before / After** picture of a sample cabinet and **Cabinets now / After the change** totals with the difference and percentage (planning prices or the dealer price book). "Job-wide changes also clear the same choices set on single cabinets and zones." **Apply** (Undo puts them back).

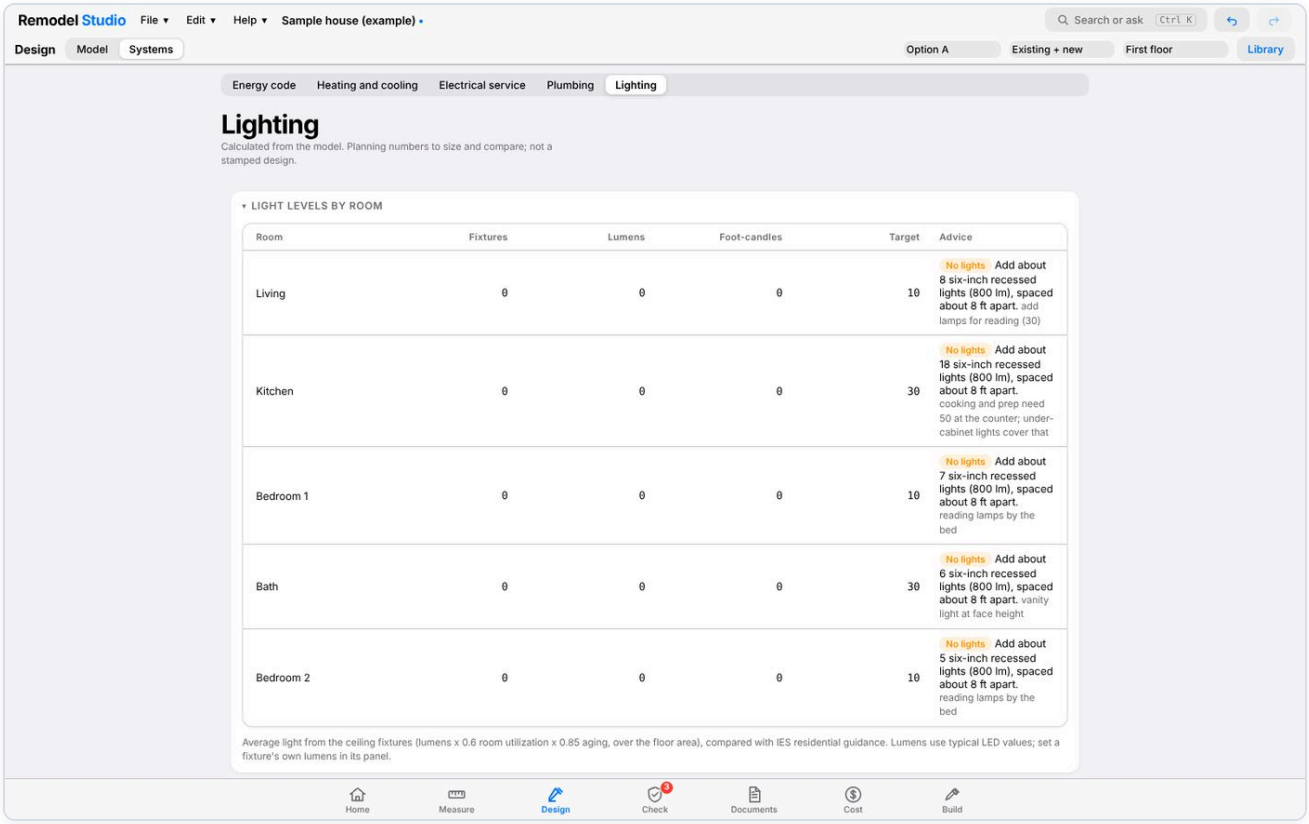
8 Countertop slab layout dialog



Countertop slab layout.

Designer menu › **Countertop slab layout...**. Counters are cut at their seams into pieces (pieces longer than a slab are split again) and laid on slabs. **Slab size** (defaults from the material, see above); **Slab photo...** (a photo of the real slab shows under the pieces; Remove photo); **Auto-layout** (longest pieces first, grain along the slab). Drag a piece to match the veining; double-click to turn it 90°; the line on each piece is the grain direction. Pieces that overlap or run off the slab show red; "N pieces are longer than a slab: add a seam on the countertop shop drawing." Stats: slabs used (size and material), countertop square feet, piece count and yield %. The layout saves with the project and prints as sheet **A5.4 Countertop slab layout** (slabs with numbered pieces, a pieces list, slab count and yield).

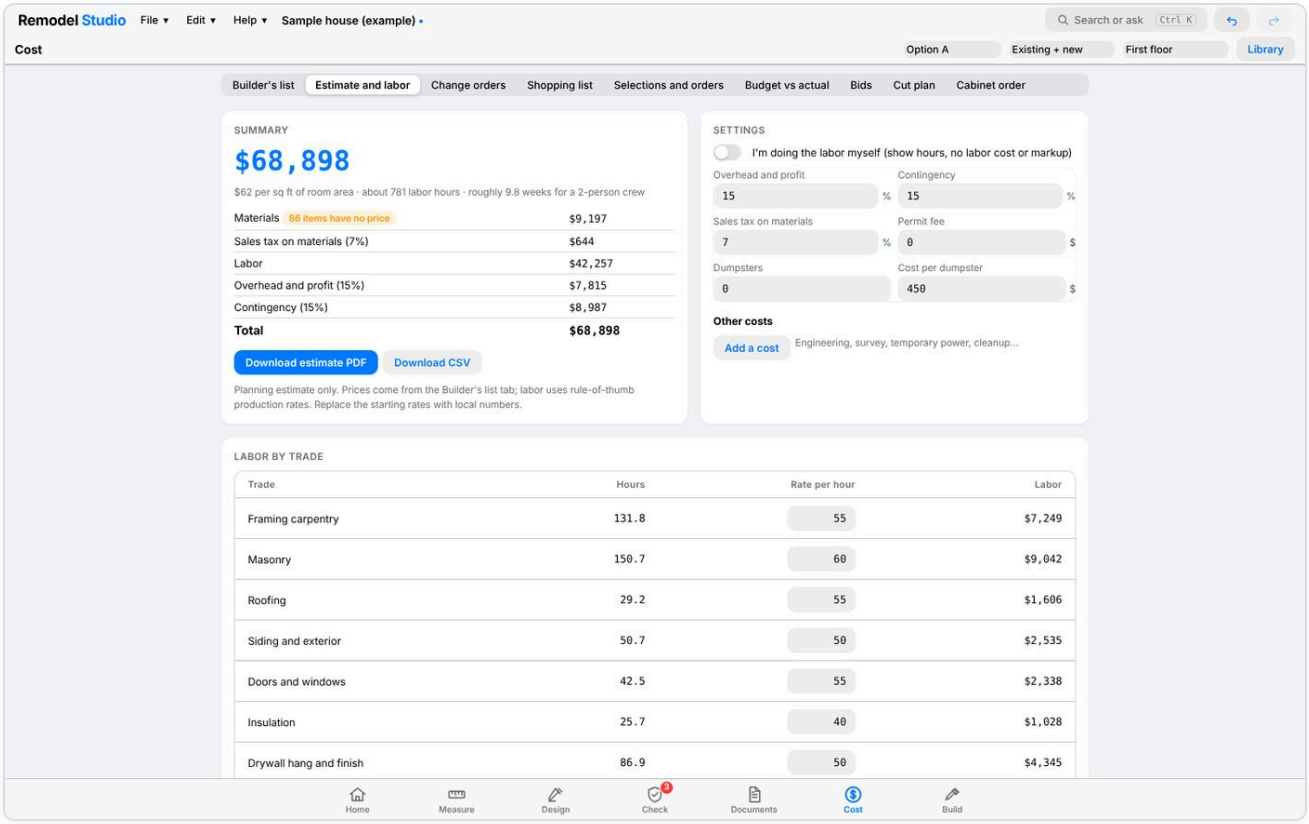
9 Lighting plan and schedule



Systems › Lighting.

Systems › Lighting tab: **Light color and marks** — fixture types get marks (A, B, C…) by type, lumens, color temperature and dimmable; **Set every light to** 2200 K (candlelight), 2700 K (warm white), 3000 K (soft white), 3500 K (neutral), 4000 K (cool white), 5000 K (daylight) › Apply; **Open the lighting schedule** (Documents › Schedules › Lighting, editable, with CSV columns Mark, Fixture, Qty, Lumens, Kelvin, Watts (LED est.), Dimmable, Switches, Rooms). **Switch groups** lists each switch (S1, S2…) with its type, room, linked light marks, estimated LED watts and "dimmer" ("Electrical › Link switches ties lights to the nearest switch"). The 3D view tints room light by the average color temperature of the lights in it.

10 Cabinet order (Cost › Cabinet order)



Cost › Cabinet order.

"Grouped by manufacturer, with modifications under each cabinet, the way a 20/20 order report reads. List prices come from your price books where loaded, otherwise planning prices." Settings: Dealer multiplier (0.45), Margin % (35), Freight % of net (3), Install $ per box (85), Sales tax % (7). For each manufacturer a table (#, Qty, SKU, Description with size and style/finish and price book, Hinge, List) with modification rows beneath; then **Moldings, panels and hardware** (qty, code, item, list); totals: List total, Net (x multiplier), Sell at margin, Freight, Installation (boxes x rate), Tax, Total. **CSV** and **PDF** buttons export the report ("Planning prices. List prices come from your price books where loaded; confirm with the manufacturer or dealer before ordering."). KraftMaid accessories added from the spec book appear as loose items.

11 Hardware whole-house finish

Set under Cost › Selections and orders › Door hardware, whole house (Finish and Brand). Every door's hardware, hinges, stops and kick plates follow it until overridden in the door's own Hardware section; window lock and crank finishes use the same finish list. Cabinet hardware finish is separate (Kitchen style › Hardware finish).

PART

Measuring

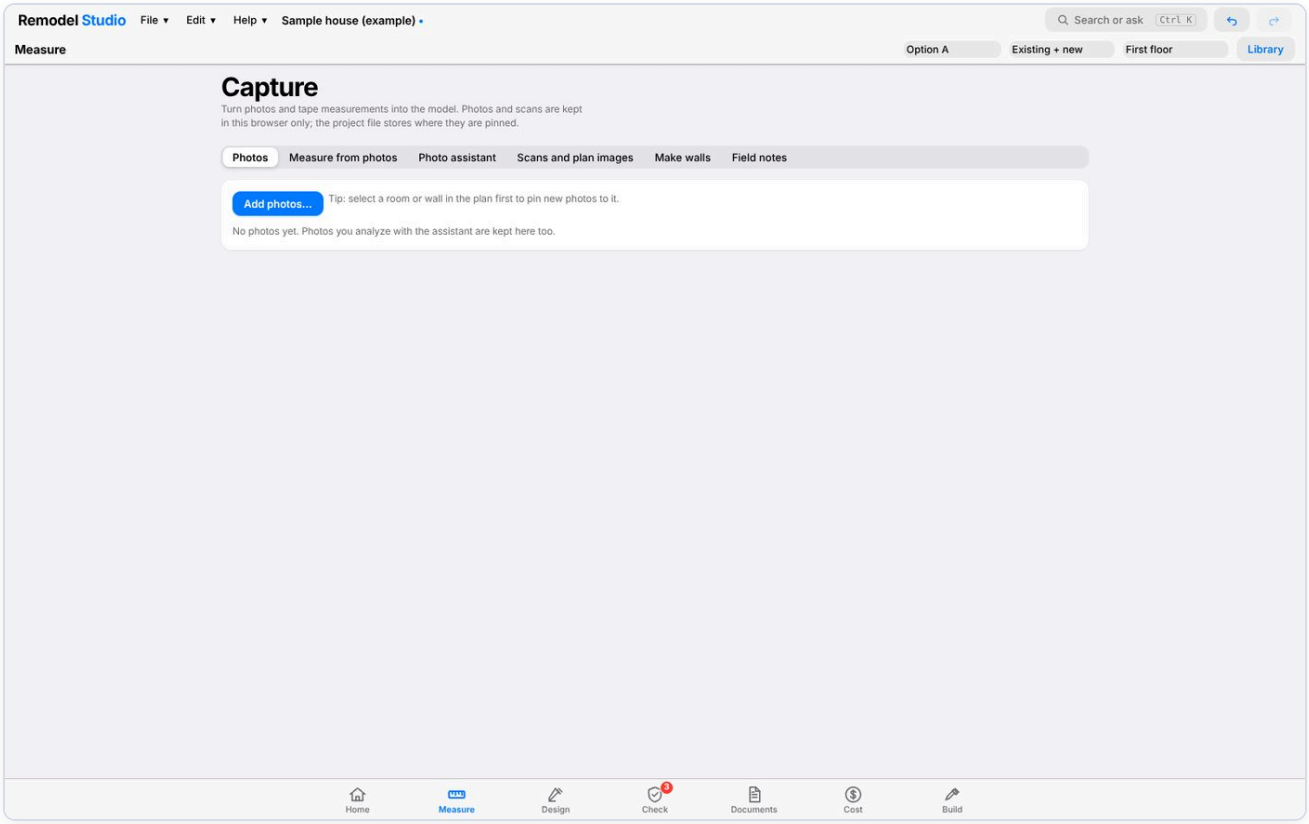
Photos, tape measurements, scans and plan images, making walls and field notes.

CHAPTER 13

The Measure workflow

Open with the **Measure** mode ("Photos and measurements"; "Photos, tape measurements, scans, underlays and field notes") or **File › Import › Photos and scans…**. The page is headed **Capture**: "Turn photos and tape measurements into the model. Photos and scans are kept in this browser only; the project file stores where they are pinned." Tabs: **Photos**, **Measure from photos**, **Photo assistant**, **Scans and plan images**, **Make walls**, **Field notes**. If the browser blocks storage: "This browser is not letting the app store photos (private window or storage turned off)."

1 Photos

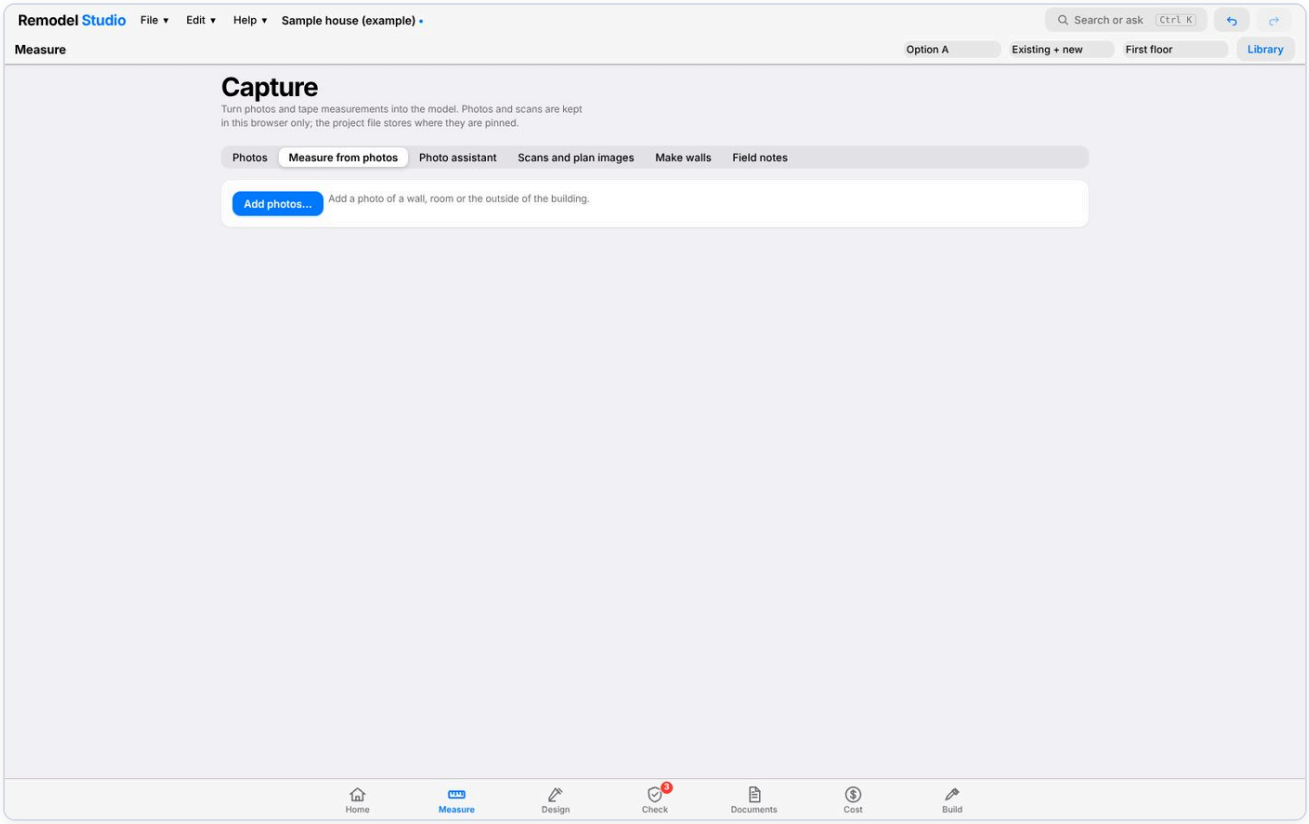


Measure › Photos.

- **Add photos…** (several at once; photos are reduced to 2000 px and stored in this browser). New photos are pinned to the item selected in the plan ("Tip: select a room or wall in the plan first to pin new photos to it").
- Gallery of photos, newest first, each with its pin label (room name, "exterior wall, 24'", a door/window mark, or "Not pinned" / "Pinned to a deleted item"), date, a **Note** field, **Pin to selection** and **Delete**.

- Pinned photos also show in the selected item's panel under **Photos**.

2 Measure from photos



Measure from photos.

Add or pick a photo (thumbnails; "scaled" pill once a scale is set). Modes above the photo: **1 · Set scale**, **2 · Measure**, **Pick color**, **Trim and finishes**, **Move**. Scroll to zoom, right-drag or Alt-drag to move, drag any dot to fine-tune; a magnifier shows where a click lands; a note reports the precision at the pointer ("2 pixels is about 1/8" on the scaled surface").

- **Set the scale** — **Known rectangle (best)**: click the four corners of a flat rectangle of known size in order top-left, top-right, bottom-right, bottom-left; this corrects the camera angle so everything on that same flat wall measures true. Pick a **Known size** preset (Standard door 36" x 80"; Door 32" x 80"; Door 30" x 80"; Garage door 16' x 7'; Plywood sheet 48" x 96"; Brick, 3 courses (8" tall) x 8" long; 8×16 concrete block 15-5/8 × 7-5/8; Outlet cover plate 2-3/4" x 4-1/2"; Letter paper 8-1/2" x 11"; Tape or stick laid flat 24" x 1") or type Width and Height (default 36 × 80). **Known length**: click both ends of something of known length (default 36"); best when the surface faces the camera straight on. Points status, **Clear scale**, **Next: measure**.
- **Measure** — click two points; the length appears on the photo. Each measurement has a name, a value, **Use for...** and delete. Use for... writes the value (rounded to 1/4") to the selected plan item: wall — Wall length, Wall height; door/window — width, height, sill (windows); fixture — Width, Depth, Height; always — Match baseboard by height, Match casing by width, Match

crown by height (picks the closest trim profile and applies it to the selected room or the whole house: "Closest match: Colonial baseboard 3-1/4", used for Kitchen."). "Measure on the same flat surface you scaled from; things closer or farther from the camera measure wrong."

- **Pick color** — click the wall, trim, door or siding; a small patch is averaged. Shows the hex, the closest library color, and the same apply buttons as a Library color (Room walls, Room ceiling, ..., All cabinets, All doors), plus **Use the library color instead**. **Correct lighting: click a white object** sets a white point so colors from that photo are corrected ("lighting corrected"); **Clear correction** removes it.
- **Trim and finishes** — when an AI service is available, **Identify trim and finishes** sends the photo (reduced to 1600 px) to it, and the assistant matches what it sees to the library: baseboard, casing, crown, chair rail, wainscot, door style and type, window operation and grilles, ceiling, wall finish, floor, siding, roofing, and estimated colors for walls, trim, ceiling, floor, door, siding, roof and cabinets, with a summary, notes (e.g. popcorn ceiling, water stains) and confidence (trim, finishes). Each row has a picture and up to three apply buttons (Whole house / This room only, Selected room / All rooms, etc.). When the program runs outside an AI service the panel explains that identification is unavailable and that trim can still be matched by measuring it.
- **Move** — drag to pan, scroll to zoom, **Fit photo**. **Delete photo** removes it.

3 Photo assistant

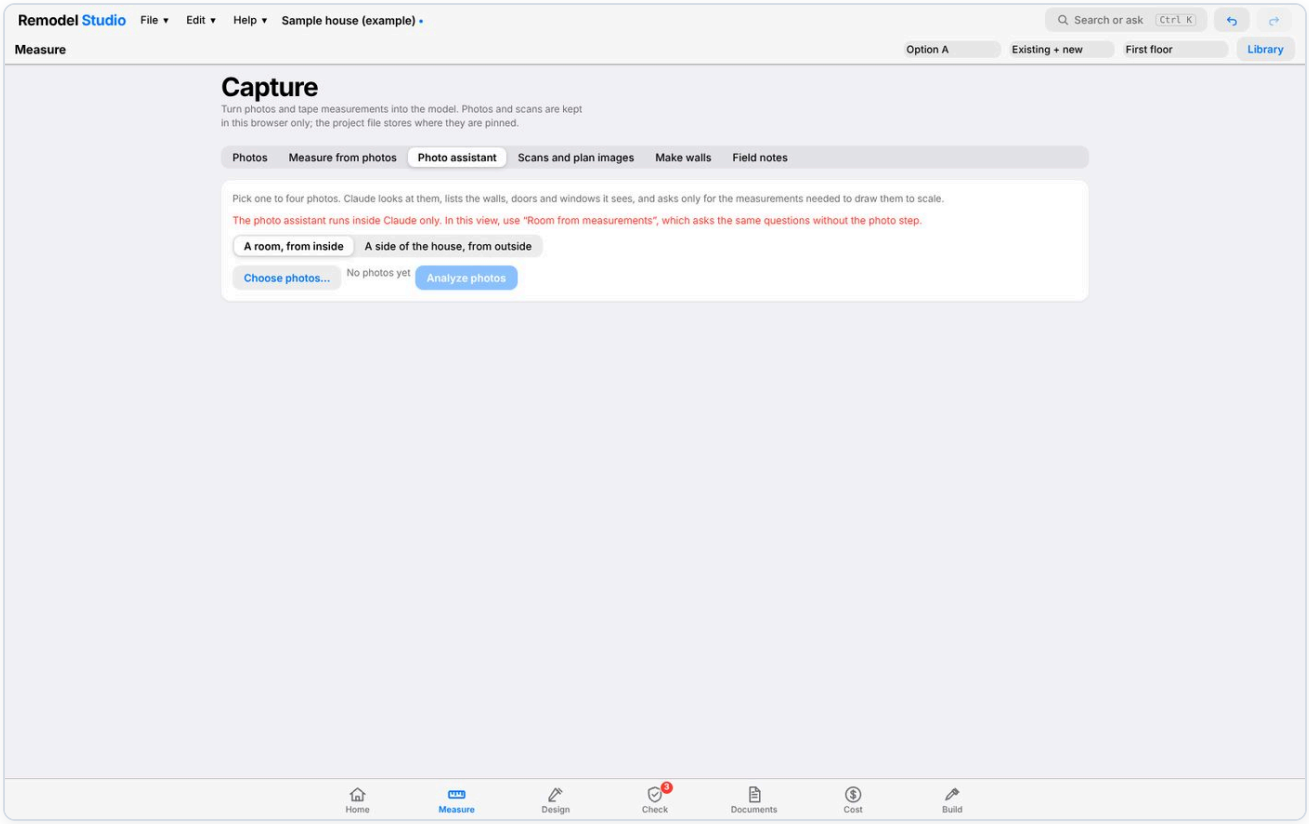
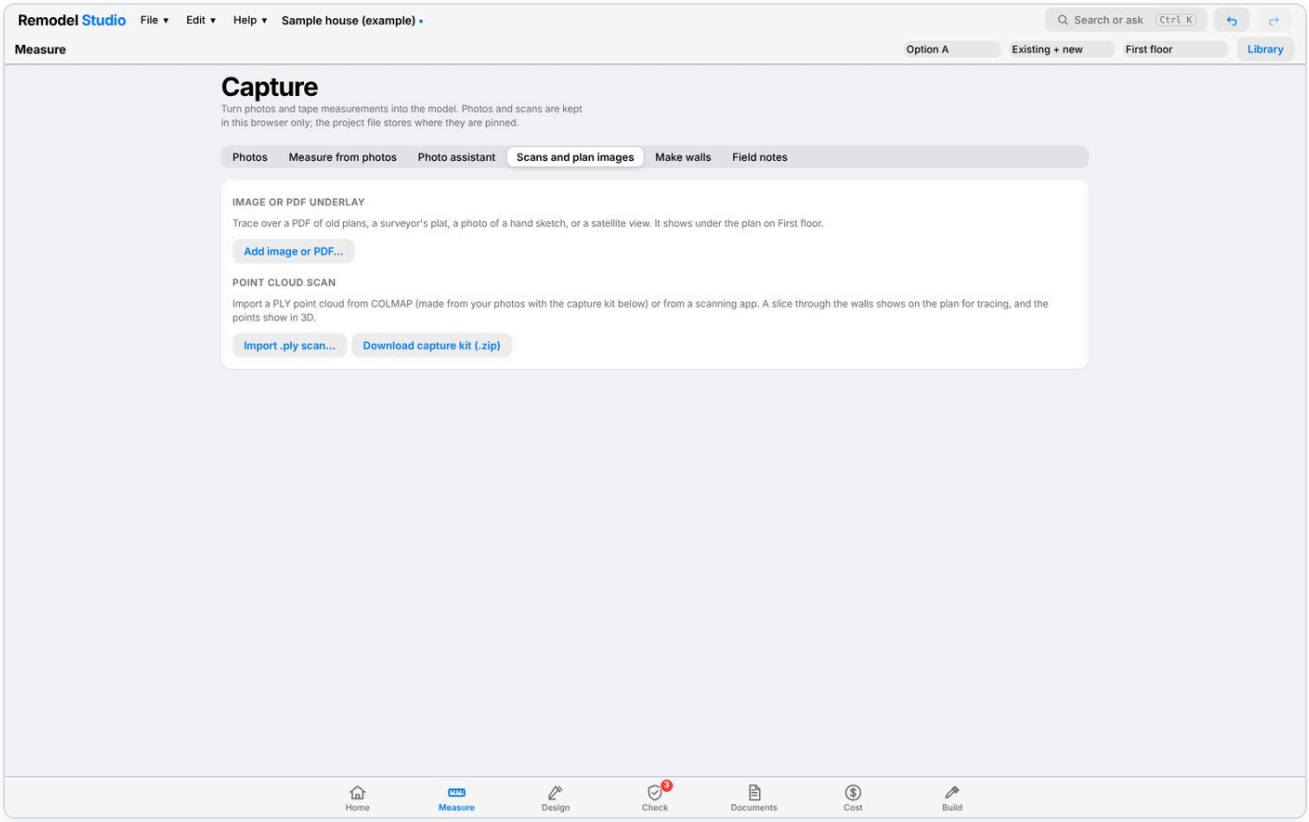


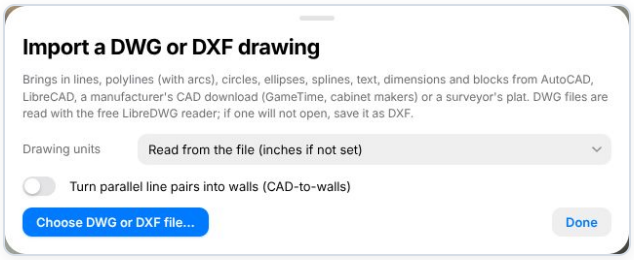
Photo assistant.

"Pick one to four photos. The assistant looks at them, lists the walls, doors and windows it sees, and asks only for the measurements needed to draw them to scale." Choose **A room, from inside** or **A side of the house, from outside** (for a side, select the outside wall in the plan first; openings are added to it). **Choose photos...** (up to 4), **Analyze photos** (**Stop** cancels). The result lists the room name, a summary and notes, then **Measurements needed**: room width (back wall), depth (left wall), ceiling height, and for each door/window its offset from the left corner, width, height and (windows) sill — "Offsets and sills left blank are treated as 0 and 36"." **Add room to plan** builds the room as existing work with the back wall at the top of the plan; **Add openings to wall** adds them to the selected wall. Error texts cover not granted, rate limited, stopped, unreadable photo. Outside Claude: "The photo assistant runs inside Claude only. In this view, use "Room from measurements"."

4 Scans and plan images (underlays)



Scans and plan images.

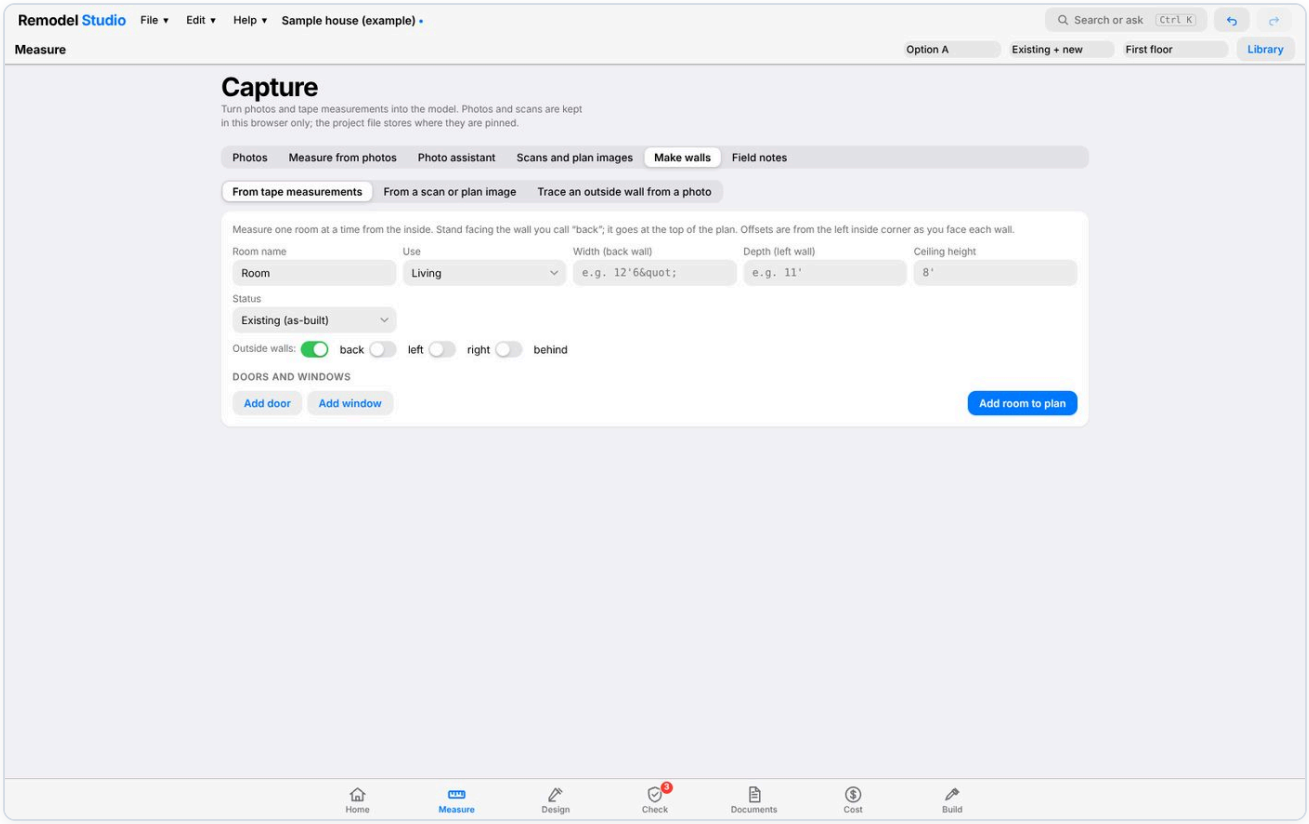


Import a DWG or DXF drawing.

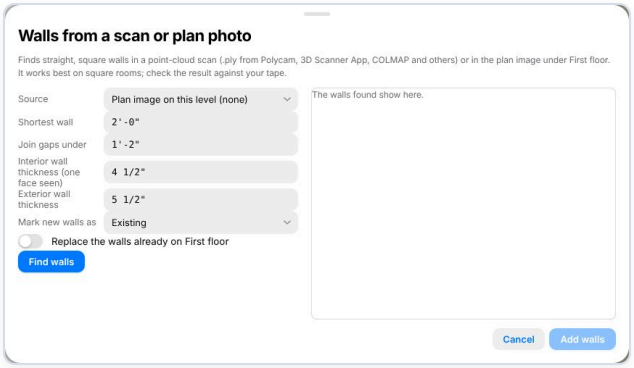
Image or PDF underlay — "Trace over a PDF of old plans, a surveyor's plat, a photo of a hand sketch, or a satellite view." **Add image or PDF...** (images are stored at up to 3000 px). For a PDF: pick the **Page** (1 to N) and the **Drawing scale** — "I will calibrate with two points", 1/4" = 1'-0", 3/16" = 1'-0", 1/8" = 1'-0", 3/8" = 1'-0", 1/2" = 1'-0", 1" = 10' (site), 1" = 20' (site), 1" = 30' (site) — then **Place page on the plan** (a printed scale places it at true size; otherwise it is fitted to the drawing). Calibration: click two points on the image, enter the **Distance between the two points**, **Set scale**. Controls: **Opacity** slider (default 0.5), **Width on plan**, **Rotate (degrees)**, **Open plan to trace** (use "Move underlay" in the plan to drag it), **Replace** / **Remove underlay**. One underlay per project, shown under the level it was added on.

Point cloud scan — **Import .ply scan...** (PLY from COLMAP, Polycam, 3D Scanner App and others; large clouds are thinned to about 150,000 points). Settings: **Scan units** Meters (default), Feet, Inches, Calibrated; **Up axis** Y up, Y down (COLMAP), Z up; **Rotate** 0-270° in steps; **Slice from floor** (12") and **Slice to** (72") — points between those heights are drawn on the plan as a red slice for tracing, and all points show in 3D. **Rescale scan**: enter a length measured on the plan with the Measure tool and its real length. **Remove scan**. **Download capture kit (.zip)** gives a README and a Python script that runs COLMAP on 40-150 overlapping photos to make cloud.ply (COLMAP 3.9+ with CUDA and Python 3.10+; `--quick` for a sparse cloud).

5 Make walls



Make walls.



Walls from a scan or plan photo.

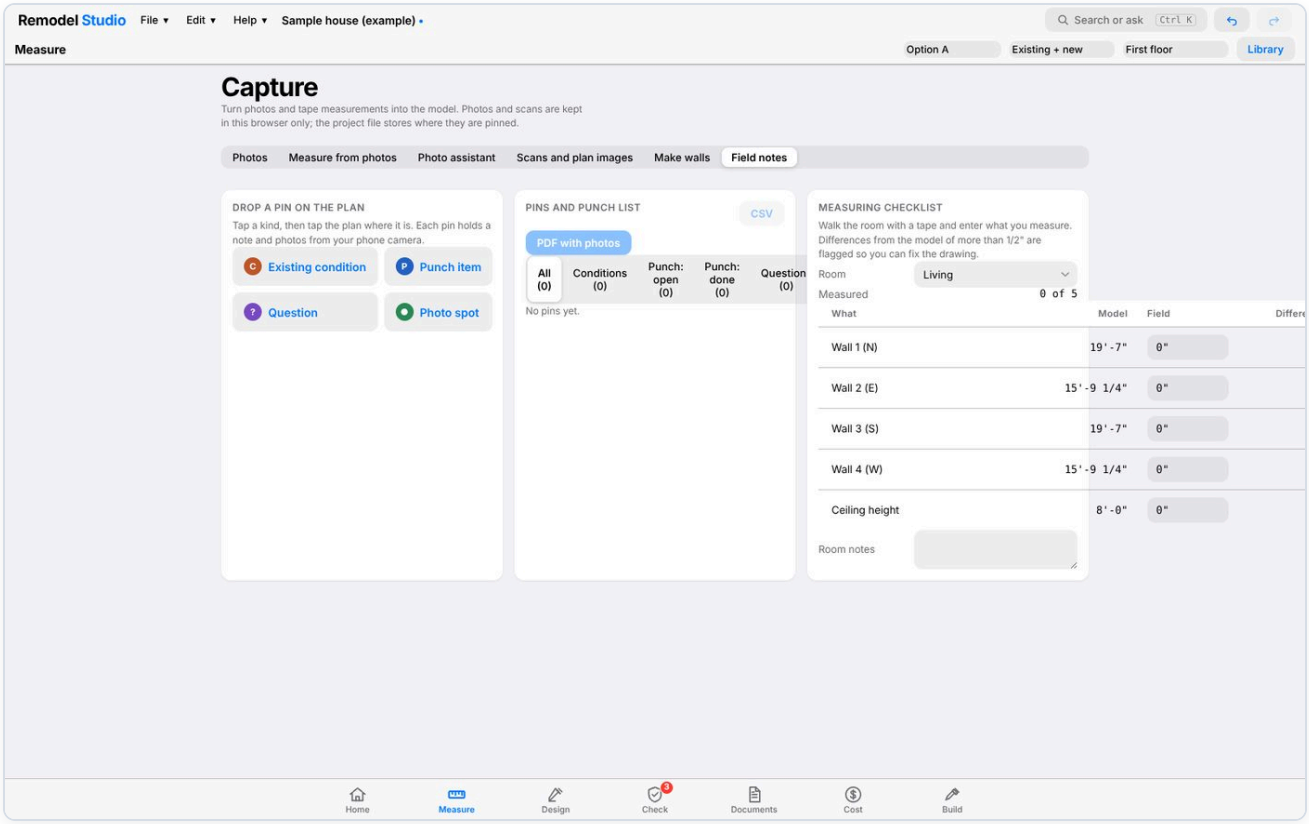
Three ways:

1. **From tape measurements** (the room wizard): "Measure one room at a time from the inside. Stand facing the wall you call "back"; it goes at the top of the plan. Offsets are from the left inside corner as you face each wall." Fields: Room name, Use, Width (back wall), Depth (left wall), Ceiling height, Status (Existing (as-built) / New construction), Outside walls (back, left, right, behind checkboxes; outside walls are 2×4 exterior 5-1/2", inside walls 2×4 partitions 4-

1/2"). **Add door** / **Add window** rows: Wall (Back, Right, Behind, Left), From left corner, Width (36), Height (80 / 48), Sill (36). **Add room to plan** places it to the right of existing walls (or sets the level height on an empty level), selects it and offers **Open the plan**.

2. **From a scan or plan image** — **Find the walls…** opens the dialog "Walls from a scan or plan photo": "Finds straight, square walls in a point-cloud scan or in the plan image under . It works best on square rooms; check the result against your tape." Source: Point-cloud scan or Plan image on this level; Shortest wall (24"); Join gaps under (14"); Interior wall thickness (one face seen) (4-1/2"); Exterior wall thickness (5-1/2"); Mark new walls as Existing/New; **Replace the walls already on** . **Find walls** reports "Found N walls, N doors and N windows. Main direction X°." with a preview (points, walls — exterior dark, interior blue — and openings). From a scan, gaps at both heights become doors (24"-76" wide) and gaps only up high become windows (16"-110"); from a plan image, door-sized gaps (24"-50") in a wall line become doors. **Add N walls** then "Check them against your tape measurements; then use Rooms › Auto rooms."
3. **Trace an outside wall from a photo** — "Stand back and take the photo as straight-on as you can. Pick the wall, then click its four corners on the photo (floor line at the bottom, top plate or soffit line at the top). The photo is straightened to the wall's real size, and you can box the windows and doors on it." Pick the **Wall** (each exterior wall by compass direction and length), **Take or choose photo…** (phone camera allowed) or use a saved photo. Step 1: click bottom left, bottom right, top right, top left (**Start over**). Step 2: on the straightened wall, drag a box over each window and door; choose **Windows**, **Doors** or **Doors if they touch the floor**; yellow dashes show openings already in the model; **Undo last box**; **Add N openings to the model** (as existing, doors swing, windows double-hung; check swing and style in each panel); **See it in 3D**. Needs outside walls drawn first.

6 Field notes

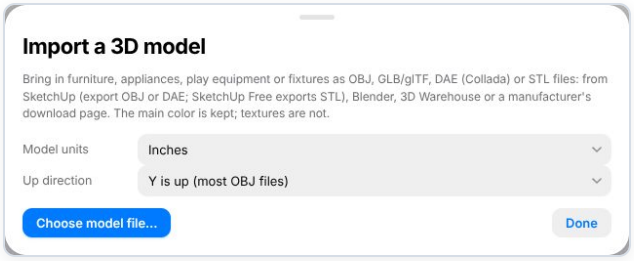


Field notes.

- **Drop a pin on the plan**: tap a kind, then tap the plan. Kinds: **Existing condition** (C, orange), **Punch item** (P, blue), **Question** (?, purple), **Photo spot** (●, green). The Field pin tool is also in the drafting tools ("tap where something is: a crack, a soft floor, a leak, a punch item or a question").
- Pin panel: Type; Note; punch items add Status (Open, In progress, Done — done date recorded), Trade (Any trade or a trade from the schedule's trade list), Due date, Assigned to; conditions add Severity (Note, Needs repair, Safety hazard); **Take or add photo** (phone camera; photos are pinned to the spot and show in the panel). Done punch pins turn gray with a check mark. Pins can be hidden with the field "hide pins" setting.
- **Pins and punch list**: filter All / Conditions / Punch: open / Punch: done / Questions (with counts); click a pin to jump to it in the plan; Done checkbox; **CSV** (#, Type, Room, Note, Status, Trade, Assigned, Due, Date); **PDF with photos** ("Field notes and punch list", each pin with its note, trade, assignee, due date and up to three photos).
- **Measuring checklist**: pick a room; the table lists each wall (numbered, with compass direction, inside face to face), ceiling height, and each door/window width, height and sill with the model value; type the field value; differences over 1/2" are flagged ("ok" otherwise); a Measured count and Room notes. "Draw rooms first (or use Room from measurements)."

- Voice notes are not part of the field tools; notes are typed.

7 Import a 3D model (File › Import › 3D model (OBJ, GLB, DAE, STL)...)



Import a 3D model.

"Bring in furniture, appliances, play equipment or fixtures as OBJ, GLB/glTF, DAE (Collada) or STL files: from SketchUp (export OBJ or DAE; SketchUp Free exports STL), Blender, 3D Warehouse or a manufacturer's download page. The main color is kept; textures are not." **Model units**: Inches, Feet, Meters, Centimeters, Millimeters. **Up direction**: Y is up (most OBJ files), Z is up (SketchUp, some CAD). **Choose model file...** (.obj, .glb, .gltf, .dae, .stl; a glTF that references separate .bin or texture files must be re-exported as one GLB). Very large models are thinned to about 200,000 triangles. The model is placed at the center of the plan as an "Imported 3D model" fixture, sized from its bounding box (rounded to 1/4"), selected, and reported ("Placed "chair": 24" x 26" x 34", 12,340 triangles. Drag it into place; size and color are in its panel."). A model smaller than 1 × 1 × 1/2 inch is refused with "Check the units." Models are stored in this browser with the project's assets.

8 Photo match (Designer menu / assistant: "Photo match")

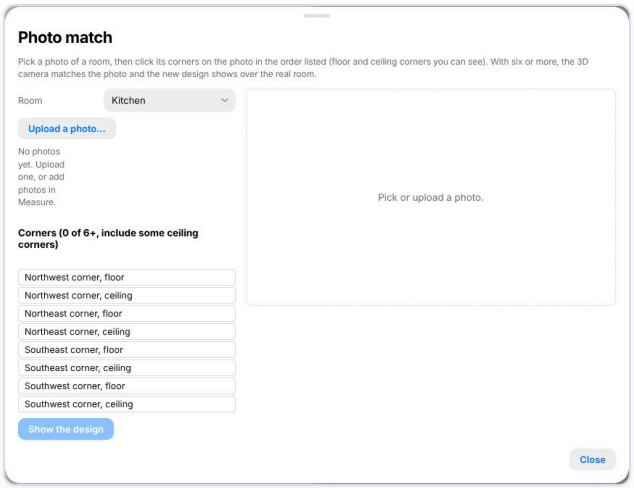


Photo match.

"Pick a photo of a room, then click its corners on the photo in the order listed (floor and ceiling corners you can see). With six or more, the 3D camera matches the photo and the new design

shows over the real room." Choose the **Room**, **Upload a photo…** or pick a saved one. The corner list names each room corner by direction ("Northwest corner, floor", "Northwest corner, ceiling", …); click a name then click it on the photo (the next unplaced corner becomes active; ✕ clears one). Include some ceiling corners (all points on one plane cannot be solved). A **Fit** line gives the average error in pixels ("recheck a corner or two" over 12 px). **Show the design** renders the model from the matched camera; view **Blend** (Design opacity slider), **Design**, **Photo**; **Save as PNG** exports the composite. The photo, room and corner points save with the project.

PART

Checking

The Check page and every code, accessibility, NKBA, structure, clash, aging-in-place and energy rule.

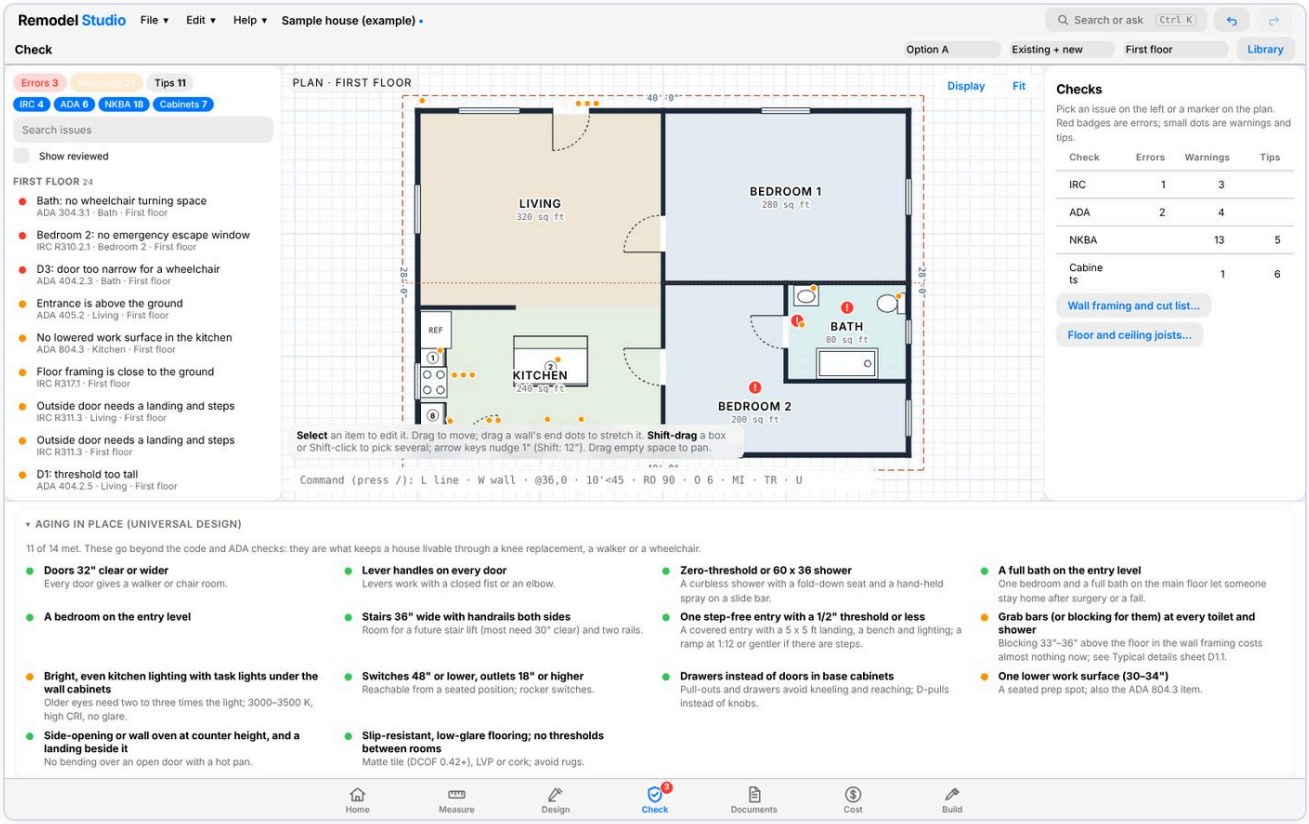
CHAPTER 14

Check workflow

1 Where the checks live

- **Check tab** in the tab bar at the bottom of the window. It wears a live badge: a red count of errors or, when there are none, an amber count of warnings. Its single page is **Issues**.
- **Issues dock** in the Designer (Design step): Home ribbon tab › Panels › **Issues** toggles a strip docked under the views. Its header has a segmented filter **Errors / Warnings / Tips / All** (each with its count), an **Open the Check page** button and a close (×) button. The list shows up to 200 issues; clicking one jumps to the item (sets the level, selects the target, opens its Inspector). Each row shows the severity dot, the title, and a small line with the code reference (set and section, e.g. "IRC R311.7.5.1") or the category, plus the room and level.
- **Plan markers** in the Designer plan views: the **Issues shown / hidden** chip above the plan toggles code-check markers. Markers are drawn at each issue's point on the current level; tips are not drawn. Red badges are errors, small dots are warnings. Clicking a marker goes to the item.
- **Plan check panel** (an Inspector panel): a "Summary" with pills "N fails" and "N to check", a sentence naming the code basis, a filter **All / Fails / To check / Tips**, issues grouped by level (or "Whole house"), each row labelled **Fails**, **Check** or **Tip** with title, detail and the rule reference ("Model check" when there is no code rule). Below it a **Structure** section lists every checked member as a card with a capacity bar (green under 85 %, amber 85–100 %, red over 100 %), its size, spacing, span, load, governing check (bending, shear or deflection), a suggested size, and an expandable **Load breakdown**. A **Show rule values / Hide rule values** button opens the editable rule table (see *Adjustable rule values* below).
- **Checks also appear** on the cover sheet A0.1 ("Code check summary", first 8 items), in the Board presentation (Code slide and Structure slide), in Inspection prep (open errors marked ⚠), and in the permit checklist.

2 The Check page (Issues)



The Check page.

Three columns:

Left column (issue list) - Severity chips: **Errors**, **Warnings**, **Tips**, each with its total count; click to show/hide that severity. Errors and warnings start on, tips start off. - Category chips, one per category present, with a count; click to toggle a category in or out. Categories, in display order: **IRC, IBC, ADA, NKBA, Structure, Framing, Electrical, Plumbing, Appliances, Cabinets, Energy, Zoning, Landscape, Model**, followed by any other code set that appears (for example NEC is shown under Electrical; CPSC, ASTM, Site, Practice appear by their own name). An issue is filed under its rule's code set; issues with no rule are sorted by keywords in their text (notch/bore/clash → Framing; joist/rafter/beam/header/span → Structure; outlet/GFCI/circuit/smoke → Electrical; drain/vent/pipe/DFU → Plumbing; R-value/insulation/U-factor → Energy; anything else → Model). - A **Search issues** box (matches title, detail and code section) and a **Show reviewed** checkbox. - The list is grouped by level in building order, with a final **Whole job** group for issues that have no level. Each row: severity dot, title, and a sub-line with "SET CODE · Room · Level". Reviewed issues are hidden unless Show reviewed is on.

Middle column: the plan view of the current level with the filtered issues drawn as markers. Clicking a marker selects that issue.

Right column (detail) when an issue is selected: - Pill: "Error / Warning / Tip · Category". - Title, the rule line "SET CODE · rule title", the full explanation, the room and level. - **Show it in Design** (switches to the Design workflow, sets the level, selects the item) and **Select it**. - **Reviewed, leave as is** checkbox. Reviewed issues (keyed by title + target + level) hide from the list and the plan "until something about them changes"; they still print in the permit checklist. The date reviewed is shown.

When nothing is selected the right column shows a **Checks** table (category × Errors / Warnings / Tips counts) and two shortcut buttons: **Wall framing and cut list…** and **Floor and ceiling joists…**.

Below the three columns, the **Aging in place (universal design)** panel (see *Aging in place* below).

Empty state text: "No issues. The model passes every check that is turned on." or "Nothing matches these filters."

3 Code settings (where they are set)

With nothing selected in the Designer, the Inspector shows the **Project** section:

- **Building code**: *House: 2021 IRC* (default) or *Commercial or city building: 2021 IBC*.
- When IBC is chosen: **Occupancy group** (B: Business (offices, city hall, clinics) — default; A-3: Assembly (meeting rooms, community center); M: Mercantile (stores); E: Educational; S-1: Storage or shop (public works garage)) and **Fire sprinklers throughout** checkbox. ADA checks are always on under IBC.
- When IRC is chosen: **Accessible design (2010 ADA checks)** checkbox (on by default).
- **NKBA kitchen and bath planning guidelines** checkbox (on by default). When on: **Cooks at once** (*One (42" aisles) / Two or more (48" aisles)*) and **Use Access Standard values** checkbox.
- Under IBC a **Code analysis (2021 IBC)** summary block appears: occupancy (and "sprinklered"), occupant load per level, exits required on the first floor, travel distance limit, toilets · lavatories needed (and drawn), drinking fountains · service sink.
- The NKBA legend on the plan also offers a **Turn on NKBA checks** button when they are off.
- Energy code items are included only when **Include energy code items in the code check list** is ticked on the Systems › Energy code tab.
- **Rule values** (Plan check panel › Show rule values): every adjustable rule for the active code set (IRC rules under IRC, IBC rules under IBC, ADA rules when ADA is on) is listed as "SET CODE — title" with an editable value (length input for inch rules, number input otherwise) and a **Reset** button when changed. Changed values are saved with the project and marked. Note: "Change a value to match a local amendment."
- **My standards** (File dialog) saves code/ADA/NKBA/energy settings as the defaults for new projects.
- Ground height for egress and ramp checks comes from the project's grade offset (first floor above grade, default 24").

4 Adjustable rule values

Set	Code	Rule	Default
IRC	R305.1	Ceiling height, habitable rooms	84 in
IRC	R305.1 Exc.	Ceiling height, baths and laundry	80 in
IRC	R304.1	Habitable room floor area	70 sf
IRC	R304.2	Habitable room, least dimension	84 in
IRC	R311.6	Hallway clear width	36 in
IRC	R311.7.5.1	Stair riser height, maximum	7.75 in
IRC	R311.7.5.2	Stair tread depth, minimum	10 in
IRC	R311.7.1	Stair width, minimum	36 in
IRC	R310.2.1	Egress window net opening	5.7 sf
IRC	R310.2.1 Exc.	Egress net opening at grade floor	5.0 sf
IRC	R310.2.1	Egress net clear width	20 in
IRC	R310.2.1	Egress net clear height	24 in
IRC	R310.2.2	Egress sill height, maximum	44 in
IRC	R311.2	Egress door clear width	32 in
IRC	R311.2	Egress door clear height	78 in
IRC	R312.2.1	Upper-floor window sill without opening limit	24 in
ADA	404.2.3	Door clear width	32 in
ADA	404.2.4.1	Latch-side clearance, pull side	18 in
ADA	304.3.1	Turning space diameter	60 in
ADA	403.5.1	Accessible route clear width	36 in
ADA	405.2	Ramp run per inch of rise (1:12)	12
ADA	405.6	Ramp rise per run, maximum	30 in
IBC	1208.2	Ceiling height, occupiable spaces	90 in
IBC	1010.1.1	Egress door clear width	32 in
IBC	1010.1.1	Egress door height	80 in
IBC	1020.3	Corridor width, 50 or more occupants	44 in
IBC	1020.3	Corridor width, under 50 occupants	36 in
IBC	1006.2.1	Most occupants in a room with one exit door	49 people
IBC	1005.3.2	Door egress width per occupant (no sprinklers)	0.2 in
IBC	1005.3.2 Exc.	Door egress width per occupant (sprinklered)	0.15 in
IBC	1005.3.1	Stair egress width per occupant (no sprinklers)	0.3 in
IBC	1011.5.2	Stair riser height, maximum	7 in
IBC	1011.5.2	Stair tread depth, minimum	11 in

Set	Code	Rule	Default
IBC	1011.2	Stair width, minimum	44 in
IBC	1007.1.1	Exit separation, fraction of the diagonal (no sprinklers)	0.5
IBC	1007.1.1 Exc. 2	Exit separation, fraction of the diagonal (sprinklered)	0.333

A door's clear opening is taken as its leaf width minus 2". A window's net clear opening is estimated from its rough size: casement/awning = (W − 4) × (H − 4); double-hung = (W − 4) × (H − 6)/2; slider = (W − 6)/2 × (H − 4); fixed = none. Rectangular room clear dimensions subtract 4.5" for wall finish.

5 Complete rule table: IRC house checks

Run when the code is 2021 IRC (most are skipped under IBC).

Code	What is checked	Threshold	Severity
— (Model)	Wall with no length	length < 1"	Warning
— (Model)	Door/window wider than its wall	opening width > wall length − 1	Error
— (Model)	Opening tight to a corner	within 1.5" of a wall end ("leave about 3"")	Warning
— (Model)	Window head above the wall top	sill + height > wall height − 4	Warning
— (Model)	Two openings overlap on a wall	overlap > 0.5"	Error
— (Model)	Room has no way in	no door on its boundary and no open side (gap > 24") and not a closet	Warning
R305.1 / R305.1 Exc.	Ceiling too low (all rooms except closet, garage)	84" habitable; 80" bath and laundry	Error
R304.1	Room too small (living, bedroom, dining)	70 sf	Error
R304.2	Room too narrow (rectangular living, bedroom, dining)	84" clear in every direction	Error
R311.6	Hallway too narrow (rectangular hall)	36" clear	Error
ADA 403.5.1	Accessible route too narrow (hall passes IRC but not ADA)	36"	Error
R310.2.1 / R310.2.2	Bedroom has no emergency escape window (exterior windows only)	5.7 sf net (5.0 sf when the floor is 44" or less above grade), 20" wide, 24" high, sill ≤ 44". Detail names the best window and suggests "36" × 60" double-hung or 30" × 48" casement with a sill at 44" or lower"	Error
ADA 304.3.1	No wheelchair turning space in a bath or kitchen (largest clear circle around floor-standing fixtures and cabinets, lavatories ignored)	60" diameter	Error
ADA 404.2.3	Door too narrow for a wheelchair	32" clear (34" door)	Error

Code	What is checked	Threshold	Severity
R311.2	No exterior door on the first floor	at least one	Error
R311.2	No exterior door wide or tall enough for egress	32" clear, 78" high	Error
R311.7.5.1	Stair risers too tall	7.75"	Error
R311.7.5.2	Stair treads too shallow	10"	Error
R311.7.1	Stair too narrow (not spiral)	36"	Error
R311.7.10.1	Spiral stair width / riser / tread at walkline	26" clear; 9.5" riser; 6.75" tread at 12" from the narrow edge	Error
R311.7.5.2.1	Winder treads at the walkline (L stair with winders)	10" at 12" from the narrow end	Error
R311.7.2	Stair headroom (floor above not cut away)	80" (6'-8")	Error
R311.7.8	Stair needs a handrail	flights with 4 or more risers and rails set to none	Error
R312.2.1	Low sill on an upper floor (exterior window)	drop outside > 72" and sill < 24"	Warning
R312.2.1 (site version)	Window fall protection using the graded site (non-fixed exterior window)	sill < 24" and sill more than 72" above the ground outside	Warning
ADA 405.2 / 405.6	Entrance is above the ground (first floor above grade)	grade offset > 0.5"; reports a 1:12 ramp length and the number of 30" runs with 60" landings, or a lift	Warning
R307.1	Toilet too close to the side (centerline to wall or fixture)	15"	Error
R307.1	Not enough room in front of the toilet	21"	Error
ADA 604.2	Toilet centerline to side wall	16" to 18"	Error
ADA 604.3.1	Toilet clear floor space blocked	60" wide × 56" deep from the side wall (lavatories and wall-hung items allowed)	Error
ADA 606.3	Lavatory or vanity rim too high	34"	Error
ADA 305.3	Tight clear floor space in front of a refrigerator, range, washer or dryer	48" (forward approach)	Warning
IRC good practice / ADA 804.2.1	Kitchen aisle too narrow (opposing fronts within 120")	36" (IRC); 40" with ADA on (60" for a U kitchen is noted)	Error
ADA 804.3	No lowered work surface in the kitchen	one base counter 30" wide at 34" max	Warning
E3901.2.1	Receptacles too far apart (living, bedroom, dining, kitchen, hall, office, etc.)	no point on a wall 24" or longer more than 6'-0" from a receptacle (72.5" allowance)	Error
E3901.6	No receptacle near a bathroom sink	GFCI within 36" of the basin edge	Error
R303.3	No exhaust fan in a bath	fan required unless 3 sf openable window	Warning

Code	What is checked	Threshold	Severity
E3901.4.1	Not enough countertop receptacles	about one per 48" of counter (counter ≥ 12", receptacles at 30"+ height)	Error
E3902.6	Countertop receptacles need GFCI (plain receptacles at counter height in a kitchen)	all	Warning
E3903.2	Room has no light (all rooms but closet and storage)	one switched lighting outlet	Warning
R314.3	No smoke alarm in a bedroom; no smoke alarm on a floor with rooms	one per bedroom, one per story	Error
R315.3	No carbon monoxide alarm (house has smoke alarms and bedrooms but no CO or combo)	one outside each sleeping area	Warning
E3901.7	No laundry receptacle	one on a 20-amp circuit	Error
E3901.8	No outdoor receptacles (house has exterior walls)	GFCI front and back, ≤ 6'-6" above grade	Warning
NEC 210.70(A)	Room has no wall switch; switch not linked to a light	one switch per habitable room/bath/hall/garage; links drawn	Warning
ADA 405.2	Ramp too steep	1:12	Error
ADA 405.6	Ramp run rises too much	30" between landings	Error
ADA 405.5	Ramp too narrow	36" between handrails (3.5" deducted when rails are present)	Error
ADA 405.7.3	Ramp landings too short	60"	Error
ADA 405.8	Ramp needs handrails	rise > 6" and rails turned off	Error
ADA 405.9	Add ramp edge protection	rise > 6" and no edge set	Warning
R310.1	Basement with habitable space needs an emergency escape opening	one exterior door or window ≥ 5.7 sf	Error
R310.2.3	Window well too small	9 sf, 36" each way	Error
R310.2.3.1	Window well needs a ladder	depth > 44"	Error
R1003.9 / R1005.3	Chimney too short	top 3 ft above the roof where it passes through and 2 ft above anything within 10 ft	Error
R1001.6	Masonry firebox may be too shallow	20" (12" Rumford)	Warning
R308.6	Skylight is not on a roof	must sit over a roof plane	Warning
R304.1 (sloped ceilings)	Too little floor area with headroom (attic/vaulted bedroom, living, office, dining, other)	70 sf with 5 ft or more headroom	Error
R305.1 (sloped ceilings)	Not enough area at 7 ft	35 sf at 7 ft	Error
R311.7	Attic rooms need a stairway	a stair on the floor below	Error
R502.3	Check the attic floor framing (attic living space)	30 or 40 psf joists	Warning
R401.3	Ground slopes toward the house	lot uphill on a side; keep 6" fall in 10 ft	Warning

Code	What is checked	Threshold	Severity
ADA 403.3 / 405.2	Walkway too steep	> 8.33 % (error with ADA on, otherwise warning); 5–8.33 % "walkway is a ramp" warning with ADA	Error / Warning
Site guideline	Steep driveway	> 15 %	Warning
R403.1.5	Step the footings (site falls across the house)	> 10 % slope	Warning
R404.1	Tall foundation wall on the low side (crawl or stem wall)	> 48" floor to ground	Warning
R606	Tall piers on the low side	> 72"	Warning
R317.1	Floor framing close to the ground (crawl or piers)	< 18" grade to framing (12" for girders)	Warning
R408.1	Crawlspace needs vents or a conditioned crawl	1 sf per 150 sf, one within 3 ft of each corner	Warning
R408.4	Crawlspace needs an access opening	18" × 24" through floor or 16" × 24" through wall	Warning
R311.7.5.1 / R311.7.5.2 / R311.3 / R311.7.1	Exterior steps: riser, tread, landing depth, width	7.75"; 10"; 36" landing; 36" wide	Error
R311.7.8 / R312.1.1	Steps need a handrail (4+ risers) / landing needs a guard (over 30")	rails set to none	Error
R312.1.1	Deck or porch needs a guard rail	floor > 30" above ground	Error
R312.1.2	Guard rail too low	36"	Error
R507.6	Deck joists may be over-spanned	Table R507.6 span for the joist size and spacing (Southern pine)	Warning
—	Porch beam is low	< 6'-8" above the porch floor	Warning
R302.5.1	Door to the garage is not fire resistant	solid wood/solid-core/steel ≥ 1-3/8", or 20-minute rated	Error
R302.5.1	Door to the garage needs a closer	self-closing	Error
ADA 309.4	Knob hardware on a door	levers, push/pull or loop handles	Warning
ADA 404.2.5	Threshold too tall (standard 1" sill on a swing or double door)	1/2" (3/4" existing sliders)	Warning
Practice: Security	Outside door does not lock (passage hardware)	keyed entry and deadbolt	Warning
Practice: Privacy	Bathroom door has no privacy lock	privacy set	Warning
R502.3 (joist layout)	Floor joists overspanned (any level)	allowable span of the chosen joist at its spacing	Warning
R806.1	Attic is not vented (ridge and soffit vents both off)	1/150 or 1/300 with balanced vents	Warning

Code	What is checked	Threshold	Severity
N1102.1.2	Energy item below the energy code (only when turned on in Systems)	prescriptive table for the climate zone	Warning
Zoning: Local	Setback, lot coverage, impervious surface or height does not meet zoning (values entered in the plan set panel)	as entered	Error
Landscape: Hardiness	Plant not hardy in the project's USDA zone	species range	Warning
Landscape: Spacing	Large tree (30 ft or taller) within 10 ft of the house; shrub too close to the wall (center closer than half its spread + 6"); same-kind plants crowded (closer than 60 % of their combined spread)	as stated	Warning
CPSC 5.2	Play item use zone not covered by safety surfacing	whole use zone inside a play area	Error
ASTM F1292	Surfacing too thin for the fall height	surface rating ≥ fall height	Error
CPSC 5.3.8	Swing use zone overlaps another	none	Error
CPSC 5.1	Equipment too close (overlapping use zones)	6 ft, or 9 ft when either play surface is over 30"	Error
ADA 1008.2.6	Play area surface not wheelchair accessible	ASTM F1951 surface	Warning
ADA 240.1 / 1008.2	No walkway reaches the play area	walk within 36"	Warning
ADA 240.2.1.2	Too few ground-level play components for the elevated ones	Table 240.2.1.2	Warning
ADA 240.2.2	Elevated components: 20 or more need 25 % by ramp; otherwise 50 % on an accessible route	as stated	Warning
CPSC 2.3	2-5 and 5-12 equipment together	separate	Warning

6 Appliance and cabinet checks

Set	What is checked	Threshold	Severity
Appliances	Opening between cabinets too narrow for the range, dishwasher or refrigerator (library appliance with published cut-out)	cut-out width (refrigerator: width + 1/4" + side clearances)	Error
Appliances	Gap beside a range or dishwasher	opening more than 0.6" over the cut-out → add a filler	Warning
Appliances	Cabinet too close above a range or cooktop	the model's published clearance above the cooking surface	Warning
Appliances (IRC M1503.6)	Range hood over 400 CFM needs makeup air	> 400 CFM	Tip
Cabinets: Blind corner	Cabinet butts straight into a blind corner cabinet	add a 3" filler and set the pull distance	Warning
Cabinets: Corner	Dishwasher tight to an inside corner	3" filler or a cabinet between	Warning

Set	What is checked	Threshold	Severity
Cabinets: Corner	Doors and drawers blocked at an inside corner	use blind/lazy Susan or fillers	Warning
Cabinets: Clearance	Refrigerator tight to a wall	leave 2" to 4"	Warning
Cabinets: Finished ends	Exposed cabinet end is unfinished	order FEL/FER or an end panel	Tip

7 Structure checks (from the span analysis)

Code	What is checked	Threshold	Severity
R502.3 (floor joists), R802.5 (ceiling joists), R802.4 (rafters), R602.7 (headers) — IBC: 1607.1 floors, 2308.4 headers/beams, 2308.7 others	Member overstressed	capacity ratio > 100 % (bending, shear or deflection) with a suggested size	Error
same	Member near limit (not headers)	ratio > 95 %	Warning
R301.1.3	New beam needed where a bearing wall is removed	span and plf load; suggested beam with posts and footings, or "Engineer to size"	Warning

8 Clash and notching checks (Framing category)

Code	What is checked	Threshold	Severity
R502.8	Pipe below the floor runs through a girder	drop below the girder: joist depth + 9.25 + 0.75 + half the pipe	Error
R502.8.1 (open-web floor trusses)	Pipe needs the truss chase	hole (outside diameter + 1/4") > 5"	Warning
R502.8.1 (I-joists)	Pipe too big for the I-joist webs	hole > depth − 3.5" (min 1.5")	Error; otherwise Tip ("use the knockouts, never cut the flanges")
R502.8.1 (sawn joists)	Pipe needs too big a hole	hole > 1/3 of the joist depth	Error
R502.8.1	Hole too close to the joist edge	< 2" from top or bottom (suggests the centered height)	Warning; otherwise Tip
R502.8.1	Crawlspace duct crosses joists (first floor, not slab)	any duct, unless truss floor and duct ≤ 10" (then Tip: use the center chase)	Error / Tip
R602.6	Pipe inside a wall bores studs (runs ≥ 14" along the wall)	hole > 40 % of the stud (bearing/exterior) or 60 % (non-bearing); 2×4 = 3.5", 2×6 = 5.5"	Error; otherwise Tip (nail plates where within 1-1/4" of the face, P2603.2.1)
Coordination	Pipe hits a duct at the same height	within half the duct height + half the pipe	Warning

2.375, 3" = 3.5, 4" = 4.5; gas 1/2" = 0.84, 3/4" = 1.05, 1" = 1.315. Ducts are taken as 12" below the floor in a crawlspace, or ceiling + 10" in an attic.

9 Aging in place (universal design) checklist

Shown at the bottom of the Check page as "N of 14 met", each item with a green (met) or amber (to do) dot, a title and advice. Not counted as errors.

Item	Met when
Doors 32" clear or wider	every non-demolished door is ≥ 32" wide (34" door gives 32" clear)
Lever handles on every door	every door's hardware name includes "lever"
Zero-threshold or 60 × 36 shower	no shower, or a roll-in shower, or a shower ≥ 60 × 36
A full bath on the entry level	a bath on the first level (or no baths)
A bedroom on the entry level	a bedroom on the ground level (or no bedrooms)
Stairs 36" wide with handrails both sides	every stair ≥ 36" wide
One step-free entry with a 1/2" threshold or less	an exterior door with threshold ≤ 1/2"
Grab bars (or blocking) at every toilet and shower	grab bars ≥ 2 × toilets
Bright, even kitchen lighting with task lights	≥ 3 light fixtures in each kitchen
Switches 48" or lower, outlets 18" or higher	every switch/dimmer at ≤ 48"
Drawers instead of doors in base cabinets	drawer-base cabinets ≥ half the base cabinets
One lower work surface (30–34")	a countertop cabinet ≤ 34" high
Side-opening or wall oven at counter height	always met (informational)
Slip-resistant, low-glare flooring	every bath, kitchen and laundry floor is tile, LVP or concrete

10 IBC checks (when the code is 2021 IBC)

Occupant loads use IBC Table 1004.5 by room use (office 150 gross; assembly/dining 15 net; seating 7 net; classroom 20 net; retail 60 gross; storage 300 gross; shop 100 gross; commercial kitchen 200 gross; living 150 gross; bedroom 200 gross; garage 200 gross; other 150 gross; occupancy default when the use is not listed: B 150, A-3 15, M 60, E 20, S 300). Halls, corridors, baths, restrooms, closets and laundries carry no load. Travel distance limits (Table 1017.2): B 200 ft / 300 sprinklered; A-3, M, E, S 200 / 250. One exit allowed for ≤ 49 occupants (S: 29) and ≤ 75 ft travel (S: 100 ft), single story. Required exits: 2 up to 500 occupants, 3 to 1,000, 4 above. Plumbing fixture ratios per Table 2902.1 by occupancy.

Code	What is checked	Threshold	Severity

Code	What is checked	Threshold	Severity
1208.2	Ceiling too low (occupiable rooms; not closet, storage, garage, bath, restroom)	90"	Error
1006.2.1	Room needs two exit doors	more than 49 occupants with one door	Error
1020.3	Corridor too narrow (rectangular hall/corridor)	44" clear with 50+ total occupants, else 36"	Error
1010.1.1	Egress door too narrow / too short (every door)	32" clear (use a 36" door); 80" high	Error
1006.3.3	Needs N exits to the outside (first-floor exterior doors)	per occupant load and single-exit rule	Error
1007.1.1	Exits too close together	≥ 1/2 of the building diagonal (1/3 sprinklered)	Error
1005.3.2	Exit doors too narrow for the occupant load	0.2" per occupant (0.15" sprinklered) of clear door width	Error
1017.2	Travel distance to an exit too long (far room corner, walking along x then y)	200/250/300 ft by occupancy; warning at 85 % of the limit	Error / Warning
1006.3.3	Upper floor needs N exit stairs	1 for ≤ 49, else per load	Error
1011.2 / 1005.3.1	Exit stair too narrow	greater of 44" and 0.3" per occupant divided among stairs	Error
1011.5.2	Stair risers too tall / treads too shallow	7" / 11"	Error
2902.1	Not enough toilets / lavatories	per occupancy table (split between sexes when over 15 occupants)	Error
2902.1	Drinking fountain and service sink	count needed; "not drawn as fixtures yet"	Info
2902.2	Separate restrooms required	over 15 occupants and fewer than 2 toilets	Warning

11 NKBA checks (how they surface)

NKBA kitchen and bath guideline checks run only when the NKBA checkbox is on. They appear in the **NKBA** category with references such as "NKBA K1 Door/entry", "NKBA K3 Work triangle", "NKBA K6 Work aisle" (kitchen guidelines numbered K1…) and "NKBA B-…" items for baths; nearly all are Warnings, with a few Tips (grab-bar backing, shower controls). The **Cooks at once** setting changes the work-aisle target (42" for one cook, 48" for two) and **Use Access Standard values** raises door and seating clearances (for example 34" door clear instead of 32"). The rule details themselves are documented with the Design workflow.

PART

Structure and systems

Spans and loads, framing, wall removal, electrical, plumbing, heating and cooling, energy and lighting.

CHAPTER 15

Structure

1 Where structure tools live

Structure ribbon tab in the Designer: - **Framing** group: *Roof wizard*, *Floor and ceiling joists*, *Wall framing*, *Remove or open a wall* (enabled only when a wall is selected; tooltip "Select a wall first"). - **Show** group: *Floor framing plan* (plan tab), *Roof framing plan* (plan tab), *Structure check in 3D* (toggle). - **Posts** group: the *Column* tool.

Also: the 3D view's **Loads** chip ("Color members by how much of their capacity they use") and **Framing** chip; the wall Inspector buttons **Remove this wall (wizard)…** / **Removal: beam, posts and footings…** and **Framing and cut list…**; a level's **Floor and ceiling framing** section in the Project properties with a **Joist layout, I-joists and floor trusses…** button; the roof's **Framing and covering** section; the opening's **Header** picker (bearing and exterior walls): *Size automatically* or any beam from the list.

2 Span and load analysis (planning level)

Every joist, rafter, header and removed-wall beam is checked as a simple span for bending, shear, live-load deflection and total-load deflection using approximate NDS reference values. Result: a capacity ratio (100 % = at the limit) and the governing check.

- **Lumber species/grades**: Southern Pine #2 (Fb 1100/1000/925/800/750 psi for 2×4/2×6/2×8/2×10/2×12, Fv 175, E 1.4e6), Spruce-Pine-Fir #1/#2 (Fb 875 × size factor 1.5/1.3/1.2/1.1/1.0, Fv 135, E 1.4e6), Douglas Fir-Larch #2 (Fb 900 × same size factors, Fv 180, E 1.6e6). LVL: Fb 2600 (depth-adjusted), Fv 285, E 2.0e6, 1-3/4" plies.
- **Beam and header choices** (smallest first): (2) 2×6, (2) 2×8, (2) 2×10, (2) 2×12, (3) 2×12, (2) 1¾" × 9¼" LVL, (2) 1¾" × 11⅞" LVL, (2) 1¾" × 14" LVL, (3) 1¾" × 14" LVL, (3) 1¾" × 16" LVL, (3) 1¾" × 18" LVL, (4) 1¾" × 18" LVL. Beyond that: "Engineer to size".
- **Live loads (IRC)**: floors 40 psf, bedrooms 30 psf, roof 20 psf, attic 10 psf (20 psf with attic storage). **IBC** by room use: office 50 (+15 psf partitions as dead load), assembly/retail/corridor/hall/living/dining 100, classroom 40, storage/shop 125, kitchen/bath/other/closet/laundry 50, garage 40, bedroom 40.
- **Dead loads**: floors 10 psf plus finish (hardwood +1, tile +5); ceilings 5 psf (10 with storage); roof covering asphalt 10, metal 6, clay/concrete tile 20; walls above 10 psf × height; floors above 10 psf.
- **Deflection limits**: floor joists and headers L/360 live, L/240 total; ceiling joists L/240; rafters L/240 live, L/180 total (with a 1.25 load-duration factor); repetitive-member factor 1.15 for joists and rafters.
- **Floor joists** per room: span measured across the room between bearing walls below (or exterior walls), in the chosen direction or the shorter way; on a crawlspace first floor the span

is capped at the girder spacing. Garages and slab floors are skipped.
- **Ceiling joists** on the top level (skipped when the roof is trussed).
- **Rafters**: horizontal span = half the roof depth (full depth for a shed); skipped for trusses.
- **Headers**: span = opening width + 3"; load = line load on the wall from floors, walls and roof above (tributary half-spans). Non-bearing walls get "Flat 2x / No load". A header chosen by hand on the opening is checked; "auto" picks the smallest that passes.
- **Removed bearing walls** (status "demolish", bearing): beam sized from the wall's load over its full length.

Results show as: the Structure cards in the Plan check panel (sorted worst first, with load breakdown); errors and warnings in Check; red sizes on the S1.x framing plans; colors in 3D (Loads chip): green under 85 %, amber 85–100 %, red over capacity, blue bearing walls (legend shown in the view); headers drawn as colored blocks; the header schedule.

3 Remove or open a wall (dialog)



Remove or open a wall.

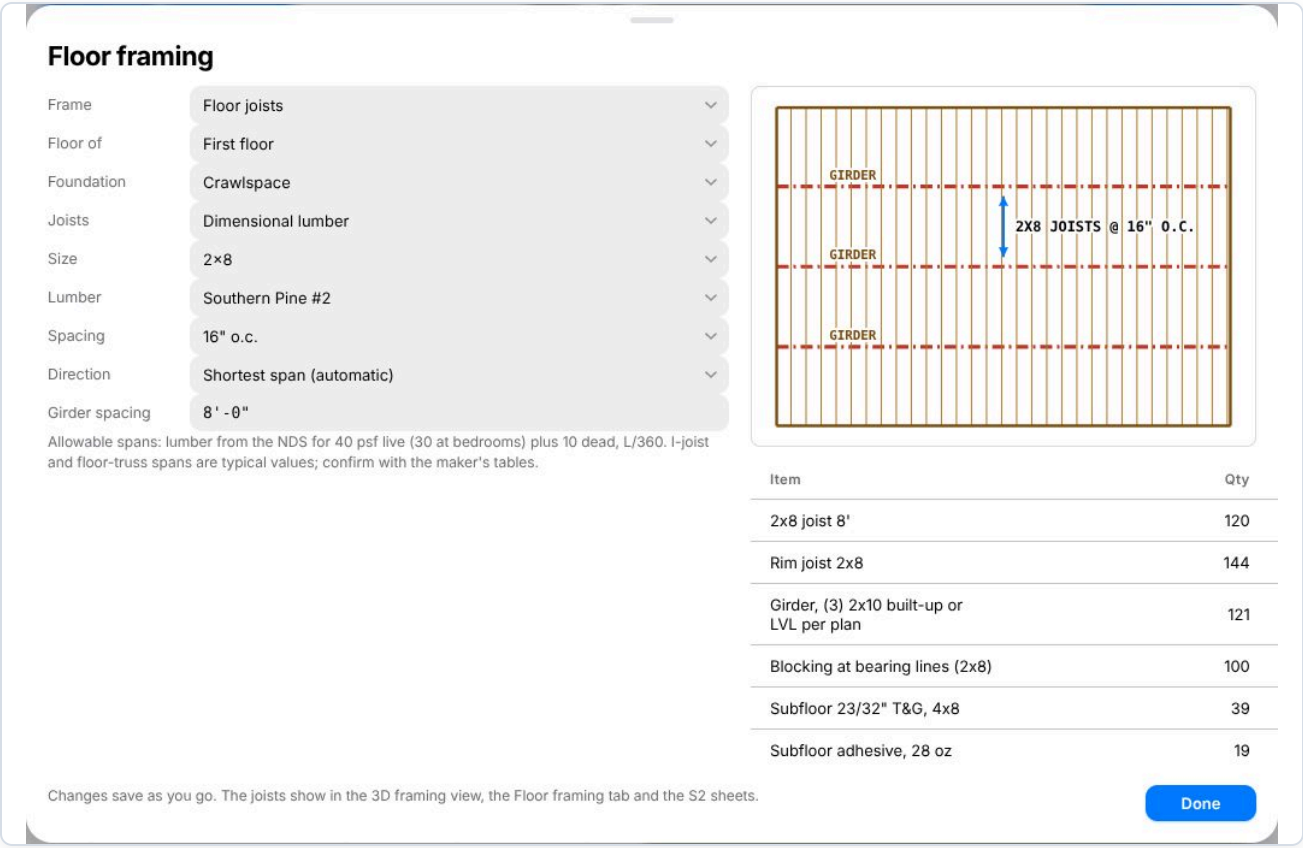
Opened from Structure › Remove or open a wall, or the wall panel. Needs a selected wall. Shows "L inside/outside wall on Level".

1. **Is it load-bearing?** Verdict "Probably load-bearing" / "Probably not load-bearing" with reasons: exterior walls always bear; joists over an adjacent room that run across the wall and whose span grows by more than 12" without it ("span X; without it about Y"); a wall on the level above within 8"; a trussed roof on the top level (not bearing unless a level is above); otherwise "Nothing in the model lands on this wall. Still look in the attic or crawlspace...". Checkbox **Treat it as load-bearing** (pre-set from the guess or the wall's bearing flag).
2. **Beam** (when bearing): load on the beam in plf with its breakdown; a **Beam** drop-down listing every stock beam with "N % of capacity (too small)"; "No stock beam in the list carries this" message; radio **Flush beam (in the ceiling, joist hangers)** / **Dropped beam (below the ceiling)**. Beam span is wall length + 6"; 12 plf self-weight added.
3. **Posts and footings**: load at each end (lb); post: ≤ 3,000 lb (2) 2×4 jack studs; ≤ 5,500 (3) 2×4 jack studs or 4×4 post; ≤ 9,000 4×6 post or (3) 2×6 studs; above that 6×6 post (engineer to confirm). **Soil bearing**: 1,500 psf (clay, untested; IRC default), 2,000 (sandy or silty clay), 3,000 (sand, gravel). Footing: square side = load/soil, rounded up to even inches, minimum 16"; 10" thick (12" when 24" or larger); note "(cut the slab)" / "on a pier in the crawlspace" / "carried down to the foundation".
4. **Before you cut**: checkbox **Temporary shoring walls on both sides (length each)** and a reminder list (wiring/plumbing/gas/ducts; permit and framing inspection; patch floor, walls,

ceiling). Non-bearing: "Remove it … comes out as demolition."

Mark wall for removal (with this beam) sets the wall to demolish and records the beam, posts, footing, reaction, shoring and soil. The builder's list then gets the LVL or lumber beam plies, posts, post footings (CY), joist hangers (flush), shoring (LF) and patching (LF).

4 Floor and ceiling joists (Floor framing dialog)



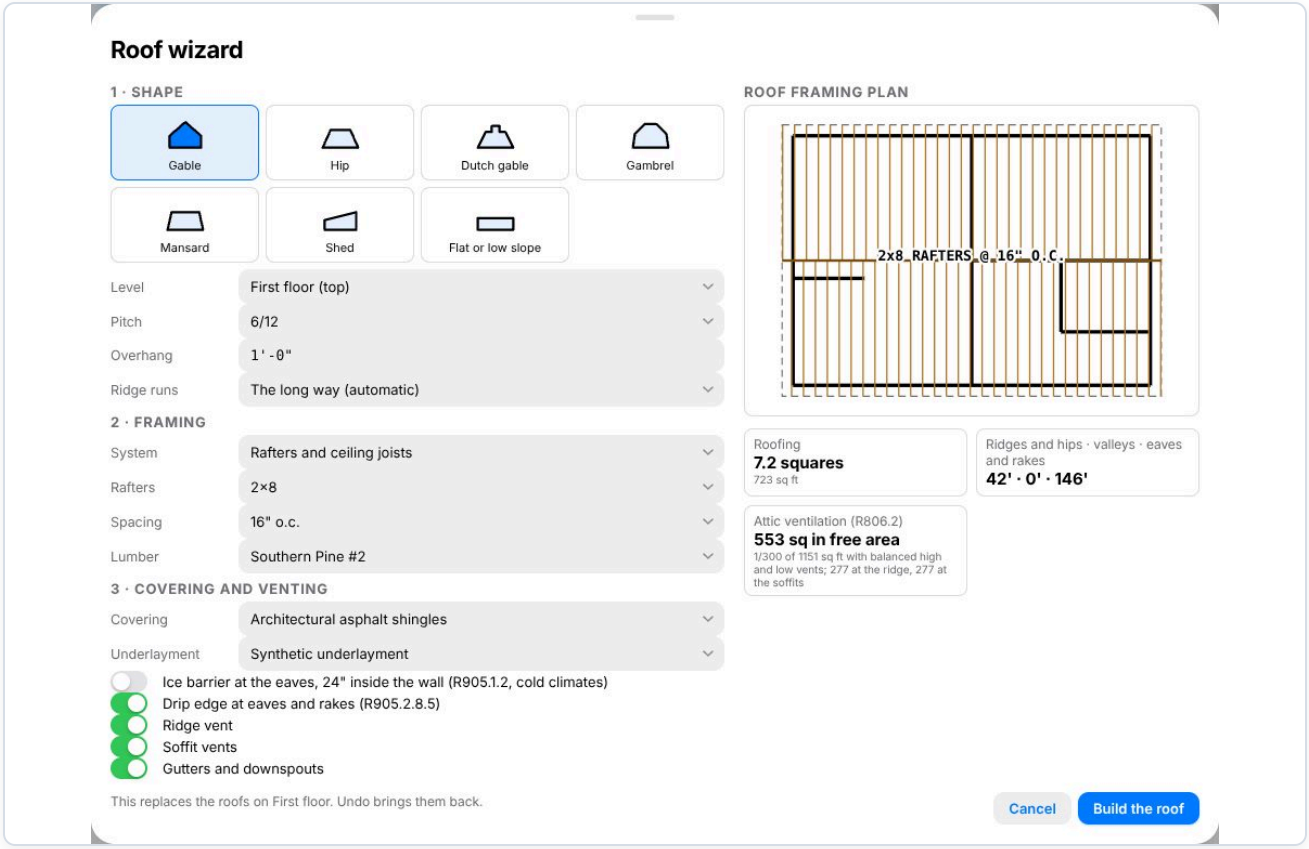
Floor and ceiling joists.

- **Frame**: *Floor joists* or *Ceiling joists (under a rafter roof or a flat ceiling)*; **Floor of / Ceiling of** level picker.
- Floor, first level: **Foundation** *Crawlspace / Basement / Slab on grade (no joists)*. A slab shows "A slab floor has no joists."
- **Joists**: *Dimensional lumber* (Size 2×6–2×12; Lumber species), *I-joists* (Depth 9-1/2", 11-7/8", 14", 16"), *Open-web floor trusses* (Depth 12", 14", 16", 18", 20", 24").
- **Spacing** 12 / 16 / 19.2 / 24" o.c.; **Direction** *Shortest span (automatic) / East to west / North to south*; **Girder spacing** (ft) on a non-slab first floor.
- Ceiling: Size 2×4–2×10, Spacing 12/16/24, Direction, **Attic used for storage (20 psf)**.
- Allowable spans: lumber from the NDS check for 40 psf live (30 at bedrooms) + 10 dead at L/360; I-joist spans (inches) 9-1/2": 208/193/181/162 at 12/16/19.2/24; 11-7/8":

248/229/215/193; 14": 281/259/243/218; 16": 311/287/269/241; floor trusses 12": 216, 14": 240, 16": 264, 18": 288, 20": 312, 24": 360 (×1.1 at ≤16", ×1.05 at 19.2").

- The layout draws joists clipped to the rim, split at bearing walls and girders, trimmed around stair and chimney openings (2" chimney clearance), with doubled trimmers and headers at openings over 48" (R502.10) and a warning when an opening header spans over 72". Over-spanned joists are red. Warnings list: "N joist spans longer than ... allow (longest ...)".
- Right side: framing plan preview and a **schedule** table (Item / Qty / Unit): joists by stock length (even feet, 1-1/2" bearing each end), trimmers and headers, rim joist or LSL rim board (LF), joist hangers, girder (LF), blocking at bearing lines, subfloor 23/32" T&G sheets and adhesive (or 1/2" ceiling drywall for ceilings).
- Changes save as you go; joists show in the 3D framing view, the Floor framing plan tab and the S2 sheets.

5 Roof wizard



Roof wizard.

Structure › Roof wizard (also Architecture › Roof, and the roof panel). Builds roofs over a level's exterior walls, replacing that level's roofs (undo restores them).

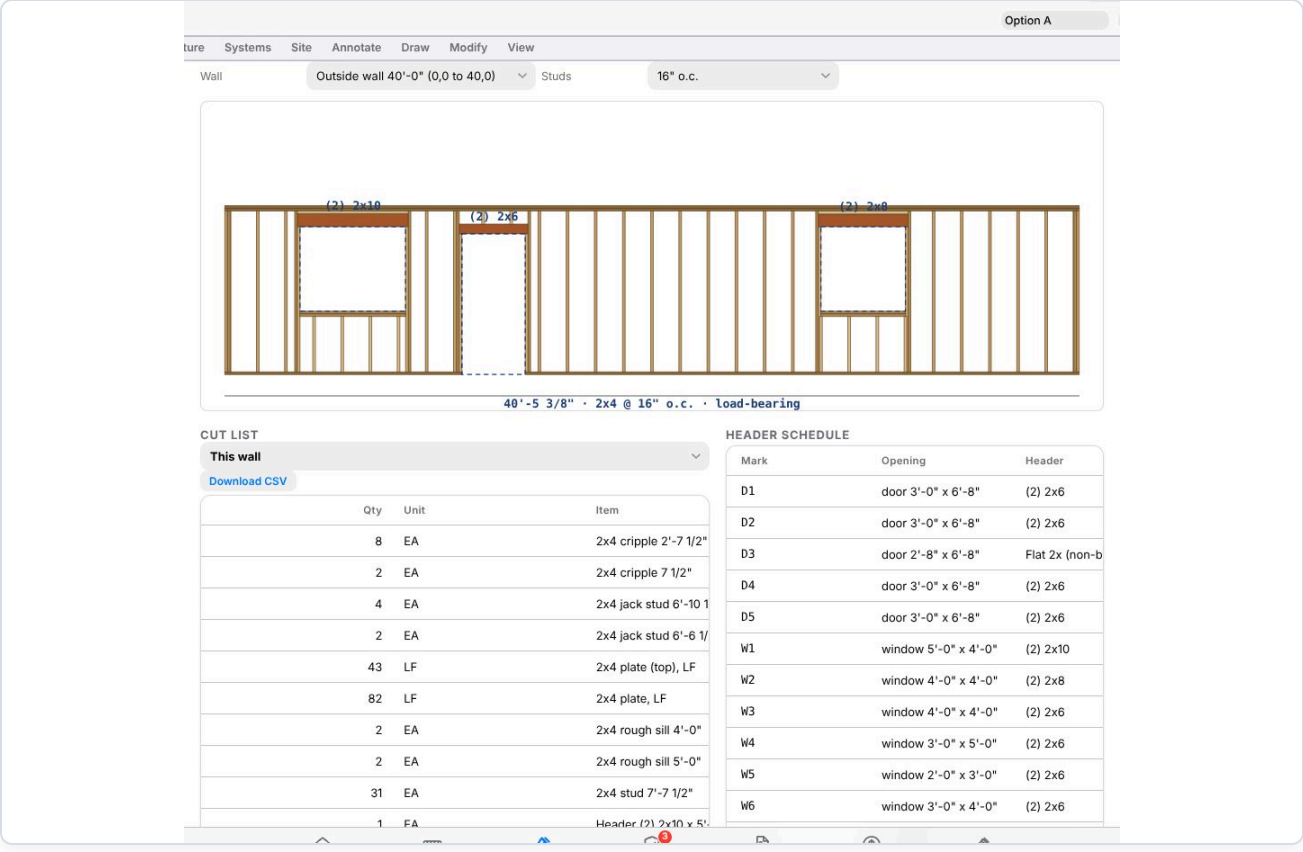
1. **Shape**: picture buttons for every roof style (gable, hip, dutch, gambrel, mansard, shed, flat); **Level**; **Pitch** 2–18/12 (2 and 3 marked low slope); **Overhang**; **Ridge runs** (gable/gambrel/shed):

long way automatic, east–west, north–south.

2. **Framing**: **System** *Engineered trusses* or *Rafters and ceiling joists*. Trusses: **Truss type** Common (Fink W) / Energy heel (raised heel) / Scissor (vaulted ceiling) / Attic (room in the roof); **Spacing** 16/19.2/24; for hip, dutch, mansard: **Hip girder setback** (ft) and **Girder plies** 2/3. Rafters: **Rafters** 2×4–2×12, **Spacing** 12/16/24, **Lumber** species.
3. **Covering and venting**: Covering (asphalt shingles, metal panels, clay or concrete tile); Underlayment (synthetic, #30 felt, ice and water shield full roof); checkboxes Ice barrier at the eaves 24" inside the wall (R905.1.2), Drip edge (R905.2.8.5), Ridge vent, Soffit vents, Gutters and downspouts.

Right side: roof framing plan preview (trusses or rafters), a **Typical truss** elevation (chords and webs, span), and statistics: roofing in squares and sq ft; ridges and hips · valleys · eaves and rakes in feet; **Attic ventilation (R806.2)** net free area = 1/300 of the attic area with half at the ridge and half at soffits; count of trusses and jacks with marks (GE1 gable ends, T1 commons, G1 girders, H1… step-down hips, EJn/CJn jacks, M1 mono, F1 flat).

6 Wall framing and cut list (dialog)



Wall framing and cut list.

Structure › Wall framing, or Check page shortcut, or the wall panel. - **Wall** picker (walls on this level, labelled "Outside/Inside wall L (x,y to x,y)"); **Studs** 12 / 16 / 24" o.c. (advanced framing). -

Framing elevation drawing: single bottom plate (cut at doors), double top plate, studs 16" from the wall start, king and jack studs at openings, headers (sized by the structure check in bearing/exterior walls; flat 2x otherwise), rough sills (doubled when the sill is above 4"), cripples, blocking, three-stud corners. Header size labelled over each opening; caption "length · stud size @ spacing · load-bearing/non-bearing". - **Cut list** scope *This wall* / *All new walls*, with **Download CSV**: studs as 92-5/8" or 104-5/8" precuts (or the exact length), jack studs and cripples by length, plates in LF (+5 %), rough sills, headers "size x length". - **Header schedule** (Mark · Opening type and size · Header (non-bearing noted) · Jacks): two jacks each side for openings over 6 ft; rows that fail the check are flagged. Note: "Headers come from the structure check (IRC R602.7)."

7 Framing plans (Designer tabs and sheets)

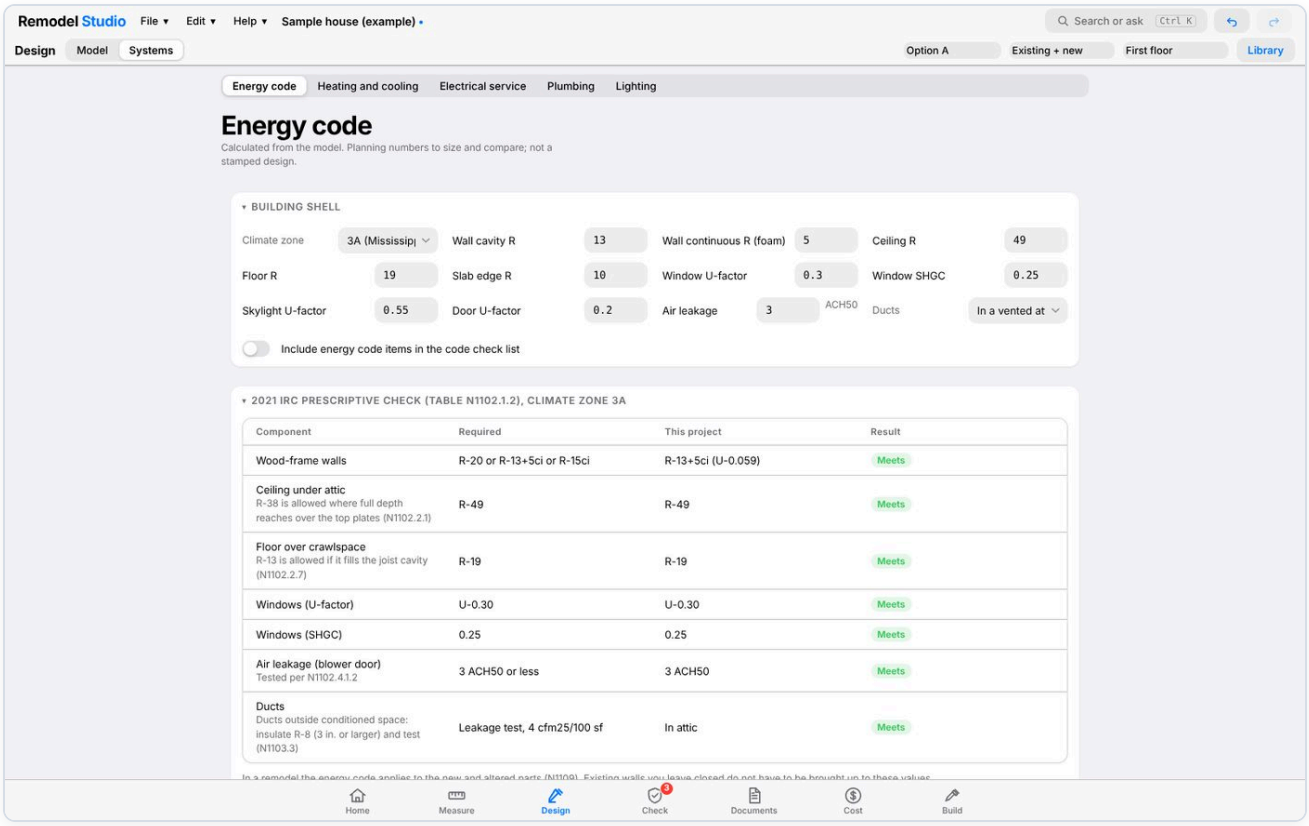
- Plan tabs **Floor framing** and **Roof framing** overlay the layout on the plan (joists with a span arrow and label "2×10 JOISTS @ 16" O.C.", GIRDER labels, OPEN crossed openings, red over-spanned members; truss lines with marks every few trusses, or rafter lines with ridges/hips).
- 3D **Framing** chip shows studs, plates, headers (LVL colored for bearing headers), king/jack/cripple studs, floor and ceiling joists, rim, rafters or truss chords and webs.
- Sheets (see *Every sheet the plan set can produce* in the Documents part): S1.x framing plan per level (member sizes, headers, new beams), S2.x joist layout per level, S2.9 ceiling joists, S3.0 roof truss layout with truss schedule, S3.1 roof rafter framing with rafter schedule, S4.1 wall framing elevations with header schedule.

CHAPTER 16

Systems page

Design workflow › **Systems** page (also Systems ribbon tab › Calculations › Systems page). Tabs: **Energy code**, **Heating and cooling**, **Electrical service**, **Plumbing**, **Lighting**. Header note: "Calculated from the model. Planning numbers to size and compare; not a stamped design."

1 Energy code



Systems › Energy code.

Building shell inputs: **Climate zone** 1A (south Florida), 2A (Gulf coast, south Louisiana), 3A (Mississippi, Alabama, Georgia — default), 4A (Tennessee, Kentucky, Virginia), 5A (Ohio, Illinois, Pennsylvania); choosing a zone also resets the design temperatures. Wall cavity R, Wall continuous R (foam), Ceiling R, Floor R, Slab edge R, Window U-factor, Window SHGC, Skylight U-factor, Door U-factor (default 0.20), Air leakage (ACH50), **Ducts** (In a vented attic / In a vented crawlspace / Inside the conditioned space), and **Include energy code items in the code check list**.

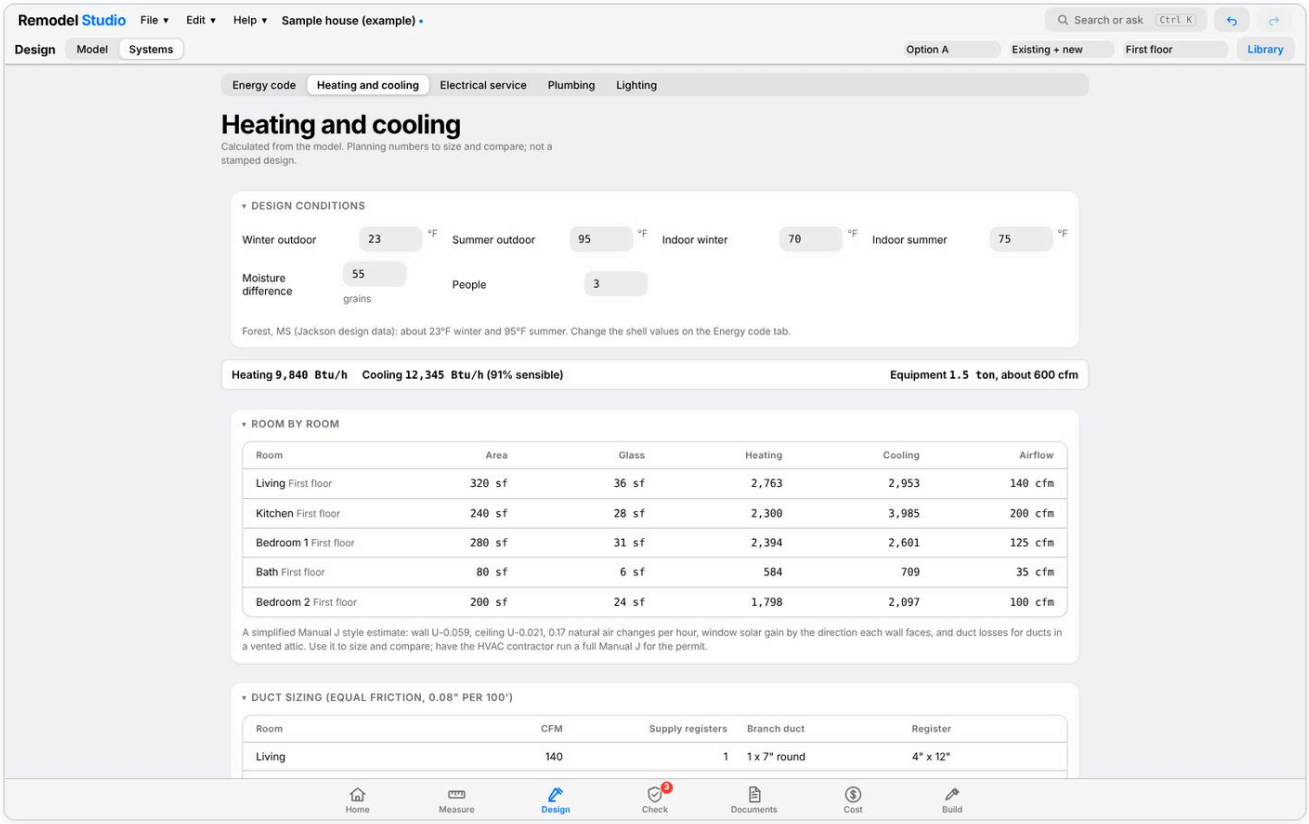
Defaults by zone (2021 IRC Table N1102.1.2):

Zone	Window U	Skylight U	SHGC	Ceiling R	Wall options	Floor R	Slab	ACH50	Design °F heat/cool
1A	0.50	0.75	0.25	30	R-13 or R-10ci	13	none	5	45 / 94
2A	0.40	0.65	0.25	49	R-13 or R-10ci	13	none	5	32 / 95
3A	0.30	0.55	0.25	49	R-20, R-13+5ci or R-15ci	19	R-10, 2 ft	3	23 / 95
4A	0.30	0.55	0.40	60	R-20+5ci, R-13+10ci or R-15ci	19	R-10, 4 ft	3	15 / 93
5A	0.30	0.55	none	60	R-20+5ci, R-13+10ci or R-15ci	30	R-10, 4 ft	3	5 / 90

2021 IRC prescriptive check table (Component / Required / This project / Meets or Below): wood-frame walls (or a maximum U: 0.084 zones 1–2, 0.060 zone 3, 0.045 zones 4–5), ceiling under attic (R-38 allowed at full depth over plates), floor over crawlspace (R-13 allowed when it fills the cavity), slab edge, windows U-factor, windows SHGC, skylights U-factor (when skylights exist), air leakage (blower door), ducts (R-8 and leakage test 4 cfm25/100 sf when outside conditioned space). Note: the energy code applies to new and altered parts (N1109).

Total UA trade-off (IECC R402.1.5, the REScheck method): table by assembly (Walls, Windows, Doors (one 24 SF door exempt), Ceilings, Floors over crawlspace, Skylights) with area, this project's UA and the code house UA (reference U: zone 1 fen 0.50/sky 0.75/ceil 0.035/wall 0.084/floor 0.064; zone 2 0.40/0.65/0.026/0.084/0.064; zone 3 0.30/0.55/0.026/0.060/0.047; zone 4 0.30/0.55/0.024/0.060/0.047; zone 5 0.30/0.55/0.024/0.045/0.033). Result **Passes/Fails** with "N % better/worse than code", SHGC mandatory check, glass area as % of floor area, and a slab-edge reminder. Window U-factors use each window's own value when set.

2 Heating and cooling



Systems › Heating and cooling.

Design conditions: Winter outdoor, Summer outdoor, Indoor winter (70), Indoor summer (75), Moisture difference (55 grains), People (default bedrooms + 1, minimum 2). Note: "Forest, MS (Jackson design data): about 23°F winter and 95°F summer."

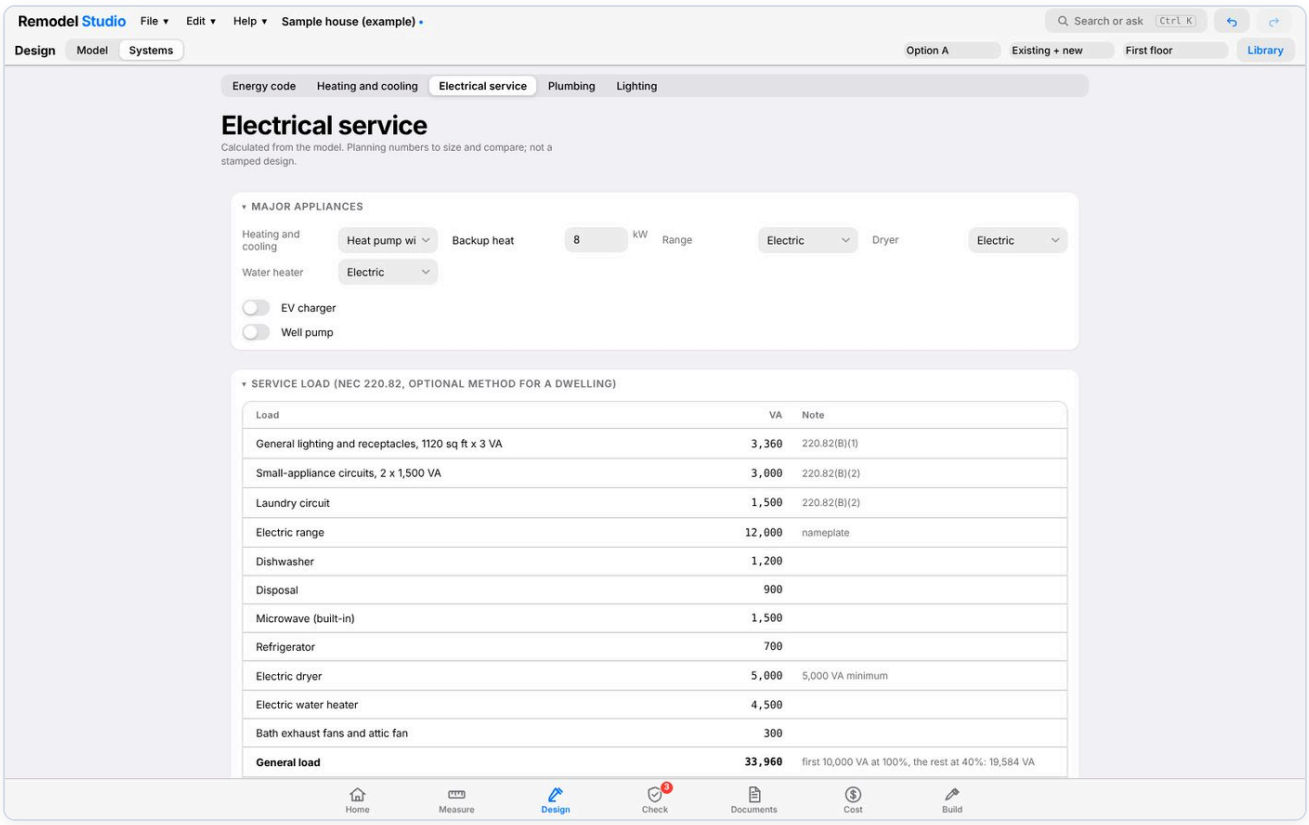
Method: a simplified Manual J. Wall U from the cavity and continuous R with 25 % framing; ceiling U = 1.07/(R+1.4); floor U = 1/(R+2.6); crawlspace floors at 60 %; slab edge 0.54 or 0.73 Btu/h·ft·°F (insulated/not); infiltration from ACH50 ÷ 18 (÷ 16 for two or more stories), cooling at 60 %; window solar gain by wall facing (N 35, NE/NW 90, E/W 140, SE/SW 110, S 70 Btu/h per sf × SHGC; skylights 200); kitchen +1,200 Btu/h; people 230 sensible + 200 latent each; duct losses ×1.15 heat/×1.20 cool in an attic, ×1.10/×1.08 in a crawlspace. Equipment = cooling ÷ 12,000 rounded up to the half ton (minimum 1.5 ton); airflow 400 cfm per ton.

Outputs: totals bar (Heating Btu/h, Cooling Btu/h with % sensible, Equipment tons and cfm); **Room by room** table (Room/level, Area, Glass, Heating, Cooling, Airflow cfm).

Duct sizing (equal friction, 0.08" per 100'): per room CFM, number of supply registers (one per 150 cfm), branch duct round size (6" ≤ 100 cfm, 7" ≤ 140, 8" ≤ 200, 9" ≤ 270, 10" ≤ 350, 12" ≤ 550, 14" ≤ 850, 16" ≤ 1250, 18" ≤ 1750, 20" ≤ 2400, else 22"), register (4" × 10" ≤ 100 cfm, 4" × 12" ≤ 150, else 6" × 12"); system note: tons, total CFM, supply trunk round and rectangular (8" ×

14" ≤ 600 cfm, 8" × 18" ≤ 800, 8" × 22" ≤ 1000, 10" × 22" ≤ 1200, 10" × 28" ≤ 1600, else 12" × 30"), return free area (sq in at about 400 fpm) and filter grille (20 × 20 ≤ 800 cfm, 20 × 30 ≤ 1200, else two 20 × 25).

3 Electrical service



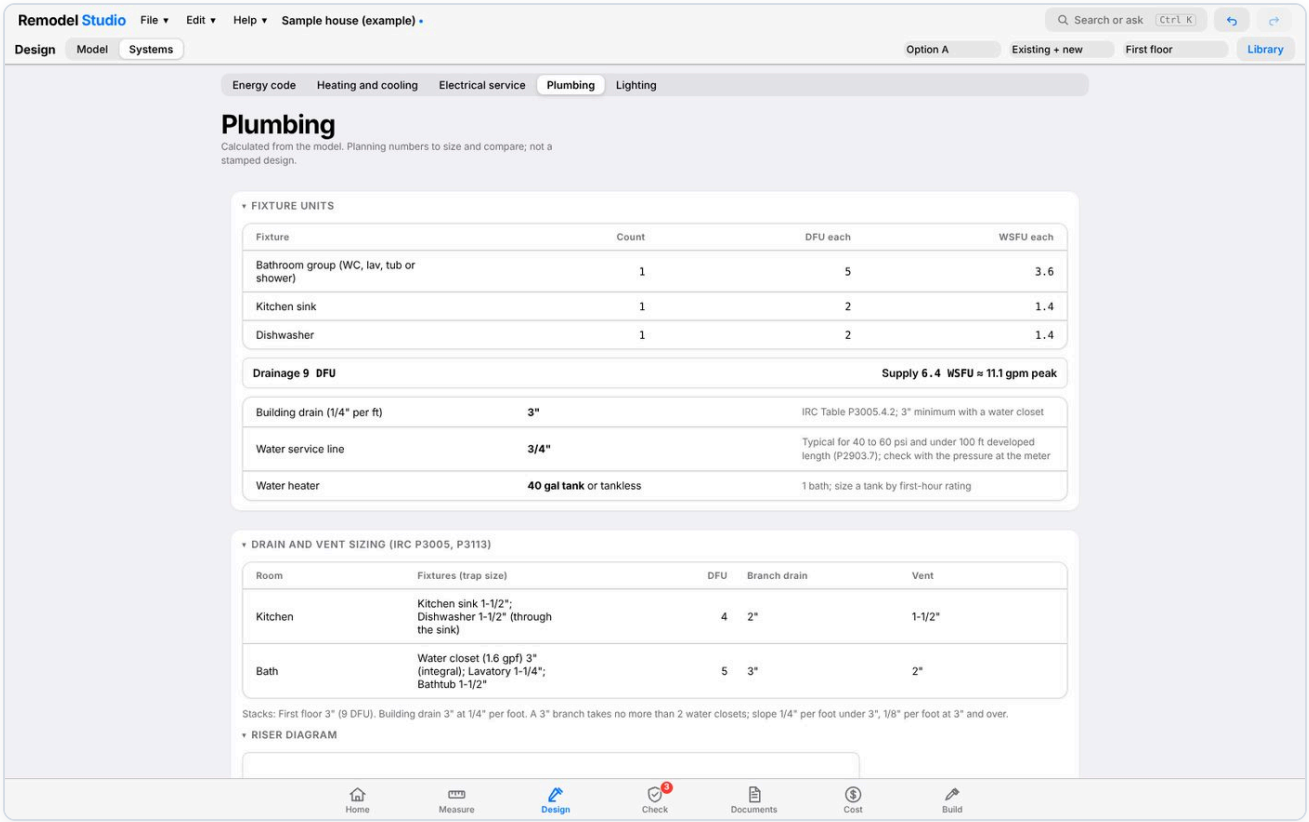
Systems › Electrical service.

Major appliances: Heating and cooling (*Heat pump with electric backup strips* (default), *Gas furnace with central AC*, *Electric furnace with central AC*, *Ductless mini-splits*); Backup heat kW (heat pump and electric furnace; default 8); Range (Electric/Gas); Dryer (Electric/Gas); Water heater (Electric/Gas/Heat pump water heater); EV charger; Well pump.

Service load (NEC 220.82 optional method) table (Load, VA, Note): general lighting and receptacles at 3 VA/sq ft of conditioned rooms; small-appliance circuits 2 × 1,500; laundry 1,500 (when a washer or laundry room exists); electric range 12,000; dishwasher 1,200 (when drawn); disposal 900; microwave 1,500; refrigerator 700; electric dryer 5,000; electric water heater 4,500; fans 300; EV charger 7,680; well pump 2,000; any extra appliances. General demand: first 10,000 VA at 100 %, remainder at 40 %. Heating/cooling: compressor = tons × 1,400 + 600 VA; heat pump = compressor + strips at 65 %; electric furnace = larger of AC and strips at 65 %; mini-split = tons × 1,200; gas = AC only. Totals bar: calculated VA, amps at 240 V, **Service size** (100, 125, 150, 200, 225, 320 or 400 A).

Panel schedule ("N A main, N spaces"): columns Space, Circuit, Breaker (A, 2P), Wire (copper NM-B: 15 A 14 AWG, 20 A 12, 25–30 A 10, 35–40 A 8, 45–60 A 6, 70–80 A 4, 90–100 A 3), Protection (GFCI/AFCI), VA, Notes. Dedicated circuits: two 20 A kitchen countertop (GFCI/AFCI), range 50 A 2P, dishwasher 20 A GFCI, disposal 20 A GFCI, microwave 20 A AFCI, refrigerator 20 A, laundry 20 A GFCI/AFCI, dryer 30 A 2P, water heater 30 A 2P, outdoor unit (breaker from MCA = tons × 5.6 + 3, × 1.5), air handler with strips (kW × 1000/240 × 1.25), furnace blower 15 A, mini-split 2P, EV charger 40 A 2P GFCI, well pump 20 A 2P, each bathroom 20 A GFCI, garage 20 A GFCI, outdoor 20 A GFCI; general lighting/receptacle circuits grouped by about 450 sq ft per 15 A AFCI circuit ("Lights and receptacles: room, room..." with receptacle and light counts); kitchen and bath lights with smoke/CO alarms 15 A AFCI. Spaces numbered odd left / even right, balancing VA per bus; panel size 12, 16, 20, 24, 30, 40 or 42 spaces (4 spare). Footnote on NEC 210.12 AFCI, 210.8 GFCI, dual-function breakers, 60 °C copper.

4 Plumbing



Systems › Plumbing.

Fixture units table (Fixture, Count, DFU each, WSFU each) built from placed fixtures: Water closet 1.6 gpf (3 DFU, 2.2 WSFU), Lavatory (1, 0.7), Bathtub (2, 1.4), Shower (2, 1.4), Kitchen sink (2, 1.4), Dishwasher (2, 1.4), Clothes washer standpipe (2, 1.4), Laundry tub (2, 1.4), Bar sink (1, 1.0), Floor drain (2, 0), Hose bibb (0, 2.5); a toilet + lavatory + tub/shower in one room counts as a **Bathroom group** (5 DFU, 3.6 WSFU). Totals: Drainage DFU, Supply WSFU ≈ peak gpm (IRC curve:

3 gpm at 1 WSFU, 5 at 2, 6.5 at 3, 8 at 4, 9.4 at 5, 10.7 at 6, 12.8 at 8, 14.6 at 10, 16 at 12, 17.5 at 15, 19.6 at 20, 21.5 at 25, 23.3 at 30, 26.3 at 40, 29.1 at 50, 32 at 60, 38 at 80, 43.5 at 100).

Results: **Building drain** (3" ≤ 42 DFU, 4" ≤ 216, else 5" or larger with a water closet; 2" ≤ 8, 2-1/2" ≤ 21, 3" otherwise without; Table P3005.4.2, 1/4" per ft); **Water service line** (3/4" ≤ 12.5 gpm, 1" ≤ 20, 1-1/4" ≤ 32, else 1-1/2"; P2903.7 at 40–60 psi under 100 ft); **Water heater** tank (40 gal for 1 bath, 50 for 2, 65 for 3, 80 above) or tankless.

Drain and vent sizing (IRC P3005, P3113): per room — fixtures with trap sizes (WC 3" integral, lav 1-1/4", tub 1-1/2", shower 2", kitchen sink 1-1/2", dishwasher 1-1/2" through the sink, washer 2", laundry tub 1-1/2", bar 1-1/2"), DFU, branch drain (with a WC: 3" ≤ 20 DFU else 4"; without: 1-1/2" ≤ 3, 2" ≤ 6, 2-1/2" ≤ 12, 3" ≤ 20, 4" ≤ 160, 5" ≤ 360, else 6"), vent (1-1/4" for 1-1/2" drains, 1-1/2" for 2" and 2-1/2", 2" for 3" and 4", 2-1/2" for 5"). Stack size per level (3" ≤ 48 DFU with WCs, else 4"); building drain; vent through roof (3" main vent with a WC, else 2"). A **riser diagram** (SVG) draws each level, room branches, fixture drops, vents, the stack and the building drain. Notes: a 3" branch takes no more than 2 water closets; slope 1/4" per foot under 3", 1/8" at 3" and over.

Water supply branches (P2903) and water heater: per room WSFU (bathroom group credit), peak gpm, cold and hot branch size in PEX (hot at 75 % of WSFU; cumulative WSFU ≤ 1.5 → 1/2", ≤ 9 → 3/4", ≤ 16 → 1", ≤ 30 → 1-1/4", else 1-1/2"); building WSFU, gpm, service and main; note on 3/8" PEX to single lavatories/toilets and upsizing over 60 ft or under 40 psi. Water heater: **First-hour rating needed** (people × 20 gal + 6 if dishwasher + 15 if washer + 8; people = bedrooms + 1), **Tank size** (40 gal ≤ 50 FHR, 50 ≤ 65, 65 ≤ 80, else 80), **Tankless** peak gpm (2.0 per bath up to two + 1.5 kitchen sink) at a temperature rise of 120 °F minus ground water (1A 75°, 2A 70°, 3A 62°, 4A 55°, 5A 47°), with a unit suggestion (≥ 5 gpm two units or 199k BTU gas; ≥ 3.5 gpm 180–199k BTU gas; else 150k BTU gas or 27 kW electric).

5 Lighting

Light levels by room table (Room, Fixtures, Lumens, Foot-candles, Target, Advice): lumens default per fixture type (ceiling light 1,600, recessed can 800, pendant 800, fan light 1,200, sconce 450, under-cabinet 300, exterior 700, or the fixture's own lumens); foot-candles = lumens × 0.6 utilization × 0.85 aging ÷ floor area; targets kitchen 30 (50 at counters), bath 30, living 10, dining 20, bedroom 10, hall 5, laundry 30, office 30, garage 20, closet 10, other 10; pills **No lights**, **Dim** (< 80 % of target), **Bright** (> 2.5× target), **Good**; advice "Add about N six-inch recessed lights (800 lm), spaced about H ft apart" and a per-room tip.

Lighting layout and foot-candles (point-by-point planner): pick a room and a recessed fixture from the library (4" LED wafer 650 lm, 6" LED wafer 1,000 lm, 6" retrofit can 900, 4" gimbal 700, flush-mount 1,600, drum pendant 1,200, mini pendant 500, ceiling fan with light 1,200, wall sconce 450, vanity bar 1,500, under-cabinet LED per 24" 350 — each with beam angle, watts and a planning price). A color map on a 1 ft grid at 30" work-plane height (blue dim, yellow/red bright at twice the target) with existing lights as white dots and a suggested grid as blue rings; stats Average fc, Lowest/highest, Target, Evenness (avg/min); button **Place N recessed lights in Room**

(replaces that room's non-existing cans/lights). The fixture library table is shown below. Note: 2,700–3,000 K, 90+ CRI.

Light color and marks: fixture type marks (A, B, C... per type/lumens/color/dimmable), **Set every light to** a Kelvin preset (2200 candlelight, 2700 warm white, 3000 soft white, 3500 neutral, 4000 cool white, 5000 daylight) with **Apply**, **Open the lighting schedule**; **Switch groups** list (S1, S2...: switch type, room, the marks it controls, LED watts, dimmer).

Lighting scenes: named dimmer levels per switch group (0–100 % sliders) used by Render; presets **Cooking**, **Dinner**, **Night light**, **Everything on**. Requires lights linked to switches (Electrical › Link switches).

6 Pipes and ducts on the plan

Systems ribbon › Plumbing and HVAC: chain-click tools **Cold water pipe**, **Hot water pipe**, **Drain pipe**, **Gas pipe**, **Supply duct**, **Return duct** (click points, Enter/Esc to finish). Each run's panel: **System** (Cold water, Hot water, Drain and waste, Vent, Gas) with sizes (cold 1/2", 3/4", 1" default 1/2"; hot 1/2", 3/4"; drain 1-1/2", 2", 3", 4" default 2"; vent 1-1/2", 2", 3" default 1-1/2"; gas 1/2", 3/4", 1" default 3/4", black steel or CSST), **Height above the floor** (defaults: water and gas 10" below the floor, drain 16" below, vent 60" above), run length and the 1/4" per foot drain slope note. Ducts: **Type** Supply (6", 8", 10", 12" round, 14" × 8", 20" × 8"; default 8" round) or Return (12", 14", 16" round, 20" × 10"; default 14" round); **Runs in** Attic or Crawlspace/below the floor; length and R-8 note. Pipes draw as colored lines labelled "CW 1/2"" etc. (blue cold, red hot, green drain/vent, gold gas); ducts as shaded strips with the size. They appear in 3D, on the plan sheets, in the builder's list (pipe LF, elbows, hangers; duct LF and elbows) and in the clash checks (1.8).

7 Systems sheets

Sheet **M0.1 Energy and systems calculations** (energy code table, room loads with totals and equipment, service load table, fixture units with drain/service/water heater), **E0.1 Electrical panel schedule** (panel header "PANEL A: N A MAIN BREAKER, 120/240 V 1-PHASE 3-WIRE, N SPACES", the schedule, grounding and tamper-resistant/smoke-alarm notes), **P1.1 Plumbing riser and duct schedule** (DWV per room, stacks, building drain and VTR; duct schedule per room with trunk and return).

PART

Documents

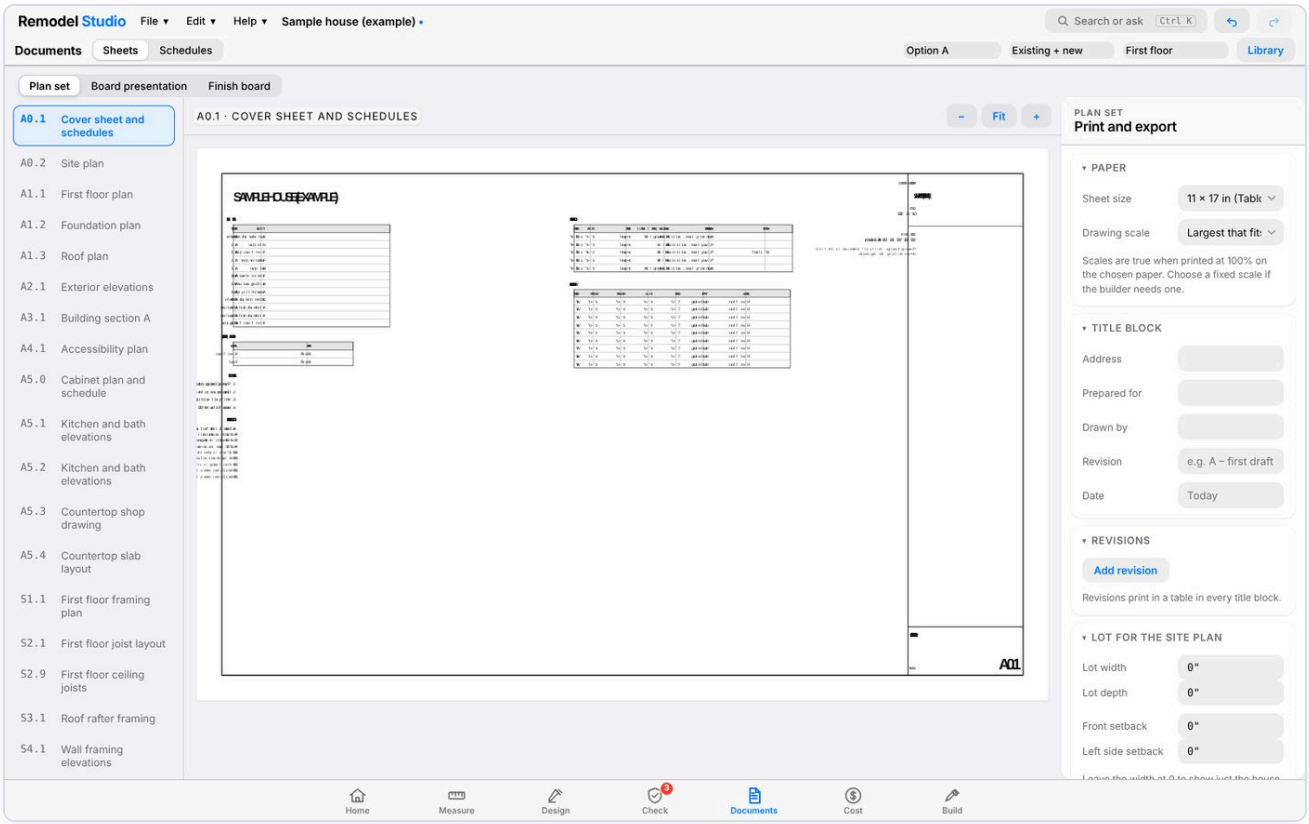
The plan set, schedules, keynotes, details, revisions, presentation, finish board, renders and exports.

CHAPTER 17

Documents workflow

Documents tab › pages **Sheets** and **Schedules**. The Sheets page has tabs **Plan set**, **Board presentation**, **Finish board**.

1 Plan set page



Documents › Sheets.

Layout: a sheet list on the left (sheet number and title; also a drop-down), the sheet preview in the middle with a title bar "No · Title · scale" and **–**, **Fit**, **+** zoom (up to 6×), and a **Print and export** panel on the right:

- **Paper**: **Sheet size** *24 × 36 in (ARCH D)*, *11 × 17 in (Tabloid)* (default), *8.5 × 11 in (Letter)*; **Drawing scale** *Largest that fits* (default) or fixed 1/2", 3/8", 1/4", 3/16", 1/8", 3/32", 1/16" = 1'-0". Site sheets also use 1" = 20', 30', 40', 60', 100'. "Scales are true when printed at 100 % on the chosen paper."
- **Title block**: Address, Prepared for, Drawn by, Revision, Date (blank = today).
- **Revisions**: rows of number, date, description; **Add revision**; revisions print in a table in every title block (last 6).

- **Lot for the site plan**: Lot width, Lot depth, Front setback, Left side setback (0 width hides the lot).
- **Zoning and permit packet**: Permit office, Parcel number, Front/Side/Rear setback (min), Max height, Max lot coverage %, Max impervious %; a live line "Heated N sq ft · footprint N sq ft · lot · coverage % · impervious %" and per-rule OK/NO; **Permit packet (PDF)** button.
- **Export**: **Download PDF (N sheets)** (vector, one page per sheet at the chosen paper size), **Download this sheet as SVG**, **Download DXF floor plans (zip)**, **Download all sheets as DXF (zip)**, **Download this sheet as DXF**.

Title block fields (every sheet): "REMODEL STUDIO"; project name (up to 3 lines); DESIGN OPTION (when options exist); ADDRESS; PREPARED FOR; DRAWN BY; DATE; REVISION; the revisions table; "CODE BASIS 2021 IRC, 2021 IBC, 2010 ADA Standards"; the disclaimer "Planning drawings. Verify all dimensions in the field before building. Not engineered."; bottom block with the sheet title, "SCALE: ...", the sheet number in large type and "SHEET". Drawing titles carry a bubble, the title, scale, a north arrow (rotated to the site bearing) and a graphic scale bar. Dimension strings are drawn on all four sides of plans (overall, wall breaks, opening centerlines).

2 Every sheet the plan set can produce

Sheets appear only when the model has the content; numbering is automatic.

Sheet	Name	What it shows
A0.1	Cover sheet and schedules	Project name, address, client, option; **Sheet index**; **Floor area (from rooms)** per level and total; **General notes** (planning drawings; dimensions to outside face of framing and centerlines; verify existing; code basis); **Code check summary** (fail/review counts and up to 8 items); **Door schedule** (Mark, Size, Door type/style, Handing (Res / ANSI), Hardware, Notes: 32" clear?, 20 MIN, CLOSER); **Window schedule** (Mark, Width, Height, Sill, Head, Type, Level); under IBC a **Code analysis (2021 IBC)** table
A0.2	Site plan	Site features, grading, decks and steps, first-floor exterior walls, roof outline, "HOUSE" label with area and overall size, overall dimensions, lot line with lot dimensions and setback dimensions when a lot is entered (site scales)
L1.0	Landscape and play area plan	Site, house, play areas with surfacing labels, equipment with use zones, plant canopies with two-letter keys; **Plant schedule** (Key, Common / Botanical, Qty, Size), **Playground schedule** (Model, Item, Qty, Fall ht), notes (plant size shown, Call 811, CPSC/ASTM, ADA surfacing)
A0.3	Keynote legend	Table No., Note, Builder's list item, Qty, Where (levels) for every keynote
A1.n	*Level* existing and demolition plan	For remodels with existing/demolished walls: existing walls gray, walls to remove dashed
A1.n	*Level* (proposed) plan	Floor plan: rooms tinted with name, size and SF, walls (new dark, existing gray), doors with swings and marks (circles), windows (hexagon marks), stairs (UP/DN), fixtures and cabinets, all registered plan objects, drafting, dimension strings, the section mark A-A when no sections are drawn

Sheet	Name	What it shows
A1.n	Foundation plan	Bearing/exterior walls with footing lines; slab note "4" CONCRETE SLAB ON GRADE OVER 6 MIL VAPOR RETARDER / THICKENED EDGE...", or crawlspace girders and 16" × 16" piers at about 8' with the joist note
A1.n	Roof plan (or *Level* roof plan)	Roof edges, ridges/hips/valleys, dormers, slope arrows with pitch "n:12", covering, overhang and framing note
A2.1	Exterior elevations	Four elevations (named by compass from the site bearing) with grade line (sloped when graded), finished floor / top of plate / ridge level markers, window grilles and arches, door styles
A3.n	Building section A (B, C...)	One per drawn section cut (or an automatic cut across the ridge): walls, floors, ceilings, roof, footings, room labels, level markers
A4.1	Accessibility plan	First-floor plan with 60" turning circles in baths and kitchens (red when short), door maneuvering clearances, toilet 60" × 56" clear floor, and the **Accessibility requirements (2010 ADA Standards)** table (door clear width, latch side, threshold 1/2", turning space, accessible route, ramp slope, toilet centerline 16–18", toilet clearance, lavatory rim 34", grab bars 33–36", work surface 34"/30", reach range 15–48")
A5.0	Cabinet plan and schedule	Numbered cabinet plan of the cabinet level with a schedule (#, SKU, Type, W x H x D, Style, Price)
A5.n	Kitchen and bath elevations	Interior elevations, four per sheet, of every wall carrying fixtures or cabinets: cabinets with numbers, counters and splash, appliances, plumbing fixtures, tile, bottom dimension string and heights; titled "Room: direction wall"
A5.n	Countertop shop drawing	Counter pieces with dimensions, "W" wall sides, sink cut-outs with centerline distance, suggested seams, piece table (PC, Size, SF, Cutouts, Seams), material/edge/splash/slab notes
A5.4	Countertop slab layout	Pieces laid on slabs with grain lines, piece list, slab count and yield
A6.1	Schedules	**Room finish schedule** (Level, Room, Floor, Base, Walls, Ceiling, Clg ht, Area) and **Plumbing, appliance and mechanical schedule** (Type, Item, Qty, Location)
S1.n	*Level* framing plan	Bearing walls shaded, joist span arrows with size and spacing ("CJ" for ceiling joists), header labels "D1 HDR: (2) 2×10", new beams "NEW BEAM: ... WITH POSTS AT EACH END"; red where the suggested upgrade is needed
S2.n	*Level* joist layout	Joists, rim, girders, trimmers/headers, openings; **Floor framing schedule**; notes on bearing, doubled trimmers, blocking
S2.9	*Top level* ceiling joists	Ceiling joist layout with its schedule
S3.0	Roof truss layout	Trusses with marks, **Truss schedule** (Mark, Qty, Type, Span, Pitch, heel, ply), truss notes
S3.1	Roof rafter framing	Rafter lines, ridges/hips, **Rafter framing schedule** (rafters by length, ridge/hip boards one size deeper, collar ties, hurricane ties)
S4.1	Wall framing elevations	Up to six new exterior wall framing elevations at 1/4" and the **Header schedule**

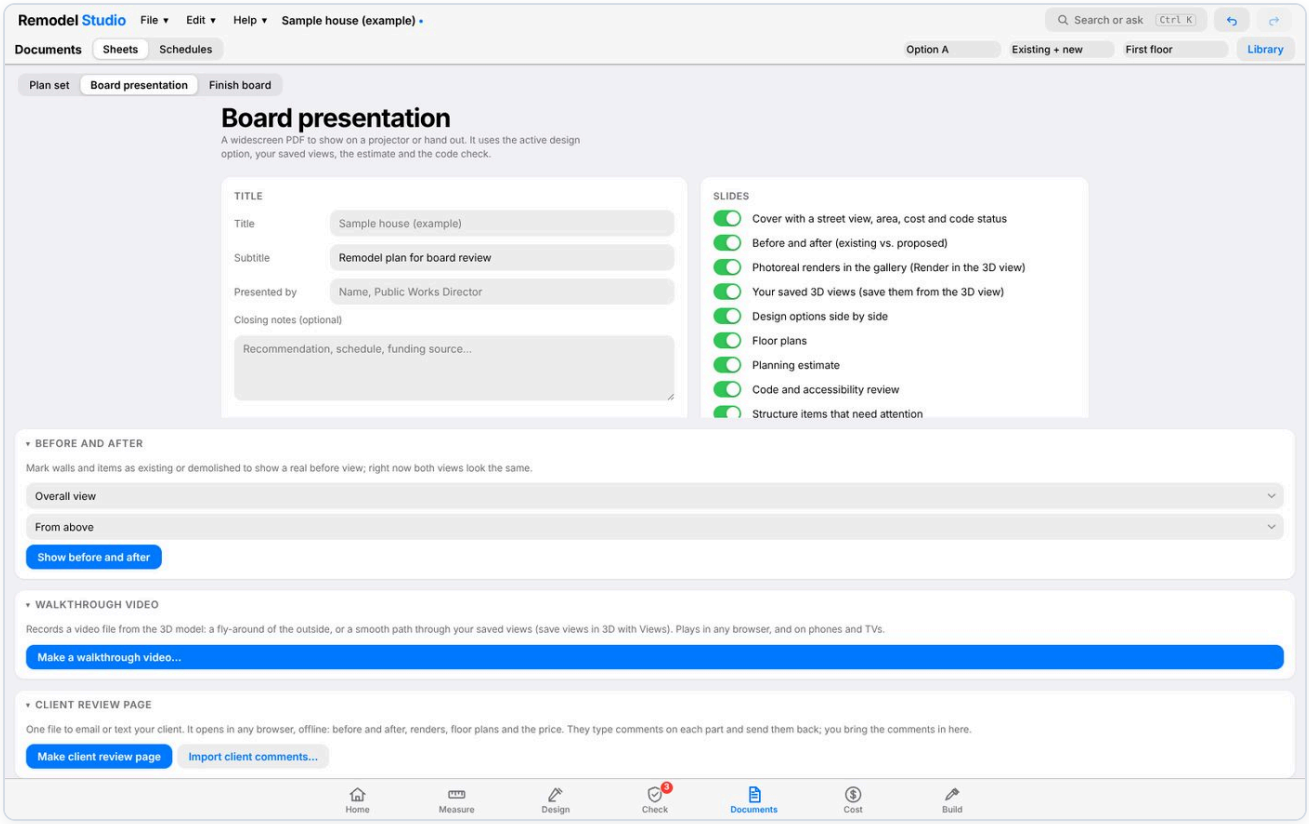
Sheet	Name	What it shows
D1.1	Typical details	Six details drawn from the project: Typical wall section (foundation type, joists, studs, insulation R-values, ceiling joists, roof), Opening framing (largest header), Floor finish assembly (tile or other), Shower pan and walls (when a tub/shower exists) or Cabinet and grab bar backing, Stair handrail and guard (when stairs exist), Deck ledger (when decks exist); each with leader notes, code references and a detail bubble
M0.1	Energy and systems calculations	See *Systems sheets* below
P1.1	Plumbing riser and duct schedule	See *Systems sheets* below
E1.n	*Level* electrical plan	Plan with devices, dashed switch legs, an **Electrical legend** with counts, and notes (receptacle and switch heights, GFCI, AFCI, alarms)
E0.1	Electrical panel schedule	See *Systems sheets* below

Revision clouds: when the contract is locked and a change order is created (Cost › change orders), the parts that changed are clouded on the plan sheets with a red cloud and a revision triangle numbered with the change order's revision; void change orders lose their clouds. A manual **Revision cloud** drawing tool (Annotate › Markup) also exists. Change orders add rows to the title-block revisions automatically ("Change order N: …", marked (signed)/(void) when the status changes).

3 Permit packet (PDF)

A letter-size cover letter and data package: date, "Building Official", the permit office, "Re: Building permit application, address"; a paragraph (remodel vs remodel and addition, 2021 IRC, ADA where applicable, number of sheets); **Project data** (project, address, parcel, owner, prepared by, occupancy and construction "R-3 one- and two-family dwelling, Type V-B", code basis "2021 IRC (, 2021 IBC), 2020 NEC", foundation); **Areas** (heated per level, total heated, garage, porches and decks, footprint, total under roof); **Site data** (lot size and area, lot coverage under roof, impervious sq ft and %, front/left/right/rear distances, building height average grade to peak); **Zoning compliance** rows (requirement · actual · complies / DOES NOT COMPLY); **Sheet index**; **Inspections** checklist (footing and foundation; under-slab plumbing; framing with rough plumbing/electrical/mechanical; insulation; final).

4 Board presentation



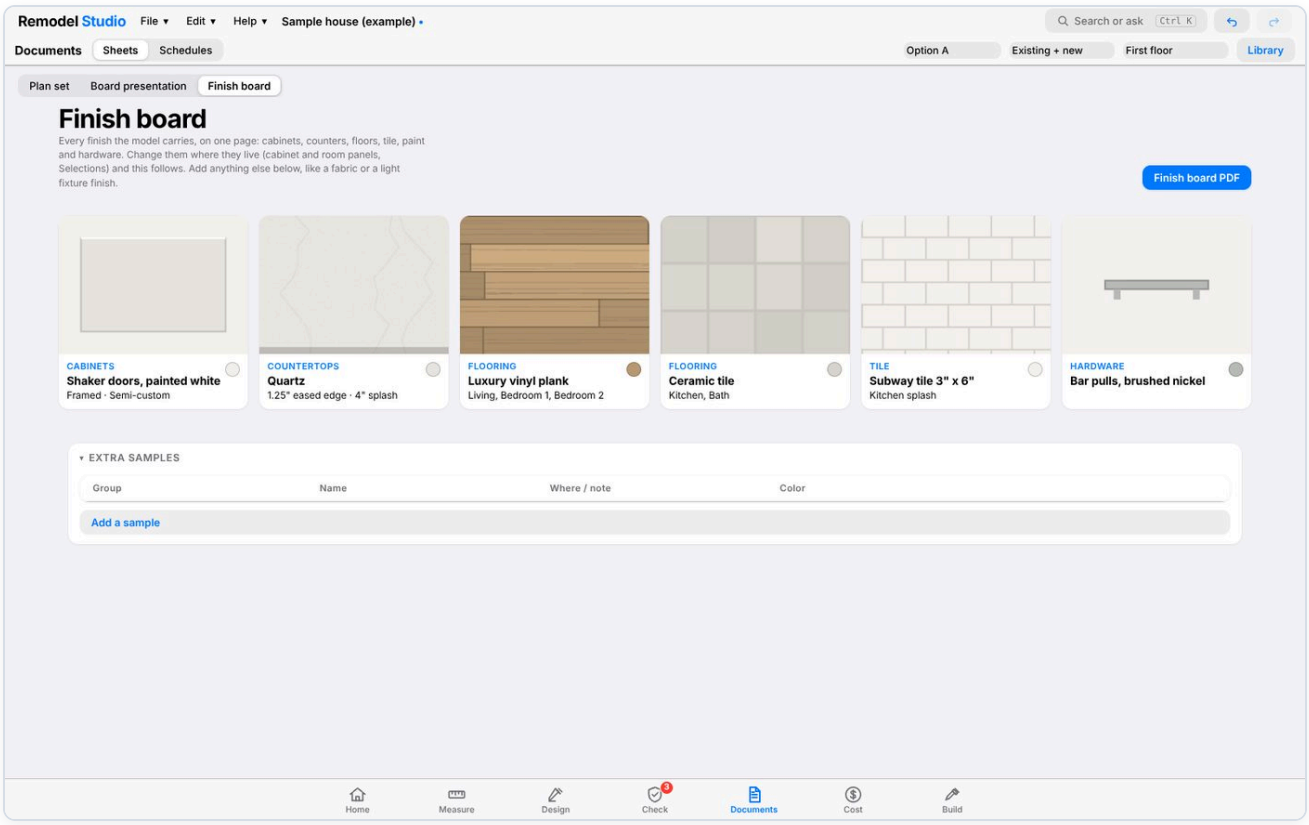
Documents › Board presentation.

Sheets › **Board presentation**: a 13.333 × 7.5 in widescreen PDF. **Title** fields: Title (defaults to the project name), Subtitle ("Remodel plan for board review"), Presented by, Closing notes. **Slides** checkboxes: - Cover with a street view, area, cost and code status (hero 3D image; "N sq ft · $ planning estimate · N code items to resolve / no code failures found") - Before and after (existing vs. proposed) — dollhouse views; skipped when nothing is marked existing - Photoreal renders in the gallery - Your saved 3D views (two per slide; or an overall and dollhouse pair when none are saved) - Design options side by side (up to 3, with area, estimate, labor hours, code items) - Floor plans (one per level) - Planning estimate (materials, tax, labor, dumpsters, permit, extras, overhead and profit, contingency, total; per-option totals) - Code and accessibility review (counts to resolve/review and up to 12 items) - Structure items that need attention (beams and overstressed members) - Finish board (cabinets, counters, floors, tile, paint, hardware) - Closing notes page when notes are entered. Buttons **Make presentation PDF** (with progress) and **Preview the cover image**.

Presentation extras on the same page: **Before and after** interactive slider (view: overall or a saved view; angle From above / From the street / Dollhouse); **Walkthrough video** (opens the Video dialog, see *Walk-through video* in the next chapter); **Client review page**: one self-contained HTML file (before/after slider or the design image, up to 8 saved views, last 6 gallery

renders, up to 3 plan sheets, price table) with a comment box per section and a **Send comments** button that saves a JSON file and opens an email; **Import client comments…** brings the JSON back as a checklist (who, section, date, text, done checkbox).

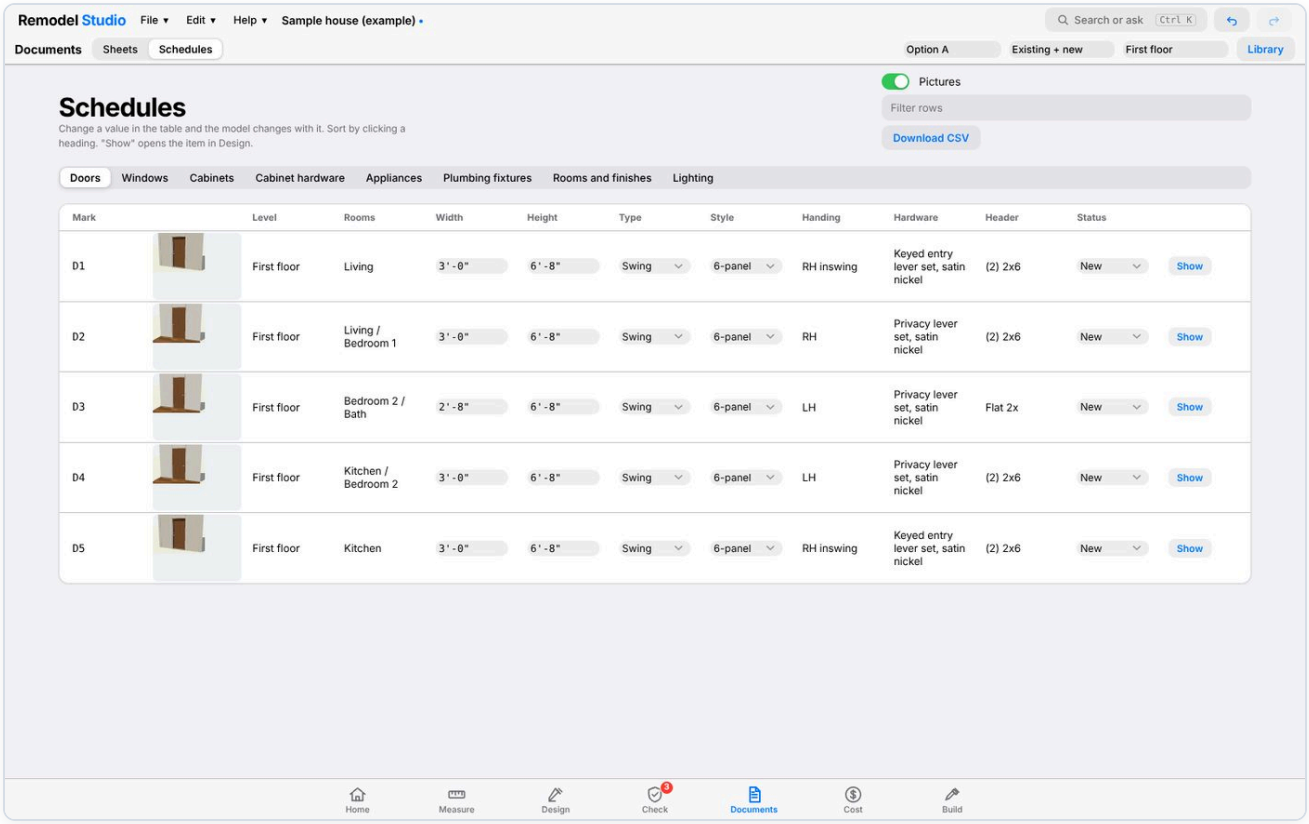
5 Finish board



Documents › Finish board.

Sheets › **Finish board**: tiles for every finish the model carries — Cabinets (door style, finish, framed/frameless, line, vendor), Countertops (material, thickness, edge, splash, brand), Flooring (per material/color with rooms), Tile (per material/color with where), Paint (per color with rooms), Wall finish, Hardware (pull style and finish, brand) — each with a generated swatch picture and color chip. **Extra samples** table (Group, Name, Where/note, Color) with **Add a sample**. **Finish board PDF** (landscape letter). The finish board also becomes a presentation slide.

6 Schedules page



Documents › Schedules.

Documents › **Schedules**: interactive tables ("Change a value in the table and the model changes with it"). Controls: **Pictures** toggle (thumbnail column), **Filter rows** search, **Download CSV**, click a heading to sort, **Show** on each row opens the item in Design. Types and columns:

Schedule	Columns
Doors	Mark, picture, Level, Rooms, Width, *Height*, Type *(door type), Style*, Handing, Hardware, Header (from the header schedule, "(check)" when failing), Status* (Existing/New/Remove)
Windows	Mark, picture, Level, Rooms, Width, *Height*, Sill, *Head, Operation* (double-hung, casement, slider, awning, fixed), Net clear opening (sf and W × H, or "fixed"), Header, Status*
Cabinets	#, picture, Code, Room, Width, *Depth*, Height, *Door style*, Finish, *Hinge* (Auto/Left/Right), Price, Status*
Cabinet hardware	#, Cabinet code, Hardware, Finish, Pieces (e.g. "3 × Bar pull 5-1/16" (128 mm)"); plus an **Order totals** table by piece and finish. Pull sizes: drawers ≤ 15" 3-3/4" (96 mm), ≤ 24" 5-1/16" (128 mm), ≤ 33" 6-5/16" (160 mm), else 7-9/16" (192 mm); doors 3-3/4", tall doors 48"+ appliance pull 12"; cup pulls 3"; edge pulls 3"/6"; knobs 1-1/4"
Appliances	picture, Appliance (brand model), Type, Room, Size W x H x D, Cut-out, Electrical or fuel, List price, Spec (link), Status*

Schedule	Columns
Plumbing fixtures	picture, Fixture, Room, Level, Width, *Depth*, Status*
Rooms and finishes	Level, Room, *Area*, *Floor*, Base, *Walls*, Wall color, *Ceiling*, Ceiling height, *Notes*
Lighting	Mark, Fixture, Qty, Lumens, *Color* (Kelvin), Dimmable*, Watts (LED), Switches, Rooms

* editable in the table.

7 Keynotes and tags

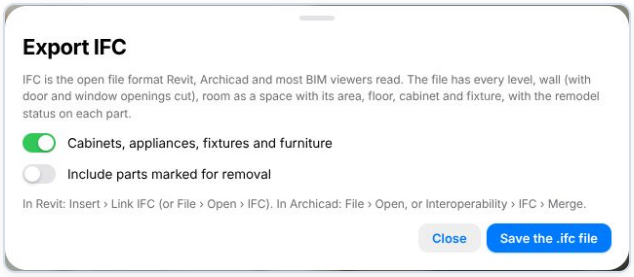
- **Keynote** tool (Annotate › Text and tags): click to place a numbered diamond; it reuses the last note. Its panel: **Note** picker (existing keynotes or "+ New keynote"), **Text** (every tag with this number changes), **Number**, **Builder's list item** link. Keynotes print as numbered diamonds and are listed on **A0.3 Keynote legend** with the linked item, quantity and the levels where they are used.
- **Tag by example** tool: click a cabinet, appliance, door, window, light or room; every item like it on the level gets a tag (cabinet code, appliance model, door/window mark, light mark, room name and SF). Tags follow their item; drag to offset (a leader is drawn when offset more than 6"). Panel button **Remove every tag made with this one**.

8 DXF / DWG export and import

Export - **DXF floor plans (zip)**: AutoCAD R12 ASCII DXF, full size in inches, one file per level of the proposed design plus a README. Layers (US National CAD Standard names): A-WALL, A-WALL-EXST, A-GLAZ, A-DOOR, A-AREA (room outlines), A-AREA-IDEN (room name and SF), A-ANNO-TAGS (door/window marks), A-FLOR-FIXT, A-FLOR-CASE-UPPR (wall cabinets, dashed), A-FLOR-STRS. - **This sheet as DXF / All sheets as DXF (zip)**: every line, shape and text of the sheet in paper inches on layers A-ANNO, A-ANNO-TEXT, A-ANNO-HIDN (dashed), A-PATT (filled). - **This sheet as SVG**: the sheet at its paper size.

Import a DWG or DXF drawing (dialog): **Drawing units** (Read from the file (inches if not set), Inches, Feet, Millimeters, Centimeters, Meters); **Turn parallel line pairs into walls (CAD-to-walls)** (pairs 3.5"–12.5" apart with 50 % overlap become walls on the centerline; ≥ 5.4" become exterior/bearing; status existing); **Choose DWG or DXF file…**. Reads LINE, LWPOLYLINE and POLYLINE (with bulges/arcs), CIRCLE, ARC, ELLIPSE, SPLINE, SOLID/3DFACE, TEXT/MTEXT/ATTRIB, INSERT (scaled, rotated, arrayed, nested blocks) and DIMENSION blocks. Items land on "DXF " layers as existing drafting; text becomes notes; the drawing is moved next to the house if it lands far away. DWG files are converted with the free LibreDWG reader (about 11 MB, downloaded once). Named blocks can be saved with **Save its N blocks to my Blocks** for the Block tool.

9 IFC export



Export IFC.

View ribbon › Export › **IFC for Revit or Archicad**. Writes an IFC2X3 Coordination View file in meters: storeys, walls (exterior/interior, with openings cut, IsExternal and LoadBearing properties, assembly name, status and thickness), doors and windows (marks, type/style or operation, width/height/sill), rooms as spaces (use, area, floor and ceiling covering) with floor slabs, and optionally cabinets/appliances/fixtures/furniture as furnishing elements (vendor, model, size, status). Options: **Cabinets, appliances, fixtures and furniture** (on) and **Include parts marked for removal** (off). The result message counts walls, doors, windows, spaces and items. Notes: Revit Insert › Link IFC; Archicad File › Open or IFC › Merge.

CHAPTER 18

Render and views

1 3D view controls

Chips over the 3D view: **Walk through** (W/A/S/D or arrows, drag to look, Shift to hurry, Esc stops; doors let you through), **Views (N)**, **Paint**, **Framing**, **360°**, **Video**, a section-cut drop-down (No section cut / Cut height / Cut left to right / Cut front to back / Section A…) with **Flip** and a position slider; corner buttons **Show roof**, **Cut at level**, **Sun**, **Render**, **AR / Export**, **Loads**, **Fit**.

2 Saved views

Views chip › name field (or "Name this walk view") › **Save view**. Each saved view lists with a go-to button (🚶 for walk views) and a delete (×). Saved views feed the Board presentation, the client review page, the Render studio camera list ("Saved view: …"), the render kit camera presets and the walkthrough video path. Project browser › 3D views also lists them.

3 Sun study

Sun chip opens the Sun study panel: date picker, time slider (5:00 AM to 9:00 PM in 5-minute steps), **Play day / Pause**, quick dates Mar 20, Jun 21, Sep 22, Dec 21; readout "N° up, from the (az°)" or "Sun is down". **Location and orientation**: Place (default Forest, MS), **Top of plan faces** (N/NE/E/…; elevations and the north arrow follow it), Latitude (32.364), Longitude (−89.474), UTC offset (standard, −6), US daylight saving. Sun position uses the NOAA low-precision formulas; shadows update in the 3D view.

4 Render studio

Render button (3D view or View ribbon › Render) opens the Render dialog: GPU path tracing in the app ("real sun and sky, glass, reflections and light bouncing between rooms"; engine three-gpu-pathtracer). Options: - **View**: *The 3D view as it is now*, *Street view* (faces the first exterior door), *Aerial view*, four corner views named by compass (e.g. "Northwest corner"), and *Saved view: …* for each saved view. - **Light and sky**: *Sunny, at the sun study time*; *Golden hour (late afternoon)* (first time after noon the sun drops to 12°); *Overcast, soft shadows*; *Dusk with the lights on*; *Night with the lights on*. - **Lights on** (N fixtures from the electrical plan; or one ceiling light per room at about 20 lm/sf when none are drawn); **Lighting scene** (when scenes exist: All lights at full or a named scene); **Light brightness** 0.25× to 4×. - **Auto exposure (like a camera)** and an **Exposure / Brighter or darker** slider −3 to +3 EV (live while rendering). - **Image size**: Draft 960 × 540, HD 1920 × 1080, Large 2560 × 1440, Square 1600 × 1600, Portrait 1350 × 1688, Plan sheet 3300 × 2100. - **Quality**: Preview (16 passes), Quick (64), Good (256), Best (700), Final (2,000). Best and Final use 8 light bounces, otherwise 6. - **Smooth out grain** (denoise, live),

Camera depth of field (f/4 instead of f/22). - Buttons **Render / Render again / Stop**, **Save image** (PNG), **Add to gallery** (JPEG kept with the project; gallery renders go on the presentation board). Progress bar shows passes done, elapsed time and the estimate left. A **Gallery** of saved renders with remove buttons; click to enlarge. Settings are remembered with the project.

The **AR / Export** dialog also offers **Open Render**, and on the desktop build a **Photoreal render with Blender** section (finds blender.exe; Camera: Street, Aerial, Every view, or a saved view; Quality Draft 1280 × 720 / Good 1920 × 1080 / Best 2560 × 1440; renders with Cycles and shows the images; files saved in Documents\Remodel Studio\Renders). A **render kit** zip (model.glb, scene.json with sun and cameras, render.py, README) can be produced for Blender 4.2+ with command-line options --camera, --all, --samples, --size, --open.

5 AR view, 3D export, panorama, video

AR / Export dialog: - **View in AR**: builds a USDZ of the proposed design (demolished work left out); on an iPhone in Safari it opens AR Quick Look; elsewhere a zip with the .usdz to AirDrop/open in Files. - **3D print or open in another CAD program**: **Scale and units** *1:48 (1/4" = 1'), millimeters*, *1:96 (1/8" = 1'), millimeters*, *Full size, millimeters*, *Full size, meters (CAD)*; **Include the roof**; buttons **STL for 3D printing** (binary STL, Z up), **OBJ**, **GLB with colors**; each in a zip with a README. Glass is left out so openings stay open. **Export 3D** on the View ribbon opens the same dialog. - Imported models for the library accept OBJ, GLB, glTF (single file), DAE and STL.

360° panorama (chip or View ribbon): captures a cube map from the current 3D camera (walk-through spot recommended) and writes a 4096 × 2048 equirectangular JPEG with a drag-to-look, scroll-to-zoom viewer; **Save image**, **Add to gallery** (also goes to the presentation).

Walkthrough video (chip, View ribbon or presentation page): **Path** *Fly around the outside* (orbit of 8 keyframes) or *Through my saved views* (two or more saved views, eased), **Seconds between views** 2/3/4/6, **Size** HD 1280 × 720, Full HD 1920 × 1080, Square 1080; checkboxes to include each saved view; records in real time with a progress bar to WebM (or MP4 where supported) at 8 Mbps; preview player, **Record again**, **Save video**.

6 Photo match (reference)

View ribbon › Render › **Photo match** lines the 3D camera up with a real photo by clicking six or more room corners (floor and ceiling); the design then renders from that camera and blends over the photo. Documented with the Measure workflow.

PART

Cost

Builder's list, estimate, change orders, shopping, selections and orders, budget, bids, cut plan and cabinet order.

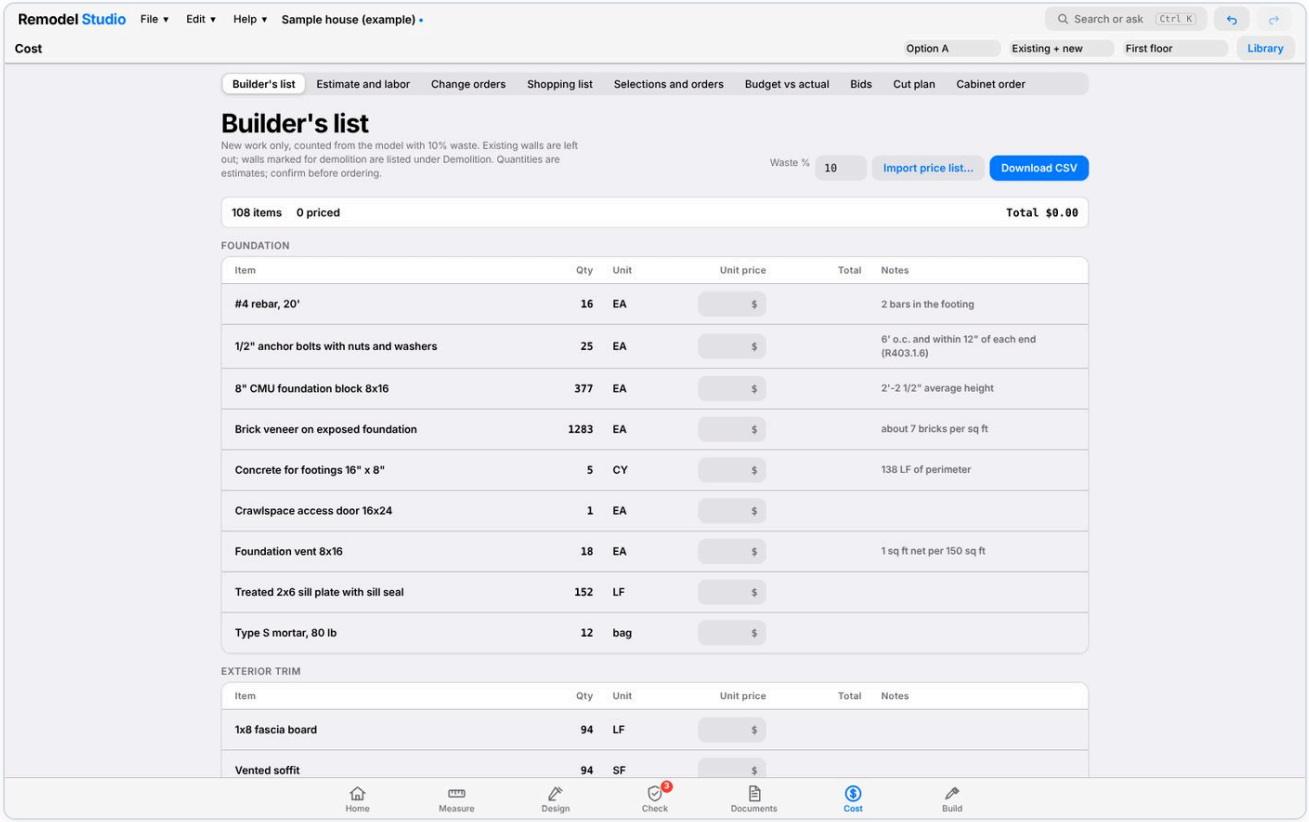
CHAPTER 19

Cost workflow

Cost has one page with a row of tabs, in this order:

1. Builder's list
2. Estimate and labor
3. Change orders
4. Shopping list
5. Selections and orders
6. Budget vs actual
7. Bids
8. Cut plan
9. Cabinet order

1 Builder's list (takeoff)



Cost › Builder's list.

A material takeoff counted from the model. The header note reads: "New work only, counted from the model with N% waste. Existing walls are left out; walls marked for demolition are listed under Demolition. Quantities are estimates; confirm before ordering."

Header controls - **Waste %** — a number box, default **10**. Accepts 0–50. Saved with the project. Waste is applied as a multiplier (1 + waste/100) to most quantities; it is not applied to count-type items (one door is one door) or to things like thinset bags, joint compound or paint. - **Import price list...** — opens a paste box (see "Pricing sources" below). - **Download CSV** — saves `<project> materials.csv` with columns Category, Item, Quantity, Unit, Unit price, Total, Notes and a final Total row.

Totals bar: `N items · N priced · Total $x,xxx.xx` (total counts only priced lines).

Tables, one per category. Columns: Item, Qty, Unit, Unit price (editable), Total, Notes. Type a price in the Unit price box to set it; clear the box to remove your price. A price you type is saved with the project and wins over every other price source.

Units used: EA (each), LF (linear feet), SF (square feet), SY (square yards), CY (cubic yards), bag, box, roll, square (100 sq ft of roofing or siding), bundle, brick, pair, set, gal, lb, ton, bale, pallet, LS (lump sum), test, note, SF face. LF, roll, square and gallon quantities are rounded up to whole numbers.

Categories, in the order they appear, and how each is counted (plain words):

Category	What is counted
Demolition	Walls marked *demolish*: LF of wall to remove (bearing walls are noted "shore first and install the new beam") and SF of drywall to remove (one face for exterior walls, two for interior). Fixtures, cabinets and electrical devices marked demolish: one "Remove ..." line each.
Masonry	CMU walls: 8" or 12" block at 1.125 blocks per sq ft, Type S mortar (1 bag per 30 blocks), #4 rebar, grout for filled cells. ICF: blocks at 5.33 sq ft each plus core concrete. Glass block: 2.25 per sq ft.
Wall framing	For each new wood or metal-stud wall: studs at 16" o.c. plus 1 end stud, 2 per opening and 2 per wall join (precut 92-5/8", 104-5/8" or 116-5/8" when the wall height matches; otherwise stock lengths); plates (double top plate and bottom plate as 16' pieces; on a slab the bottom plate is listed separately as pressure-treated); metal stud and track for metal walls; SIP panels (32 sq ft each); post-frame posts at 8' o.c. and 2×4 girts at 24" o.c.; flat 2x headers for openings that do not need a structural header. Brick or stone veneer walls add bricks (6.75 per sq ft) or SF of stone.
Headers and beams	One line per opening header sized by the structure check: LVL plies (1-3/4" x depth x stock length), dimensional lumber plies, or "Engineered header, size by engineer".
Floor framing	Per room on a framed floor: floor joists (room width ÷ spacing + 1, at the size the structure check suggests) and 3/4" T&G subfloor sheets (32 sq ft each).
Roof and ceiling framing	Ceiling joists per room; rafters per roof (run × pitch factor, both sides unless shed) when the roof is not trussed; ridge board (2×10 or 2×12) by ridge length; dormer framing packages.
Sheathing and wrap	7/16" OSB wall sheathing (32 sq ft per sheet) on exterior wood walls; house wrap (9' x 150' rolls, 1,350 sq ft).

Category	What is counted
Roofing	7/16" OSB roof sheathing; synthetic underlayment (10 sq rolls); the roof cover by type: shingles in bundles (3 per square) with starter strip along the eaves, metal in SF, tile/slate/shake in squares, low-slope membranes in squares; ridge and hip cap (LF), drip edge (10' pieces), ice and water barrier rolls when turned on, ridge vent (80% of ridge) and continuous soffit vent when turned on, gutters (5" K-style, color noted) and downspouts (2 per eave run).
Siding	Brick veneer (bricks), fiber cement / wood / SmartSide planks (8-1/4" x 12', 7" exposure), vinyl or Dutch lap (squares), other sidings in SF. Veneer walls (brick, stone, CMU, ICF) get no separate siding line.
Doors and windows	One line per door with its full description (prehung interior/exterior, style, type, size, handing with ANSI code, material, sidelites, transom, 20-minute rating); pocket door frame kits, barn door and bifold track sets. One line per window (operation, size, casement hinge side; frame, glass and screen in the notes).
Insulation	R-13 batts (2×4 walls) or R-21 (2×6) by net exterior wall area; R-38 blown attic insulation by room area on the top floor.
Drywall	1/2" 4×12 sheets: one face of exterior walls, both faces of interior walls, ceilings by room area; all-purpose joint compound (4.5 gal boxes) and paper tape (500' rolls) from the sheet count.
Ceilings	By the room's ceiling type: acoustic tile (2×2 or 2×4) with grid tees, cross tees and wall angle; T&G, plank, shiplap or beadboard boards (SF); tin tiles; box beams (one about every 4 ft); coffer beam material (4' grid); tray framing; ceiling texture bags; a "Popcorn ceiling: asbestos test before scraping" line for popcorn ceilings.
Wall finishes	Tile (backsplash height in kitchens, wet-area height in baths, full height elsewhere), FRP panels, wallpaper (56 sq ft rolls), shiplap/beadboard/paneling/stone/brick (SF), wainscot by its height.
Flooring	By room finish: carpet and pad (SY), floor tile (SF) with thinset bags, hardwood, luxury vinyl plank, or any other library finish in SF (porcelain, stone and brick pavers also get thinset). Rooms default to tile in baths, kitchens and laundries, concrete in garages, LVP elsewhere.
Trim and molding	From the whole-house (or room) trim choices: base, shoe and chair rail along the room perimeter less door widths; crown by perimeter; casing per door and window (stool and apron when a stool profile is chosen); totals per profile in LF with "about N pieces of 16'"; one box of 2" finish nails.
Paint	Wall paint by room (2 coats at 350 sq ft per gallon, color named), ceiling paint (2 coats flat at 400), trim paint (semi-gloss), exterior paint on paintable siding (2 coats at 300).
Fixtures and cabinets	Plumbing fixtures, generic cabinets and other fixtures one each; countertop LF (depth noted) for counter-bearing cabinets.
Cabinets	One line per cabinet (SKU, door style and finish; price book named in the note). Each carries its own price (see Pricing sources).
Cabinet trim and accessories	Crown, light rail, toe skin, scribe, finished ends, panels, legs, corbels and hardware counted from the cabinet run (planning prices attached).
Countertops	Slab material from the countertop layout.
Appliances	Library appliances one each.
Electrical	Devices by type, new-work boxes, old-work (remodel) boxes for existing walls, fan-rated ceiling boxes, 14/2 and 12/2 NM-B cable (about 25 ft per device, 250' rolls), AFCI/dual-function breakers (about 10 devices per 15 A circuit, 4 per 20 A), wire connectors, cover plates.
Lighting fixtures	Chosen light fixtures by model with lumens in the note.

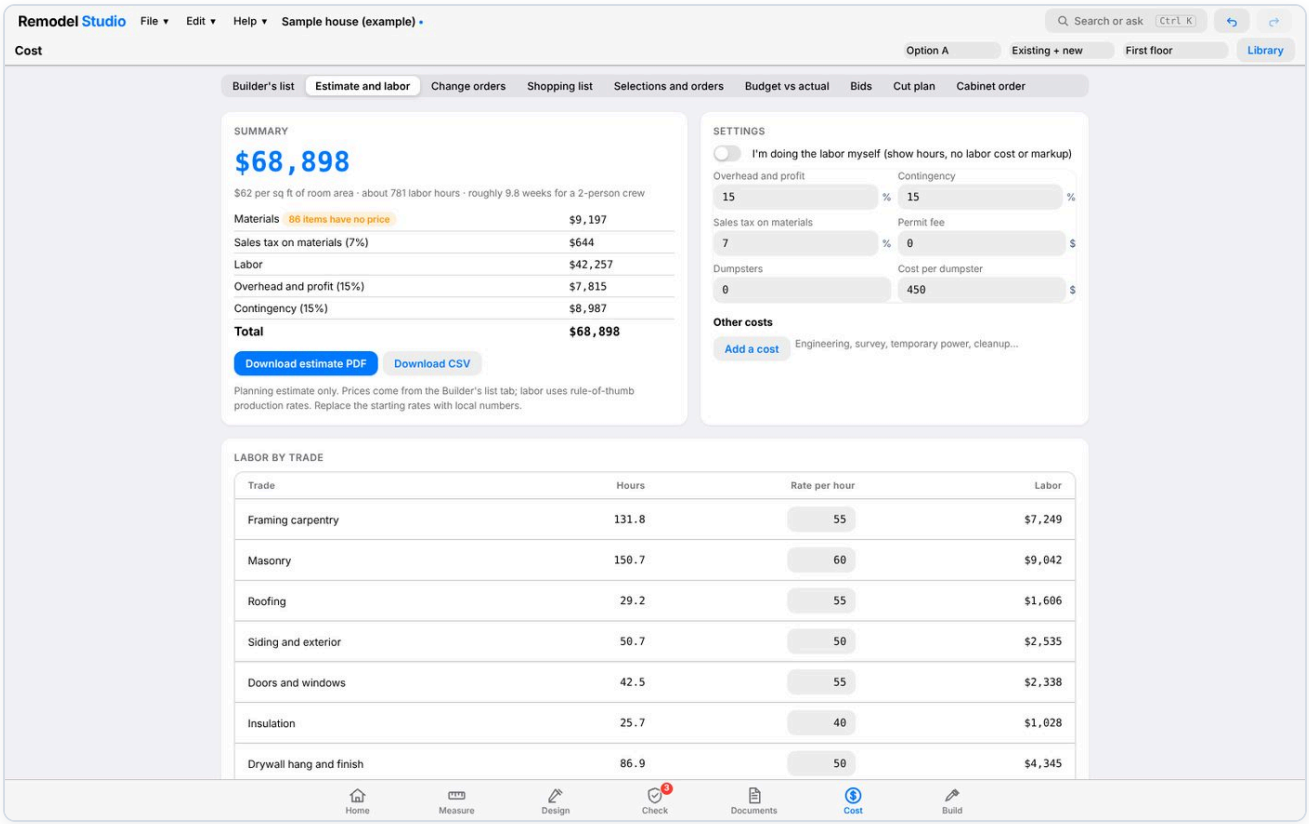
Category	What is counted
Door hardware	Levers, knobs, handlesets, pulls, deadbolts, flush bolts, with finish and brand, keyed to the door mark.
Plumbing / HVAC	Plumbing and HVAC symbols one each; pipe runs by system and size (LF); duct runs by size (LF).
Foundation	Footings, stem walls, slab concrete, block, anchor bolts and the like from the Foundation panel.
Decks, porches and steps	Decking, framing, posts, railing (LF), columns, step and landing material (wood or concrete).
Fireplace and chimney, Skylights, Tile, Framing (trusses and joists), Windows, Sitework	Added by their own tools when present.
Site and landscape	Driveways, patios and walks (concrete CY, asphalt SF, gravel tons, pavers SF plus base), mulch/pine straw/river rock for beds, sod pallets or seed, straw over seed, fence LF and posts, retaining wall SF face, trees and shrubs, edging, landscape fabric, irrigation heads/rotors/drip and valves, controller and backflow preventer, and a "Call 811 before digging" reminder line.
Landscape plants	Each species and container size, with the plan key and a planning price in the note; tree staking kits and planting soil for 15-gallon and larger trees.
Playground	Equipment by model and color scheme ("price by quote"), swing seats, surfacing (CY for loose fill, SF for unitary), drainage aggregate and geotextile under loose fill, wear mats, accessible entry ramp, borders, certified installation and signage.
Ramps and site	Concrete or wood ramps: concrete CY and forming, or treated decking, joists and posts; handrail LF; edge curb LF.
Fasteners	16d framing nails (about 5 lb per 10 studs) and 1-1/4" drywall screws (about 5 lb per 8 sheets).

Items are tagged "Selected: *brand product*" in the Notes column when a brand has been chosen on the Selections tab (roofing, siding, flooring, countertops, windows, exterior doors, interior doors) and cabinet lines are tagged with the cabinet maker.

Pricing sources (lowest to highest priority): 1. **Built-in planning prices.** Cabinets are priced from the generic line (stock / semi-custom / custom multipliers, door style, painted finish and drawer layout factors, rounded to $5); cabinet trim and accessories, chosen light fixtures and brand appliances carry planning prices. These show in the Unit price box as a starting value. 2. **Price-book prices.** If a cabinet dealer price book has been imported (Library › Cabinets) and chosen under Cabinet maker, each cabinet is priced by its SKU from the book. 3. **Store prices import** (Estimate and labor › Store prices › *Import store price list...*). Reads a CSV, Excel or JSON price list or cart export (Lowe's, Home Depot, 84 Lumber, Menards or any supplier), or rows pasted from a web page or spreadsheet. Columns are recognized by header (SKU / Item # / Model, Description / Product / Name, Price / Unit price / Each / Cost, UOM / Unit); without headers the longest text cell is the description and the last money cell is the price. Each builder's-list item is matched by size and description (fractions, inches, feet and dimensions like `2 in x 4 in x 8 ft`

file..., paste box, **Read pasted rows**, **Only items without a typed price** (checkbox, on). **Apply checked prices** writes the prices and records the store, SKU, description, unit and date for each; the Estimate tab then lists how many prices came from each store. 4. **Import price list...** on the Builder's list tab: paste lines like `2x4 stud 92-5/8" precut, 3.98`; items are matched by name (exact, or 75% of the words) and the matched prices are written. "Matched N of M items from K prices." 5. **Your own price edits** in the Unit price box always win.

2 Estimate and labor



Cost › Estimate and labor.

A planning estimate built from the builder's list plus labor from production rates. The page is a grid of cards.

Summary card (title shows the active design option name) - Large total. - Line: `$N per sq ft of room area · about N labor hours · roughly N weeks for a 2-person crew` (weeks = hours ÷ 80, or ÷ 20 when you do the labor yourself). - Table: Materials (with a warning pill "N items have no price"), Sales tax on materials (N%), Labor (or "Labor (self-performed)"), Dumpsters (N × $cost) when any, Permit when non-zero, each Other cost by name, Overhead and profit (N%) (hidden when self-performed), Contingency (N%), Total. - **Download estimate PDF** — `<project> estimate.pdf`: page 1 Planning estimate with project, address, date, Summary lines, Total, per-sq-ft and hours, Labor by trade (hours at rate), Design options comparison if more than one option; following pages list every material by category with quantity, item and extended price.

Footer: "Planning estimate made with Remodel Studio. Quantities and labor hours are estimates; get quotes before committing." - **Download CSV** — `<project> estimate.csv` with Section, Item, Hours, Rate, Amount rows for labor, materials, tax, dumpsters, permit, extras, markup, contingency and total. - Note: "Planning estimate only. Prices come from the Builder's list tab; labor uses rule-of-thumb production rates. Replace the starting rates with local numbers."

Settings card (these are the estimate settings; they live here on the Estimate and labor tab and are saved with the project)

Setting	Default	Notes
I'm doing the labor myself (show hours, no labor cost or markup)	off	When on, labor cost, overhead/profit and trade risk are zero; hours still show.
Overhead and profit %	15	Applied to the subtotal (materials + tax + labor + dumpsters + permit + other costs).
Contingency %	15	Applied to subtotal plus markup.
Sales tax on materials %	7	On priced materials only.
Permit fee $	0	
Dumpsters	automatic	Computed from demolition quantities (about one 2,500-unit load; SF counts 1, LF counts 8, each counts 20; at least 1 when there is any demolition, 0 otherwise). Type a number to override.
Cost per dumpster $	450	
Other costs	none	**Add a cost** adds a row (name, amount, remove ×); the new row is named "Equipment rental". Suggested uses: engineering, survey, temporary power, cleanup.

Labor by trade card. Table of Trade, Hours, Rate per hour (editable), Labor, with a Total row. Hours come from production rates applied to each builder's-list line (for example studs 0.12 h each, drywall sheets 1.1 h, interior doors 2.5 h, windows 3 h, paint 3.5 h per gallon, cabinets 1.5 h each, tile 0.12–0.15 h per sq ft, roofing 0.33 h per bundle). Trades and default rates:

Trade	Default $/h
Demolition and haul-off	35
Framing carpentry	55
Masonry	60
Roofing	55
Siding and exterior	50
Doors and windows	55
Insulation	40
Drywall hang and finish	50
Ceilings	50
Tile	60
Flooring	50

Trade	Default $/h
Finish carpentry and trim	55
Painting	45
Cabinets and counters	55
Plumbing	95
Electrical	90
General labor	35

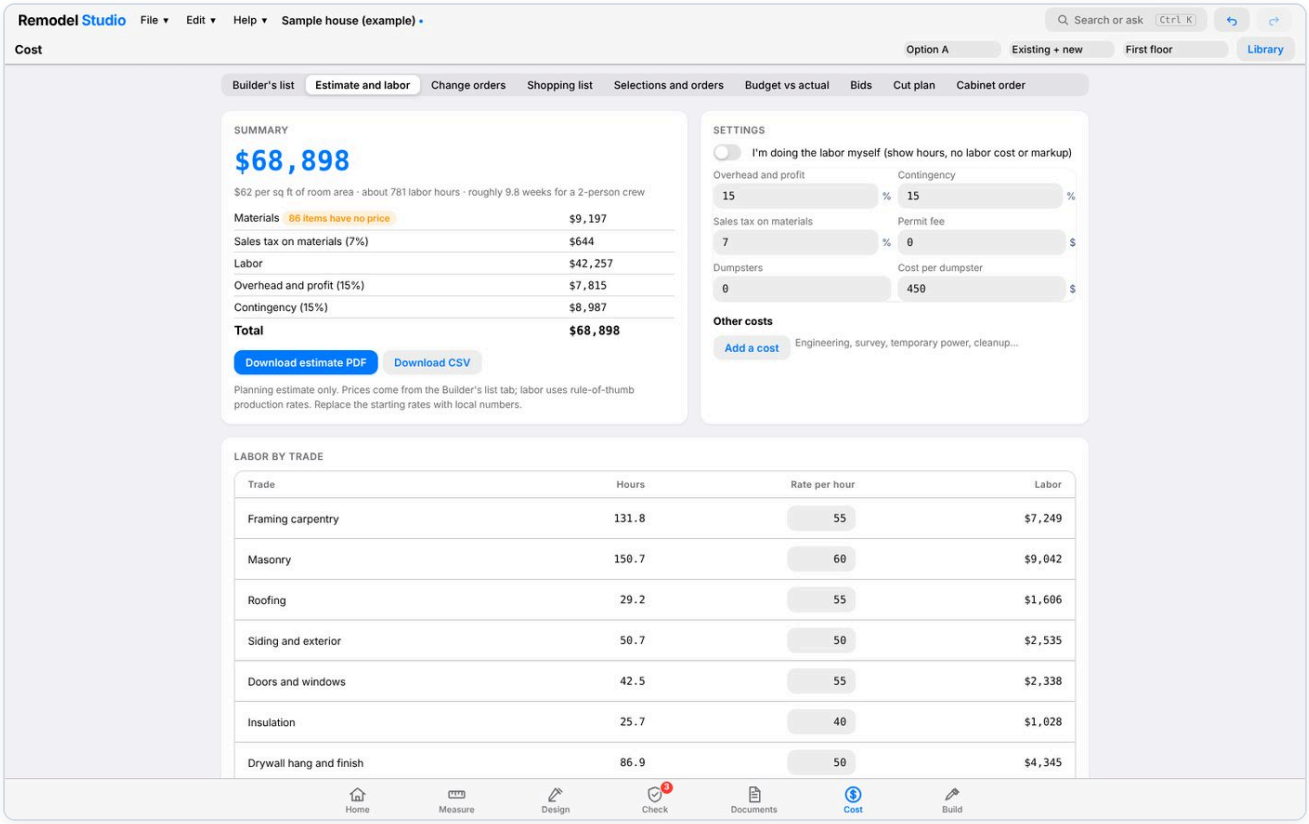
Compare design options card (only when the project has more than one design option). Rows per option: Area, Materials, Labor, Hours, Total, Per sq ft; the active option is highlighted. Note: all options use the same prices, rates and markups; unpriced items count as $0.

Store prices card. "N of M builder's list items have a price. From: Lowe's (12), Home Depot (4)..." plus a note that cabinet trim, moldings and modifications carry planning prices until priced. Button **Import store price list...** (dialog described above).

Risk by trade card. An extra contingency percentage on any one trade's labor, on top of the overall contingency. Table: Trade, Labor, Extra % (box per trade, default empty = 0, capped at 100). When any are set, a **Trade risk allowance** total appears and is added to contingency and the total. Ignored when labor is self-performed.

What changes between options card (more than one option). Pick **From** and **To** options; shows `To compared with From: ±$N`, then a table of every line whose quantity differs (Item, qty in A, qty in B, Cost change or "not priced") sorted by biggest change, followed by labor hour differences by trade. Checkbox **List the other options as priced alternates in the proposal** (off) — when on, the Proposal PDF lists each non-active option as "Alternate: *name* (add/deduct) ± $N".

3 Change orders



Cost › Change orders.

Lock the contract when the client signs; afterwards every change to the model is detected against the locked estimate.

Before locking: a card "Contract and change orders" with the button **Lock the contract at today's estimate**. Locking stores a snapshot of today's estimate (date, total, materials, labor, every line with quantity and price) and a snapshot of the model.

After locking, four statistics: **Original contract (date)**, **Signed change orders** (sum of signed change-order amounts; count), **Contract now** (original + signed), **Unrecorded changes** (difference between today's estimate and the last snapshot, red when positive; count of changed lines).

How changes are detected and priced: today's builder's list is compared line by line (category + item + unit) with the last snapshot. Lines whose quantity changed by 0.01 or more are listed as **Added**, **Removed** or **Changed**, with Was and Now quantities and the material cost difference (quantity change × unit price; blank when unpriced). The change-order amount is the difference in the full estimate total, so it includes labor, markup and contingency.

Creating a change order: with changes present, type **Describe the change for the client** and optionally **+ days** (working days added or saved), then **Create change order**. This numbers the

change order (#1, #2…), dates it, records the lines and amounts, sets its status to *Waiting for signature*, advances the baseline to today's snapshot, adds a revision row to the drawing revision history ("Change order N: description"), and clouds the changed parts on the plan sheets (revision clouds). Message: "Change order #N recorded as revision R; N changed parts are clouded on the plan sheets. Print it for signatures."

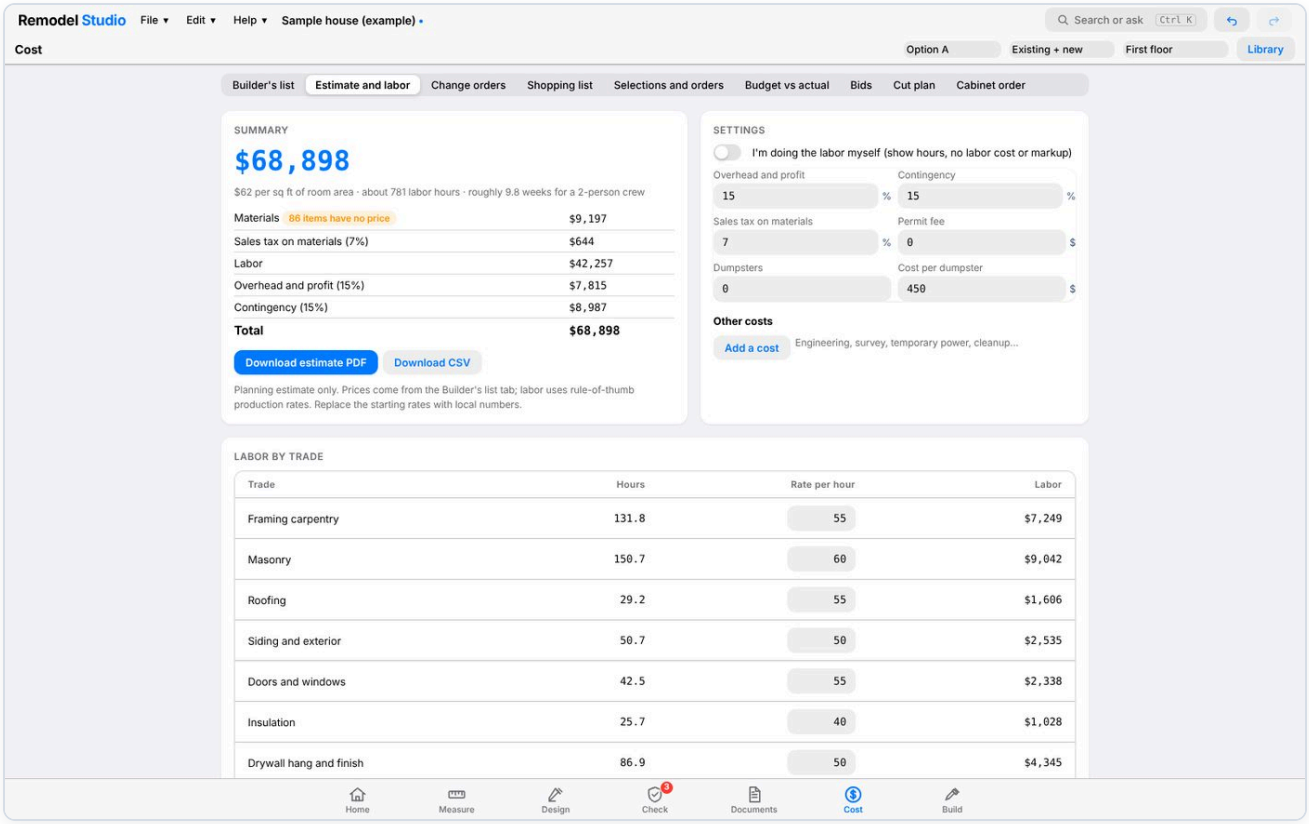
Change orders table: #, Date, Description, Amount, Status (select), PDF button. Status options and client signature flow:

Status	Meaning
Waiting for signature	Just created; printed and sent to the client.
Signed	Client signed; the amount is added to "Contract now" and the revision row is marked "(signed)".
Void	Withdrawn; the revision row is marked "(void)".

Change order PDF (`<project> change order N.pdf`): "CHANGE ORDER #N" and date; Project name; address and owner if entered; **Description of the change**; a table of Item (prefixed + added, – removed, ~ changed), Was, Now, Amount ("labor and markup" when the line has no unit price); Contract before this change; This change order (materials, labor, markup and contingency); New contract amount; "Schedule: adds/saves about N working days" when days were entered; the sentence "The work above is added to or removed from the contract at the amount shown. All other terms stay the same."; Owner and Contractor signature and date lines.

Unlock the contract (link at the bottom) removes the lock and all change orders.

4 Shopping list



Cost › Shopping list.

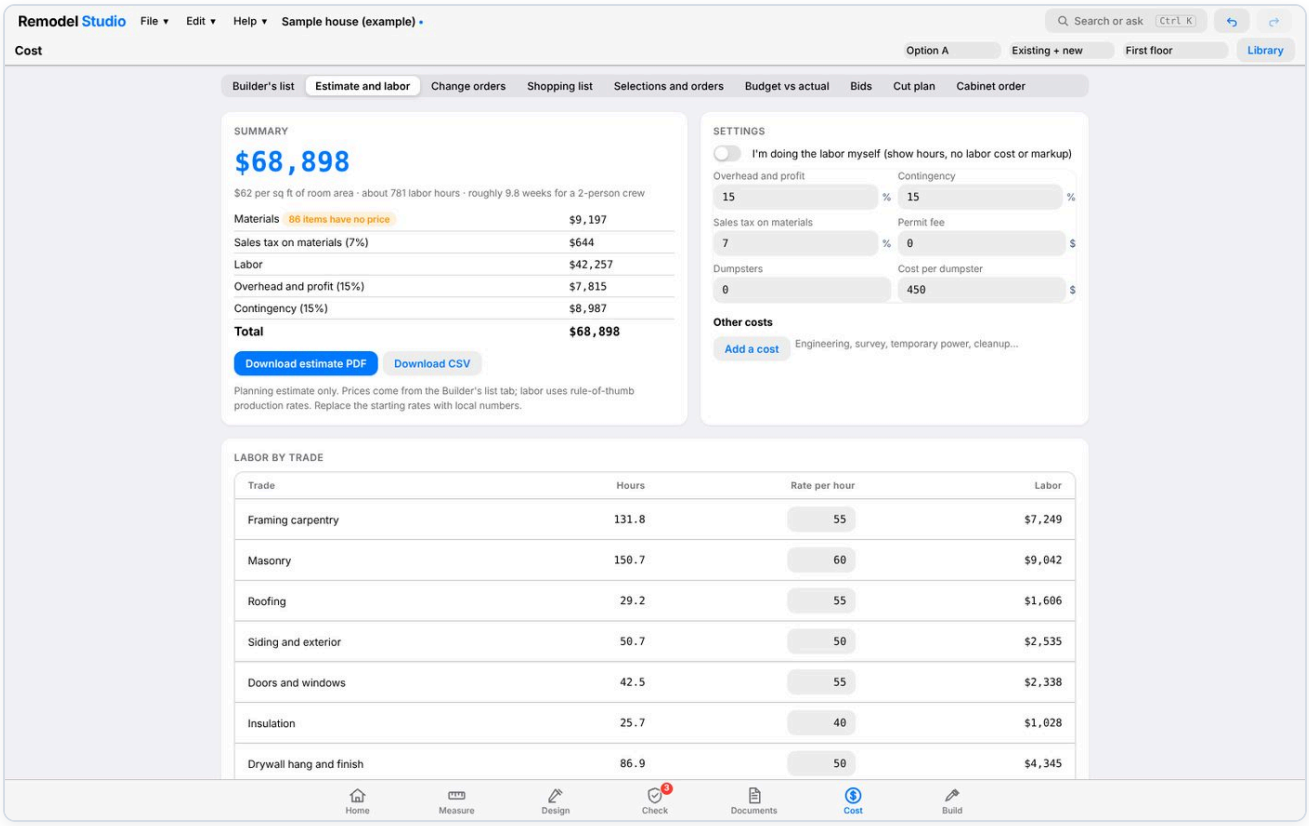
Everything on the builder's list (demolition lines left out), sorted the way you buy it, with check-off boxes saved in the project.

- **Group by**: **By job phase** (default) or **By store department**.
- Job phases, in order: Demolition · Foundation and site · Framing and dry-in · Rough-in · Insulation and drywall · Exterior finish · Interior finishes · Fixtures and final · Landscape.
- Store departments: Tool rental and disposal · Concrete and masonry · Masonry and specialty · Lumber and building materials · Hardware and fasteners · Roofing · Doors and windows · Plumbing · Heating and cooling · Electrical · Insulation · Drywall · Siding and exterior trim · Treated lumber and decking · Flooring · Tile · Wall paneling and finishes · Paint · Millwork and trim · Kitchen and bath · Kitchen and bath fixtures and appliances · Garden center · Other. (There is no aisle-level grouping.)
- **Hide items I have** checkbox.
- **CSV** — `<project> shopping list.csv` (Phase or Department, Item, Qty, Unit, Unit price, Total, Got it).
- **Print list (PDF)** — `<project> shopping list.pdf`, one section per group with check boxes (already-checked items are drawn checked), quantity, item and the other grouping as a right-

hand note.

- Totals bar: `N items · N still to get` and **Clear checks** when any are checked.
- Each group shows its priced subtotal; rows show a checkbox, qty and unit, item, the other grouping, and extended price. Checked rows are struck through.

5 Selections and orders



Cost › Selections and orders.

One tab with two halves. **Selections** (brands and products) is on top, **Selections and orders** (the tracker) below.

Selections half - Header: "Pick the brands and products for this job. They go on the materials list, the schedules and the proposal. N cabinet makers and N product brands are in the library." Button **Import vendor catalog…**. - **Door hardware, whole house**: **Finish** (default Satin nickel; choices: Satin nickel, Matte black, Oil-rubbed bronze, Aged bronze, Polished chrome, Stainless steel, Satin brass, Polished brass, Antique brass, White) and **Brand** (default Any brand; Kwikset, Schlage, Baldwin, Emtek, Weiser, Defiant, Yale, Delaney, Nostalgic Warehouse). Every door uses these unless changed in its own panel; hinges, stops and kick plates follow the finish. - **Cabinets**: shows the chosen cabinet maker, line, door and finish (or "Generic cabinets."); change them in any cabinet's panel (Kitchen style) or the Cabinet maker panel (Brand, Line, Door style, Finish, Price book). - One section per product category — **Windows**, **Exterior doors**, **Interior doors**, **Roofing**, **Siding**, **Flooring**, **Countertops**, **Plumbing fixtures**, **Appliances** — each with **Brand** (Not

chosen / brand list; "(confirm with dealer)" marks unverified entries), then **Product line** (or **Color** for countertops, **Product** for appliances), then **Color** where the line has colors. Notes under the pickers show frame, material or catalog notes; windows show the ENERGY STAR 7.0 Southern-zone U-factor and SHGC; catalog links are listed. - **Paint colors**: **Paint brand**, then a per-room table of **Wall color** and **Trim and ceiling** color from that brand's palette (choosing a color also sets the room's screen colors), a swatch strip, and the note "Screen colors are close, not exact; check a physical sample."

Orders half (the tracker) - Header: "Track each purchase from choosing to installed. Allowance is what the contract carries; price is what you are actually paying. Lead times of items not yet delivered hold their phase in the job schedule." - **Show** filter: Everything · Not delivered yet · Choosing · Approved · Ordered · Delivered · Installed. - Statistics: **Allowances** (sum), **Prices** (sum), **Over or under allowance** (price – allowance for items with both; red when over), **Items** (count; N delivered). - Rows are generated from the model, grouped by category: **Appliances** (brand and model when a library appliance), **Plumbing fixtures**, **Furniture and decor**, **Other** fixtures; **Cabinets** (one row for the whole set: count, style, finish, maker, rooms, planning price); **Countertops** ("template, fabricate and install"); **Lighting** (one row per light mark: type, Kelvin, dimmable, rooms, count); **Doors and windows** (one row for all windows, one for all doors); **Flooring** and **Tile** (one row per room with finish and sq ft). - Columns: picture · Item (label, sub-line, Qty) · Room · **Status** (select) · **Vendor** · Estimate (from the builder's list) · **Allowance $** · **Price $** (with the over/under difference shown under it) · **Lead (workdays)** (placeholder is the typical lead) · **Ordered** (date) · **Arrives** (computed) · **Note**. Text boxes save when you leave them or press Enter.

Tracking statuses, in order:

Status	What happens when you pick it
Choosing	Default.
Approved	Makes the item eligible for a purchase order (needs a vendor).
Ordered	Fills in today's **Ordered** date if empty.
Delivered	Records today's delivered date; the item no longer holds the schedule.
Installed	Records today's installed date; starts the appliance warranty clock in the binder.

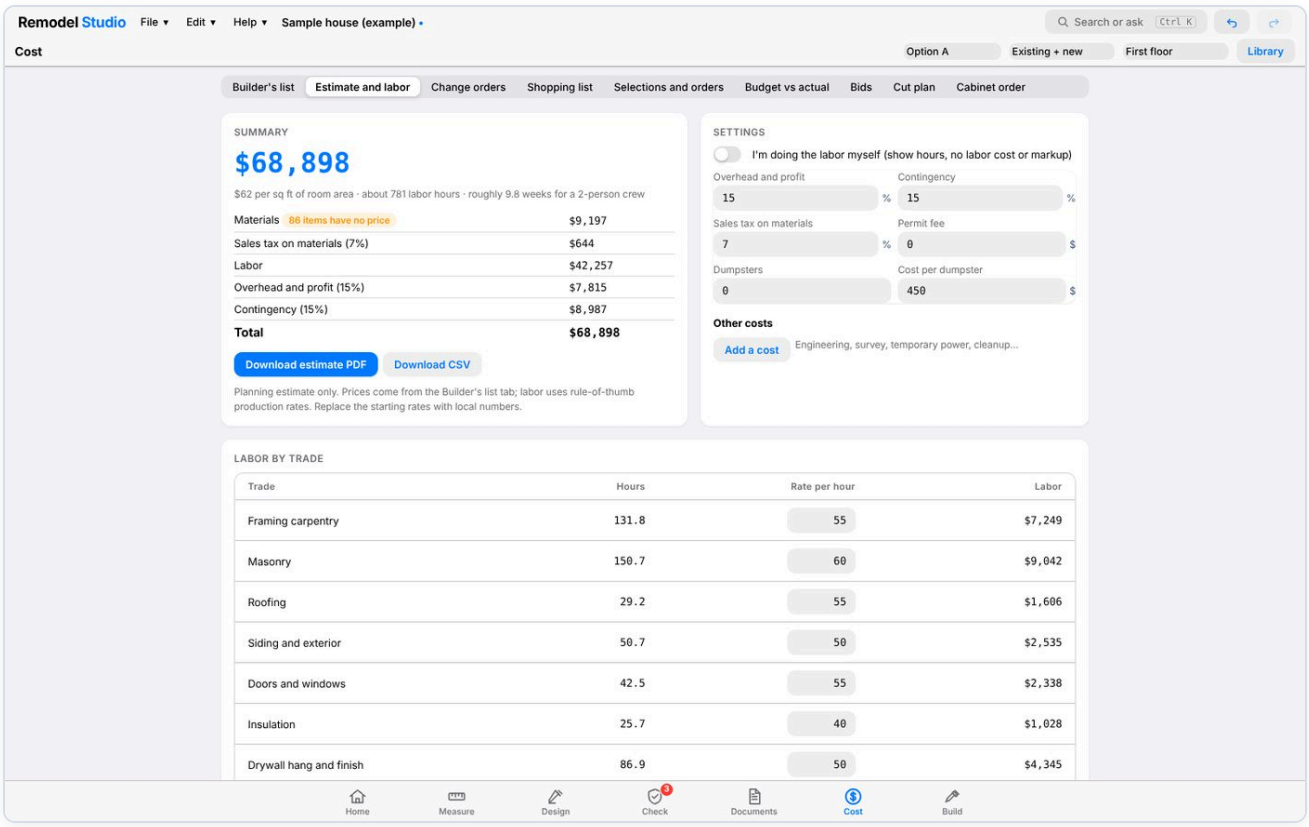
ETA / lead times: typical leads are 14 workdays for refrigerators and ranges, 7 for other appliances and plumbing fixtures, 10/30/45 for stock/semi-custom/custom cabinets, 10 for countertops, 5 for lighting, 25 for windows, 14 for doors, 8 for hardwood, 5 for other flooring and tile. **Arrives** = Ordered date + lead in weekdays, unless you type an ETA. Items not yet delivered feed the job schedule: their remaining lead (lead minus weekdays since ordering) can push back the phase they belong to (appliances, plumbing fixtures and lighting → trim-out; cabinets → Cabinets; countertops → Countertops; doors and windows → Windows; flooring → Flooring; tile → Tile).

Order-by dates vs the schedule: the Job schedule's long-lead table and the Dashboard show, for each lead item, the date its phase needs it and the latest **Order by** date (needed date minus lead

in workdays). If that date is before the schedule start it says "order at signing"; if the lead actually delays the phase it says "holds the schedule".

Purchase orders: items set to *Approved* with a vendor are grouped by vendor with the count and total. **PO PDF** prints a purchase order; **PO PDF and mark ordered** also sets each item to Ordered today with the PO number. PO numbers run `PO-<first four letters of the project>-001`, `-002` … **Purchase order PDF** contents: "PURCHASE ORDER" and number; Vendor; Job (name · address); Date; Ordered by (designer); a table of Item (label — sub-line (room)), Qty, Price (tracker price, or the estimate), Need by (from the tracker's *need* field); Total; "Please confirm prices, lead times and delivery dates in writing."

6 Budget vs actual



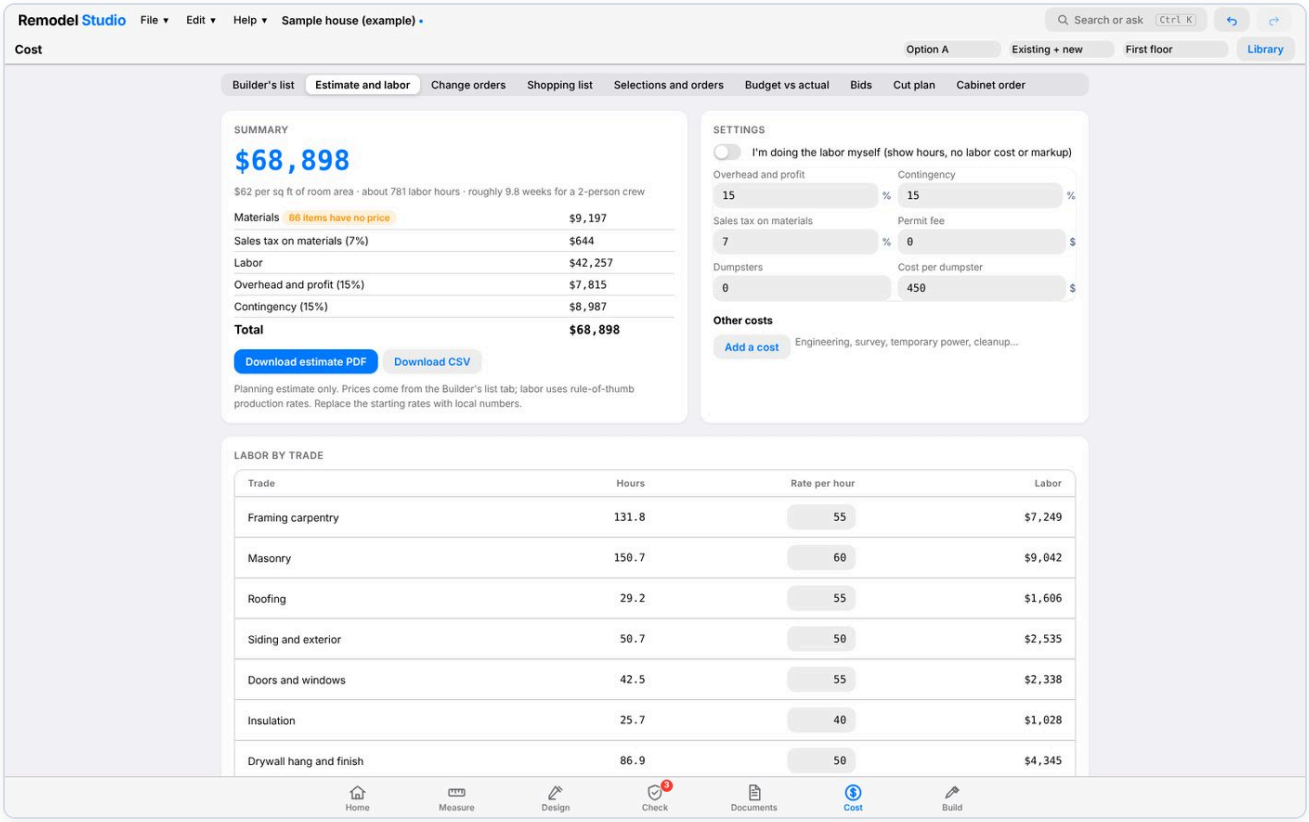
Cost › Budget vs actual.

Log receipts against the estimate. "Over-runs come out of the contingency first."

- **Budget buckets** (automatic): one per material category on the builder's list (priced total), one per labor trade (hidden when self-performed), Sales tax, Dumpsters, Permit, each Other cost, and **Other (not in the estimate)** at $0. Grouped as Materials, Labor, Other.
- Statistics: **Budget (before markup)**, **Spent** (N receipts), **Ordered, not yet paid** (tracker items Ordered/Delivered/Installed at their price or estimate), **Contingency left** (contingency carried minus the sum of over-runs by bucket; red when exhausted).

- **Add a receipt** form: **Date** (today), **Store**, **Amount**, **For** (budget line select grouped Materials / Labor / Other), **Note**, **Photo of the receipt** (camera or file; stored with the project assets). **Add receipt**.
- **By budget line** table: Line, Budget, Spent, Left (red and negative when over), and a bar (green under 85%, amber to 100%, red over).
- **Receipts** table (newest first): Date, Store, Amount, For (changeable), Note, Photo thumbnail (opens full size), delete ✕.
- **Download receipts (CSV)** — `<project>-receipts.csv` (Date, Store, Amount, For, Note).

7 Bids



Cost › Bids.

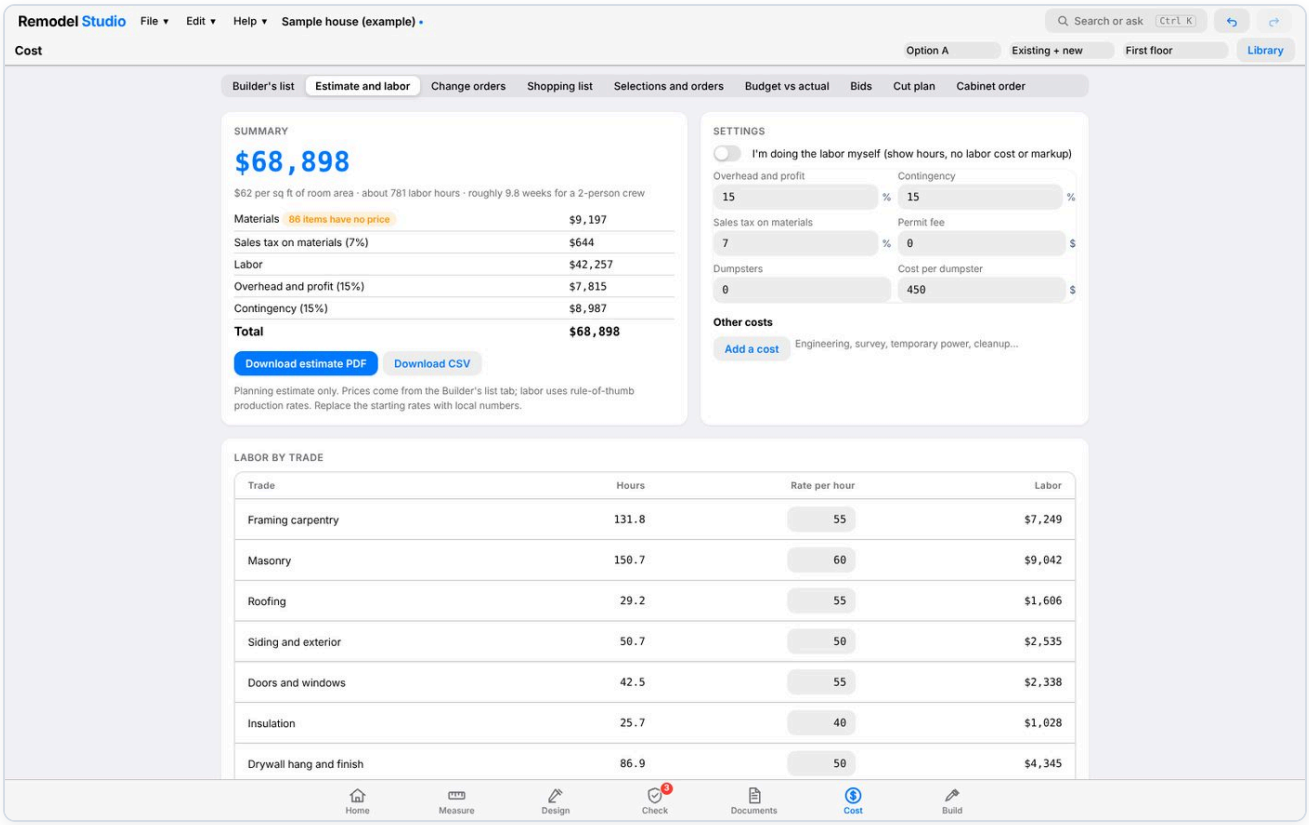
One scope of work per trade, built from the builder's list; request bids, enter quotes, compare with the plan estimate.

- A tab strip of trades present in the job (a ✓ after a trade that has at least one quote). Each trade's scope groups the builder's-list items whose labor belongs to that trade, with their material total and labor hours.
- **Scope of work** text box (default text: "*Trade* for the *project* remodel as drawn. / Approximately N labor hours by the plan estimate. / Protect finished surfaces; clean up daily; haul your own

debris. / Work to the 2021 IRC and the drawings; call for your own inspections."). Edit freely; saved per trade.

- **Bidder supplies materials too ($N of materials on the list)** checkbox (off). When off, the request says materials are furnished by the owner. The comparison estimate is labor only, or labor plus materials when on.
- **Bid request PDF** — `<project> bid request <trade>.pdf`: "REQUEST FOR BID" and trade; Job (name · address); Contact (designer); Date; "Please bid labor [and materials] ..."; **Scope of work** bullets; **Quantities from the plan** (Item, Qty, Unit, up to 80 lines); fill-in lines for Bid amount, Days to complete, Start available; "Include: license number, insurance certificate, and what is excluded."; Signature and Date.
- Note beside the button: `N line items · plan estimate $N (N h labor [+ materials])`.
- **Quotes** table with three rows: **Bidder**, **Amount**, **Days**, **Start** (date), **Notes**, **vs estimate** (+/− against the plan estimate, red when higher), and a **Choose** button (becomes **Chosen**; click again to unchoose). Below: "The plan estimate for this trade is $N. Chosen bid: *who* at $N."
- **All trades** table (once any quote exists): Trade, Estimate, Low bid, Chosen.

8 Cut plan (sheet-goods cut optimizer)



Cost › Cut plan.

Nests pieces from the plan onto sheets with a cut diagram. Pieces come from the model: drywall by wall (horizontal 4-ft courses, full-length strips, one side for exterior walls and two for interior,

openings reduce the count) and by ceiling, subfloor by room (not on a slab), sheathing on exterior walls, cement backer board under tile floors and in tile zones, plus any pieces you add. Nesting is shelf-cut (guillotine) with a 1/8" kerf; pieces may be rotated unless grain is locked.

- **Material** select: 1/2" drywall 4 × 12 · 1/2" drywall 4 × 8 · 3/4" plywood 4 × 8 · 7/16" OSB 4 × 8 · 3/4" T&G subfloor 4 × 8 · 1/2" cement board 3 × 5 ("(none)" after materials with no pieces).
- **Use 4 × 8 sheets instead** checkbox (shown for 4 × 12 drywall) — re-cuts the drywall strips to 8-ft lengths and nests on 4 × 8 sheets.
- **Cut list CSV** — `<project>-cutlist-<material>.csv` (Sheet, Piece, Width, Height, X, Y).
- Statistics: **Pieces**, **Sheets of** , **Yield %** (piece area ÷ sheet area; sq ft of pieces), **Waste** sq ft.
- **Your own pieces (plywood, shelves, boxes)** table: Label, Sheet (select), Width, Height, Qty, **Grain fixed** (checkbox: do not rotate), remove ×. **Add a piece** adds "Shelf", 3/4" plywood, 11-1/4" x 30", qty 4.
- Cut diagrams: up to 40 sheets drawn with each piece's size; caption "Sheet N · N% used". Pieces too big for the sheet are drawn red.

9 Cabinet order

An order report grouped by manufacturer with modifications under each cabinet, "the way a 20/20 order report reads". List prices come from imported price books where loaded, otherwise planning prices. Empty state: "Place cabinets from the Library (Cabinets) or with the cabinet run tool, and the order report fills in here."

- **CSV** — `<project> cabinet order.csv` (Manufacturer, #, Qty, SKU, Description, Attributes, Hinge, List each, List ext; modification rows indented; moldings and accessories; totals).
- **PDF** — `<project> cabinet order.pdf` with the same content and the note "Planning prices. List prices come from your price books where loaded; confirm with the manufacturer or dealer before ordering."
- Order settings (saved with the project):

Setting	Default
Dealer multiplier	0.45
Margin %	35
Freight % of net	3
Install $ per box	85
Sales tax %	7

- One table per manufacturer: **#** (cabinet number as on the elevations), **Qty**, **SKU** (with hinge tag), **Description** (layout name, W x H x D; attributes line: door style · finish or "custom color" · price book), **Hinge**, **List**; modification rows (code, label, extended price) under each cabinet.
- **Moldings, panels and hardware** table: Qty, Code, Item (with note), List.
- Totals: List total · Net (× multiplier) · Sell at N% margin · Freight N% · Installation, N boxes (fillers are not boxes) · Tax N% · Total.

PART

Building

The dashboard, the job schedule, field notes, wall photos, inspection prep and the homeowner binder.

CHAPTER 20

Home dashboard

The Home workflow has one page, **Dashboard** ("The job at a glance"). It is the first thing you see when a project opens.

1 Header

- **Project name** (large), then a line of facts: the address, "for *client*" if a client is entered, and `N rooms · N sq ft · N cabinets`. Square feet is the sum of all room areas; the cabinet count leaves out cabinets marked for demolition.
- **Open the design** button opens the Design › Model page.
- **New or open a project…** opens the New project dialog.

2 Cards

Each card has a title, a button in its header that opens the related page, and a summary.

Checks (Open → Check page). Green when there are no errors, red when there are. - Big number: count of **errors**; beside it, the count of **warnings**. - Up to four error titles as links. Clicking one switches to the error's level, opens Design, and selects the item the error is about. - If there are no errors: "No errors. *N* warnings to review." or "Nothing to review."

Cost (Open → Cost › Estimate and labor). - Big number: the **estimate total** (materials, tax, labor, other costs, markup and contingency). Beside it, the total **spent** from receipts (Budget vs actual). - A progress bar of spent against budget. - Note line: `$N budget before markup · $N contingency` and, when some builder's-list items have no price, `· N items not priced`. - If change orders exist: "N change orders, N waiting for signature." - If there are no walls yet: "Draw walls to get an estimate."

Schedule (Open → Build › Job schedule). - Big date: the **finish** date, with `N workdays from <start date>`. - **Now:** the phase in progress today and the date it runs through. - **Next:** the next phase and its start date. - With no schedule: "The schedule builds from the estimate's labor hours."

Orders (Open → Cost › Selections and orders). - Chips with the count of tracked items in each status (Choosing, Approved, Ordered, Delivered, Installed) — only statuses with items are shown. - Red note when items are past their expected arrival date: "N items are past the expected arrival date." - **Order by** list: the four soonest long-lead order-by dates, each as `date item (what it covers)`; a row is red if it should already have been ordered. "No long-lead items yet." when there are none.

Field (Open → Build › Field notes and punch list). - Big number: **open punch items**; beside it `N questions · N wall photos`. - Up to four open punch items as links (text, and `due <date>` if

set). Clicking one opens Design on that level and selects the pin.

Recent projects (Open… → the Open project dialog). - Up to six recently saved projects with the save date; click a name to open it. "Projects you save show here." when empty.

3 Footer

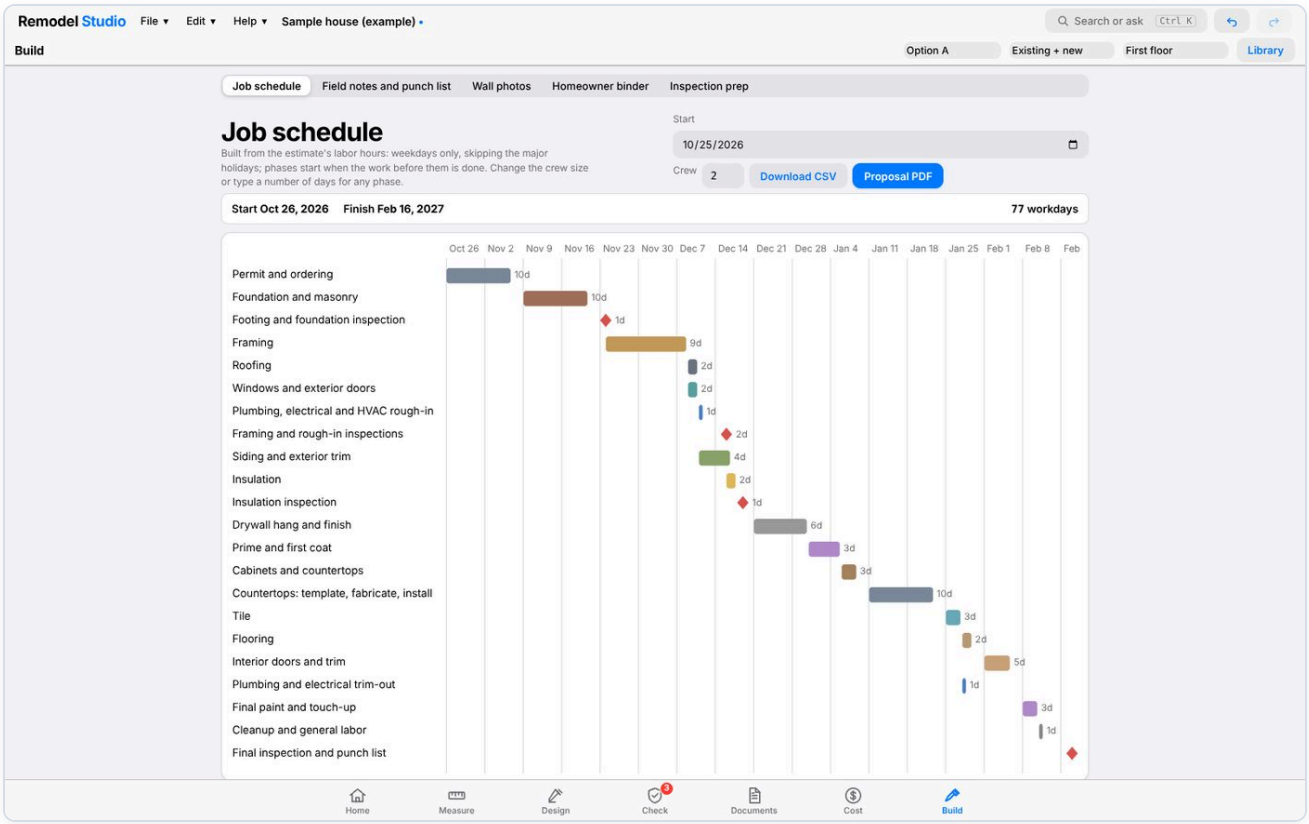
"Measure → Design → Check → Documents → Cost → Build. Press Ctrl+K anywhere to find a command or ask the project a question."

CHAPTER 21

Build workflow

Build has one page ("Job") with these tabs, in order: **Job schedule · Field notes and punch list · Wall photos · Homeowner binder · Inspection prep**.

1 Job schedule



Build › Job schedule.

Built from the estimate's labor hours: weekdays only, skipping the major holidays; phases start when the work before them is done.

Header controls - **Start** (date; default today). The schedule begins on the first workday on or after it. - **Crew** (number, default 2) — the default crew size for every phase. - **Download CSV** — `<project> schedule.csv`: Phase, Start, Finish, Workdays, Crew, Labor hours; then Payment, Percent, Amount, Due. - **Proposal PDF** — see below.

How durations are computed: each phase adds up the labor hours of its trades (some phases take a share: windows 60% of door/window hours, rough-in 70% of plumbing and electrical, trim-out 30%, first paint 50%, final paint 50%, trim 40% of door hours). Days = hours ÷ (crew × 8), divided by the number of trades for phases whose trades work in parallel, rounded up. Phases

with no hours are skipped. Inspections and a few others have fixed day counts. A phase starts on the workday after its latest dependency finishes; long-lead items (ordered at signing) can hold a phase back.

Workdays and holidays: Saturdays, Sundays and these holidays are skipped: New Year's Day, Memorial Day, July 4, Labor Day, Thanksgiving and the day after, Christmas Eve and Christmas Day.

Phases, in order:

Phase	Duration from	Starts after	Notes
Permit and ordering	10 days fixed	start	
Demolition	Demolition hours	Permit	
Foundation and masonry	Masonry hours	Demolition	then footing and foundation inspection
Footing and foundation inspection	1 day, milestone	Foundation	only when there is foundation work
Framing	Framing hours	the inspection	
Roofing	Roofing hours	Framing	
Windows and exterior doors	60% of Doors and windows hours	Framing	
Plumbing, electrical and HVAC rough-in	70% of Plumbing + Electrical, in parallel	Roofing, Windows	
Framing and rough-in inspections	2 days, milestone	Rough-in	
Siding and exterior trim	Siding hours	Windows	
Insulation	Insulation hours	Inspections	then insulation inspection
Insulation inspection	1 day, milestone	Insulation	only when there is insulation
Drywall hang and finish	Drywall + Ceilings hours	Insulation inspection	
Prime and first coat	50% of Painting	Drywall	
Cabinets and countertops	Cabinets hours	First coat	
Countertops: template, fabricate, install	10 days fixed	Cabinets	only when the model has countertops
Tile	Tile hours	First coat, Countertops	
Flooring	Flooring hours	Tile	
Interior doors and trim	Finish carpentry + 40% of Doors	Flooring, Cabinets	
Plumbing and electrical trim-out	30% of Plumbing + Electrical, in parallel	Cabinets, Countertops, Tile	
Final paint and touch-up	50% of Painting	Trim, Trim-out	
Cleanup and general labor	General labor hours	Final paint	

Phase	Duration from	Starts after	Notes
Final inspection and punch list	3 days, milestone	Cleanup	

Summary bar: Start, Finish, N workdays. **Gantt chart**: week lines with dates, a bar per phase colored by trade, diamonds for milestones, day counts at the right.

Phases table: Phase (with "then " where one follows), Start, Finish, Labor hours, **Crew** (per-phase override; placeholder is the default crew; blank for fixed phases), **Days** (type a number to override the computed days; clear to go back).

Payment schedule: "Draws on the estimate total of $N, due when the phase finishes." A warning pill shows when the percentages do not add up to 100. Table: **Draw** name, **Percent**, Amount, **Due after** (phase select), Date, remove ×. **Add draw** (adds "New draw", 0%, after Final inspection). **Reset to the standard draws**. Standard draws:

Draw	%	Due after
Deposit at signing	10	Permit and ordering
Framing complete	25	Framing
Dried in, rough-ins passed	20	Framing and rough-in inspections
Drywall finished	20	Drywall hang and finish
Cabinets and trim set	15	Interior doors and trim
Final inspection and punch list	10	Final inspection and punch list

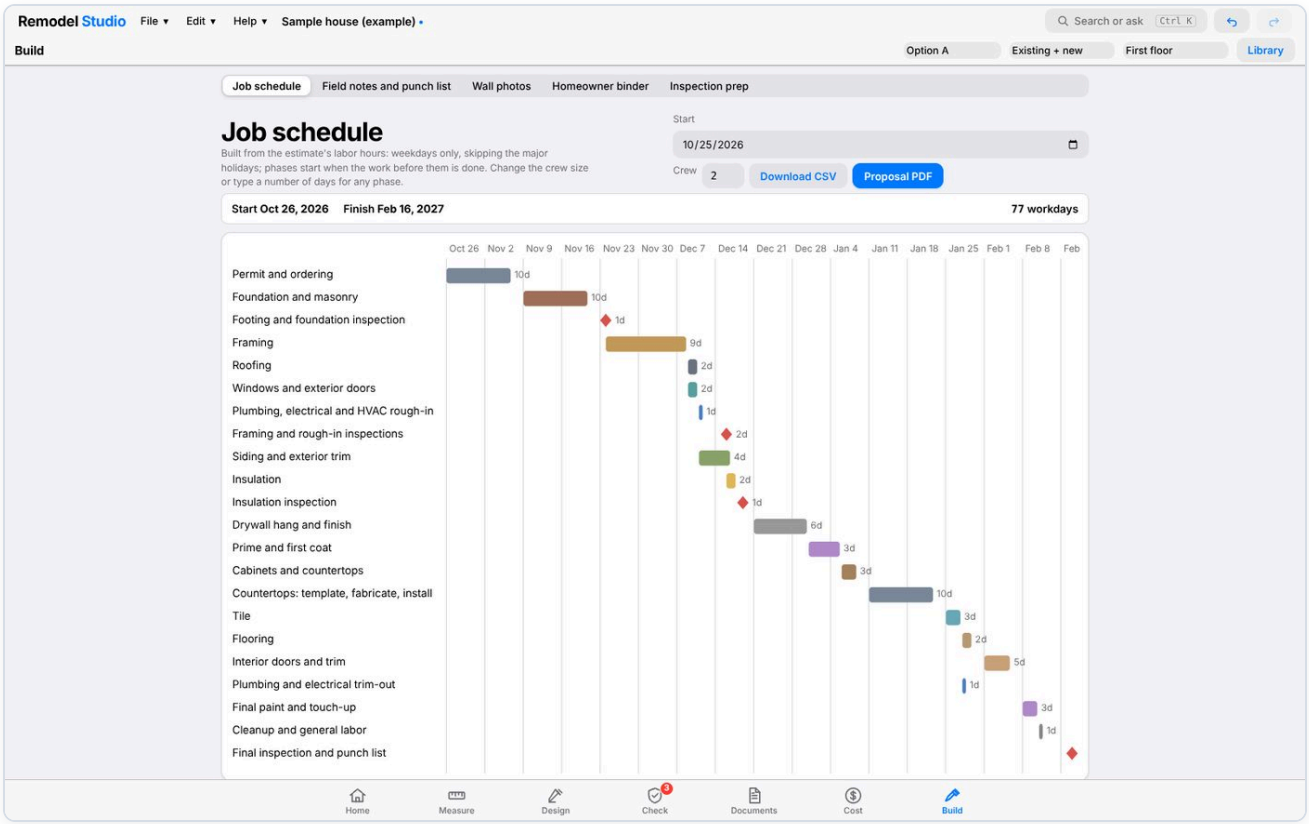
Order dates for long-lead items: "Lead times are typical workdays from order to delivery; type your supplier's numbers. Items are assumed ordered at signing, so a long lead can hold a phase back." Table: Item (with what it covers), **Lead (workdays)** (editable; placeholder is the default), Needed, Order by, and a pill "holds the schedule" or "order at signing". Long-lead items detected from the model and their default leads: Windows and exterior doors 25; Cabinets 10 (stock) / 30 (semi-custom) / 45 (custom); Appliances 10; Plumbing fixtures 7; Roof trusses 15; Tile 5; Hardwood flooring (plus 3 days to acclimate) 8; plus any item from the order tracker that is not yet delivered. **Inspections** list below it: the date of each milestone.

Put it on your calendar (.ics export): "An .ics file opens in Google Calendar, Outlook, iPhone and Android." - **Schedule and order dates (.ics)** — `<project> schedule.ics`: an all-day event for every phase (description: N workdays, crew of N, or "Inspection or milestone"; milestones get a reminder the day before) and an "Order: " event on each order-by date (description: what it covers, needed date, lead time; reminder the day before). - **Maintenance reminders (.ics)** — `<project> maintenance.ics`: recurring reminders (monthly alarm/GFCI/filter check; HVAC filter every 2 months; range hood/under-sink/gutters every 6 months; yearly smoke-alarm batteries, water heater flush, dryer vent, re-caulk, HVAC service) and a "Warranty ends in 30 days" event for each installed library appliance using the binder's warranty years.

Proposal PDF (`<project> proposal.pdf`): "Remodeling proposal"; project name; address · Prepared for client; date · Prepared by designer. **Scope of work**: sq ft of finished rooms across N

walls removed. **Product selections** (cabinets and each chosen brand). **Price**: Materials and sales tax (N of M items priced), Labor, Permit, Debris removal, each other cost, Overhead and profit, Contingency, **Total contract price**; alternates when enabled. **Schedule**: Start, Substantial completion (N workdays), each phase with dates. **Payment schedule**: each draw with percent, amount and "due about ". **Terms**: changes by written change order; owner provides access, utilities and staging, hidden conditions priced as changes; work to the 2021 IRC as adopted locally, proposal good for 30 days. Owner and Contractor signature and date lines. Footer: "Prepared with Remodel Studio. Quantities, labor and schedule are planning estimates."

2 Field notes and punch list



Build › Field notes and punch list.

(Also available under Measure › Photos and measurements as "Field notes".)

Drop a pin on the plan: four big buttons; tap one, then tap the plan where it is. Pin kinds (letter on the pin, color):

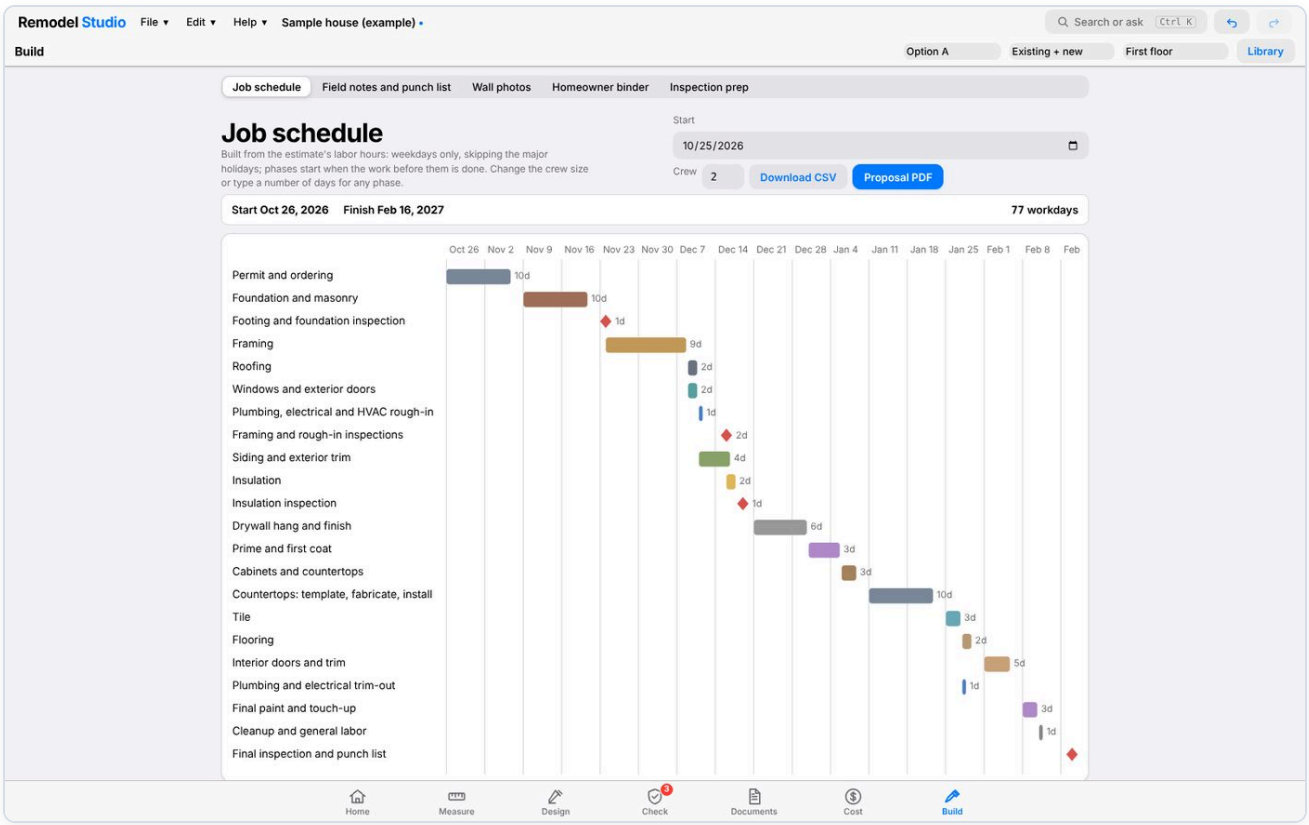
Kind	Marker	Color
Existing condition	C	rust
Punch item	P	blue
Question	?	purple
Photo spot	●	green

The **Field pin** tool is also on the Design › Annotate ribbon (Markup group). Each pin's panel (Properties) has: **Type**; **Note** (text); for punch items **Status** (Open · In progress · Done), **Trade** (Any trade or one of the 17 trades), **Due** (date), **Assigned to**; for conditions **Severity** (Note · Needs repair · Safety hazard); **Take or add photo** (camera or files, several at once; photos are pinned to the spot and shown below); a 🎤 **Dictate** button that adds speech to the note (browser speech recognition, when available). Done punch items turn gray with a ✓ on the plan.

Pins and punch list card - **CSV** — `<project> punch list.csv` (#, Type, Room, Note, Status, Trade, Assigned, Due, Date). - **PDF with photos** — `<project> punch list.pdf`: each shown pin numbered with kind, status, room, note, trade · assignee · due, and up to three photos. - Filter: **All · Conditions · Punch: open · Punch: done · Questions** (counts in each). - The list: marker, note (click to select the pin on the plan), kind · room · trade · assignee · due; punch items have a **Done** checkbox.

Measuring checklist: pick a **Room**; the table lists each wall side (inside face to face, with compass direction), the ceiling height, and every door and window width and height (and window sill) with the model value, a **Field** box to type your tape measurement, and the **Difference** ("ok" within 1/2", otherwise ±). Rows off by more than 1/2" are flagged. "Measured N of M". **Room notes** text box per room.

3 Wall photos



Build › Wall photos.

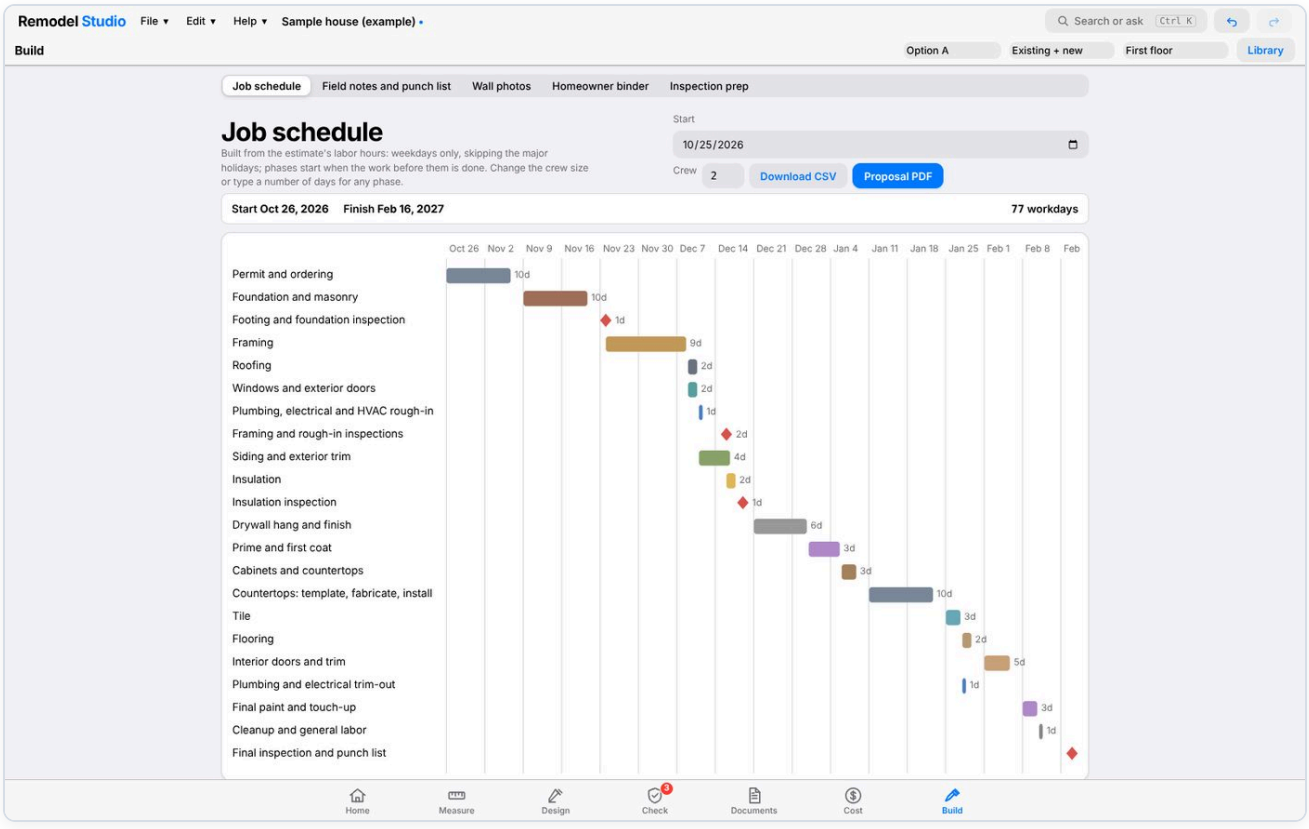
Photos of what is inside each wall, pinned where they were taken, so they appear on the plan, the interior elevations and the binder for years. Place one with the **Wall photo** tool (Design › Annotate › Markup): click the wall the photo shows, on the side you stood. The mark sits just off the wall face.

Wall photo panel: position note ("N along this wall, one/the other side, N up"); **Stage** (Before drywall · Rough-in inspection · During work · Finished; default Before drywall); **Height of the camera mark** (default 48"); **Date** (default today); **What it shows** (text, e.g. "Drain at 18 in, 12/2 to the island at 24 in, blocking for the TV"); **Other side of the wall** button; **Add photos...** (camera or files); thumbnails; a Dictate button.

On elevations: each mark prints on the interior elevation of the wall it is pinned to as a small camera symbol at its height with the label `PHOTO <date>`, in purple.

Wall photos tab: a warning box **Walls with new wiring or piping and no photo yet** ("Take these before the insulation and drywall go up.") listing room · wall length · level for every wall that has new electrical devices or pipe runs on it and no photo mark; then a card per mark: room, stage pill, date, **Show on the plan**, the note, and photo thumbnails ("No photos added to this mark yet.").

4 Homeowner binder



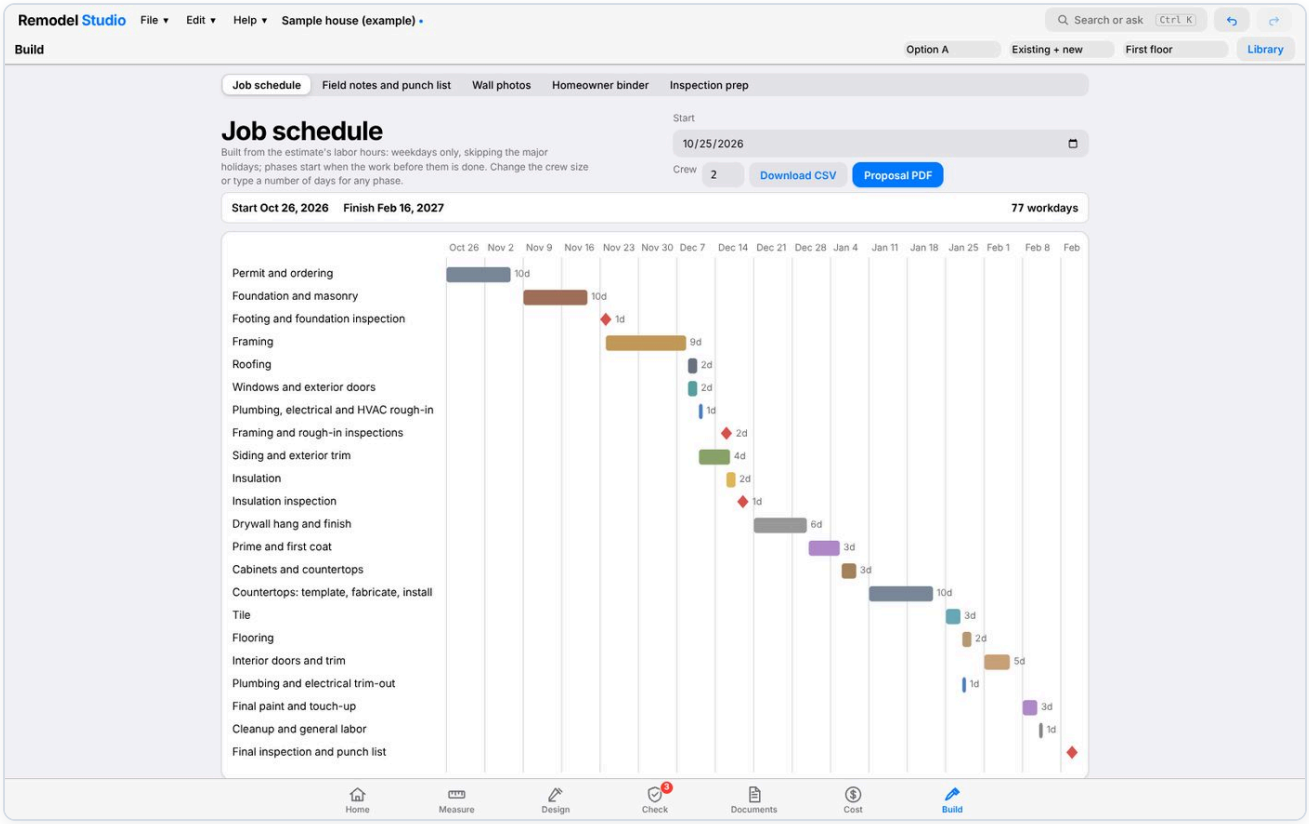
Build › Homeowner binder.

One PDF for the owner at the end of the job. Most of it comes from the model; the rest you fill in here. Also reachable from File › Export › Homeowner binder (PDF)... as a dialog.

Make the binder (PDF) — `<project> homeowner binder.pdf` (progress shows each section as it is built). Sections, in order: 1. **Cover**: "Homeowner binder", project name, address, Prepared for client, Prepared by designer, date, and a contents line. 2. **Where things are**: a table of the items you filled in below (only filled items; otherwise a prompt to fill them in). 3. **Paint and finishes by room**: Room, Floor, Walls (finish), Wall color (paint brand + color, or the screen color name), Trim and ceiling, Ceiling type. 4. **Cabinets** (when any): maker · door style · finish · hardware and finish; table of #, Code, Size W x H x D, Room; a care note. 5. **Appliances and warranties** (when any): Appliance, Model, Installed (from the order tracker's installed or delivered date), Warranty to (installed date + years, or "N yr"), Manual or spec page (link). 6. **Plumbing fixtures** (when any): Fixture, Room, Notes; a note about shutoff valves and aerators. 7. **Electrical panel (N A main)**: Space, Circuit, Breaker, Protection from the panel schedule. 8. **Lighting**: Mark, Fixture, Qty, Color (K), Lumens, Rooms; a note to match Kelvin when replacing lamps. 9. **What is inside the walls**: each wall photo mark (room · date · stage: note) with one photo. 10. **Who did the work**: your contacts text plus "Suppliers: ..." from the vendors in the order tracker. 11. **Maintenance calendar**: Monthly (test smoke/CO alarms and GFCIs; check the HVAC filter) · Every 1 to 3 months (replace the filter) · Every 6 months (range hood filter, under sinks, gutters) · Yearly (alarm batteries, flush water heater, dryer vent, refrigerator coils, re-caulk, exterior caulk and paint, HVAC service) · Every 10 years (replace smoke alarms).

Fields you fill in - **Where things are**: Main water shutoff · Gas shutoff · Electrical panel location · HVAC filter size and location · Water heater (make, model, size) · HVAC equipment (make, model) · Attic access · Crawlspace access · Other notes for the owner. (Text boxes; saved when you leave the box.) - **Appliance warranties** (for each library appliance): years of coverage — 1 (default), 2, 3, 5 or 10 — counted from the installed or delivered date in Cost › Selections and orders. - **Who did the work**: a free text box ("Electrician: name, phone, license. Plumber: ... Cabinets: ...").

5 Inspection prep



Build › Inspection prep.

What the inspector checks at each visit, with this job's own sizes filled in. Tick items as you verify them (saved with the project, with the date); **Print** prints the page. Items marked ⚠ are open errors from the Check page that match the phase.

Phases and how their items are generated: - **Footing and foundation** (shown unless the house is on a slab with no new exterior walls): footings below frost depth and the footing type for the foundation chosen (R403.1); rebar and anchor bolt layout (R403.1.6); vents or conditioned crawl and 18 × 24 access (R408); grade sloped 6" in 10 ft (R401.3); plus ⚠ footing/foundation/crawl errors. - **Framing**: headers per the header schedule (first six marks, R602.7); floor joist size and spacing with rim and blocking, holes and notches (R502.8); fire blocking and draft stopping (R302.11); hurricane ties and truss layout (R802.11); stair geometry (R311.7); bedroom egress windows (R310); plus ⚠ header/joist/rafter/stair/egress/span/beam errors. - **Rough plumbing, electrical and mechanical**: DWV sizes and the building drain size from the riser (P3005); water or air test (P2503.5); nail plates (P2603.2.1, E3802.1); receptacle spacing, GFCI/AFCI, interconnected smoke and CO alarms (E3902, R314, R315); bath fans and makeup air (M1503.6); dryer duct (M1502); plus ⚠ outlet/GFCI/AFCI/drain/vent/duct/smoke errors. - **Insulation**: the wall cavity and continuous R, ceiling R and floor R from the energy settings (N1102.1.2); baffles and air barrier contact (N1102.4.1.1); air sealing before drywall; vapor retarder class (R702.7). - **Final**: handrails

and guards (R311.7.8, R312); tempered glass locations (R308.4); alarms, GFCI/AFCI test, panel labeled (E3706.2); water heater T&P and expansion tank (P2804); address numbers and landings (R319, R311.3); each library appliance "installed with its listed clearances" (up to four); plus ⚠ landing/threshold/handrail/guard/tempered errors.

Each phase header shows "N of M" ticked; a ticked item shows the date it was verified.

PART

Site and settings

Site and landscape tools, and every project setting in one place.

CHAPTER 22

Site and outdoors

Site tools are on the Design ribbon under **Site › Site and landscape**. Site items sit on the ground, draw under the plan, and show on the site plan sheet; they are left off the floor-plan sheets. Each site item has a status (new or existing); existing items are not counted in the builder's list.

1 Site plan tools

Area tools (click the corners, click the first point to close; at least three points): **Driveway**, **Patio**, **Planting bed**, **Lawn area**, **Property line**. Line tools (click points along the line; Enter or Esc to finish): **Walkway**, **Fence**, **Retaining wall**. Point tools (click to place; keep clicking to add more): **Tree**, **Shrub**, **Plant**, **Play equipment**. Area tool: **Play area and surfacing**. All snap to the grid.

Panel fields by item (the first field is **Surface**, **Style** for fences, or **Kind** for trees and shrubs):

Item	Choices (default first)	Other fields
Driveway	Concrete · Asphalt · Gravel · Pavers	Color; Area (sq ft)
Patio	Pavers · Concrete · Flagstone · Brick	Color; Area
Walkway	Concrete · Pavers · Stepping stones · Gravel · Brick	Color; **Width** (36"); Length
Planting bed	Mulch · Pine straw · River rock	Color; Area; **Mass planting** (None, or a groundcover/perennial/grass/shrub species) with **Spacing on center** (default 75% of the plant's spread, at least 6") and "Plants to fill it"; **Edging** (Steel (default) · No edging · Aluminum · Plastic · Brick soldier course · Natural stone); **Landscape fabric under the mulch** (on); **Irrigation** (No irrigation · Spray heads · Rotors · Drip line)
Lawn area	Sod · Seed	Color; Area; **Grass** (Bermuda (Tifway 419) · St. Augustine · Zoysia · Centipede · Tall fescue); **Irrigation**
Fence	Wood privacy · Picket · Chain link · Split rail · Ornamental iron · Vinyl privacy	Color; **Height** (72"); Length. Note over 72": many cities limit back-yard fences to 6 ft and front-yard fences to 3 or 4 ft.
Retaining wall	Segmental block · Stone · Timber · Concrete	Color; **Height above grade** (30"); **Thickness** (12"); Length. Note over 48": needs an engineered design (IRC R404.4).
Property line	(dashed lot line)	Area. Property lines draw as a long-dash lot line on the site plan and are not built in 3D.

Item	Choices (default first)	Other fields
Tree	Shade tree (oak, maple) · Evergreen (pine, cedar) · Ornamental (crepe myrtle) · Palm	Color; **Canopy spread** (20 ft; bare numbers are feet); **Height** (auto)
Shrub	Rounded shrub · Boxwood · Ornamental grass	Color; **Spread** (42")

Setbacks and the lot are entered on the Documents › Sheets panel: **Lot for the site plan** (Lot width, Lot depth, Front setback, Left side setback — "Leave the width at 0 to show just the house. Setbacks are measured from the top and left edges of the plan to the lot line.") and **Zoning and permit packet** (Permit office, Parcel number, Front / Side / Rear setback (min), Max height, Max lot coverage %, Max impervious % — each defaults to Not set; a summary line gives heated area, footprint, lot area, coverage and impervious %; failing rules appear as Zoning errors in Check; **Permit packet (PDF)** prints the cover letter, project data, areas, site data, zoning compliance and sheet index).

Decks, porches, steps, ramps, columns and railings are Exterior tools (Architecture › Exterior) and are covered elsewhere; the Foundation panel has **Add steps at every outside door**.

2 Trees and plants library (Landscaping)

The **Plant** tool places the species chosen in the Library under **Landscaping** (sub-groups: Shade trees, Ornamental trees, Evergreen trees and screens, Palms, Shrubs, Ornamental grasses, Groundcovers, Perennials and color). 56 species for the Gulf South (USDA zones 6–10), each with common and botanical name, mature height and spread, zone range, sun, water, growth rate, native flag, evergreen/deciduous, form, foliage and flower color and bloom season. Examples: Live oak, Willow oak, Shumard oak, Red maple, Southern magnolia, River birch, Bald cypress, Lacebark elm, Ginkgo; Crape myrtles Natchez/Muskogee/Dynamite, Flowering dogwood, Eastern redbud, Japanese maple, Little Gem magnolia, Chaste tree, tree-form Yaupon; Loblolly pine, Eastern red cedar, Leyland cypress, Nellie R. Stevens holly, Green Giant arborvitae, Wax myrtle; Sabal and Windmill palms; Encore and Formosa azaleas, Wintergreen boxwood, Indian hawthorn, Loropetalum, Gardenia, Bigleaf and Oakleaf hydrangea, Dwarf yaupon, Glossy abelia, Knock Out rose, Distylium, Dwarf Burford holly, Sasanqua camellia, Sweet viburnum, Nandina Firepower; Pink muhly, Fountain grass Hameln, Maiden grass; Liriope Big Blue, Mondo grass, Asiatic jasmine, Blue Rug juniper; Daylily Stella de Oro, Hosta, Purple coneflower, Black-eyed Susan, Lantana New Gold, Salvia Mystic Spires, Seasonal color (annuals).

Plant panel: picture; **Plant** (grouped select); a fact line (botanical name, mature size, growth, sun, water, evergreen/deciduous, flowers, native, zones); a warning when the species is not rated for the project's zone; **Size to buy** (4" pot · 1 gallon · 3 gallon · 7 gallon · 15 gallon · 30 gallon · 2" caliper B&B · 3" caliper B&B — each with a size at planting and a planning price); **Mature spread**; **Mature height**; **Show size** (At planting · In 5 years · Mature) — this is the project-wide landscape display setting; "Shown now" readout; **Foliage color**.

Plants draw as canopy symbols with a two-letter key (first letters of the common name), a flower dot when the species blooms, and in 3D as trunk-and-canopy, cone, palm, fountain or mat forms

container size; in 5 years grows 30/50/75% of the way for slow/medium/fast growers; mature uses the full size.

Landscape checks (Check page, category Landscape): not hardy in the project zone; large tree (30 ft or taller) within 10 ft of the house; shrub that will grow into the wall (center closer than half its spread plus 6"); plants of the same kind crowded (closer than 60% of their combined spread).

3 Playground equipment (GameTime catalog)

The **Play equipment** tool places the item chosen in the Library under **Playground** (sub-groups by brand and category; **Safety surfacing** entries start the Play area tool with that surface). The catalog holds 7 generic items (2-bay belt swing frame, backyard wood swing set with fort and slide, composite 3-deck structure, freestanding slide, spring rider, 6 × 6 sandbox, 6 ft park bench) and 128 GameTime products compiled from gametime.com product pages (October 2026) across the lines PrimeTime, PowerScape, GTEvents, Xscape, Inclusive Play, Challenge Course, GT Jams, GTfit, THRIVE, VistaRope, KidNetix, Modern City, Altus Towers, Sensory Wave, Sensory Dome, TotStuff, Tuffclad, Duraclad, TuffCrete Fun Forms, Nature climbers, Shade (QuickRelease), Swing Shade, Spring riders, Site furnishings and Freestanding packages. Categories: Play structures, Swings, Slides, Climbers, Spinners, Rockers and spring riders, Freestanding play, Musical and sensory, Inclusive play, Fitness, Shade, Site furnishings, Other. Each item carries model number, name, line, age group (2-5, 5-12, 2-12), footprint, height, fall height, published use zone, capacity, accessibility (ground-level / transfer / inclusive), accessible-component counts, product URL and notes. Footprints marked "estimated" were taken from the published use zone.

Play equipment panel: picture; **Item** (grouped select); a fact line (brand · line · ages · users · accessibility, notes, **Product page** link, and warnings "Size is estimated..." or "Fall height equals the deck it attaches to"); **Width**, **Depth**, **Fall height**, **Rotation** (degrees); **Colors** (schemes: Primary (default) · Earthtone · Coastal · Woodland · Bold, plus GameTime palettes: Metal Blue/Plastic Yellow/Plastic Red; Metal Brown/Lumber Sand/Plastic Green; Metal Ocean/Metal Vanilla/Plastic Azure; Metal Dark Green/Timber Cedar/Plastic Spring Green; Metal Royal Purple/Plastic Orange/Metal Spring Green); for swing frames **Seats** (Belt seats (standard), or GameTime seats: 1481 Belt Swing Seat, 5293 Tot Seat 8', 4952 Nexgen Swing Seat Package, 5295 Expression Swing 8', 5375/5376 Zero-G, 5299 Expression Swing with Adaptive, 5297 Expression Swing Tandem); for structures **Elevated play events**, **Ground-level play events**, **Elevated reached by ramp or transfer**, **Ground-level types**; the **Use zone** size; **Show use zones on the plan** (on, project-wide). Use zones: 6 ft around most items, twice the pivot height in front of and behind swings, an exit zone for slides, none for benches and tables.

Play area panel: **Surfacing** (Engineered wood fiber (EWF) · Wood chips · Wood mulch (non-CCA) · Shredded rubber mulch · Sand · Pea gravel · Poured-in-place rubber · Rubber tiles · Synthetic turf with pad); for loose fill **Installed depth** (12" default; 6" for rubber mulch) with the note that loose fill packs down about 25%; for unitary surfaces **Tested critical fall height** (8 ft default) and **Thickness**; **Border** (Recycled plastic timbers (default for loose fill) · Plastic border · Concrete curb · None (default for unitary) · GameTime PlayCurb 12" (#4862) · GameTime PlayCurb 8" (#4850));

a GameTime surfacing option link (EWF, GT Impax PIP, tiles, turf, loose-fill rubber); **Area**; **Highest fall height inside**; **Surfacing protects to** (red when less than the highest fall height; "not enough depth" when loose fill is under 9" compressed); **Wheelchair accessible surface** (yes for EWF, PIP, tiles and turf per ASTM F1951); **Color**.

Playground checks (Check page): use zone not covered by safety surfacing (CPSC 5.2); surfacing too thin for the fall height (ASTM F1292); swing use zones overlapping other use zones (CPSC 5.3.8); equipment closer than 6 ft (9 ft when a play surface is over 30") (CPSC 5.1); play area surface not wheelchair accessible (ADA 1008.2.6); no walkway reaching the play area (ADA 240.1 / 1008.2); too few ground-level components for the number of elevated ones (ADA Table 240.2.1.2); elevated components not reached by ramp or transfer (ADA 240.2.2); 2-5 and 5-12 equipment mixed in one area (CPSC 2.3).

4 Drainage, swales and downspouts

Drawn automatically on the site plan sheet from the house outline and the lot slope: - A **drainage arrow** away from each side of the house, labeled **5% MIN.**; on a side where the lot is uphill from the house the label reads **SWALE** and a dashed swale line with arrows at both ends is drawn around that side. - **Downspouts** (circle marked **DS** with a dashed splash line) at every outside corner of the house when any roof has gutters. - **Walkway and driveway slopes** as percentages along each run (red over 8.33%). - **Spot grades** at each house corner (finished floor = 100.00') and **FF 100.00'** at the center; on sloped lots a dashed slope arrow labeled **SLOPE N% DOWN**.

Related Check items: "Ground slopes toward the house" (IRC R401.3, with a swale recommendation); "Walkway too steep" over 8.33% (error with ADA checks on, otherwise warning) and "Walkway is a ramp" over 5% (ADA 403.3 / 405); "Steep driveway" over 15%; and "Low window sill high above the ground" (IRC R312.2.1) when an operable window sill under 24" is more than 72" above the ground outside.

5 Grade and terrain

Set in the **Project › Foundation** panel (select nothing in the Design view to see the Project panel): - **Type**: Crawlspace: raised floor on a foundation wall (default) · Piers and beams: raised floor on piers · Stem wall with a filled slab · Slab on grade · Basement. - **Finished floor above the ground** (on a sloped lot: "Floor above the ground (high side)") — a new project starts at 24". - **Ground falls across the house** — total drop in inches across the house footprint (blank = level lot). - **Downhill toward** (only when a fall is set): North (top of the plan) · Northeast · East (right) · Southeast · South (bottom of the plan) (default) · Southwest · West (left) · Northwest. A note gives the floor height on the high and low sides and the slope percentage; the foundation, piers, steps, decks, site items, plants, play equipment and the 3D ground follow the slope, and the foundation wall steps down with the ground. - Also in this panel: **Exposed foundation finish** (Brick (default) · Painted block (CMU) · Parged or stuccoed · Stone veneer · Siding carried down), **Skirting between piers** (Lattice (default) · Vinyl or board skirting · Brick curtain wall · Open), **Pier size** (16"), **Piers every** (96"), **Foundation color**, **Foundation vents (R408.1)** (on), **Crawlspace access**

door (R408.4) (on), **Footings**: Width (16"), Thickness (8"), Bottom below grade (12"); "Ground to bottom of floor framing" readout; **Add steps at every outside door**.

6 Landscape sheet (L1.0 Landscape and play area plan)

Added to the plan set after sheet A0.2 whenever the project has plants, play equipment or play areas. It shows: the site items (driveways, walks, patios, beds, lawns, fences, retaining walls, lot line) with surface labels; the house footprint shaded and labeled HOUSE; play areas with "PLAY AREA · [N" DEEP]"; play equipment footprints with model numbers and dashed use zones (when use zones are on); every plant as a canopy circle (sized at planting, 5 years or mature per the landscape setting) with a two-letter key (numbered when two species share letters). On the right: a **PLANT SCHEDULE** (KEY, COMMON / BOTANICAL, QTY, SIZE), a **PLAYGROUND SCHEDULE** (MODEL, ITEM, QTY, FALL HT) and **NOTES** (what the circles show; call 811, keep trees 10 ft from the foundation; use-zone and surfacing standards when relevant). Uses the site-plan scales.

CHAPTER 23

Settings and project info

Settings are saved inside the project file. They are edited in several places; this section lists every one, its default and where to change it.

1 Project info (title block and documents)

Edited in **Documents › Sheets** (Inspector, **Title block**) and **Zoning and permit packet**; the project **Name** is in the Project panel (Design, nothing selected) or File › Rename project....

Field	Default	Where	Used in
Project name	"Untitled house"	Project panel; File › Rename project...	everywhere
Address	blank	Sheets › Title block	title blocks, estimate, proposal, POs, binder, permit letter, dashboard
Prepared for (client)	blank	Sheets › Title block	title blocks, proposal, binder, permit letter, dashboard
Drawn by (designer)	blank	Sheets › Title block	title blocks, proposal, POs ("Ordered by"), bid requests ("Contact"), binder, permit letter; also saved in My standards
Revision	blank	Sheets › Title block	title block
Date	blank (today)	Sheets › Title block	title block
Permit office (jurisdiction)	blank	Sheets › Zoning and permit packet	permit cover letter
Parcel number	blank	Sheets › Zoning and permit packet	permit packet
Revision history	from change orders	Sheets › Revisions	revision table and clouds

There is no separate permit-number field; the permit packet records the permit office and parcel.

2 Project panel (Design view with nothing selected)

Setting	Default	Choices / notes
Snap grid	6"	1" · 3" · 6" · 1'
Building code	House: 2021 IRC	or Commercial or city building: 2021 IBC
Occupancy group (IBC only)	B	IBC occupancy list
Fire sprinklers throughout (IBC only)	off	
Accessible design (2010 ADA checks) (IRC only)	on	
NKBA kitchen and bath planning guidelines	on	

Setting	Default	Choices / notes
Cooks at once (NKBA)	One (42" aisles)	or Two or more (48" aisles)
Use Access Standard values (NKBA)	off	
Foundation panel	see Grade and terrain above	
Landscape: USDA zone	8a	6a · 6b · 7a · 7b · 8a · 8b · 9a · 9b · 10a ("Forest, Mississippi is in zone 8a")
Landscape: Show plants	Mature	At planting · In 5 years · Mature
Landscape: Show flowers in 3D	on	
Trim (whole house)	program defaults per slot	Baseboard, Door and window casing, Crown, Chair or picture rail, Base shoe, Wainscot, Window stool — library profiles; **Trim color**
Automatic roof settings	Hip, 6/12, 12" overhang	Style (Gable · Hip · Dutch · Gambrel), Pitch, Overhang
Levels	one level	name, floor elevation, wall height; framing per level: Foundation (first level), Floor joists, Spacing (12 · 16 · 19.2 · 24), Joists run, Girders every, Lumber species, Ceiling joists (top level), Attic storage
Code rule values (Check page › rule list)	code values	any rule can be changed to a local amendment; changed values are saved with the project

3 Display popover (plan view: "Display" button, or type `osnap` in the command line)

- **Show in the plan** (all on by default): Room names and areas · Wall lengths · Fixtures and cabinets · Electrical · Roof lines · Dimensions · Notes · Drafting lines · Site and landscape · Decks, porches and steps · Code-check markers · Floor below (ghosted) · Cabinet numbers · Switch legs (off by default).
- **Snaps**: Endpoint (on) · Midpoint (on) · Intersection (on) · Center (on) · Perpendicular (on) · Nearest (off) · Grid (on). The command line accepts `osnap on`, `osnap off`, `osnap all` or a list such as `osnap end mid`.
- **Drafting layers**: current layer (default L-DRAFT), color, On/Off, Lock, Print, **New layer**.
- Command-line settings: `AR n` sets the Array count (default 3); `O n` sets the Offset distance (default 6").

4 Units and precision

Lengths are shown in feet and inches to 1/16" everywhere; there is no unit or precision setting to change. Type metric values and they are converted. Bare numbers are inches except in wall-length and level boxes (feet) and boxes labeled in feet (tree spread and height, girder spacing, hip girder setback).

5 Estimate settings

On **Cost › Estimate and labor › Settings** (and Labor by trade, Risk by trade): self-performed labor (off), Overhead and profit 15%, Contingency 15%, Sales tax 7%, Permit fee $0, Dumpsters (automatic), Cost per dumpster $450, Other costs (none), labor rates per trade (table above), extra contingency per trade (none), list alternates in the proposal (off). **Waste %** (10) is on the Builder's list tab. **Cabinet order** settings (multiplier 0.45, margin 35%, freight 3%, install $85 per box, tax 7%) are on the Cabinet order tab.

6 Energy settings (Design › Systems › Energy code and Heating and cooling)

Setting	Default	Notes
Climate zone	3A (Mississippi, Alabama, Georgia)	1A · 2A · 3A · 4A · 5A; changing it resets the outdoor design temperatures and the prescriptive R and U values
Wall cavity R	20 (zone 3A)	
Wall continuous R (foam)	0	
Ceiling R	49	
Floor R	19	
Slab edge R	10	
Window U-factor	0.30	
Window SHGC	0.25	
Skylight U-factor	0.55	
Door U-factor	0.20	
Air leakage (ACH50)	3	
Ducts	In a vented attic	or In a vented crawlspace · Inside the conditioned space
Include energy code items in the code check list	off	
Winter outdoor / Summer outdoor °F	23 / 95 (zone 3A)	
Indoor winter / Indoor summer °F	70 / 75	
Moisture difference (grains)	55	

Setting	Default	Notes
People	auto	

7 Other settings saved with the project (and where they are set)

Setting	Default	Where
Schedule: start, crew, per-phase crews and days, draws, lead times	today, 2, none, standard draws, typical leads	Build › Job schedule
Shopping check-offs	none	Cost › Shopping list
Inspection ticks	none	Build › Inspection prep
Reviewed issues	none	Check page ("Reviewed, leave as is")
Next PO number	1	advances when you print a PO and mark items ordered
Selections (brands, paint brand)	none	Cost › Selections and orders
Door hardware finish / brand	Satin nickel / Any brand	Cost › Selections (Door hardware, whole house)
Cabinet maker, line, door, finish, price book	Generic	any cabinet's panel or the Cabinet maker panel
Countertop material, edge, thickness, overhang, splash	quartz, eased, 1-1/4", 1", 4"	Countertop panel; Countertop slabs dialog adds slab size, kerf 1/4" and edge allowance 1"
Electrical load assumptions (HVAC, strips, range, dryer, water heater, EV, well)	heat pump, 8 kW strips, electric range/dryer/water heater, no EV, no well	Design › Systems › Electrical service
Lighting scenes	none	Lighting scenes panel (Render studio uses them)
Render options (smooth, auto, camera, size, quality, sky, lights)	on, on, screen, HD, 256, day, on	Render studio
Playground: show use zones, default color scheme	on, Primary	Play equipment panel
Revision clouds for change orders	on	internal; turned off only by project text
NKBA drawing standard for cabinet plans	on	NKBA drawings
Hide field pins	off	internal
Zoning limits	not set	Documents › Sheets › Zoning and permit packet
Lot size and setbacks for the site plan	0	Documents › Sheets › Lot for the site plan
Sheet paper and scale	Tabloid, largest that fits	Documents › Sheets › Paper
Saved view setups	none	Project browser › Saved view setups

8 My standards (File › My standards… or Help › My standards…)

Defaults that every new project starts with (kept in this browser or desktop install, not in the project). **Save this project's settings as my standards** stores: cabinet brand/line/door/finish, trim profiles, estimate rates, markup, contingency, tax, dumpster cost and self-labor, the first level's framing (joists, spacing, ceiling joists, foundation), energy settings, NKBA settings, ADA on/off, building code, NKBA drawing standard, waste %, grid, hardware finish and brand, product selections and the designer name, with the save date. The dialog summarizes them (Cabinets, Trim, Framing, Estimate, Code, Energy, Saved). **Apply my standards to this project** copies them into the open project (cabinets without their own style follow the new defaults). **Clear** removes them. The sample house, lesson rooms and picture previews do not receive standards.

9 Appearance (Help › Appearance (light or dark)…)

Light or dark colors for the whole program including the plan and 3D views; "Follow Windows" uses the Windows Personalization › Colors setting.

10 Backups, versions and autosave

- **File › Backups…** (desktop): every save keeps a dated copy in `Documents\Remodel Studio\Backups`, and an autosave copy is written every 5 minutes while there are unsaved changes (only when the project has walls). The newest 25 saves and 40 autosaves per project are kept. The dialog lists copies by project with time, Saved/Autosave, size and an **Open** button that opens the copy as an unsaved project named "… (restored)"; **Open the backups folder**. In the web version the dialog points to File › Versions.
- **File › Versions…**: named snapshots stored in the browser with the project's assets; save one with a name (default "Version "), compare its summary with the current model, and restore it (Undo brings back what you had).
- **File › Download a copy (.json)** (web) and **Show file in folder** (desktop); **Copy or paste project text…** moves a project as text.

11 Performance

Whole-project calculations (code checks, countertop geometry, cabinet numbering, structure analysis, room walls, header schedule) are cached per model version, so switching tabs and reopening panels does not recompute them until the model changes. There is nothing to configure; this is why the Dashboard, Check page and Cost tabs stay responsive on large projects.

PART

Appendices

File formats, troubleshooting, where to get help, the glossary and the index. Every keyboard shortcut is listed in the chapter on the window chrome.

CHAPTER 24

File formats

Format	Direction	Where	Notes
`.rsp`	Open, save	File menu (Windows program)	A Remodel Studio project: the whole model, options, revisions, settings, selections, orders, field notes, bids and schedule in one file. Older `.json` projects open too.
`.json`	Open, save	File menu (web version), *Copy or paste project text*	The same project as plain JSON text.
`.pdf`	Export	Documents, Cost, Build, Render	Plan set, permit package, presentation, finish board, estimate proposal, change orders, bid requests, purchase orders, cabinet order, inspection prep, homeowner binder.
`.dxf`	Import, export	File › Import › DWG or DXF; Documents › Print plans and DXF	Export writes layers for walls, openings, rooms, fixtures, dimensions, notes and drafting. Import reads lines, polylines, arcs, circles and text and can turn walls into walls.
`.dwg`	Import	File › Import › DWG or DXF	AutoCAD drawings (converted on import).
`.ifc`	Export	File › Export › IFC	IFC 2×3 walls, openings, slabs, roofs, spaces and items for Revit, Archicad and IFC viewers.
`.obj`, `.glb`, `.gltf`, `.dae`, `.stl`	Import	File › Import › 3D model	Furniture, fixtures and site models placed as items.
`.usdz`, `.glb`, `.stl`	Export	File › Export › AR view and 3D model	The model for AR on an iPhone or iPad (USDZ), for other 3D programs (GLB) or for printing (STL).
`.csv`	Export	Cost (builder's list, estimate, cut list), Schedules	Opens in Excel.
`.xlsx`, `.csv`, `.pdf`, `.json`	Import	File › Import › Vendor catalog; Store prices	Vendor catalogs and price books; store price lists.
`.ics`	Export	Build › Job schedule	Calendar file of the schedule phases and the maintenance reminders for Outlook, Google Calendar or the iPhone.
`.png`, `.jpg`	Export, import	Render studio, Views, Measure	Renders and saved views out; photos, scans and plan images in.
`.pdf` (plans)	Import	Measure › Scans and plan images	A PDF plan placed as an underlay.
`.ply`	Import	Measure › Scans and plan images, Walls from a scan	Point clouds from COLMAP, Polycam, 3D Scanner App and other LiDAR apps; large clouds are thinned to about 150,000 points.

CHAPTER 25

Troubleshooting

The 3D view is blank or very slow. The graphics driver may not support WebGL 2. Update the display driver from the computer maker's site. On a laptop with two graphics chips, set Windows *Graphics settings* to use the high-performance chip for Remodel Studio. Use *Cut at level* to hide the floors above and *One view* while drawing; the model only rebuilds after you stop editing.

A tool key (W, R, D…) does nothing. The plan does not have the keyboard focus; click once on the plan (on empty space with Select) and try again. Keys are ignored while the cursor is in a text box.

My walls will not join at the corner. Snapping is off or the ends are not on the same grid point. Turn **Snap** on in the Display popover, or drag one wall's end dot onto the other wall's end; the endpoint snap marker (■) shows when it will join.

The room tool does not trace the room. There is a gap in the walls. Zoom in at each corner and drag the wall end dots together until the endpoint snap (■) shows; or click the corners yourself with the Room tool and close on the first point. Doors and windows count as wall.

The Check tab shows errors I disagree with. Select the issue on the Check page and tick **Reviewed, leave as is**; it leaves the list but still prints on the permit checklist with your note. Check that the **Building code**, **ADA** and **NKBA** switches in the Project panel match the job.

Prices show "no price". Those items have no price in the built-in list, your edits, a store price list or a vendor catalog. Type a price in the **Price** column of the builder's list (kept with the project), import a store price list or a vendor catalog, or set your usual prices in **My standards**.

The vendor catalog import found nothing. The file's first row must contain column names the importer recognizes (see *Vendor catalog import*). Download the template from the import dialog, paste the maker's data into it and import that. PDF price books work when the text is selectable; a scanned PDF must be run through OCR first.

A PDF will not download in the web version. Pop-ups or downloads are blocked for the page. Allow downloads for the site in the browser's address-bar icon, or use the Windows program.

The project is gone from the browser list. The browser's site data was cleared or a different browser profile is in use. Open the `.json` copy you saved, or restore a **Version** if one exists. Keep a file copy of every job with **Save as… › Save to a file on this computer**.

The sample house opens every time. The last project is remembered only after it is saved once. Save your project with **Ctrl+S**.

Dark mode does not follow Windows. Help › Appearance is set to *Light* or *Dark*; choose **Follow Windows**.

Voice notes do not record. The microphone is blocked. In Windows Settings › Privacy › Microphone allow desktop apps; in the browser allow the microphone for the page.

The photo assistant says it is not available. The assistant needs an AI service. In the Windows program it uses a local model when Ollama is installed and running (see *Command palette*); otherwise the photo tools that do not need AI (measuring, color picking, underlays) still work.

Where are my backups? File › Backups… lists them with dates; *Restore* opens one as a new unsaved project so nothing is overwritten.

CHAPTER 26

Getting more help

- **Help › Lessons** walks through drawing, cabinets, checks and documents on a practice room.
- **Help › Keyboard shortcuts** lists every key.
- **Ctrl+K** and type a question in plain words: *how many bedrooms*, *square feet of the kitchen*, *what is the panel size*.
- The cover of every PDF the program prints carries the program version; quote it when you report a problem.

CHAPTER 27

Glossary

Terms used in the program and in this manual. Construction and code terms are defined the way the program uses them.

ADA 2010 Standards — The 2010 ADA Standards for Accessible Design. The program checks door clear widths, turning space, ramp slopes, reach ranges and knee space when *ADA* is on in the Project panel (always on for IBC projects).

AFCI — Arc-fault circuit interrupter; a breaker that trips on arcing. Required for most 120-volt living-area circuits (IRC E3902.17).

Aging in place — Design choices that let someone stay in the house as they age: zero-step entries, 36" doors, lever handles, blocking for grab bars, a bedroom and full bath on the entry level. The Check page lists them as a separate review.

Appliance — A catalog item with a brand, a model, a published size, electrical and gas requirements and clearances. Appliances are placed like fixtures and listed on the appliance schedule and the orders list.

Backer board — Cement board (CBU) under tile on floors and walls. Counted on the builder's list and nested on the cut plan.

Backsplash (splash) — The vertical finish behind a countertop: a 4" strip of the counter material, or tile up to the wall cabinets (full height).

Bearing wall — A wall that carries floor, ceiling or roof load from above. The program decides bearing from joist direction, girders and the walls above; *Remove or open a wall* sizes the beam that replaces one.

Bid — A subcontractor's price for a trade's scope. The Bids tab makes the request, records up to three quotes per trade and compares them with the estimate.

Binder (homeowner binder) — The PDF hand-over book: drawings, selections, appliance models, paint colors, warranties, maintenance reminders, shut-off locations and contacts.

Block — A saved group of items (a bath layout, a kitchen island with stools) that can be placed again with one click. Edit › Blocks.

Builder's list — The quantity take-off: every material the model needs, by category, with waste, unit, quantity and price.

Budget buckets — The estimate grouped into spending categories (framing, cabinets, electrical…) so receipts can be compared with the plan on *Budget vs actual*.

Cabinet nomenclature — The industry code for a cabinet: letter for type and numbers for size. *B36* is a 36" base; *W3030* a 30" wide, 30" high wall cabinet; *SB36* a sink base; *BBC39* a blind base corner; *T2484* a tall cabinet. The catalog and the schedules use these codes.

Change order — A priced record of a change made after the contract was locked, with a before/after list and a signature line.

Clash — A pipe or duct that runs through a joist, girder or stud where it must not (notch and bore limits of IRC R502.8 and R602.6). Listed on the Check page under *Framing*.

Clear opening — The usable width of a door with the door open 90°, measured from the face of the door to the stop. ADA needs 32"; the IRC needs one 32" exit door.

Climate zone — The IECC zone (1A–7) that sets insulation and window requirements on the Energy code tab. The default is 3A.

Color by — A view mode that tints the plan and the 3D model by a property: remodel status, item category, vendor or brand, order status, lead time, cost, issues, bearing walls, room use or floor finish.

Command palette — The **Search or ask** box (**Ctrl+K**): finds commands, tools, pages, levels and catalog items, places items by name and answers questions about the project.

Construction line — An endless guide line that never prints; used to line things up.

Countertop — The work surface on base cabinets. Material, edge, thickness, overhang and splash are job settings; each counter piece gets a number on the shop drawing and a seam suggestion at inside corners.

Crawlspace / slab / basement — The three foundation types in the Project panel. They change the floor framing, the insulation checks and the take-off.

Cut plan — Sheet goods (drywall, subfloor, sheathing, backer, plywood) nested on standard sheets with a diagram, yield and a CSV cut list.

Dashboard — The Home tab: checks, cost, schedule, orders, field items and recent projects on one page.

Demolish / Existing / New — The three remodel statuses of a wall, opening or fixture. *Existing* is what is there and stays; *Demolish* is removed; *New* is built. The layer control shows any combination; the take-off counts demolition and new work.

Design option — A complete alternate version of the model (Option A, B, C...) inside one project, for showing a client two layouts with their own prices.

Dimension string — A line of dimensions along a wall: overall, corners, door and window centers. *Automatic dimensions* draws them for every exterior wall.

Display (plan) — The popover on the plan that turns layers on and off (rooms, wall lengths, electrical, roof lines, dimensions, notes, drafting, site), sets object snaps and picks drafting layers.

Dormer — A roofed projection out of a sloped roof with a window in its face.

Drafting layer — A named layer for drafting lines, notes and hatches that can be hidden or printed separately.

Egress window — A bedroom window large enough to climb out of: 5.7 sq ft clear opening (5.0 at grade), 24" high, 20" wide, sill no more than 44" above the floor (IRC R310).

Elevation (drawing) — A straight-on view of a wall: exterior elevations (A4 sheets) show the outside; interior elevations and kitchen elevations (A5 sheets) show cabinet walls.

Estimate — Materials plus labor, tax, overhead and profit, contingency, permit and extras. A planning number, printed as a proposal.

Field pin — A note pinned to a spot on the plan from the site: an existing condition, a punch item, a question or a photo spot, with a category, a status, a due date, photos and a voice note.

Finish board — A one-page sample board of the job's cabinets, counters, floors, tile, paint and hardware, with swatches.

Fixture — Anything placed in a room: cabinets, plumbing fixtures, appliances, furniture, built-ins. Fixtures are listed on the fixture and cabinet schedules.

Fixture unit — Water supply (WSFU) and drainage (DFU) units that size pipes: each fixture type has a value from the IRC tables and the totals pick the pipe sizes.

Footprint wizard — Draws the outside walls of a house from overall dimensions and a shape (rectangle, L, T, U).

Framing plan — Sheets that show joists, girders, headers, beams and rafters with sizes and spacing (S sheets).

GFCI — Ground-fault circuit interrupter protection, required at counters, baths, garages, outdoors and laundry (IRC E3902).

Girder / beam — A horizontal member that carries joists or a wall. Sized by the span tables on the Structure tab.

Grade — The ground level around the house; set per side in the Project panel so foundation walls, steps and ramps are the right height.

Header — The beam over a door or window opening. The header schedule lists each opening's size and the header the span table gives it.

IBC 2021 — The International Building Code, used when the project is a commercial or city building. Adds occupancy groups, exit counts, corridor widths and accessible route checks.

IFC — Industry Foundation Classes, the open BIM file format. *File › Export › IFC* writes the model for Revit, Archicad or any IFC viewer.

Inspector — The right panel. It shows the properties of whatever is selected, or the Project panel when nothing is.

Inspection prep — Checklists for each inspection (footing, framing, rough-in, insulation, final) filled with the job's own sizes and the open check errors.

IRC 2021 — The International Residential Code, the default code for houses. Chapters R (building), N (energy), E (electrical), P (plumbing), M (mechanical) are checked.

Issue — One finding of the checks: a severity (error, warning, tip), a title, the rule, the detail and the item it points to.

Joist — A floor or ceiling framing member. The joist dialog sizes them from span, spacing, species and load.

KCD — The NKBA's Kitchen and Bath Certified Designer credential; the program's kitchen drawings follow the NKBA graphics standard a KCD uses.

Keynote — A numbered note tag on the plan tied to a project-wide keynote list printed on its own sheet.

Lead time — The weeks between ordering and delivery for windows, cabinets, counters and appliances; the schedule turns it into an order-by date.

Level — A story of the building (basement, first floor, second floor, attic) with its own height and plan.

Lighting scene — Named dimmer levels for each switch group (Cooking, Dinner, Night light, Everything on), set up on Systems › Lighting and used by the Render studio.

My standards — Your saved defaults: cabinet line and finish, hardware, counters, estimate rates, framing, energy, code settings. New projects start from them.

NKBA guidelines — The NKBA Kitchen and Bath Planning Guidelines (work aisle, landing areas, clearances, seating, bath spacing). Shown as tips and warnings when NKBA is on.

Object snap — Snapping to endpoints, midpoints, intersections, centers, perpendiculars and nearest points while drawing.

Option bar — The strip above the plan that changes with the tool (wall type, door type, fixture type, 45° snap).

Order-by date — The latest date a long-lead item can be ordered and still arrive before its schedule phase.

Panel schedule — The electrical panel's circuits: space, breaker size, wire, what it feeds. Built from the devices on the plan.

Permit package — The checklist and the set of sheets a building department asks for, printed together.

Photo match — Lining up the 3D model with a real photograph of the house so the proposed work is drawn over the photo.

Plan set — The numbered sheets (A0 cover through S, E, P, L sheets) printed as one PDF with a title block.

Presentation — A widescreen PDF for client meetings: cover, before and after, renders, views, options, plans, cost, code, structure, finish board, notes.

Project panel — The Inspector with nothing selected: project name, snap grid, building code, ADA and NKBA switches, foundation, framing, energy, cabinets, estimate and sheet settings.

Punch list — The list of items to finish or fix before the job is done; field pins with the *Punch* category.

Purchase order — The PDF sent to a supplier for a tracked order, listing the items, sizes and the delivery date.

Rafter / truss — Roof framing members; the roof wizard sizes rafters or lays out trusses.

Remodel layer — The *Existing + new* / *Existing only* / *New only* control that filters the plan and 3D by remodel status.

Revision — A named snapshot of the model. A later revision can be compared with it and the changed areas clouded on the sheets with a revision number.

Ribbon — The toolbar on the Design page with tabs Home, Architecture, Interiors, Structure, Systems, Site, Annotate, Draw, Modify and View. It can be pinned open, collapsed to its tabs or shown as icons only.

Riser diagram — The plumbing drawing of drains, vents and supplies by size (P sheets).

Schedule (document) — A table of items of one kind with their properties: doors, windows, rooms, cabinets, hardware, lighting, fixtures, finishes, appliances.

Schedule (job) — The construction timeline: phases with durations from the labor hours, crew size, start date and lead times.

Section cut — A line through the plan that becomes a building section sheet.

Selections — The brands and products chosen for each category of the job (windows, roofing, flooring, counters, fixtures, appliances, paint), stamped on the builder's list and the schedules.

Sheet — One page of the plan set, with a number (A1.1), a name, a scale and a title block.

Snap grid — The invisible grid free clicks round to (6" by default).

Span table — The IRC tables (R502.3, R802.4, R602.7) the program uses to size joists, rafters and headers.

Standards (My standards) — See *My standards*.

Tape wizard — The *Make walls › From tape measurements* wizard that turns the tape measurements of a room (wall lengths and the offsets of doors and windows) into walls and openings.

Takeoff — See *Builder's list*.

Tempered glass — Safety glass required near doors, tubs, showers and low windows (IRC R308.4).

Title block — The box on every sheet with the project name, address, client, designer, sheet number and date.

Underlay — A scan, a plan image or a PDF placed under the plan at a known scale to trace over.

Vendor catalog — A manufacturer's lines, door styles, finishes, colors and SKUs with prices, imported from Excel, CSV, PDF or JSON and shared by every project.

Wall photo — A photo taken before drywall, pinned to a wall so the pipes and wires inside it can be found later; shown on the interior elevations and in the binder.

Wainscot — Tile or beadboard on the lower part of a wall to a cap height.

Workflow — One of the seven tabs in the tab bar (Home, Measure, Design, Check, Documents, Cost, Build), each with its own pages.

WSFU — Water supply fixture units; see *Fixture unit*.

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